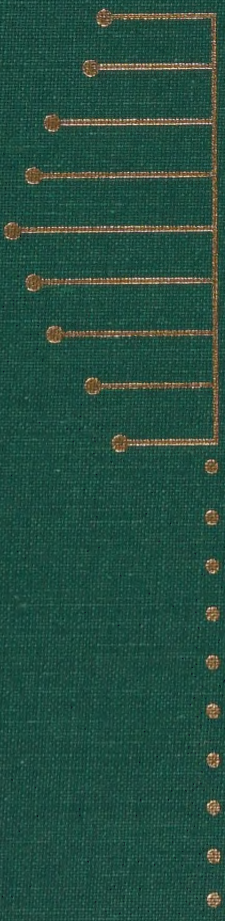
the CG-FFT Method

Application of Signal Processing Techniques to Electromagnetics



by

Manuel F. Cátedra
Rafael P. Torres
José Basterrechea
Emilio Gago

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To
Virginia, Begoña, and José (M.F.C.)
Africa and Isabel (R.P.T.)
Marta and Héctor (J.B.)
My sister Ada and my parents (E.G.)
And all our good friends

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Preface

In the last few years, many numerical methods have appeared that are capable of solving a variety of problems in applied electromagnetics. As a result, the range of problems that can be treated with such methods has broadened significantly. These advances have benefited applications in many areas, including microwaves, antennas, radar cross section (RCS) analyses, and studies of the biological effects of electromagnetic fields. In fact, in many of these areas numerical simulation has begun to replace the costly experimental studies required until now.

One of the most effective techniques that have appeared in recent years is the conjugate gradient method in combination with the fast Fourier transform (i.e., the CG-FFT method). Since the CG-FFT method has aroused a great deal of interest, there are many papers describing the fundamentals and different applications of the method. Indeed, an excellent monographic book on this subject, edited by Sarkar, has been published (*Application of Conjugate Gradient Method to Electromagnetics and Signal Analysis, PIER 5,* Elsevier, 1991.) This book presents a collection of valuable papers on CG-FFT techniques covering a wide array of applications. Among the topics treated in this monograph or in other papers are the following applications: scattering and RCS, transient electromagnetic problems, the inverse problem, analysis of antennas and microwave devices, frequency-selective surfaces (FSS), and optimum array processing. The use of the CG-FFT appears to be quite interesting in all of these applications.

In spite of the relatively ample amount of literature on the CG-FFT method available today, the authors felt that there was a gap. In our opinion, what was lacking was a book addressed to the reader who is not necessarily an expert on CG-FFT. This book should introduce him or her to the fundamentals and applications of the method, beginning with the basic concepts but leading on to sophisticated analyses of some engineering problems. The lack of such a text is no doubt partly due to the limited space usually available for technical papers, which allows researchers merely to report briefly the different advances of the CG-FFT method.

This book aims to fill this gap. Thus, special attention has been paid to the combination of the two theories that are the foundations of the method: electromagnetics and signal processing. Many readers are probably familiar with both theories separately. However, possibly due to inertia in teaching practice, no connection is usually made between these two theories. Consequently, to many potential users, CG-FFT may appear to be an intricate and cumbersome method. However, as will be seen throughout this book, the application of signal processing theories to electromagnetics can be a useful and quite straightforward approach. In fact, the authors believe that the CG-FFT method would be more widely used if many of these potential users realized how simple and beautiful this method can be. We will be proud if this book contributes in some way to promoting the application of signal processing techniques to electromagnetics.

The book is organized into four major parts. The first part, which comprises Chapters 1 to 4, presents a general introduction to the main CG-FFT topics. Chapter 1 outlines the main characteristics and history of the CG-FFT method, and situates it among other techniques in computational electromagnetics. Chapter 2 reviews the continuous and discrete Fourier transforms, emphasizing the main properties of the transforms that are used most frequently in the CG-FFT method. Chapter 3 employs a simple one-dimensional static case to summarize the three main ways of applying the CG-FFT method: discretization in the space domain of problems with finite-sized structures, discretization in the spectral domain in the case of infinite-sized problems with periodic geometry, and discretization in the spectral domain of finite-sized problems. Chapter 4 presents an overview of the different conjugate gradient algorithms.

The second part of the book, Chapters 5 to 8, describes in some detail various applications of the CG-FFT technique in which finite-sized problems are discretized in the space domain by obtaining equivalent periodic problems. Chapter 5 analyzes conducting plate problems. The analysis of three-dimensional bodies with arbitrary geometry and material composition is undertaken in Chapter 6. The case of finite-sized plane multilayer structures is treated in Chapter 7. Chapter 8 considers the analysis of metallic plates that can be conformed to surfaces with symmetry of revolution. This case is quite interesting because the CG-FFT method allows the analysis of incomplete bodies of revolution (BOR), such as those with apertures. Other methods, such as the moment method, have special formulations for the analysis of BORs, but they become inefficient when the BORs are incomplete.

The analysis of periodic problems is considered in the third part of the book, Chapters 9 and 10. This kind of problem is especially suitable for treatment with the CG-FFT method because the functions that characterize them have a discrete nature in the transformed domain. In this case, it is best to perform most of the discretization operation in the spectral domain. Chapter 9 begins this part with a review of Floquet's theory and the treatment of periodic conducting patches located in free space. Chapter 10 presents the more general case of conducting patches in a dielectric multilayer medium. In this case, the CG-FFT method shows an additional

advantage: it is possible to work directly in the spectral domain with analytical expressions for the Green's function. These functions can be found easily in the spectral domain, while it is nearly impossible to obtain closed analytical expressions in the spatial domain.

Chapters 11 and 12, the last part of the book, use a combination of the above techniques. Chapter 11 analyzes finite-sized conducting patches in dielectric multilayer structures by employing the technique developed in Chapter 10 for infinite-sized periodic problems. In order to do this without large aliasing errors, the Green's function is processed in the spectral domain to obtain an equivalent periodic problem. Finally, Chapter 12 outlines a method for the analysis of problems that are infinite-sized and periodic in one direction dimension, but finite sized in the others. In particular, volumetric problems are analyzed by using a combination of the two techniques mentioned for discretization; that is, discretizing in the spatial domain in the directions in which the bodies are finite-sized, and in the spectral domain for the other direction.

The authors would like to acknowledge the invaluable help provided by many people in the development and execution of this book. In the first place, we wish to recognize the assistance of our linguistic adviser, Professor Edward Dalley, whose help has been essential in expressing our ideas in English. We thank Dr. Robert C. Hansen for reviewing the book and for all his suggestions and comments. We also wish to acknowledge the contribution of the researchers and students of the Department of Señales, Sistemas y Radiocomunicaciones of the Polytechnic University of Madrid; particularly, Francisco J. Martín, Luis Nuño, Luis Eusebio, Fernando Ruiz, Juan C. Arredondo, Antonio Lopez, José A. Alcaraz, and Pedro Bravo deserve our thanks. Special mention must be made of Jesús G. Cuevas, who made excellent contributions to the development of most of the techniques presented in the book. The authors are grateful to Professor Jesús Sanchez-Miñana for encouraging us when our research on the CG-FFT technique was still a venture. Mr. Jose M. Desiré from the Dean's Office of Cantabria University deserves recognition for his support of research activities here. Finally, the authors wish to express their gratitude to all the members of the Systems and Radio Group at Cantabria University for their assistance and technical support in many areas, including the artwork for the book.

Manuel F. Cátedra
Rafael P. Torres
José Basterrechea
Emilio Gago
Santander, December 1994

Chapter 1

Introduction to the Conjugate Gradient Fast Fourier Transform Method

1.1 INTRODUCTION

The purpose of this chapter is to introduce the conjugate gradient fast Fourier transform (CG-FFT) method, and particularly to situate it within the field of applied computational electromagnetics (ACE). The CG-FFT method was conceived primarily for solving resonant-sized problems, as will be seen in the survey of ACE methods given in Section 1.2. Attempts to solve these problems by means of other methods led, after applying some discretization procedures, to ill-conditioned systems of linear equations. As we will point out in the brief outline of Section 1.3, the CG method was used to find accurate solutions for these ill-conditioned systems of equations. At approximately the same time, it was recognized that many ACE problems require the computation of potential functions given in terms of convolution integrals which can be calculated very efficiently by using the FFT. Section 1.4 will discuss how the CG-FFT emerged when it was realized that instead of computing and storing the matrices of these systems of equations, the convolution property of the FFT could be used to solve the operator equation of an ACE problem by means of a CG algorithm.

1.2 BRIEF SURVEY OF ELECTROMAGNETIC COMPUTATIONAL METHODS

In the last 30 years, significant advances have been made in the field of ACE. Most of these advances have been motivated by the need to analyze problems of an increasing degree of difficulty, such as calculating radar cross sections (RCS) or designing increasingly sophisticated circuit components, microwave devices, antennas, and so on. The increasing density of components and systems, higher frequency applications, and more demanding design specifications required the flexibility and speed of analysis provided by computer-aided design (CAD).

Another factor that has contributed to the rapid progress of computational electromagnetics is the spectacular evolution of computers in the last three decades. Now a relatively cheap desktop computer can offer performance comparable to the large mainframes of the 1960s, giving enough computational power to satisfy the requirements of most of the electromagnetic computer tools available today. In addition to these dramatic advances in computer hardware in recent years, many new software techniques have become available that greatly facilitate the difficult computational tasks that appear in ACE. In particular, the development and popularization of signal processing techniques have helped to improve the speed and accuracy of many electromagnetic methods.

Excellent works can be found on the advances in computational electromagnetics. For instance, there are the classical monographic texts [1–3] or the more recently published books [4–6] that provide an update on the latest advances. Indeed, extensive reference lists could be given of introductory or general works about ACE; [1–6] provide only a small sample of the dynamism of ACE in the last few years. Although ACE is already a relatively mature discipline, new and original findings continue to revolutionize the field, further increasing the efficiency and usefulness of this technology. Roughly speaking, ACE techniques can be classified into two generic categories: rigorous methods and asymptotic methods. Rigorous methods involve solving an operator equation obtained by imposing the Maxwell equations and the appropriate boundary conditions of the problem. A discretized version of the operator equation is usually employed. The most popular rigorous method is probably the moment method (MM), introduced and popularized in numerous works; see, for instance, those of [7–9].

In asymptotic methods, a guess or assumption is made about the shape of the solution when solving the problems. In a sense, this approach combines analytical solutions and numerical procedures. Asymptotic methods can be found for both low- and high-frequency problems [3]. The geometrical theory of diffraction (GTD) method and its derivatives, such as the uniform theory of diffraction (UTD), are undoubtedly the most widely used asymptotic methods for electrically large problems [10–12].

In general, we can say that the MM is applicable to electrically small problems and the GTD to electrically large ones. For problems of the size of a few wavelengths (the so-called resonance size), neither one of these methods works very well. The MM fails because the size of its characteristic system of equations is too large to be affordable with moderate computer resources. The GTD is deficient because its assumptions for the solution become inaccurate. In order to overcome this lack of an adequate method for the analysis of resonant-sized problems, a great deal of research in recent years has been devoted to considering combinations of the GTD with the MM [4] and other methods. Those that have received the most attention from researchers and that appear to be very promising are the finite difference–time domain (FD-TD) and the finite element–time domain (FE-TD) methods [6,13], the generalized multipole technique (GMT) [14–16], and the measured equation of invariance (MEI) [17]. To be complete, the list would have to include many other approaches,

but for the sake of conciseness we will end this section by referring to the technique that is the subject of this book, the CG-FFT method [18–20], which was also introduced in ACE for the computation of resonance-sized problems.

All the methods mentioned above have their strong and weak points, and it is practically impossible to establish which one of them is the best in general. All of them incorporate important features to reduce dramatically the computational cost involved in analyzing resonance-sized problems, especially if they are compared to traditional MM schemes. The performance of any one of these methods will depend on the particular problem being considered. From a very general point of view, the CG-FFT method is ideal for periodic problems, because in these cases the discrete nature of the operator equation in the spectral domain greatly simplifies the computation of the Fourier transform by means of the FFT. Problems having geometries with a body of revolution (BOR) symmetry will be included in the class of periodic problems because BORs are periodic for the revolution angle (the azimuthal coordinate if the BOR axis is located on the z-axis of a cylindrical coordinate system). The CG-FFT is also well suited to volumetric or surface problems that fill completely (or nearly) a rectangular parallelepiped in the case of three-dimensional (3D) problems, or a rectangle in the case of plate problems. For the CG-FFT to be efficient, the geometry of the problem must be well represented by means of a regular mesh.

1.3 CG METHODS

The CG method was introduced by Hestenes and Steifel for the solution of linear systems of equations [21]. It is basically an iterative optimization procedure for which many applications have been found in areas such as optimal control, signal processing, numerical matrix algebra, inverse problem analysis, computational geometry, and electromagnetics (see [18,19,22,23] for general references). The CG method is well suited to the treatment of operator equations, such as the linear system of equations,

$$[\mathbf{Y}] = [\mathbf{Z}][\mathbf{X}] \tag{1.1}$$

where $\mathbf{X}$ and $\mathbf{Y}$ are the unknown and excitation vectors, respectively, and $\mathbf{Z}$ is a matrix. The most general case is when $\mathbf{Z}$ is a nonsymmetric, complex value matrix (i.e., a non-Hermitian matrix). In this case, $\mathbf{Z}$ becomes a non-self-adjoint matrix.

There are several algorithms that are used to apply the CG method (see Chapter 4). All of these algorithms require the computation at any iteration of $\mathbf{P}$, the application of $\mathbf{Z}$ over a known vector $\mathbf{W}$,

$$[\mathbf{P}] = [\mathbf{Z}][\mathbf{W}] \tag{1.2}$$

or of $\mathbf{Q}$, the application of the adjoint of $\mathbf{Z}$ over $\mathbf{W}$,

$$[\mathbf{Q}] = [\mathbf{Z}^a][\mathbf{W}] \tag{1.3}$$

where $[\mathbf{Z}^a]$ is the adjoint matrix of $\mathbf{Z}$ (i.e., the transposed complex conjugate matrix). The CG method is an iterative approach that has been shown to converge to the solutions in a finite number of iterations if all the computations are performed exactly. However, due to the round-off errors introduced by the use of a finite-length computer, convergence cannot be guaranteed in a finite number of steps. Nevertheless, even in this case, CG is one of the fastest and most powerful methods for solving systems of linear equations with ill-conditioned matrices.

Many electromagnetic problems (e.g., in MM applications) give rise to systems of equations with large, dense matrices that are also non-Hermitian. In most cases, these matrices are very ill-conditioned and standard Gaussian inversion procedures often fail. In such cases, the CG method provides a way of finding solutions with acceptable accuracy. Some pioneering works that recommended CG algorithms or actually applied them to these kinds of ill-conditioned matrices can be found in [3,24–28]. Some of these references report finding solutions for problems that could not be solved in an affordable way by other methods.

The CG method also presents advantages for the solution of the linear systems of equations that arise when solving electromagnetic problems formulated in terms of an integral equation, as usually happens with the MM. The CG technique is advantageous because it is not necessary to save all the terms of the $\mathbf{Z}$-matrix in the main computer memory. This was especially important 10 years ago when the available computer memory was smaller by several orders of magnitude than it is now. As was reported in [27], recomputing the matrix $\mathbf{Z}$ in each iteration makes it possible to compute the vector $\mathbf{P}$ or $\mathbf{Q}$ in expressions (1.2) and (1.3) by storing only a part of $\mathbf{Z}$ at each moment. This is impossible or extremely difficult when using noniterative methods.

1.4 TOEPLITZ SYMMETRIES AND THE CG-FFT METHOD

Nearly all the problems of interest in ACE are linear and space-invariant. Linearity implies that the superposition principle is applicable; that is, the fields created by a complex source can be computed as the summation of the fields originated by each one of the elemental sources into which the complex source can be split. For the sake of simplicity, we will assume that these elemental sources are infinitely small (point sources). When we say that the problems are space-invariant, we mean that the electromagnetic fields of each point source measured at an observation point depend only on the relative positions of the source point and the observation point.

When the principles of linearity and space invariance are applicable, we can write the fields as an integral of convolution between the source function and a Green's function, as the following expression shows for the case of the electric field created by the current on a conducting plate:

$$E(\mathbf{r}) = L_{Dr} \int_{S'} G(\mathbf{r} - \mathbf{r}')\mathbf{J}(\mathbf{r}')\, ds' \tag{1.4}$$

where the operator L_{Dr} can contain a derivative in the variable r, and S is the area of the plate. The Green's function $G(\mathbf{r} - \mathbf{r}')$ can also introduce some derivatives over the variable $\mathbf{r}'$ of the source function, which is the current $\mathbf{J}(\mathbf{r}')$ in (1.4). It should be pointed out that the Green's function is written only in terms of the difference between the observation and source point vectors $(\mathbf{r} - \mathbf{r}')$. Now it is easy to identify these ACE cases with the linear and time-invariant (LTI) systems of signal processing. Employing the same terminology as in signal processing, we will use LTI to refer to linear and space-invariant problems like the one of (1.4).

When an LTI problem is analyzed by using the MM with regularly spaced sampling points, it is easy to obtain a matrix with a Toeplitz (or block Toeplitz) structure; that is, all the rows (or columns) of the matrix can be obtained by a cyclic rotation of one of them [28]. This can happen if, for instance, we are considering a straight-wire problem. In this case, the MM system of equations can be solved very efficiently by using the special technique developed ad hoc for matrices with a Toeplitz structure.

Greater advantage can be taken of the special nature of LTI problems by calculating the space domain convolution that appears in (1.4) by means of the Fourier transform [29–31]. It is well known that a spatial domain convolution becomes a simple product in the spectral domain. Computing the transform by means of the FFT results in a highly efficient method. Moreover, operations like derivatives that are cumbersome in the space domain turn into simple algebraic operations in the transformed domain. For all these reasons it seems to be advantageous to work directly in the spectral domain to solve an equation like the one in (1.4). However, care must be taken in generalizing the usage of the Fourier transform, because the problems must satisfy boundary conditions defined in the spatial domain, such as surface S of the conducting plate of (1.4). Of course, an attempt could be made to transform the boundary conditions as well, but in this case a very complicated equation would be obtained in which new convolution integrals would appear. This occurs because until the problem is solved, the fields created by the unknown source are known only in the region of the boundary conditions. To overcome this difficulty, some approximate iterative methods, mainly the spectral iterative technique (SIT), have used initial assumptions for the values of the fields in the region outside the domain of the boundary conditions [32–34]. However, the convergence of these methods is highly dependent on the quality of the assumptions for the initial field values, and no general criteria have been found that always guarantee accurate solutions. Therefore, it appears to be better to apply the boundary conditions in the spatial domain.

The central idea of the CG-FFT method in ACE is to solve the operator equation of an LTI problem like that of (1.4) by imposing the boundary conditions in

the spatial domain and computing the convolutions by means of the FFT [20,35]. An initial estimate of the solutions is made in the CG-FFT method, but neither the accuracy of the method nor the speed of convergence depends on this initial guess. Like the SIT, the CG-FFT enjoys the other advantages of using the Fourier transform, such as the simplification of complex operations like derivatives. However, unlike the SIT, no divergent behavior appears when using the CG-FFT.

Probably the greatest disadvantage of the CG-FFT method is that present CG-FFT formulations require the use of a regular mesh in the representation of the geometry. Over this regular mesh the unknown functions (usually currents) are expanded in terms of only one kind of basis function. This limitation is imposed by a direct computation of the Fourier transform by means of the FFT. As is well known, the estimation of the spectrum by means of the FFT requires a set of samples located at regular intervals. However, many techniques are now being developed to perform this spectral estimation without the requirement of sampling at regular intervals [36]. The authors think that the CG-FFT method can be greatly improved by using these techniques, although at the cost of more sophisticated application schemes.

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Chapter 2
Fourier Transforms

2.1 INTRODUCTION

The CG-FFT method is largely based on Fourier transforms and their properties. Therefore, skill in using Fourier transforms is a necessary requirement for understanding the CG-FFT method and knowing how to apply it properly to real problems. Thorough consideration of Fourier transforms and their applications can be found in the excellent works listed in [1–6]. Nevertheless, for the sake of completeness of this book, we will review in this chapter the properties of Fourier transforms that are essential for CG-FFT methods. These properties will be used repeatedly to perform operations such as sampling, differentiation, smoothing, shifting, Green's function formulation, and, probably most importantly for our purposes, convolution.

In order to be able to handle continuous and discrete functions, both the continuous Fourier transform (CFT) and the discrete Fourier transform (DFT) must be employed. We shall consider a function $x(t)$ to be continuous if it is defined for any real value of the independent variable t, while a function $x[n]$ is discrete if it is only defined for integer values of the independent variable n. To avoid confusion between these two kinds of functions, we shall always represent continuous functions by enclosing the independent variable in parentheses, while discrete functions will be written with square brackets.

All the problems dealt with in this book are formulated in terms of continuous operator equations, which are obtained from fundamental physical laws (Maxwell's equations). By continuous operators we mean those that relate continuous functions. The operator equations are discretized in order to solve the problems numerically by means of computers. This can be done in two ways. We can first apply the CFT to the continuous operators and then discretize the resulting functions. In this case, we will compute the CFTs approximately by using DFTs, as will be shown in Chapter 3. The other alternative is to discretize the operators first and then apply the DFT to the resulting discrete functions. In this case, only the DFT is used in all the steps of the CG-FFT approach.

2.2 DISCRETE FOURIER TRANSFORM

2.2.1 Definition

The DFT allows any discrete periodic function to be expanded in terms of a series of complex exponentials. The discrete periodic function $x[n]$ of period N,

$$x[n] = x[n + mN] \quad \text{for} \quad m = 0, \pm 1, \pm 2, \ldots \tag{2.1}$$

can be expanded as follows:

$$x[n] = \frac{1}{N} \sum_{i=0}^{N-1} \tilde{x}[i]\, S_i[n] \tag{2.2}$$

where $\tilde{x}[i]$ is the so-called DFT of $x[n]$ and $S_i[n]$ is the ith harmonic

$$S_i[n] = \exp\left[jn \frac{2\pi}{N} i \right] \tag{2.3}$$

of the fundamental harmonic function

$$S_1[n] = \exp\left[jn \frac{2\pi}{N} \right] \tag{2.4}$$

Combining (2.2) and (2.3), we obtain the expression for the inverse DFT,

$$x[n] = \frac{1}{N} \sum_{i=0}^{N-1} \tilde{x}[i] \exp\left[jn \frac{2\pi}{N} i \right] \tag{2.5}$$

which allows us to obtain $x[n]$ from its DFT, $\tilde{x}[i]$. Taking into account the fact that the harmonic functions $S_i[n]$ are orthogonal,

$$\langle S_i, S_j \rangle = \sum_{n=0}^{N-1} S_i[n]\, S_j^*[n] = N\delta_{ij} \tag{2.6}$$

we can express the DFT directly as

$$\tilde{x}[i] = \sum_{n=0}^{N-1} x[n]\, S_n^*[i] \tag{2.7}$$

or more explicitly,

$$\tilde{x}[i] = \sum_{n=0}^{N-1} x[n] \exp\left[-ji\frac{2\pi}{N}n\right] \tag{2.8}$$

This provides a way to compute the transform function $\tilde{x}[i]$ from the values of $x[n]$. Since all the harmonic functions $S_n[i]$ are periodic, with the same period N, the DFT $\tilde{x}[i]$ is also periodic and has the same period as $x[n]$ and the harmonic functions.

If the expressions for the direct and inverse DFTs, (2.8) and (2.5) respectively, are compared, it can be seen that they are quite similar. Actually, these transforms are dual functions, as is shown in Figures 2.1 and 2.2, where the transforms of a pair of simple functions and their dual functions, respectively, are pictured. A more complete vision of the dual behavior of the transform can be obtained by examining the properties of the DFT listed in Table 2.1 or the transforms of the elementary functions in Table 2.2.

Since N terms are considered in the series of (2.8) and (2.5), N^2 multiplications must be performed in order to calculate all the values of a direct or inverse DFT if either one of these expressions is used directly. However, the direct or inverse DFT

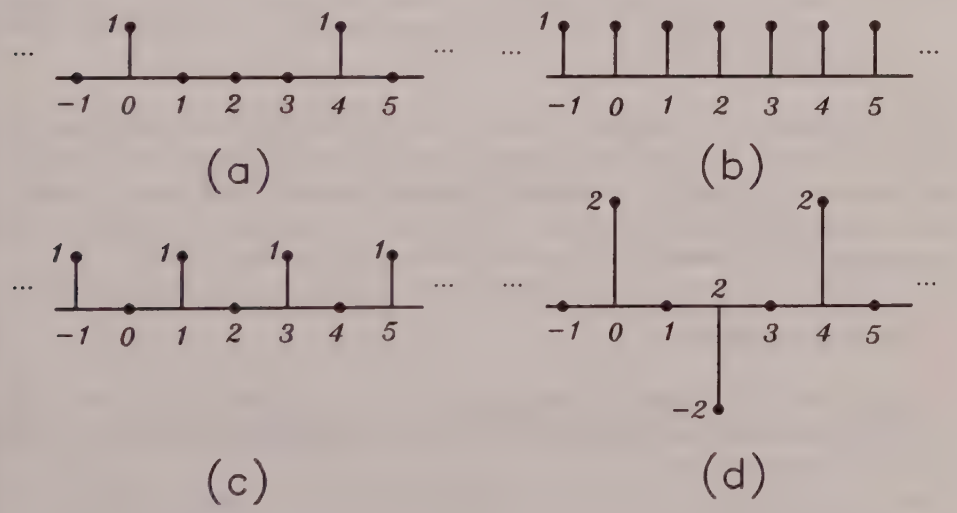


Figure 2.1 The DFT of the periodic impulse function $\delta_4^p[n]$ pictured in (a) is the constant function $S_0[n] = 1$ of (b). The DFT of the function $x_2[n]$ of (c), which is the sum of two shifted impulse functions, $\delta_4^p[n-1]$ and $\delta_4^p[n-3]$, is the discrete cosine function $(2S_1[i] + 2S_3[i] = 2\cos[i\pi/2])$ shown in (d).

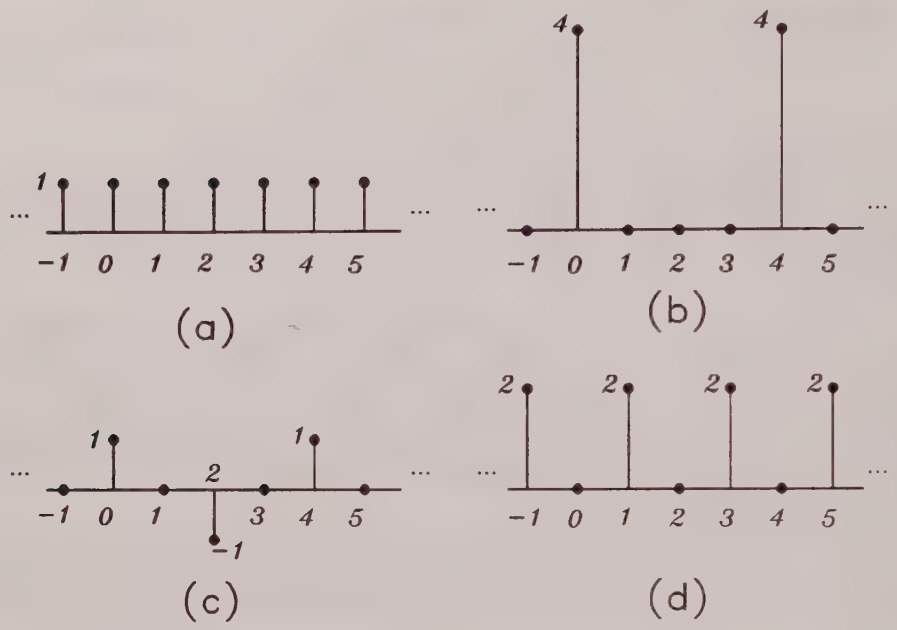


Figure 2.2 The DFT of the constant function ($S_0[n] = 1$) of (a) is the periodic impulse function ($4\delta_4^p[i]$) indicated in (b). The DFT of the discrete cosine function ($S_1[n] + S_3[n] = \cos[n\pi/2]$) pictured in (c) is the sum of the two shifted impulse functions ($2\delta_4^p[i-1] + 2\delta_4^p[i-3]$) shown in (d).

can be performed more efficiently by using the FFT, which only requires the computation of about $N \log(N)$ terms. Many studies of FFT algorithms are available in the literature; [1] and [4], for instance, present excellent descriptions of some of these algorithms. For our purposes, the FFT only represents an efficient way of computing DFTs, and we will therefore often say FFT instead of DFT.

2.2.2 DFT Properties

Although the DFT has many interesting properties [1,3,4], the most relevant for the CG-FFT method are its linearity, shift, convolution, and modulation properties, which will now be summarized.

2.2.2.1 Linearity

The DFT of a weighted sum of functions is the same weighted sum of the DFTs to each function

Table 2.1
List of DFT Properties

Property	*Space Domain*	*Transform Domain*
Linearity	$\alpha x[n] + \beta y[n]$	$\alpha\tilde{x}[i] + \beta\tilde{y}[i]$
Time shifting	$x[n-k]$	$\tilde{x}[i]\exp(-j2\pi ik/N)$
Frequency shifting	$x[n]\exp(j2\pi nk/N)$	$\tilde{x}[i-k]$
Convolution	$z[n] = \sum_{k=0}^{N-1} x[k]y[n-k] = x[n] \otimes y[n]$	
Time convolution theorem	$x[n] \otimes y[n]$	$\tilde{x}[i]\tilde{y}[i]$
Frequency convolution theorem	$x[n]y[n]$	$\frac{1}{N}\tilde{x}[i] \otimes \tilde{y}[i]$
Difference	$x[n] - x[n-1]$	$(1 - \exp(-j2\pi i/N))\tilde{x}[i]$
Summation	$\sum_{m=-\infty}^{n} x[m]$	$\frac{1}{1-\exp(-j2\pi i/N)}\tilde{x}[i] + \pi\tilde{x}[0]\sum_{k=-\infty}^{\infty}\delta\left(\frac{2\pi ik}{N} - 2\pi k\right)$
Parseval's theorem	$\sum_{n=0}^{N-1}\lvert x[n]\rvert^2 = \frac{1}{N}\sum_{i=0}^{N-1}\lvert\tilde{x}[i]\rvert^2$	

$$\mathrm{DFT}(\alpha x[n] + \beta y[n]) = \alpha\tilde{x}[n] + \beta\tilde{y}[n] \tag{2.9}$$

2.2.2.2 *Shift Property*

If $\tilde{x}[i]$ is the DFT of $x[n]$, then the DFT of the shifted function $x[n-m]$ is given by

$$\mathrm{DFT}(x[n-m]) = \tilde{x}[i]\exp\left[-jim\frac{2\pi}{N}\right] \tag{2.10}$$

From this property we can derive the following expression for the transform of a discrete difference:

$$\mathrm{DFT}(x[n] - x[n-1]) = \left(1 - \exp\left[-ji\frac{2\pi}{N}\right]\right)\tilde{x}[i] \tag{2.11}$$

Table 2.2
DFT of Elementary Discrete Functions

Time Domain $x[n]$	*Transform Domain* $\bar{x}[i]$
1	$N\delta_N[i]$
$\delta_N[n]$	1
$\delta_N[n - n_0]$	$\exp[-j2\pi in_0/N]$
$\exp[j2\pi n_0 n/N]$	$N\delta_N[i - n_0]$
$\cos[2\pi n_0 n/N]$	$\frac{N}{2}[\delta_N[i - n_0] + \delta_N[i + n_0]]$
$\sin[2\pi n n_0/N]$	$\frac{N}{2j}[\delta_N[i - n_0] + \delta_N[i + n_0]]$
1 if $\lvert n\rvert < N_1$	$\frac{2N_1 + 1}{N}$ if $i = 0$
0 if $N_1 < \lvert n\rvert < N/2$ and $x[n + N] = x[n]$	$\frac{\sin[2\pi i(N_1 + 1/2)/N]}{\sin[2\pi i/(2N)]}\delta_N[i]$ otherwise

2.2.2.3 *Convolution Property*

The DFT of the cyclic convolution of a pair of functions is the product of their transforms:

$$\mathrm{DFT}(x[n] \otimes y[n]) = \bar{x}[i]\bar{y}[i] \tag{2.12}$$

where the symbol $\otimes$ indicates cyclic convolution,

$$x[n] \otimes y[n] = \sum_{k=k_0}^{k_0+N-1} x[k]y[n - k] \tag{2.13}$$

where $x[n]$ and $y[n]$ are periodic functions, with the same period N, and k_0 is any integer. The function resulting from the convolution is also periodic and has the same period N as $x[n]$ and $y[n]$. Figure 2.3 shows an example of the convolution of a pair of functions.

The convolution is very important for our purposes because the CG method requires many convolution computations to solve the operator equations. Computing the convolution by using (2.13) directly requires N^2 operations. However, if the convolution is calculated by performing the FFT of the functions, taking the product

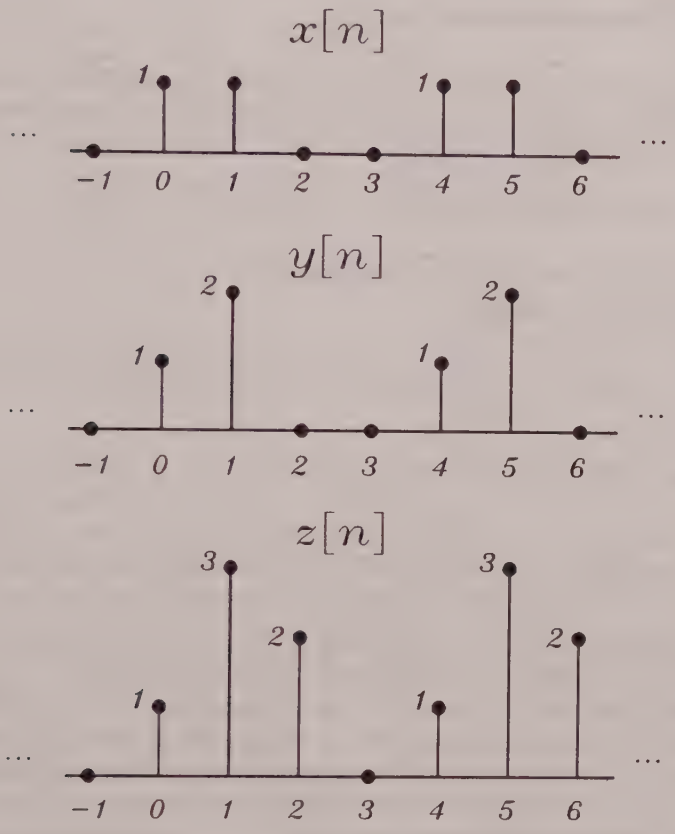


Figure 2.3 The function $z[n]$ results from performing the convolution of the functions $x[n]$ and $y[n]$.

of the transformed functions and then returning to the original domain by means of an inverse FFT, the number of basic operations is proportional to $N \log(N)$. For large values of N, the use of FFTs can reduce the number of operations by several orders of magnitude.

2.2.2.4 Modulation Property

The DFT of the product of a pair of functions is proportional to the cyclic convolution of the transforms of the functions

$$\text{DFT}(x[n]y[n]) = \frac{1}{N}\tilde{x}[i] \otimes \tilde{y}[i] \tag{2.14}$$

2.2.3 DFT of Multidimensional Functions

Discrete periodic functions of two or more variables can also be expanded easily in terms of complex exponentials with the help of the DFT. A two-dimensional (2D) periodic function can be obtained by replaying a generating rectangle,

$$x[m,n] = x[m + kM, n + pN] \quad \text{for} \quad k, p = 0, \pm 1, \pm 2, \ldots \tag{2.15}$$

where $M \times N$ is the size of the generating rectangle.

The direct DFT is given by

$$\tilde{x}[u,v] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x[m,n] \exp\left[-ju \frac{2\pi}{M} m - jv \frac{2\pi}{N} n \right] \tag{2.16}$$

and the inverse DFT by

$$x[m,n] = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} \tilde{x}[u,v] \exp\left[jm \frac{2\pi}{M} u + jn \frac{2\pi}{N} v \right] \tag{2.17}$$

$\tilde{x}[u,v]$ is also periodic and has a generating rectangle of the same size ($M \times N$) as $x[m,n]$. Figures 2.4 and 2.5 show the DFTs of a pair of functions.

The DFTs for functions of three or more dimensions are similar to (2.16) and (2.17). In fact, as has been shown in [6], all the properties applicable to the 2D case can be generalized in a straightforward way to higher dimensional problems.

As in the case of the one-dimensional (1D) DFT, there are several FFT algorithms available for multidimensional functions [6], which allow the transforms

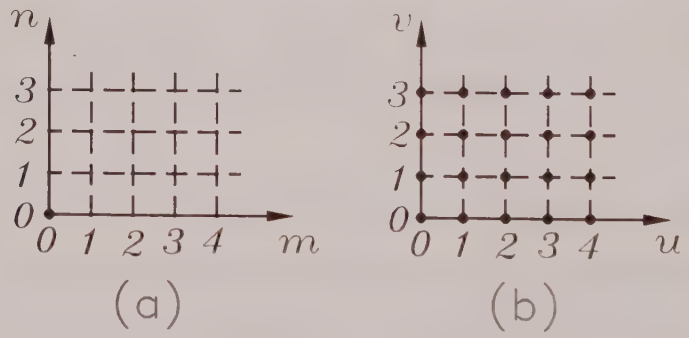


Figure 2.4 (a) Generating cell of the 2D impulse function $\delta^{p}_{3,4}[m,n]$; (b) DFT of this function ($\tilde{x}[u,v] = 1$).

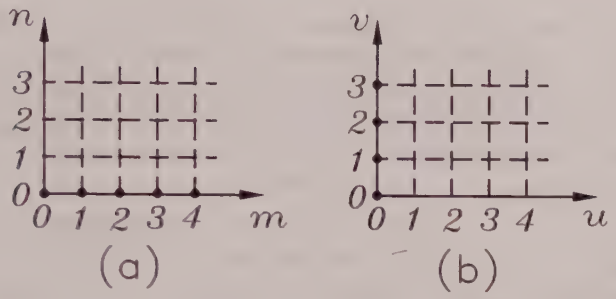


Figure 2.5 (a) Generating cell of a discrete blade function ($x[m,n] = \delta_4^p[n]$); (b) DFT of this function ($\bar{x}[u,v] = 5\delta_5^p[u]$).

to be performed with a computational cost proportional to $N_c \log N_c$, where N_c is the size of the generating cell ($M \times N$ for the 2D case).

The DFT can also be extended to handle periodic multidimensional functions formed by nonrectangular generating cells. These DFTs are useful for analyzing electromagnetic problems with this kind of periodic geometry. Nevertheless, in this book we will restrict our attention to the rectangular cell. The CG-FFT approaches developed here for rectangular periodicities can easily be extended to more arbitrary periodic shapes.

2.2.4 Properties of the Multidimensional DFT

The properties described for the 1D DFT can be extended to the multidimensional DFT. In addition, some new properties appear for the multidimensional transform. The properties of the multidimensional DFT that are most relevant to the CG-FFT method, which are summarized in Table 2.3, will now be outlined briefly.

2.2.4.1 Linearity

$$\text{DFT}(\alpha x[m,n] + \beta y[m,n]) = \alpha \bar{x}[u,v] + \beta \bar{y}[u,v] \tag{2.18}$$

2.2.4.2 Shift Property

If $\bar{x}[u,v]$ is the DFT transform of $x[m,n]$, then

$$\text{DFT}(x[m-p,n-q]) = \bar{x}[u,v] \exp\left[-jup\frac{2\pi}{M} - jvq\frac{2\pi}{N}\right] \tag{2.19}$$

Table 2.3
List of Two-Dimensional DFT Properties

Property	*Space Domain*	*Transform Domain*
Linearity	$\alpha x[m,n] + \beta y[m,n]$	$\alpha \tilde{x}[u,v] + \beta \tilde{y}[u,v]$
Space shifting	$x[m-p,n-q]$	$\tilde{x}[u,v] \cdot \exp[-j2\pi(up/M + vq/N)]$
Frequency shifting	$x[m,n] \cdot \exp[j2\pi mp/M] \cdot \exp[j2\pi nq/N]$	$\tilde{x}[u-p,v-q]$
Convolution	$z[m,n] = \sum_{k=0}^{M-1} \sum_{s=0}^{N-1} x[k,s]y[m-k,n-s] = x[m,n] \otimes y[m,n]$	
Space convolution theorem	$x[m,n] \otimes y[m,n]$	$\tilde{x}[u,v]\tilde{y}[u,v]$
Frequency convolution theorem	$x[m,n]y[m,n]$	$\frac{1}{MN} \tilde{x}[u,v] \otimes \tilde{y}[u,v]$
Difference	$x[m,n] - x[m-1,n-1]$	$(1 - \exp[-j2\pi u/M] \cdot (1 - \exp[-j2\pi v/N]) \cdot \tilde{x}[u,v]$

This property is used extensively throughout the book to compute differences. For instance, the DFT of a crossed finite difference can be computed as follows:

$$\begin{aligned} &\mathrm{DFT}[(x[m,n] - x[m-1,n])(x[m,n] - x[m,n-1])] \\ &= \left(1 - \exp\left[-ju\frac{2\pi}{M}\right]\right)\left(1 - \exp\left[-jv\frac{2\pi}{N}\right]\right)\tilde{x}[u,v] \end{aligned} \tag{2.20}$$

2.2.4.3 *Convolution Property*

If $x[m,n]$ and $y[m,n]$ have the same generating cell size, $M \times N$, then

$$\mathrm{DFT}(x[m,n] \otimes y[m,n]) = \tilde{x}[u,v]\tilde{y}[u,v] \tag{2.21}$$

where the convolution shall be performed cyclically:

$$z[m,n] = x[m,n] \otimes y[m,n] = \sum_{k=k_0}^{k_0+M-1} \sum_{p=p_0}^{p_0+N-1} x[k,p]y[m-k,n-p] \tag{2.22}$$

with k_0 and p_0 having any integer values.

The function $z[m,n]$ resulting from convolution is periodic and has the same rectangular generating cell as the functions $x[m,n]$ and $y[m,n]$.

As in the case of the 1D convolution, multidimensional convolutions can be evaluated more efficiently by using FFTs than by computing the expression (2.22) directly.

2.2.4.4 Modulation Property

The DFT of products is proportional to the convolution of the transform

$$\mathrm{DFT}(x[m,n]\,y[m,n]) = \frac{1}{MN}\,\bar{x}[u,v] \otimes \bar{y}[u,v] \tag{2.23}$$

2.2.4.5 Symmetry Property

If $\bar{x}[u,v]$ is the transform of $x[m,n]$, then

$$\mathrm{DFT}(x[n,m]) = \bar{x}[v,u] \tag{2.24}$$

2.3 CONTINUOUS FOURIER TRANSFORM

2.3.1 Definition

Any continuous function $x(t)$ can be expanded in terms of a continuous spectrum of exponentials by means of the CFT:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \bar{x}(\omega)\exp(j\omega t)\,\mathrm{d}\omega \tag{2.25}$$

where $\bar{x}(\omega)$ is the so-called spectrum of $x(t)$:

$$\bar{x}(\omega) = \int_{-\infty}^{\infty} x(t)\exp(-j\omega t)\,\mathrm{d}t \tag{2.26}$$

Expression (2.25) is the inverse CFT that allows us to go from the spectral domain (ω space) to the "real" domain (t space). The direct CFT is given by (2.26) and provides the way to obtain the spectrum of the function $x(t)$ defined in the "real" domain.

For illustrative examples, we will consider the CFTs of the pulse, triangle, and train of delta functions (see Figures 2.6 to 2.8), which will appear frequently throughout this book.

The CFT of the pulse function $P_{2a}(t)$,

$$P_{2a}(t) = \begin{cases} 1 & \text{for} \quad |t| \le a \\ 0 & \text{elsewhere} \end{cases} \tag{2.27}$$

is given by

$$\tilde{P}_{2a}(\omega) = 2a \operatorname{sinc}\left(\frac{a\omega}{\pi}\right) \tag{2.28}$$

where the sinc function is defined as

$$\operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t} \tag{2.29}$$

The CFT of the triangle function $T_{2a}(t)$

$$T_{2a}(t) = \begin{cases} 1 - \dfrac{|t|}{a} & \text{for} \quad |t| \le a \\ 0 & \text{elsewhere} \end{cases} \tag{2.30}$$

is the following squared sinc function:

$$\tilde{T}_{2a}(\omega) = \frac{2a^2}{\pi} \operatorname{sinc}^2 \frac{(a\omega)}{\pi} \tag{2.31}$$

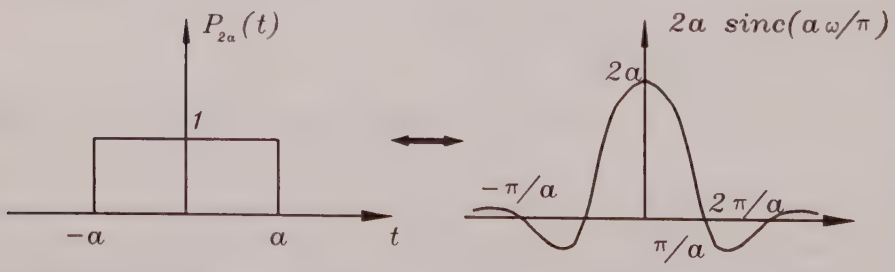


Figure 2.6 Pulse function of length 2a and its CFT.



Figure 2.7 Triangle function of length 2a and its CFT.

Finally, the CFT of a train of delta functions $\mathit{III}_a(t)$,

$$\mathit{III}_a(t) = \sum_{k=-\infty}^{\infty} \delta(t - ka) \tag{2.32}$$

is also a train of deltas,

$$\mathrm{CFT}(\mathit{III}_a(t)) = \frac{2\pi}{a}\,\mathit{III}_{2\pi/a}(\omega) = \frac{2\pi}{a}\sum_{k=-\infty}^{\infty} \delta\left(\omega - k\frac{2\pi}{a}\right) \tag{2.33}$$

where CFT(g) denotes the continuous Fourier transform of g.

Table 2.4 shows the CFTs of some other common elementary functions.

2.3.2 Properties of the CFT

The formulation of the CG-FFT method makes extensive use of the CFT properties, which are summarized in Table 2.5. These properties can help us to understand many physical phenomena and engineering problems. Some excellent discussions of the applications of the CFT properties can be found in [1,5]. Here, the properties most relevant to the CG-FFT method will be outlined.

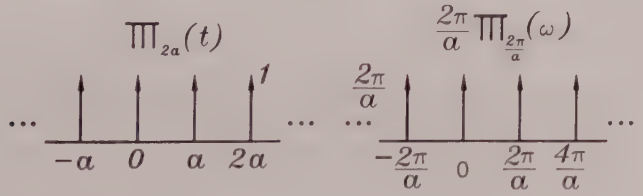


Figure 2.8 Train of delta functions with period a and its CFT.

Table 2.4
CFT of Elementary Functions

Time Domain $x(t)$	*Transform Domain $\tilde{x}(\omega)$*
1	$2\pi\delta(\omega)$
$\delta(t)$	1
$\delta(t - t_0)$	$\exp(-j\omega_0 t)$
$\exp(j\omega_0 t)$	$2\pi\delta(\omega - \omega_0)$
$\sum_{k=-\infty}^{\infty} a_k \exp(jk\omega_0 t)$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$
$\sum_{k=-\infty}^{\infty} \delta(t - kT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k/T)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$\frac{W}{\pi}\sin(Wt/\pi) = \frac{\sin(Wt)}{\pi t}$	$P_{2W}(\omega) = \begin{cases} 1 & \text{if } \lvert\omega\rvert < W \\ 0 & \text{if } \lvert\omega\rvert > W \end{cases}$
$P_{2T}(t) = \begin{cases} 1 & \text{if } \lvert t\rvert < T \\ 0 & \text{if } \lvert t\rvert > T \end{cases}$	$2T\,\mathrm{sinc}(\omega T/\pi) = \frac{2\sin(\omega T)}{\omega}$
$T_{2T}(t) = \begin{cases} 1 - \frac{\lvert t\rvert}{T} & \text{if } \lvert t\rvert \le T \\ 0 & \text{elsewhere} \end{cases}$	$\frac{2T^2}{\pi}\mathrm{sinc}^2(\omega T/\pi)$
$u(t)$	$\frac{1}{j\pi} + \pi\delta(\omega)$
$\exp(-at)u(t)$, $\mathrm{Re}[a] > 0$	$\frac{1}{a + j\omega}$

Table 2.5
List of CFT Properties

Property	*Space Domain*	*Transform Domain*
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha\tilde{x}(\omega) + \beta\tilde{y}(\omega)$
Time shifting	$x(t - t_0)$	$\tilde{x}(\omega)\exp(-j\omega t_0)$
Frequency shifting	$x(t)\exp(j\omega_0 t)$	$\tilde{x}(\omega - \omega_0)$
Convolution	$z(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)\,d\tau = x(t) \otimes y(t)$	
Time convolution theorem	$x(t) \otimes y(t)$	$\tilde{x}(\omega)\tilde{y}(\omega)$
Frequency convolution theorem	$x(t)y(t)$	$\frac{1}{2\pi}\tilde{x}(\omega) \otimes \tilde{y}(\omega)$
Time scaling	$x(kt)$	$\frac{1}{\lvert k\rvert}\tilde{x}(\omega/k)$
Frequency scaling	$\frac{1}{\lvert k\rvert}x(t/k)$	$\tilde{x}(k\omega)$
Differentiation	$\frac{dx(t)}{dt}$	$j\omega\tilde{x}(\omega)$
Integration	$\int_{-\infty}^{t} x(\tau)\,d\tau$	$\frac{1}{j\omega}\tilde{x}(\omega) + \pi\tilde{x}(0)\delta(\omega)$
Parseval's theorem	$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} \lvert\tilde{x}(\omega)\rvert^2\,d\omega$	

2.3.2.1 Linearity

$$\text{CFT}(\alpha x(t) + \beta y(t)) = \alpha\tilde{x}(\omega) + \beta\tilde{y}(\omega) \tag{2.34}$$

2.3.2.2 Shift Property

The CFT of a shifted function $x(t - a)$ is given by

$$\text{CFT}(x(t - a)) = \tilde{x}(\omega)\exp(-ja\omega) \tag{2.35}$$

where $\tilde{x}(\omega)$ is the transform of $x(t)$.

2.3.2.3 Differentiation Property

If $\tilde{x}(\omega)$ is the CFT of $x(t)$, then

$$\mathrm{CFT}\left(\frac{\mathrm{d}x(t)}{\mathrm{d}t}\right) = j\omega\tilde{x}(\omega) \tag{2.36}$$

2.3.2.4 Convolution Property

The CFT of the continuous convolution of a pair of functions

$$x(t) \otimes y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)\,\mathrm{d}\tau \tag{2.37}$$

is the product of the CFTs of the functions

$$\mathrm{CFT}(x(t) \otimes y(t)) = \tilde{x}(\omega)\tilde{y}(\omega) \tag{2.38}$$

We can use the convolution property to obtain the CFT of a periodic function. As is indicated in Figure 2.9, any periodic function $x^P(t)$ can be formed by replaying at regular intervals a generating function $x^g(t)$

$$x^P(t) = \sum_{m=-\infty}^{\infty} x^g(t - mT) \tag{2.39}$$

This expression can be written as a convolution:

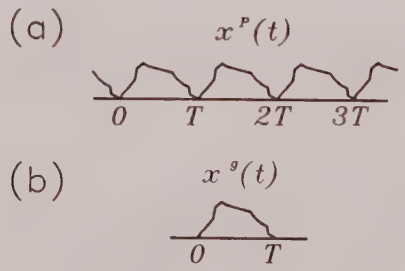


Figure 2.9 The periodic function $x^P(t)$ is formed by replaying the generating function $x^g(t)$ at regular intervals of length T.

$$x^P(t) = \Pi\!\!\Pi_T(t) \otimes x^g(t) \tag{2.40}$$

The CFT of the periodic function is the following product:

$$\begin{aligned} \tilde{x}^P(\omega) &= \frac{2\pi}{T}\, \Pi\!\!\Pi_{2\pi/T}(\omega) \otimes \tilde{x}^g(\omega) \\ &= \Delta\omega \sum_{k=-\infty}^{\infty} \tilde{x}^g(k\Delta\omega)\,\delta(\omega - k\Delta\omega) \end{aligned} \tag{2.41}$$

where

$$\Delta\omega = \frac{2\pi}{T} \tag{2.42}$$

Therefore, the CFT of a periodic function is a train of delta functions with the delta amplitudes modulated by the transform of the generating function. In other words, the CFT of the periodic function is obtained by sampling the CFT of the generating function with a train of delta functions.

2.3.2.5 Modulation Property

The CFT of the product of a pair of functions is proportional to the convolution of the transforms of the functions.

$$\mathrm{CFT}(x(t)y(t)) = \frac{1}{2\pi}\,(\tilde{x}(\omega) \otimes \tilde{y}(\omega)) \tag{2.43}$$

Since modulation is the dual property of convolution, we can state that if the CFT of a function is periodic,

$$\tilde{x}^P(\omega) = \sum_{k=-\infty}^{\infty} \tilde{x}^g(\omega - k\omega_T) \tag{2.44}$$

then the function in the time domain is a series of delta functions:

$$x(t) = \mathrm{CFT}^{-1}(\tilde{x}^P(\omega)) = \Delta t \sum_{n-\infty}^{\infty} x^g(n\Delta t)\,\delta(t - n\Delta t) \tag{2.45}$$

where

$$\Delta t = \frac{2\pi}{\omega_T} \tag{2.46}$$

and $x^g(t)$ is the inverse CFT of $\tilde{x}^g(\omega)$.

The convolution and modulation properties can be combined to show that the function resulting from sampling a periodic function at regular points has a CFT that can be obtained by sampling the CFT of the generating function and making it periodic.

To illustrate this, consider the following process. First, the periodic function is defined:

$$x^P(t) = \sum_{m=-\infty}^{\infty} x^g(t - mT) = x^g(t) \otimes \amalg_T(t) \tag{2.47}$$

Second, this function is sampled:

$$\begin{aligned} x^{PS}(t) &= x^P(t) \otimes \amalg_{\Delta t}(t) \\ &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} x^g(n\Delta t - mT)\,\delta(t - n\Delta t) \end{aligned} \tag{2.48}$$

where

$$\frac{T}{\Delta t} = N \tag{2.49}$$

We can write $x^{PS}(t)$ explicitly in terms of a discrete periodic function $a[n]$,

$$x^{PS}(t) = \sum_{n=-\infty}^{\infty} a[n]\,\delta(t - n\Delta t) \tag{2.50}$$

where $a[n]$ is obtained by calculating the series

$$a[n] = \sum_{m=-\infty}^{\infty} x^g[n\Delta t - mT] \tag{2.51}$$

Then we can apply the convolution property,

$$\tilde{x}^P(\omega) = \frac{2\pi}{T} \sum_{i=-\infty}^{\infty} \tilde{x}^g(i\Delta\omega)\,\delta(\omega - i\Delta\omega) \tag{2.52}$$

and the modulation property,

$$\tilde{x}^{PS}(\omega) = \frac{2\pi}{T}\frac{1}{\Delta t}\sum_{k=-\infty}^{\infty}\sum_{i=-\infty}^{\infty}\tilde{x}^{g}(i\Delta\omega - k\omega_T)\,\delta(\omega - i\Delta\omega) \tag{2.53}$$

This last expression can be written in terms of a periodic series of delta functions:

$$\tilde{x}^{PS}(\omega) = \frac{\Delta\omega}{\Delta t}\sum_{i=-\infty}^{\infty} b[i]\,\delta(\omega - i\Delta\omega) \tag{2.54}$$

where $b[i]$ can be obtained by using an expression analogous to (2.51):

$$b[i] = \sum_{k=-\infty}^{\infty}\tilde{x}^{g}[i\Delta\omega - k\omega_T] \tag{2.55}$$

Applying the definitions of the DFT and the Fourier series [2], we can obtain the following relation between the discrete functions $a[n]$ and $b[i]$:

$$b[i] = \Delta t\,\mathrm{DFT}(a[n]) \tag{2.56}$$

2.3.3 CFT of Multidimensional Functions

Continuous functions of two or more variables can also be expanded in terms of a continuous spectrum of exponentials. For instance, for $a(x,y)$ we have

$$a(x,y) = \frac{1}{(2\pi)^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\tilde{a}(k_x,k_y)\exp(jk_x x + jk_y y)\,\mathrm{d}k_x\,\mathrm{d}k_y \tag{2.57}$$

$$\tilde{a}(k_x,k_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}a(x,y)\exp(-jk_x x - jk_y y)\,\mathrm{d}x\,\mathrm{d}y \tag{2.58}$$

Expressions (2.57) and (2.58) are the inverse and direct 2D CFTs, respectively. In these expressions, we have considered the cartesian coordinates (x,y) as an independent pair of variables, because many times in the book we are going to use these variables and their corresponding spectral coordinates, the spatial frequency pair (k_x,k_y).

As examples of this type of CFT, Table 2.6 contains the transform pairs of some elementary 2D functions. Some of these functions are pictured in [1].

Table 2.6
CFTs of Some 2D Functions

Space Domain $a(x,y)$	*Transform Domain* $\tilde{a}(k_x,k_y)$
$\delta(x,y)$	1
$\amalg_{\Delta x,\Delta y}(x,y) = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}\delta(x-m\Delta x, y-n\Delta y)$	$\frac{4\pi^2}{\Delta x\,\Delta y}\amalg_{2\pi/\Delta x,2\pi/\Delta y}(k_x,k_y)$
$\delta(x)\amalg_{\Delta y}(y)$	$\frac{2\pi}{\Delta y}\amalg_{2\pi/\Delta y}(k_y)$
$\delta(y)\amalg_{\Delta x}(x)$	$\frac{2\pi}{\Delta x}\amalg_{2\pi/\Delta x}(k_x)$
$P_{\Delta x,\Delta y}(x,y) = \begin{cases} 1 & \text{if } \lvert x\rvert \le \Delta x/2 \\ & \quad \lvert y\rvert \le \Delta y/2 \\ 0 & \text{otherwise} \end{cases}$	$\Delta x\Delta y\ \mathrm{sinc}(k_x\Delta x/2)\cdot\mathrm{sinc}(k_y\Delta y/2)$
$W_x(x,y) = \begin{cases} 1-\frac{\lvert x\rvert}{\Delta x} & \text{if } \lvert x\rvert \le \Delta x \\ & \quad \lvert y\rvert \le \Delta y/2 \\ 0 & \text{otherwise} \end{cases}$	$\Delta x\Delta y\ \mathrm{sinc}^2(k_x\Delta x/2)\cdot\mathrm{sinc}(k_y\Delta y/2)$
$W_y(x,y) = \begin{cases} 1-\frac{\lvert y\rvert}{\Delta y} & \text{if } \lvert y\rvert \le \Delta y \\ & \quad \lvert x\rvert \le \Delta x/2 \\ 0 & \text{otherwise} \end{cases}$	$\Delta x\Delta y\ \mathrm{sinc}(k_x\Delta x/2)\cdot\mathrm{sinc}^2(k_y\Delta y/2)$
$B_x(x,y) = \begin{cases} 1 & \text{if } \lvert x\rvert \le \Delta x/2 \\ & \quad y = 0 \\ 0 & \text{otherwise} \end{cases}$	$\Delta x\ \mathrm{sinc}(k_x\Delta x/2)$
$B_y(x,y) = \begin{cases} 1 & \text{if } \lvert y\rvert \le \Delta y/2 \\ & \quad x = 0 \\ 0 & \text{otherwise} \end{cases}$	$\Delta y\ \mathrm{sinc}(k_y\Delta y/2)$

2.3.4 Properties of Multidimensional CFTs

Multidimensional transforms have all the properties corresponding to 1D CFTs, but also present some additional properties such as symmetry. The properties that are most significant for the CG-FFT method are summarized in Table 2.7 and described briefly in the following paragraphs.

2.3.4.1 Linearity

$$\mathrm{CFT}(\alpha a(x,y) + \beta b(x,y)) = \alpha \tilde{a}(k_x,k_y) + \beta \tilde{b}(k_x,k_y) \tag{2.59}$$

Table 2.7
List of Two-Dimensional CFT Properties

Property	*Space Domain*	*Transform Domain*
Linearity	$\alpha a(x,y) + \beta b(x,y)$	$\alpha \tilde{a}(k_x,k_y) + \beta \tilde{b}(k_x,k_y)$
Space shifting	$a(x - x_0, y - y_0)$	$\tilde{a}(k_x,k_y) \cdot \exp(-jk_x x_0 - jk_y y_0)$
Frequency shifting	$a(x,y) \cdot \exp(jk_{x0}x + jk_{y0}y)$	$\tilde{a}(k_x - k_{x0}, k_y - k_{y0})$
Convolution	$c(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} a(\tau,\mu) b(x-\tau, y-\mu)\, d\tau\, d\mu$ $= a(x,y) \otimes b(x,y)$	
Space convolution theorem	$a(x,y) \otimes b(x,y)$	$\tilde{a}(k_x,k_y)\tilde{b}(k_x,k_y)$
Frequency convolution theorem	$a(x,y)b(x,y)$	$\frac{1}{4\pi^2}\tilde{a}(k_x,k_y) \otimes \tilde{b}(k_x,k_y)$
Space scaling	$a(Rx,Sy)$	$\frac{1}{\lvert RS\rvert}\tilde{a}(k_x\lvert R, k_y\lvert S)$
Frequency scaling	$\frac{1}{\lvert RS\rvert}a(x\lvert R, y/S)$	$\tilde{a}(Rk_x, Sk_y)$
Differentiation	$\frac{\partial^m}{\partial x^m}\frac{\partial^n}{\partial y^n}a(x,y)$	$(jk_x)^m(jk_y)^n\tilde{a}(k_x,k_y)$
Integration	$\int_{-\infty}^{x}\int_{-\infty}^{y} a(\tau,\mu)\, d\tau\, d\mu$	$\frac{-1}{k_x k_y}\tilde{a}(k_x,k_y) + \pi^2\tilde{a}(0,0)\delta(k_x,k_y)$
Parseval's theorem	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \lvert a(x,y)\rvert^2\, dx\, dy = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \lvert\tilde{a}(k_x,k_y)\rvert^2\, dk_x\, dk_y$	

2.3.4.2 Shift Property

If $\tilde{a}(x,y)$ is the CFT of $a(x,y)$, then

$$\mathrm{CFT}(a(x - x_0, y - y_0)) = \tilde{a}(k_x,k_y) \exp(-jx_0k_x - jy_0k_y) \tag{2.60}$$

2.3.4.3 Differentiation Property

The formulations of many electromagnetic operational equations include spatial derivatives. The differentiation property makes it easy to obtain the CFT of these equations. For instance, the transform of the gradient of a scalar function,

$$\nabla\phi(x,y) = \left(\frac{\partial}{\partial x}\hat{\mathbf{x}} + \frac{\partial}{\partial y}\hat{\mathbf{y}}\right)\phi(x,y) = \frac{\partial\phi}{\partial x}\hat{\mathbf{x}} + \frac{\partial\phi}{\partial y}\hat{\mathbf{y}} \tag{2.61}$$

is given by

$$\begin{aligned}\mathrm{CFT}(\nabla\phi(x,y)) &= (jk_x\hat{\mathbf{x}} + jk_y\hat{\mathbf{y}})\tilde{\phi}(k_x,k_y) \\ &= jk_x\tilde{\phi}(k_x,k_y) + jk_y\tilde{\phi}(k_x,k_y)\end{aligned} \tag{2.62}$$

It can be seen that the differential operator

$$\nabla \equiv \left(\frac{\partial}{\partial x}\hat{\mathbf{x}} + \frac{\partial}{\partial y}\hat{\mathbf{y}}\right) \tag{2.63}$$

is transformed in the spectral domain into an algebraic operator

$$\tilde{\nabla} \equiv (jk_x\hat{\mathbf{x}} + jk_y\hat{\mathbf{y}}) \tag{2.64}$$

2.3.4.4 Convolution Property

The transform of the convolution

$$a(x,y) \otimes b(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} a(\tau_x,\tau_y)b(x - \tau_x, y - \tau_y)\,\mathrm{d}\tau_x\,\mathrm{d}\tau_y \tag{2.65}$$

is the spectral product

$$\mathrm{CFT}(a(x,y) \otimes b(x,y)) = \tilde{a}(k_x,k_y)\tilde{b}(k_x,k_y) \tag{2.66}$$

If this property is used, the CFT of a 2D periodic function,

$$a^P(x,y) = \sum_{m_x=-\infty}^{\infty} \sum_{m_y=-\infty}^{\infty} a^g(x-m_xT_x, y-m_yT_y) \tag{2.67}$$

can be written in terms of a series of delta functions modulated by the transform of the generating function:

$$\tilde{a}^P(k_x,k_y) = \Delta k_x \Delta k_y \sum_{n_x=-\infty}^{\infty} \sum_{n_y=-\infty}^{\infty} \tilde{a}^g(n_x\Delta k_x, n_y\Delta k_y)\,\delta(k_x - n_x\Delta k_x)\,\delta(k_y - n_y\Delta k_y) \tag{2.68}$$

where

$$\Delta k_\alpha = \frac{2\pi}{T_\alpha}, \quad \alpha = x \quad \text{or} \quad y \tag{2.69}$$

2.3.4.5 Modulation Property

The transform of a product is a convolution:

$$\text{CFT}(a(x,y)b(x,y)) = \frac{1}{(2\pi)^2}\,(\tilde{a}(k_x,k_y) \otimes \tilde{b}(k_x,k_y)) \tag{2.70}$$

By means of this property, it can be shown that a function that is periodic in the spectral domain,

$$\tilde{A}^P(k_x,k_y) = \sum_{l_x=-\infty}^{\infty} \sum_{l_y=-\infty}^{\infty} \tilde{A}^g(k_x - l_x\omega_x, k_y - l_y\omega_y) \tag{2.71}$$

has as its inverse CFT a series of delta functions,

$$\begin{aligned} a(x,y) &= \text{CFT}^{-1}(\tilde{A}^P(k_x,k_y)) \\ &= \Delta x \Delta y \sum_{n_x=-\infty}^{\infty} \sum_{n_y=-\infty}^{\infty} a^g(n_x\Delta x, n_y\Delta y)\,\delta(x - n_x\Delta x, y - n_y\Delta y) \end{aligned} \tag{2.72}$$

where a^g is the inverse CFT of $\tilde{A}^g$ and

$$\Delta\alpha = \frac{2\pi}{\omega_\alpha}, \quad \alpha = x \quad \text{or} \quad y \tag{2.73}$$

As in the 1D case, the modulation and convolution properties can be used to show that the CFT of a periodic function sampled at regular points is also a sampled periodic function.

If

$$a^{PS}(x,y) = \sum_{n_x=-\infty}^{\infty} \sum_{n_y=-\infty}^{\infty} a[n_x,n_y]\,\delta(x - n_x\Delta x, y - n_y\Delta y) \tag{2.74}$$

with

$$a[n_x,n_y] = \sum_{m_x=-\infty}^{\infty} \sum_{m_y=-\infty}^{\infty} a^g[n_x\Delta x - m_xT_x, n_y\Delta y - m_yT_y] \tag{2.75}$$

then

$$\tilde{a}^{PS}(k_x,k_y) = \frac{\Delta k_x \Delta k_y}{\Delta x \Delta y} \sum_{i_x=-\infty}^{\infty} \sum_{i_y=-\infty}^{\infty} b[i_x,i_y]\,\delta(k_x - i_x\Delta k_x, k_y - i_y\Delta k_y) \tag{2.76}$$

where

$$b[i_x,i_y] = \sum_{l_x=-\infty}^{\infty} \sum_{l_y=-\infty}^{\infty} a^g[i_x\Delta k_x - l_x\omega_x, i_y\Delta k_y - l_y\omega_y] \tag{2.77}$$

The discrete functions $a[n_x,n_y]$ and $b[i_x,i_y]$ of (2.75) and (2.77) are related by

$$b[i_x,i_y] = \Delta x \Delta y \ \text{DFT}\,(a[n_x n_y]) \tag{2.78}$$

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Chapter 3
Static Problems

3.1 INTRODUCTION

As was pointed out in Chapter 1, conjugate gradient algorithms require the computation of a large number of convolutions. The efficiency of the CG-FFT method is due to the fact that the integrals of the convolutions are computed by means of FFTs. This makes it possible to reduce the CPU time and memory storage requirements by orders of magnitude. In this chapter, we will lay down the guidelines for computing these integrals by using FFTs. To simplify the presentation, our treatment will be restricted to convolution integrals without derivatives. Such integrals appear when solving electrostatic and magnetostatic problems. We will consider 1D integrals. Although they are simpler than higher dimensional cases, they allow us to present the three main techniques that will be used to compute the convolutions by means of FFTs [1–5].

The first technique is applicable to finite-sized problems in which an analytical expression for the operator equation is available in the spatial domain. In this case, an equivalent discrete periodic problem will be found [6–13]. The convolutions will involve periodic discrete functions which can be computed in a cyclic sense by using the FFT.

The second technique is applicable to periodic problems. In this case, by exploiting the discrete nature of the CFT of the original problem, it is easy to find an equivalent operator equation that relates the periodic and discrete functions [14,15]. To do this, we only need to sample the original periodic problem in the spatial domain [16]. The convolutions are computed by applying the CFT to the periodic trains of delta functions found in both domains. As was shown in Chapter 2, these CFTs can be computed efficiently by means of the FFT.

The third technique is useful for finite-sized structures in which closed analytical expressions for the operator are not available in the spatial domain, but where it is assumed that analytical expressions for the Green's functions can easily be found in the spectral domain. To analyze these structures, we will define an equivalent periodic problem that is obtained by replaying a portion (window) of the original

Green's function at regular distances [17–21]. Once the equivalent periodic problem has been defined, the solution is obtained by using the second procedure.

These three ways of computing the convolution will be presented, using a generic expression for the Green's function. In order to discretize the operator, a set of basis and testing functions must be employed. Although any kind of basis or testing function can be considered in the formulations, pulse and delta functions have been considered throughout the chapter. The only reason for choosing these particular functions is to simplify the figures.

As will be discussed in the concluding remarks of this chapter, the three techniques presented in this chapter can be employed with any kind of basis or testing functions. In fact, it will be seen that it is very easy to change from one kind of basis or testing function to another.

We should also point out that although some very specific basis and testing functions (rooftops and razor blades) have been considered in the formulation of electrodynamic problems throughout the book, the use of other functions would also represent a minor change in this case.

There are two appendixes at the end of the chapter. In one of them the effect of reducing the length of the zero padding in the analysis of individual structures is considered. The other appendix outlines a trick that can be used to reduce the memory storage required to compute the FFT of zero-padded sequences.

3.2 FORMULATING A 1D CONTINUOUS CONVOLUTION PROBLEM FOR INDIVIDUAL STRUCTURES

In applied electromagnetics, problems involving individual structures arise frequently (e.g., nonperiodic problems such as computing the electrical potential due to a metallic plate having a known charge density). Such problems lead to continuous operator equations. Using signal processing terminology, we are dealing with LTI systems [22] because the source superposition principle (linearity) is assumed and the Green's function (impulse response) is invariant with the position (time) of the point source (impulse excitation). In other words, the Green's function only depends on the relative positions of the source and observation points.

The response of an LTI system to a generic excitation (arbitrary charge density) is given by the convolution of the impulse response and the excitation. By performing the CFT, this convolution becomes a product between the Green's function and the charge density. This CFT provides new intuitive interpretations of the problem and also makes it easier to perform some elementary operations like the spatial domain derivatives that appear in electrodynamics. However, a direct evaluation of the CFT can require as many computations as a direct evaluation of this convolution in the spatial domain. Of course, we will take advantage of the Fourier transform if we consider the FFT, but this transform, which is just a fast algorithm for computing

the DFT, is only applicable to discrete periodic functions. Therefore, if the computation of convolutions is to be speeded up by means of the FFT, the operator equation will have to be discretized and made periodic.

Here we propose converting the original problem of an individual structure into an equivalent period problem (EPP). The main steps of this process will be indicated for both the spatial domain (Section 3.2.1) and the spectral domain (Section 3.2.2). Once the EPP has been obtained, it will be relatively easy to discretize the operator and the functions so as to obtain a convolution between discrete and periodic functions that can be computed efficiently by means of FFTs.

3.2.1 Spatial Domain Formulation of the Continuous EPP

Our objective is to compute the function $\phi(x)$ resulting from the convolution of functions $G(x)$ and $\rho(x)$:

$$\phi(x) = \int_{-\infty}^{\infty} \rho(x')\, G(x - x')\, dx' \tag{3.1}$$

Here it can be assumed that $\phi(x)$ represents an electric potential and $G(x - x')$ is the Green's function that gives the potential at point x due to a source located at point x'. Figure 3.1 depicts an arbitrary Green's function extending over the entire range of the variable x, $(-\infty,\infty)$. For the sake of simplicity, the function pictured is symmetric with respect to the origin, $x = 0$ (i.e., it is an even function). However, this symmetry does not imply any restriction on the choice of Green's function, since the development is also applicable to nonsymmetric Green's functions.

For the sake of simplicity in the drawings, we will restrict our consideration to space-limited functions, such as the one depicted in Figure 3.2, for the source function $\rho(x)$ of (3.1). Although the source function shown in this figure is a triangle function, the following development can be applied to more complex space-limited source functions.

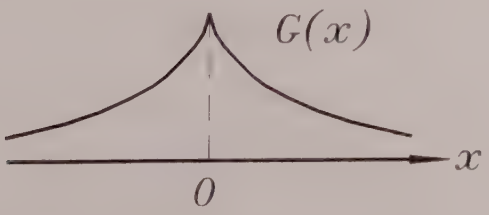


Figure 3.1 Space-unlimited Green's function $G(x)$.

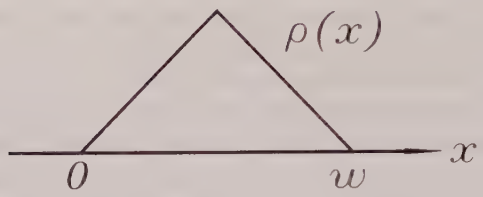


Figure 3.2 Space-limited source function $\rho(x)$.

The following graphical scheme can be used to compute the integral of (3.1).

- First, the $\rho(x')$ and $G(x - x')$ are drawn (Figures 3.3(a) and 3.3(b)). The function G is shifted to the point x where $\phi(x)$ is to be evaluated.
- Second, the functions $G(x - x')$ and $\rho(x')$ are multiplied to obtain the product function $h(x,x')$ as is shown in Figure 3.3(c).

$$h(x,x') = \rho(x')G(x - x') \tag{3.2}$$

- Third, $\phi(x)$ is computed as the area of function $h(x,x')$:

$$\phi(x) = \int_{-\infty}^{\infty} h(x,x')\, dx' \tag{3.3}$$

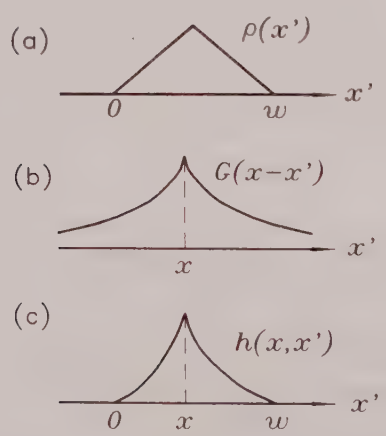


Figure 3.3 Graphical scheme for computing the convolution of functions $\rho(x)$ and $G(x)$.

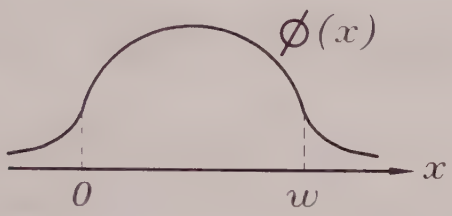


Figure 3.4 Potential function $\phi(x)$ resulting from the convolution of $\rho(x)$ and $G(x)$.

Considering different values for the parameter x in the graphical scheme, we obtain a potential function $\phi(x)$ having approximately the shape indicated in Figure 3.4.

It can be seen that the potential $\phi(x)$ is an unlimited function. But if we are interested in obtaining the exact value of this function only for the x interval $(0,w)$, we can consider the space-limited Green's function $G^L(x)$ sketched in Figure 3.5. $G^L(x)$ is obtained by multiplying the space-unlimited Green's function G by the pulse $P_W(x)$; that is,

$$G^L(x) = P_W(x)\, G(x) \tag{3.4}$$

where

$$P_W(x) = \begin{cases} 1 & |x| < w \\ 0 & \text{elsewhere} \end{cases} \tag{3.5}$$

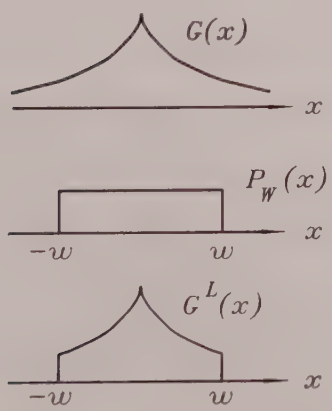


Figure 3.5 Space-limited Green's function obtained by multiplying the function $G(x)$ by the pulse $P_W(x)$.

Using signal processing terminology, it can be said that $G^L(x)$ is obtained by windowing the function $G(x)$ with $P_W(x)$. Examining Figure 3.6, it appears evident that the product function $h^L(x,x')$

$$h^L(x,x') = \rho(x')\, G^L(x,x') \tag{3.6}$$

has exactly the same values as $h(x,x')$ for x in the interval $(0,w)$. Therefore, the space-limited potential function $\phi^L(x)$

$$\phi^L(x) = \int_{-\infty}^{\infty} h^L(x,x')\, dx' \tag{3.7}$$

is completely equal to $\phi(x)$ in the interval of interest $(0,w)$, as can be seen in Figure 3.7

From a practical point of view, it is more convenient to consider $G^L(x)$ than $G(x)$ because the space-limited Green's function provides a more economical procedure for computing the potential of x in the interval $(0,w)$. The scheme can be further improved if we consider the periodic Green's function $G^P(x)$ (see Figure 3.8), which is obtained from $G^L(x)$ by using

$$G^P(x) = \sum_{m=-\infty}^{\infty} G^L(x - mT) \tag{3.8a}$$

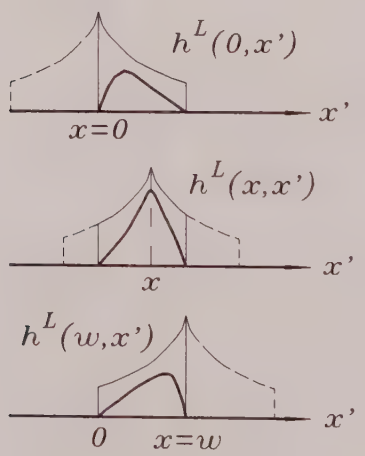


Figure 3.6 Shape of the function $h(x,x')$, considering x' as a variable, for three values of x in the interval $(0,w)$.

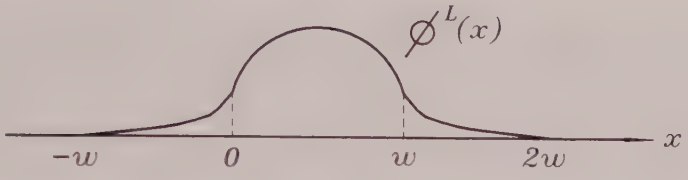


Figure 3.7 Potential function $\phi^L(x)$ resulting from the convolution of $\rho(x)$ and $G^L(x)$.

where m is an integer and T, the period of G^P, is given by

$$T = 2w \tag{3.8b}$$

Since $G^P(x)$ is identical to $G^L(x)$ in the interval $(-w,w)$, the potential $\phi^P(x)$ resulting from the convolution of $G^P(x)$ by $\rho(x)$,

$$\phi^P(x) = \int_{-\infty}^{\infty} \rho(x')\, G^P(x - x')\, dx' \tag{3.9}$$

has the same values as $\phi^L(x)$ for x in the interval $(0,w)$, as is shown in Figure 3.9. It can be seen that $\phi^P(x)$ is also periodic with the same period as $G^P(x)$. This is the EPP we were looking for, and it will prove to be very easy to discretize.

An additional step in the strategy of computing convolutions by means of the FFT consists of also considering a periodic function $\rho^P(x)$ for the source function (Figure 3.10),

$$\rho^P(x) = \sum_{m=-\infty}^{\infty} \rho(x - mT) \tag{3.10}$$

and computing the convolution in a cyclic way:

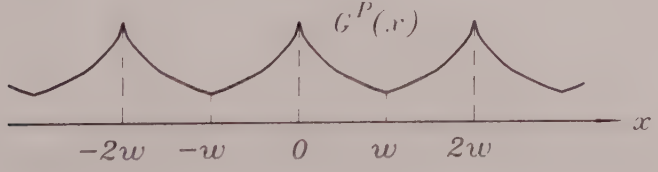


Figure 3.8 Periodic Green's function $G^P(x)$ obtained by replaying at regular intervals the space-limited function $G^L(x)$.

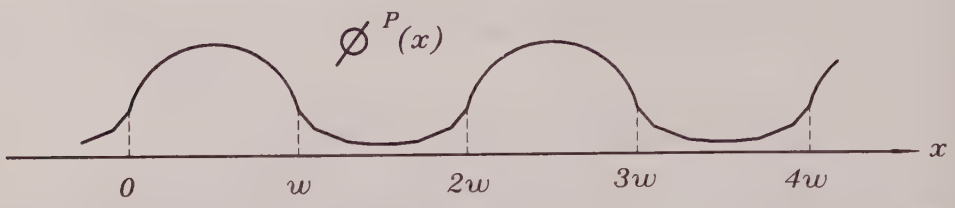


Figure 3.9 Periodic potential function $\phi^P(x)$ resulting from the convolution of $\rho^P(x)$ and $G^P(x)$.

$$\phi^P(x) = \int_0^T \rho^P(x')\, G^P(x - x')\, dx' \tag{3.11}$$

As can be seen in Figure 3.10, the periodic potential $\phi^P(x)$ is identical to $\phi^L(x)$ or $\phi(x)$ in the interval of interest $(0,w)$. The advantage of using $G^P(x)$, $\rho^P(x)$, and $\phi^P(x)$ is that these functions can be discretized and the convolutions computed by means of the FFT using (2.12) of Chapter 2, as will be shown in Section 3.3.

The period T must be at least twice as long as w to ensure that $\phi^P(x)$ is identical to $\phi^L(x)$ in the interval $(0,w)$. This period length guarantees accuracy while minimizing the cost of computing the convolutions by means of FFTs. If T is less than twice w, an error is introduced, which, depending on the shape of the Green's function, can be very significant. This point is discussed in detail in Appendix 3A. A trick that reduces the sizes of the FFT required is presented in Appendix 3B.

3.2.2 Spectral Domain Formulation of the EPP

The steps in the procedure for obtaining the potential functions $\phi(x)$, $\phi^L(x)$, and $\phi^P(x)$ have an interesting and intuitive interpretation in the spectral domain. We will

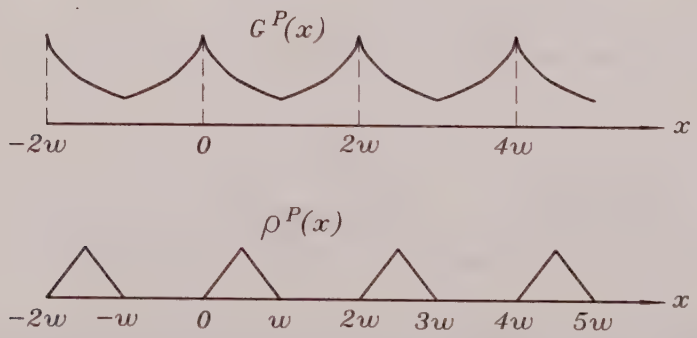


Figure 3.10 Periodic functions $G^P(x)$ and $\rho^P(x)$ for the cyclic convolution.

use $A(\tilde{k}_x)$ to represent the CFT of a spatial domain function $A(x)$, where k_x is the spectral coordinate. The direct and inverse CFTs were defined in expressions (2.25) and (2.26) of Chapter 2. Figure 3.11 shows the shape of the CFTs of the functions $G(x)$, $\rho(x)$, and $\phi(x)$. In the spectral domain, the function $\tilde{\phi}(k_x)$ is obtained by multiplying $\tilde{\rho}(k_x)$ by $\tilde{G}(k_x)$:

$$\tilde{\phi}(k_x) = \tilde{\rho}(k_x)\,\tilde{G}(k_x) \tag{3.12}$$

This equation is the CFT of the convolution of $\phi(x)$ by $G(x)$, which appears in (3.1). Using a more compact notation, we can rewrite (3.1) as

$$\phi(x) = \rho(x) * G(x) \tag{3.13}$$

where the symbol * indicates convolution. To find $\tilde{G}^L(k_x)$, we must perform the CFT of the space-limited Green's function $G^L(x)$, which is obtained by multiplying $G(x)$ by the window function $P_W(x)$ in the spatial domain. Therefore, in the spectral domain, $\tilde{G}^L(k_x)$ is obtained by means of the convolution

$$\tilde{G}^L(k_x) = \frac{1}{2\pi}\,\tilde{P}_W(k_x) * \tilde{G}(k_x) \tag{3.14}$$

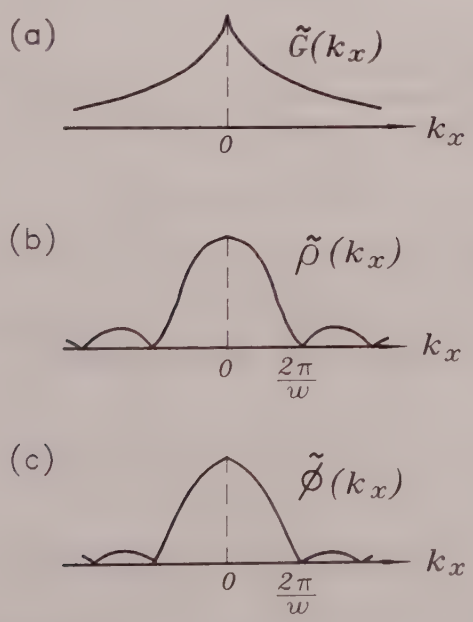


Figure 3.11 Shape in the spectral domain of the CFTs of (a) $G(x)$, (b) $\rho(x)$, and (c) $\phi(x)$.

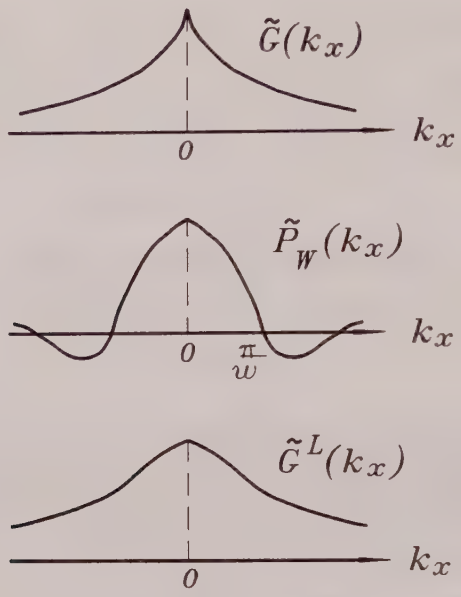


Figure 3.12 The CFT of $G^L(x)$ is obtained by a convolution in the spectral domain of $\tilde{G}(k_x)$ and $\tilde{P}_w(k_x)$.

Figure 3.12 shows the shape of $\tilde{G}^L(k_x)$. It can be observed that $\tilde{G}^L(k_x)$ is smoother than $\tilde{G}(k_x)$, as is to be expected after performing a convolution.

$\tilde{\phi}^L(k_x)$ is obtained by multiplying $\tilde{\rho}(k_x)$ by $\tilde{G}^L(k_x)$ in the spectral domain.

$$\tilde{\phi}^L(k_x) = \tilde{\rho}(k_x)\,\tilde{G}^L(k_x) \tag{3.15}$$

Figure 3.13 illustrates $\tilde{\phi}^L(k_x)$. The shape of this function is different from that of $\tilde{\phi}(k_x)$, although $\tilde{\phi}^L(k_x)$ is also an unlimited function.

The CFT of the periodic function $G^P(x)$ can be obtained easily by performing the convolution of $G^L(x)$ with the sampling function $\mathit{\Pi}_T(x)$:

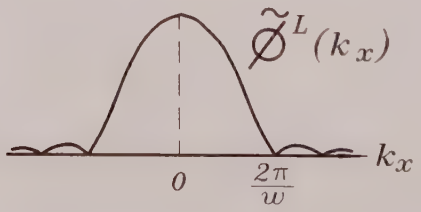


Figure 3.13 CFT of the space-limited potential $\phi^L(x)$.

$$G^P(x) = \text{Ш}_T(x) * G^L(x) \tag{3.16}$$

where $\text{Ш}_T(x)$ is the train of delta functions,

$$\text{Ш}_T = \sum_{m=-\infty}^{\infty} \delta(x - mT) \tag{3.17}$$

and T is the period of $G^P(x)$. This procedure for making the Green's function periodic is shown in Figure 3.14. In the spectral domain, the convolution of (3.16) turns into a sampling (see Figure 3.15):

$$\tilde{G}^P(k_x) = \frac{2\pi}{T} \text{Ш}_{2\pi/T}(k_x)\, \tilde{G}^L(k_x) \tag{3.18}$$

where $\text{Ш}_{2\pi/T}(k_x)$ is a train of delta functions whose impulses are spaced at intervals of $2\pi/T$. It must be recalled that the CFT of a train of delta functions is another train of deltas,

$$\text{CFT}(\text{Ш}_T(x)) = \frac{2\pi}{T} \text{Ш}_{2\pi/T}(k_x) \tag{3.19}$$

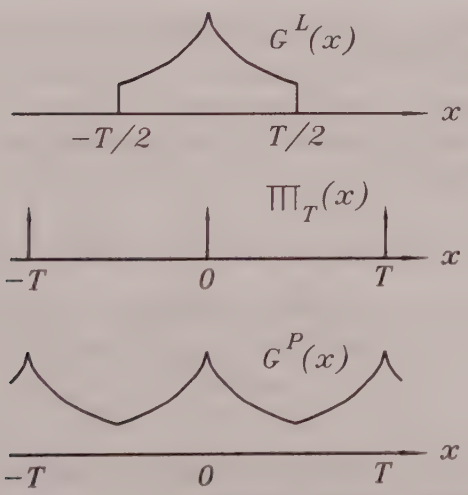


Figure 3.14 The periodic function $G^P(x)$ can be obtained by performing the convolution of the space-limited function $G^L(x)$ and the train of delta functions $\text{Ш}_T(x)$.

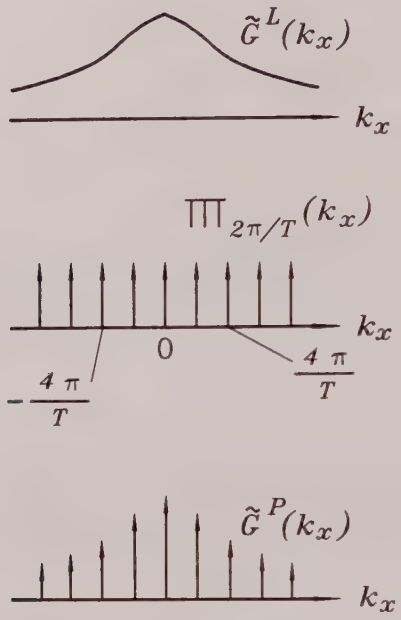


Figure 3.15 Function $\tilde{G}^P(k_x)$ is obtained by sampling $\tilde{G}^L(k_x)$ with the train of delta functions $\amalg_{2\pi/T}(k_x)$.

and therefore the spectrum of $G^P(x)$ is an impulse train whose amplitudes are modulated by $\tilde{G}^L(k_x)$ (see Figure 3.15):

$$\tilde{G}^P(k_x) = \Delta k \sum_{m=-\infty}^{\infty} \tilde{G}^L(k_{xm})\, \delta(k_x - k_{xm}) \tag{3.20}$$

where

$$k_{xm} = m\Delta k \tag{3.21a}$$

$$\Delta k = \frac{2\pi}{T} \tag{3.21b}$$

Since the spectral potential $\tilde{\phi}^P(k_x)$ is the product of the CFT of the source function $\tilde{\rho}(k_x)$ and the sampled Green's function $\tilde{G}^P(k_x)$,

$$\tilde{\phi}^P(k_x) = \tilde{\rho}(k_x)\, \tilde{G}^P(k_x) \tag{3.22}$$

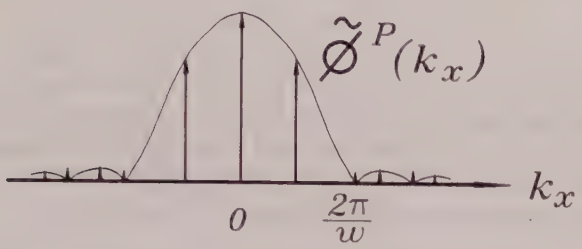


Figure 3.16 The CFT of $\phi^P(x)$ is a train of delta functions modulated by $\tilde{\phi}^L(k_x)$.

it can easily be seen that the spectral potential $\tilde{\phi}^P(k_x)$ is a train of delta functions whose amplitudes are modulated by $\tilde{\phi}^L(k_x)$:

$$\begin{aligned}\tilde{\phi}^P(k_x) &= \Delta k \sum_{m=-\infty}^{\infty} \tilde{\rho}(k_{xm})\, \tilde{G}^L(k_{xm})\, \delta(k_x - k_{km}) \\ &= \Delta k \sum_{m=-\infty}^{\infty} \tilde{\phi}^L(k_{xm})\, \delta(k_x - k_{xm})\end{aligned} \tag{3.23}$$

as is shown in Figure 3.16.

3.3 DISCRETIZATION OF THE CONTINUOUS EPP

In this section, we will proceed to discretize the EPP, which was obtained for individual problems in Section 3.2. The discretization is performed entirely in the spatial domain. Although the CFT is not required in this discretization scheme, Section 3.3.2 presents a Fourier analysis of all the steps considered in the discretization. This analysis provides some rather new and intuitive insights into the whole process.

3.3.1 Spatial Domain Formulation of the Discretization Process

We will now obtain a discrete version of the EPP defined in (3.9) or (3.11) in order to apply the FFT. First, the continuous source function $\rho(x)$ is approximated in terms of basis functions:

$$\rho(x) \approx s(x) = \sum_{i=1}^{N} a_i b_i(x) \tag{3.24}$$

where $b_i(x)$ is obtained by shifting a generating basis function $b(x)$ by $(i - 0.5)\Delta x$:

$$b_i(x) = b(x - (i - 0.5)\Delta x) \tag{3.25}$$

As this last expression indicates, the set of basis functions is obtained by replaying the generating function at regular intervals so that the FFT can be used to compute the convolutions. However, we have some degree of freedom in choosing the generating basis functions. For the sake of simplicity in the presentation, we will consider pulses as basis functions (see Figures 3.17 and 3.18).

$$s(x) = \sum_{i=1}^{N} a_i P_i(x) = \sum_{i=1}^{N} a_i P(x - x_i) \tag{3.26}$$

where

$$P(x) = \begin{cases} 1 & |x| < \dfrac{\Delta x}{2} \\ 0 & \text{elsewhere} \end{cases} \tag{3.27a}$$

and

$$x_i = (i - 0.5)\,\Delta x \tag{3.27b}$$

The approximate source $s(x)$ can be obtained by first sampling $\rho(x)$ with a train of delta functions,

$$s^a(x) = \rho(x)\, \text{Ш}_{\Delta x}(x - \Delta x/2) \tag{3.28}$$

where

$$\text{Ш}_{\Delta x}(x - \Delta x/2) = \sum_{m=-\infty}^{\infty} \delta(x - x_m) \tag{3.29}$$

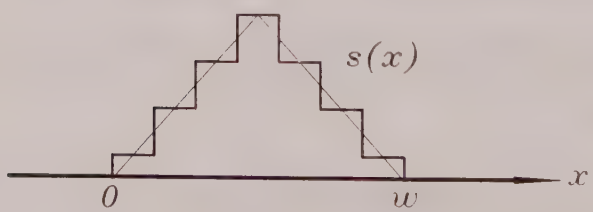


Figure 3.17 Shape of the approximate source function $s(x)$ (thick line) in comparison with the original source function $\rho(x)$ (thin line).

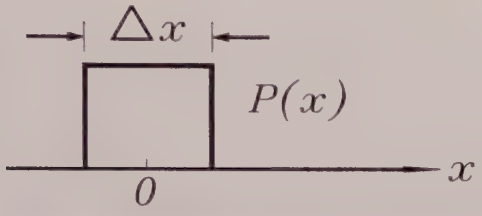


Figure 3.18 Pulse taken as basis function in the expansion of the source $\rho(x)$.

and then performing a convolution with $P(x)$:

$$s(x) = s^a(x) * P(x) \tag{3.30}$$

Figure 3.19 shows the sampling function $\mathit{III}_{\Delta x}(x - \Delta x/2)$ and the intermediate source function $s^a(x)$.

Expanding (3.30) in series form, we have

$$s(x) = \sum_{i=1}^{N} \rho(x_i)\, P(x - x_i) \tag{3.31}$$

This can be written in a more compact form as

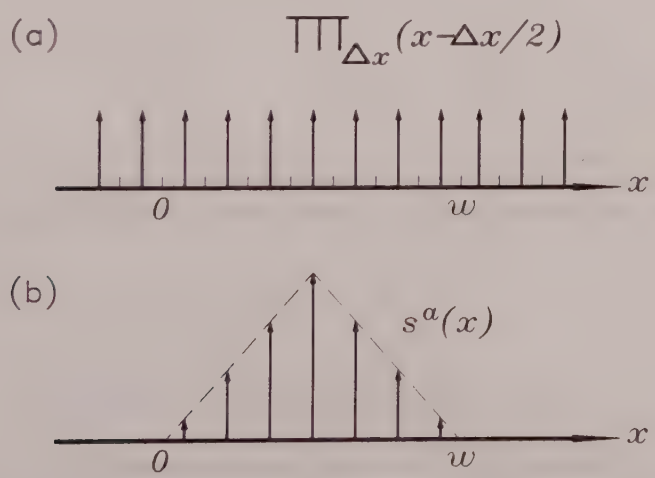


Figure 3.19 (a) Sampling function $\mathit{III}_{\Delta x}(x - \Delta x/2)$ and (b) intermediate source function $s^a(x)$ considered to obtain the approximate source $s(x)$.

$$s(x) = \sum_{i=1}^{N} s^D[i]\, P(x - x_i) \tag{3.32}$$

where

$$s^D[i] = \rho(x_i) \tag{3.33}$$

is the discrete version of $\rho(x)$ that we were looking for. By extension, we can define the discrete replica of $\rho^P(x)$:

$$s^P[i] = \rho^P(x_i) \tag{3.34}$$

Figures 3.20 and 3.21 illustrate the discrete functions $s^D[i]$ and $s^P[i]$.

Approximating the source function in (3.9) by the pulse expansion given in (3.32), we have

$$\phi(x) \approx f^a(x) = \int_{-\infty}^{\infty} s(x')\, G^P(x - x')\, dx' \tag{3.35}$$

where

$$\begin{aligned} f^a(x) &= \sum_{j=1}^{N} s^D[j] \int_{-\infty}^{\infty} G^P(x - x')\, P_j(x')\, dx' \\ &= \sum_{j=1}^{N} s^D[j]\, g_j(x) \end{aligned} \tag{3.36}$$

Figure 3.22 shows $g_j(x)$, which can be considered as a basis function for the expansion of the approximate potential $f^a(x)$ in terms of more elementary functions.

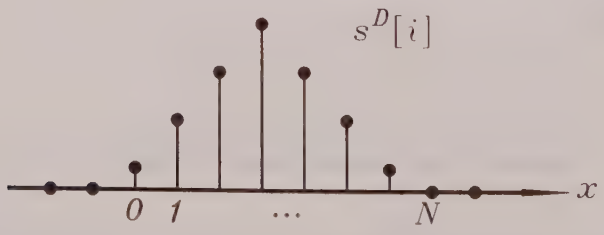


Figure 3.20 Discrete source function $s^D[i]$.

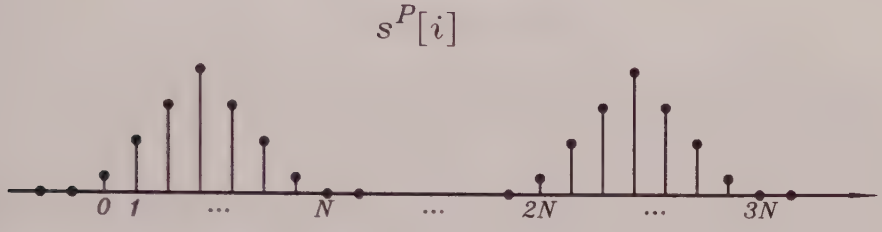


Figure 3.21 Discrete periodic source function $s^P[i]$.

Since $P_j(x)$ is a constant function over its definition interval, we can write the following expression for $g_j(x)$:

$$g_j(x) = \int_{(j-1)\Delta x}^{j\Delta x} G^P(x - x')\, dx' = G^P(x) * P(x - x_j) \tag{3.37}$$

The basis $g_j(x)$ can have sharp peaks, especially when the Green's function includes derivatives, as usually occurs in electrodynamics, which is not very suitable for expanding a smooth function like an impressed field component. To smooth g_j, we can make the convolution between this function and the window function $t(x)$ as indicated in (3.38),

$$g_j^b(x) = \int_{-\infty}^{\infty} g_j(x')\, t(x - x')\, dx' = g_j(x) * t(x) \tag{3.38}$$

and

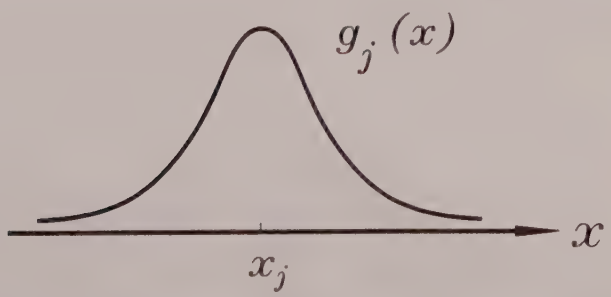


Figure 3.22 Elementary potential function $g_j(x)$ used to expand the approximate function $f^a(x)$ in the interval $(0,w)$.

$$f^b(x) = f^a(x) * t(x) \tag{3.39}$$

Expression (3.39) shows the effect of this windowing on the approximate potential function. Figure 3.23 illustrates this windowing effect. In electrostatics, where derivatives do not usually appear in the operator, the convolution of (3.39) is not required. In electrodynamics, however, these convolutions are very common. The window function $t(x)$ is usually called the testing function in the MM. When using, for instance, a Galerkin procedure in the MM with triangles, the testing function $t(x)$ is of course a triangle. In a point-matching procedure, no windowing is performed; in other words, $t(x)$ becomes a delta function, which is equivalent to saying that in (3.38), $g_j^b(x)$ and $g(x)$ would be identical.

The next step in the discretization scheme is to sample with the train of delta functions defined in (3.29) (see Figure 3.24).

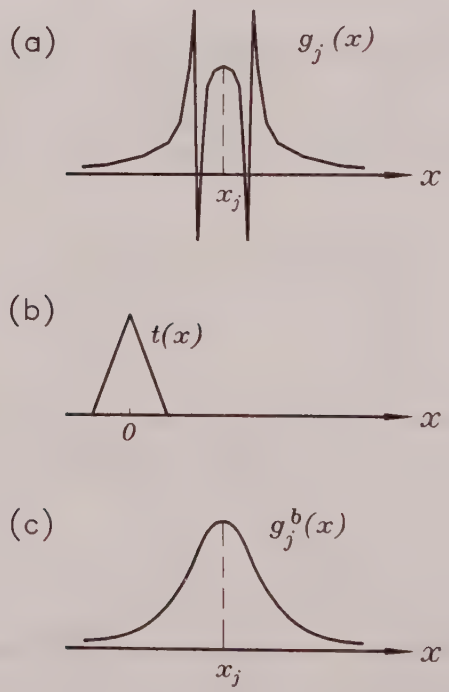


Figure 3.23 In many electrodynamic problems, the function $g_j(x)$ will have sharp peaks like those shown in (a). By performing a convolution with the testing function $t(x)$ pictured in (b), the original (sharp) function $g_j(x)$ is converted into $g_j^b(x)$, the smooth function in (c).

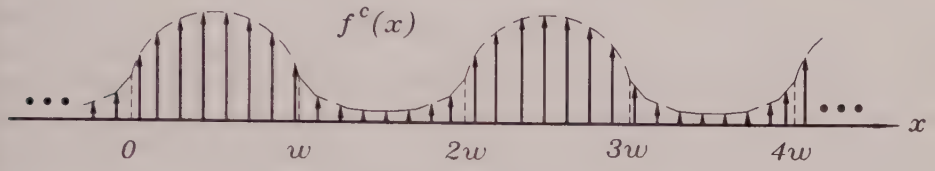


Figure 3.24 Sampled potential function $f^c(x)$.

$$f^c(x) = f^b(x)\, \text{Ш}_{\Delta x}(x - \Delta x/2) = \sum_{i=-\infty}^{\infty} f^b(x_i)\, \delta(x - x_i) \tag{3.40}$$

We can obtain a periodic discrete function $f^P[i]$, which is the replica corresponding to $\phi(x)$ in the discrete space. This function $f^P[i]$, depicted in Figure 3.25, is obtained from

$$f^c(x) = \sum_{i=-\infty}^{\infty} f^P[i]\, \delta(x - x_i) \tag{3.41}$$

$$f^P[i] = f^b(x_i) \tag{3.42}$$

Now we group the terms involved in the computation of $f^c(x)$ as follows.

$$\begin{aligned} f^c(x) &= \sum_{j=1}^{N} s^D[j] \sum_{i=-\infty}^{\infty} g_j^b(x_i)\, \delta(x - x_i) \\ &= \sum_{i=-\infty}^{\infty} \left(\sum_{j=1}^{N} s^D[j]\, g_j^b(x_i) \right) \delta(x - x_i) \end{aligned} \tag{3.43}$$

To simplify, we assume that $g_j^b(x)$ and $g(x)$ are identical; in other words, the testing function is a delta function (point matching, in MM terminology):

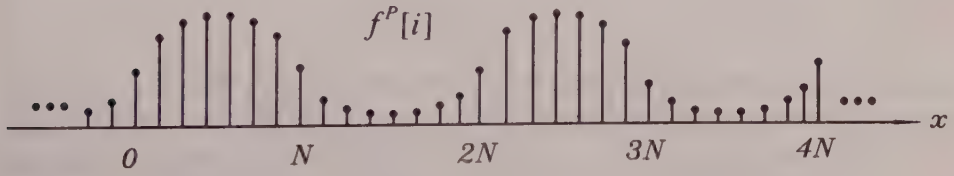


Figure 3.25 Discrete periodic function $f^P[i]$.

$$g_j^b(x) = g_j(x) * \delta(x) = g_j(x) \tag{3.44}$$

Finally, we can write a discrete version of (3.9) by relating the amplitudes of the delta function expansions in (3.41) and (3.43):

$$f^P[i] = \sum_{j=1}^{N} s^D[j]\, g^P[i-j] \tag{3.45}$$

where $g^P[i-j]$ is the discrete periodic Green's function:

$$g^P[i-j] = \int_{(j-1)\Delta x}^{j\Delta x} G^P(x_i - x')\, dx' \tag{3.46}$$

Figure 3.26 shows a graphic visualization of $g^P[i]$.
Equation (3.45) can be written as a linear convolution between $s^D[i]$ and $g^P[i]$,

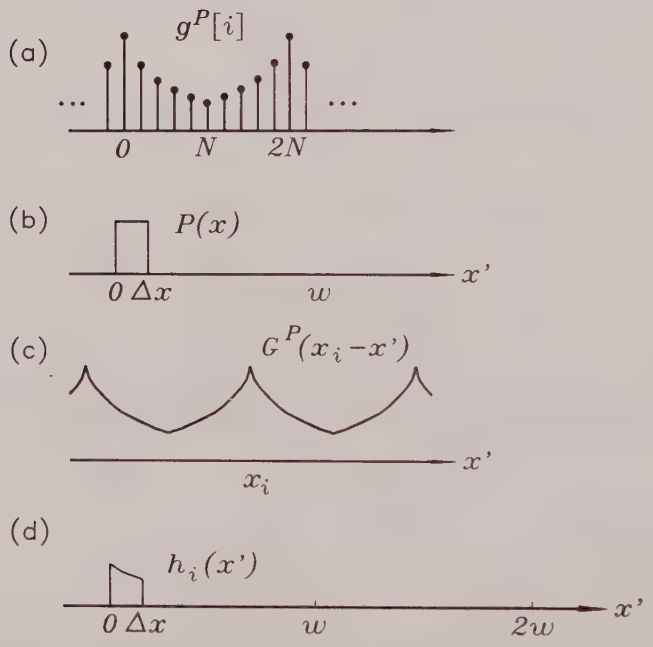


Figure 3.26 If $G^P(x)$ is considered to be an even function (i.e., $G^P(x) = G^P(-x)$), (a) the discrete periodic function $g^P[n]$ can be obtained by multiplying (b) $P(x)$ by (c) $G^P(x_i - x')$, and (d) integrating the function $h_i(x)$ resulting from this multiplication.

$$f^P[i] = \sum_{j=-\infty}^{\infty} s^D[j]\, g^P[i-j] \tag{3.47}$$

or in a more compact form,

$$f^P[i] = s^D[i] * g^P[i] \tag{3.48}$$

Assuming that we wish to compute the convolutions by means of FFTs, it is only necessary to write the discrete source as a periodic function by padding with zeros,

$$s^P[i] = \begin{bmatrix} s^D[i] & \text{for} \quad 0 \le i < N \\ 0 & \text{for} \quad N \le i < 2N \end{bmatrix} \tag{3.49}$$

in order to obtain $f^P[i]$ from a cyclic convolution,

$$f^P[i] = \sum_{j=0}^{2N-1} s^P[j]\, g^P[i-j] \tag{3.50}$$

or using the same notation for cyclic convolutions as in Chapter 2,

$$f^P[i] = s^P[i] \otimes g^P[i] \tag{3.51}$$

We have already shown how to compute this last convolution by means of FFT (Chapter 2):

$$f^P[i] = \frac{1}{2N} \mathrm{DFT}_{2N}^{-1} \left(\bar{s}^P[v]\, \bar{g}^P[v]\right) \tag{3.52}$$

where

$$\bar{s}^P[v] = \mathrm{DFT}_{2N}\left(s^P[i]\right) \tag{3.53a}$$

$$\bar{g}^P[v] = \mathrm{DFT}_{2N}\left(g^P[i]\right) \tag{3.53b}$$

and DFT_{2N} indicates a discrete Fourier transform of length $2N$ that is assumed to be performed by means of an FFT algorithm. It should be pointed out that the period N_T of all functions in (3.49) through (3.55) is

$$N_T = 2N \tag{3.54}$$

This value represents a minimum for computing $f^P[n]$ accurately by means of the FFT, assuming that the function is to be evaluated in the discrete interval $(0, N-1)$. Of course, larger values of N_T are allowable, but will involve greater computational costs. Smaller values for N_T are not recommended, because aliasing will appear in the spatial domain.

Table 3.1 summarizes the parallelism between the continuous and the discrete problems.

In order to compute the initial convolution of (3.1) accurately in the interval of interest $(0,w)$, we have to define periodic functions with a generating cell of length $2N$, which means using twice the number of sampling points considered to approximate the continuous functions in the interval $(0,w)$. Coding the scheme will involve storing matrices of length $2N$. At first glance, it may seem that this duplication in length will lead to an unnecessary increase in the CPU time and memory storage requirements. However, it will be shown graphically in Appendix 3.A that this duplication is necessary to avoid aliasing in the spatial domain when an FFT is used to compute the transform of the convolution of functions sampled at regular intervals. Nevertheless, to avoid a needless increase in memory requirements, Appendix 3.B presents a method of computing the FFT of sequences having a length of $2N$ as two FFTs of length N. This method is applicable when dealing, as in our case, with periodic functions of length $2N$ obtained by padding with zero functions initially of length N. The procedure in this appendix can be used to reduce memory requirements, but it increases the CPU time. In any case, it should be recalled that direct evaluation of (3.1) would require memory storage and CPU time proportional to N^2

Table 3.1
Continuous and Discrete Versions of the Problem

Continuous Version	*Discrete Version*
$\phi^P(x)$	$f^P[i]$
$G(x)$	$g^P[i]$
$\rho(x)$	$s^D[i]$
$\rho^P(x)$	$s^P[i]$
Continuous interval $(0,w)$	Discrete interval $[0,N-1]$
Continuous period $T \geq 2w$	Discrete period $N_T \geq 2N$
$\int_{-\infty}^{\infty} \rho(x)\, G^P(x-x')\, dx'$	$\sum_{n'=-\infty}^{\infty} s^D[n']\, g^P[n-n']$
$\int_0^T \rho^P(x)\, G^P(x-x')\, dx'$	$\sum_{n'=0}^{N_T-1} s^P[n']\, g^P[n-n']$

(assuming that, as in the MM, the discretized Green's function is stored so that it will not have to be recomputed each time a convolution is performed).

Now, assuming that function $f^P[i]$ is obtained by computing the FFT indicated in (3.52) and (3.53), we consider the final shape of the approximate potential function. The reader may be concerned about the sharp peaks in $f^c(x)$, because this function is a train of delta functions. Although the shape of this function may appear to be rather strange, $f^c(x)$ contains all the information required for an approximate solution to the EPP. For instance, if we are going to proceed to perform an inner product with $f^c(x)$, this delta function expansion is excellent for that purpose. If we wish to obtain a smooth representation of the approximate potential, we can proceed to perform a convolution of $f^c(x)$ with a suitable basis function $z(x)$:

$$f^d(x) = f^c(x) * z(x) \tag{3.55}$$

The new function $f^d(x)$ has the more common form of

$$f^d(x) = \sum_{i=-\infty}^{\infty} f^b(x_i)\, z(x - x_i) \tag{3.56}$$

Usually the testing function $t(x)$ is employed for $z(x)$. This choice is not necessary, however, nor are the last steps indicated in (3.55) and (3.56) strictly necessary, because $f^c(x)$ has as much information as $f^d(x)$.

3.3.2 Spectral Domain Representation of the Discretization Process

In the discretization process presented in Section 3.3.1, it was not necessary to use the CFT because all the steps were carried out in the spatial domain and the final discrete convolutions were computed by means of DFTs. As was pointed out in Chapter 2, the DFT should not be regarded as a special case of the CFT; although these transforms are related, each one is a separate entity. However, applying concepts and properties of the CFT to the discretization scheme of Section 3.3.1 provides an interesting perspective on many of the steps in the process and gives insight into the causes of error.

First, let us consider the approximation for the source function in terms of basis functions given by (3.34). Following the same order as in Section 3.1.1, the CFT of preliminary function $s^a(x)$ is given by

$$\begin{aligned} \tilde{s}^a(k_x) &= \frac{1}{2\pi} \mathrm{CFT}(\mathit{III}_{\Delta x}(x - \Delta x/2)) * P(k_x) \\ &= \frac{1}{\Delta x} \sum_{m=-\infty}^{\infty} \tilde{\rho}(k_x - mT_k) \exp(-j\pi m) \end{aligned} \tag{3.57}$$

with

$$T_k = \frac{2\pi}{\Delta x} \tag{3.58}$$

which can be written as

$$\bar{s}^a(k_x) = \frac{1}{\Delta x} \sum_{m=-\infty}^{\infty} (-1)^m \, \tilde{\rho}(k_x - mT_k) \tag{3.59}$$

Examination of this expression indicates that $\bar{s}^a(k)$ is periodic, which is to be expected, since $s^a(x)$ is obtained by sampling with a train of delta functions. As is shown in Figure 3.27, this periodic function is obtained by replaying the CFT of $\rho(x)$ at regular intervals of length T_k. Observing this figure, we immediately realize that aliasing is unavoidable if $\tilde{\rho}(k_x)$ is a space-unlimited function; in any spectral band there will be reiterative overlapping of successive replicas of $\tilde{\rho}(k)$. Moreover,

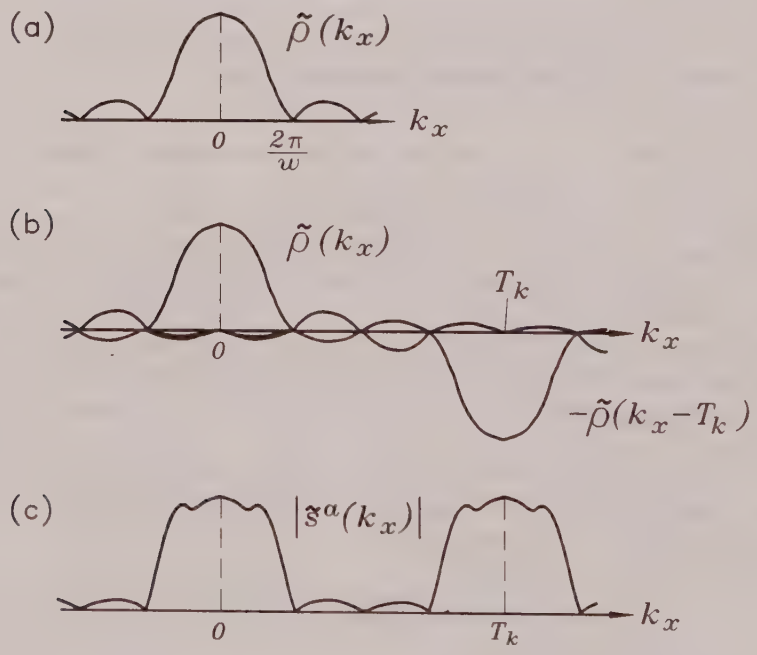


Figure 3.27 The spectral source function shown in (a) is repeated at regular intervals shown in (b), giving the periodic function $\bar{s}^a(k_x)$, shown in (c).

for any problem with a source region of finite length (i.e., any problem with individual structures) and for nearly all problems with periodic structures, $\tilde{\rho}(k)$ will extend over an unlimited spectral band, and therefore $\tilde{\rho}^a(k)$ will be an aliased signal.

The next step in obtaining the approximate source function is the convolution of $s^a(x)$ with the basis function. Here we will also use pulses as basis functions in order to simplify the figures. The CFT of (3.30) is a product whose result is

$$\tilde{s}(k_x) = \frac{\tilde{P}(k_x)}{\Delta x} \sum_{m=-\infty}^{\infty} (-1)^m \tilde{\rho}(k_x - mT_k) \tag{3.60}$$

where

$$\tilde{P}(k_x) = \Delta x \operatorname{sinc}\left(k_x \frac{\Delta x}{2\pi}\right) \tag{3.61}$$

It can be seen from (3.60) or Figure 3.28 that $\tilde{s}(k_x)$ is concentrated in the lower part of the spectral axis, approximately in the same band as the CFT of the basis function. The fact that the energy of $\tilde{s}(k_x)$ is concentrated at low frequencies simplifies the following steps in the discretization process, reducing the errors significantly. However, the error introduced by the aliasing due to the sampling in (3.28) remains unaltered after performing the convolution (3.30). In fact, to reduce the aliasing effects, the spectral band length of $\tilde{\rho}(k_k)$ must be reduced by means of windowing. This windowing would imply a convolution in the spatial domain of both the source function $\rho(x)$ and the impressed potential (the unknown function in the CG-FFT operator equation) with a window. As a result of this convolution, the windowed source function would extend over all the spatial band. Such a windowing,

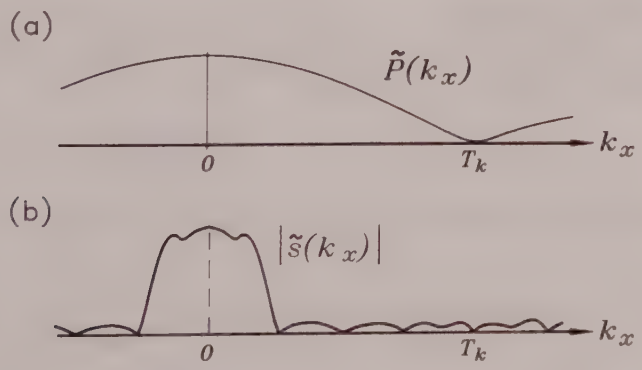


Figure 3.28 (a) The CFT of the approximate source function is obtained by multiplying $\tilde{s}^a(k_x)$ by the CFT of the basis function; (b) the resulting function $\tilde{s}(k_x)$.

which was not considered in Section 3.3.1, would probably improve the accuracy and computer efficiency of the CG-FFT or the MM. Thus, it appears that there is still room for future improvement in these methods.

From $\tilde{s}(k_x)$, we can obtain the CFT of the potential $f^a(x)$:

$$\begin{aligned}\tilde{f}^a(k_x) &= \tilde{s}_k(k_x)\,\tilde{G}^{\mathrm{P}}(k_x) \\ &= \left(\sum_{i=1}^{N} s^D[i]\,\tilde{P}(k_x)\exp(-jk_x x_i)\right)\tilde{G}^P(k_x) \\ &= \tilde{P}(k_x)\,\tilde{G}^P(k_x)\left(\sum_{i=1}^{N} s^D[i]\exp(-jk_x x_i)\right) \\ &= \tilde{g}(k_x)\,\tilde{s}^c(k_x)\exp(jk_x\Delta x/2)\end{aligned} \tag{3.62}$$

where $\tilde{g}(k_x)$ is the CFT of the elementary potential function,

$$\tilde{g}(k_x) = \tilde{G}^P(k_x)\,\tilde{P}(k_x) \tag{3.63}$$

and $\tilde{s}^c(k_x)$ is a function related to $s^D[i]$ by

$$\tilde{s}^c(k_x) = \sum_{i=1}^{N} s^D[i]\exp(-ji\Delta x k_x) \tag{3.64}$$

The term $\exp(jk_x\Delta x/2)$ appears in (3.62) due to the fact that the train of delta functions employed to expand $s^a(x)$ is shifted by $\Delta x/2$ with respect to $III_{\Delta x}(x)$.

Using these functions, we can write $\tilde{f}^b(k_x)$ as

$$\tilde{f}^b(k_x) = \tilde{g}(k_x)\,\tilde{s}^c(k_x)\,\tilde{t}(k_x)\exp(jk_x\Delta x/2) \tag{3.65}$$

Considering the expansion of $\tilde{G}^P(k_x)$ given in (3.20), we have

$$\tilde{f}^b(k_x) = \Delta k \sum_{m=-\infty}^{\infty} \tilde{G}^L(m\Delta k)\,\tilde{P}(m\Delta k)\,\tilde{s}^c(m\Delta k)\,\tilde{t}(m\Delta k)\exp\left(j\frac{m\pi}{2N}\right)\delta(k_x - m\Delta x) \tag{3.66}$$

where

$$\begin{aligned}\tilde{s}^c(m\Delta k) &= \sum_{i=1}^{N} s^D[i] \exp\left(ji\Delta x m \frac{2\pi}{2N\Delta x}\right) \\ &= \sum_{i=1}^{N} s^D[i] \exp\left(j\frac{2\pi}{2N} mi\right) \\ &= \sum_{i=1}^{2N} s^P[i] \exp\left(j\frac{2\pi}{2N} mi\right) = \tilde{s}^P[m]\end{aligned} \tag{3.67}$$

In this last expression, $\tilde{s}^P[m]$ is the DFT of the periodic function $s^P[i]$, which is obtained by padding $s^D[i]$ with zeros as indicated in (3.49).

Since $\tilde{f}^c(x)$ is obtained by the sampling indicated in (3.40), we have in the transform domain

$$\tilde{f}^c(k_x) = \frac{1}{2\pi} \tilde{f}^b(k_x) * \mathrm{CFT}(\mathit{III}_{\Delta x}(x - \Delta x/2)) \tag{3.68}$$

or more explicitly,

$$\tilde{f}^c(k_x) = \frac{1}{\Delta x} \sum_{l=-\infty}^{\infty} (-1)^l \tilde{f}^b(k_x - lT_k) \tag{3.69}$$

where $(-1)^l$ is due to the $\Delta x/2$ shift of the train of delta functions in (3.68).

Since $\tilde{f}^c(k_x)$ has been formed in the spatial domain by sampling with a train of delta functions, it is periodic. But $f^c(x)$ is also periodic because it has been generated by a convolution in the spatial domain with $G^P(x)$. Therefore, $f^c(x)$ is a periodic series of delta functions that, as was shown in Chapter 2 (Section 2.3.2.5), will have a CFT that is also a periodic series of delta functions,

$$\tilde{f}^c(k_x) = \frac{\Delta k}{\Delta x} \sum_{m=-\infty}^{\infty} \left\{ \tilde{s}^P[m] \exp\left(j\frac{m\pi}{2N}\right) \right\} \tilde{g}^P[m]\, \delta(k_x - m\Delta k) \tag{3.70}$$

$$\begin{aligned}\tilde{g}^P[m] = \sum_{l=-\infty}^{\infty} &\tilde{G}^L(\Delta k(m - l2N))\, \tilde{P}(\Delta k(m - l2N)) \\ &\cdot \tilde{t}(\Delta k(m - l2N)) = \mathrm{DFT}_{2N}(g[i])\end{aligned} \tag{3.71}$$

whose amplitudes are related to the amplitudes $f^P[i]$ of the delta series representing $f^c(x)$ in (3.41). The relation is given by (2.54) and (2.56) of Chapter 2, which in this case become

$$f^c(x) = \sum_{i=-\infty}^{\infty} (g^P[i] * s^P[i])\, \delta(x - (i - 0.5\Delta x)) \tag{3.72a}$$

$$\tilde{f}^c(k_x) = \frac{\Delta k}{\Delta x} \sum_{m=-\infty}^{\infty} \exp\left(j\frac{m\pi}{2N}\right) \tilde{g}^P[m]\, s^P[m] \tag{3.72b}$$

As has already been mentioned, the exponential in (3.72b) is due to the $\Delta x/2$ shift delay of the deltas in (3.72a) with respect to those in (2.50).

Thus, we have shown that the spatial domain discretization scheme presented in Section 3.3.1 leads to a series of delta functions whose amplitudes can be computed efficiently by means of FFTs.

It is worth pointing out that it would be preferable to obtain $g^P[i]$ in the spatial domain by using (3.46) if analytical expressions were available for the Green's function. In this way the problem would be discretized almost completely in the spatial domain, and it could be said that a spatial domain discretization was being performed. This approach will be taken in Chapters 5 to 8.

An alternative to spatial domain discretization is to obtain the discrete periodic function by computing directly from (3.71). This approach represents a spectral domain discretization. This procedure is useful for the analysis of periodic structures (as will be shown in Section 3.4 and in Chapters 9 and 10) or when there is no analytical expression available that allows the Green's function to be evaluated easily in the spectral domain, as will be discussed in Section 3.5 and in Chapter 11.

3.4 DISCRETIZATION OF PERIODIC PROBLEMS

In this section, we will take advantage of the discrete nature of the CFT of the operator equation for periodic problems (PP) to find a discrete problem whose amplitudes are related by cyclic convolutions.

We begin by rewriting (3.1) as

$$\phi(x) = \int_{-\infty}^{\infty} \rho^P(x')\, G(x - x')\, dx' \tag{3.73}$$

where $\rho^P(x')$ is a periodic function that extends over the whole infinite range of x'.

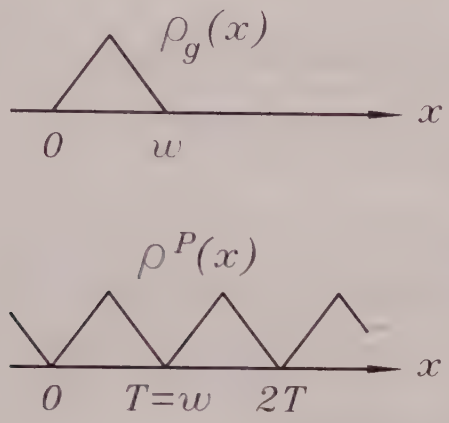


Figure 3.29 Representation of the modulus of $\rho_g(x)$ and $\rho^P(x)$. The period of $\rho^P(x)$ is exactly equal to the length w over which $\rho_g(x)$ extends.

As is shown in Figure 3.29, $\rho^P(x)$ can be formed by a generating function $\rho_g(x)$ that only extends over the period T as is indicated in the following expression:

$$\rho^P(x) = \sum_{m=-\infty}^{\infty} \rho_g(x - mT) \exp(-jk_0 x) \tag{3.74a}$$

$$T = w \tag{3.74b}$$

where we have considered the possibility that the source may have a phase shift. From Figure 3.29 it is obvious that in the PP we can go from $\rho_g(x)$ to $\rho^P(x)$ without padding with zeros.

Replacing the source in (3.73) by its expansion in terms of the generating function, we have

$$\phi(x) = \sum_{m=-\infty}^{\infty} \int \rho_g(x' - mT) \exp(-jk_0 x') \, G(x - x') \, dx' \tag{3.75}$$

Then, by making

$$z = x' - mT \tag{3.76}$$

we obtain the following alternative way to express the convolution of (3.73):

$$\phi(x) = \sum_{m=-\infty}^{\infty} \int_{-\infty}^{\infty} \rho_g(z) \exp(-jk_0 z)\, G(x - mT - z) \exp(-jk_0 mT)\, dz$$

$$= \int_{-\infty}^{\infty} \rho_g^a(z)\, G^F(x - z)\, dz \tag{3.77}$$

$$= \rho_g^a(x) * G^F(x)$$

where $\rho_g^a(x)$ is now a space-limited source function,

$$\rho_g^a(z) = \rho_g(x' - mT) \exp(-jk_0 x) \tag{3.78}$$

and $G^F(x)$ is a new periodic function,

$$G^F(x) = \sum_{m=-\infty}^{\infty} G(x - mT) \exp(-jk_0 mT) \tag{3.79}$$

Figure 3.30 shows how we have gone from the convolution with a periodic source function (3.74) to the convolution in (3.77), where the source is now space-limited, but the Green's function is periodic.

The advantage of this formulation is that we now have a convolution that is quite similar to the one in (3.9); hence, nearly all the concepts developed in Sections 3.2 and 3.3 for problems with individual structures can be applied. There are some

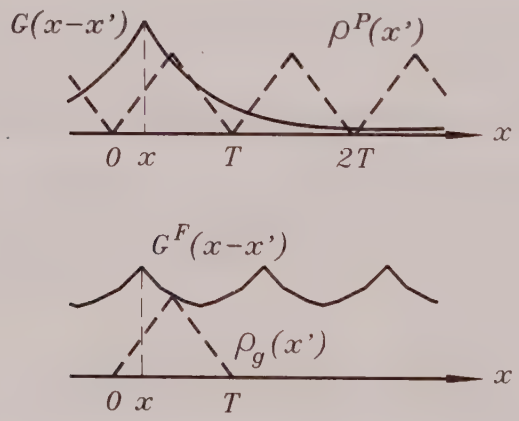


Figure 3.30 The convolution of the Green's function $G(x)$ and the periodic source $\rho^P(x)$ is equal to the convolution of the periodic Green's function $G^F(x)$ and the space-limited source $\rho_g^a(x)$.

differences however. In the first place, no zero padding will be required when forming the periodic function. This represents an advantage in both CPU time and memory storage requirements because the discrete transform will be a DFT_N, where N is the number of samples used to represent the source function, and since there is no zero padding, it is obvious that it will not be necessary to double the sizes of the matrices.

Another difference, which also represents an advantage, is that we can use the fact that $\tilde{G}^F(k_x)$ is a train of delta functions to perform the discretization directly in the spectral domain, thus avoiding cumbersome and time-consuming integrals like those of (3.37) and (3.38) to obtain the discrete Green's function. $\tilde{G}^F(k_x)$ is nearly discrete, because in the spatial domain, $G^F(x)$ is a periodic function that can be obtained by sampling $G(x)$ with a train of delta functions,

$$G^F(x) = G(x) * \amalg_T(x) \exp(-jk_0x) \tag{3.80}$$

and, therefore,

$$\tilde{G}^F(k_x) = \frac{1}{T} \sum_{m=-\infty}^{\infty} \tilde{G}(k_{xm} - k_0)\, \delta(k - k_{xm} + k_0) \tag{3.81}$$

where k_{xm} is defined in (3.21).

As is illustrated in Figure 3.31, the amplitudes of the delta functions in (3.82) are quite easy to compute. It is only necessary to particularize $\tilde{G}(k_x)$ for a series of values of k_x located at regular intervals. In almost all cases, $\tilde{G}(k_x)$ can be found easily, even more easily than $G(x)$. For further discussion of procedures for obtaining the Green's function for multilayer problems, the reader should consult [23].

Returning to the PP, the next step in the discretization procedure is to expand the source function $s_g^a(x)$ in terms of a series of basis functions. Taking pulses as basis functions, as in the case of the EPP, we have

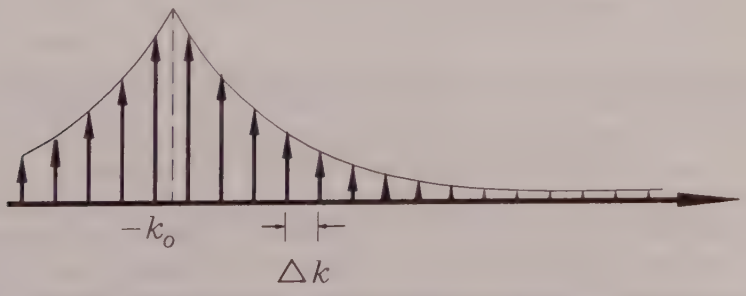


Figure 3.31 Function $\tilde{G}^F(k_x)$ is a train of delta functions whose amplitudes are modulated by $\tilde{G}(k_x)$ (dashed line). The regular interval between deltas is given by Δk.

$$\rho_g^a(x) \approx s_g(x) = \sum_{i=1}^{N} s_g^P[i]\, P(x - x_i) \exp(-jk_0 x) \tag{3.82}$$

where $s_g^P[i]$ is a discrete periodic function with a period N. In the spectral domain, (3.82) becomes

$$\tilde{s}_g(k_x) = \sum_{i=1}^{N} s_g^P[i] \exp[-jx_i(k_x + k_0)]\, \tilde{P}(k_x + k_0) \tag{3.83}$$

For the particular values of k_x for which $\tilde{G}(k_x)$ must be computed in (3.81), we have

$$\tilde{s}_g(k_{xm} - k_0) = \tilde{s}_g^P[m] \exp(j\pi m/N)\, \tilde{P}(k_{xm}) \tag{3.84}$$

where

$$\tilde{s}_g^P[m] = \mathrm{DFT}_N(s_g^P[i]) \tag{3.85}$$

Taking advantage of the similarity between the EPP and PP cases, we can move quickly through the rest of the steps to obtain the discrete operator in the spectral domain. The approximate function $f^a(x)$ for the PP is the convolution

$$f^a(x) = s_g(x) * G^F(x) \tag{3.86}$$

which is quite similar to that of (3.35): the source is a space-limited function and the Green's function is an unlimited periodic function. The sampled potential function $f^c(x)$ is once again given by a train of delta functions,

$$f^c(x) = \sum_{i=-\infty}^{\infty} f_g^P[i]\, \delta(x - x_i) \tag{3.87}$$

whose amplitudes form a periodic function $f_g^P[i]$, which can be computed from

$$f_g^P[i] = s_g^P[i] \otimes g_g^P[i] = \mathrm{DFT}_N^{-1}\,(\tilde{s}_g[m]\, \tilde{g}_g^P[m]) \tag{3.88}$$

where $\tilde{g}_g^P[m]$ can be computed from a direct evaluation of

$$\tilde{g}_g^P[m] = \sum_{l=-\infty}^{\infty} \tilde{G}(\Delta k(m - lN))\, \tilde{P}(\Delta k(m - lN))\, \tilde{t}(\Delta k(m - lN)) \tag{3.89}$$

This expression can be obtained by following a procedure very similar to the one used in the last part of Section 3.3.2 to derive (3.71). The differences are that all the DFTs are now defined over functions of period N, and the CFT of the Green's function that appears in (3.89) is very easy to find. Therefore, in the case of the PP it is preferable to discretize the problem in the spectral domain, because by doing so, and at the same time taking the periodic behavior of the source into account, we can avoid the computation of integrals like (3.46).

Examination of (3.89) suggests that an infinite number of terms must be computed. However, since $\tilde{P}(k_x)$ and $\tilde{t}(k_x)$ are selected in such a way that they present a rapidly decreasing behavior with k_x, there are two alternative methods that could be used to compute this sum. In the first place, it can be computed by means of numerical techniques [24]. This provides good results and is not expensive from the viewpoint of CPU time. The second option, which is based on the rapid decrease of $\tilde{P}(k_x)$ and $\tilde{t}(k_x)$ with k_x, is less expensive from a computational point of view. With this technique, a small number of terms of the series centered around the term with index 0 are considered. The authors' experience indicates that the use of three such terms provides sufficiently accurate results; consequently, this is the solution used in the rest of the book.

3.5 SPECTRAL DOMAIN DISCRETIZATION OF PROBLEMS INVOLVING INDIVIDUAL STRUCTURES

In Sections 3.3 and 3.4, we have shown how to discretize and compute the convolution integral by means of FFTs. In both cases, the discretization process was applied to an equivalent problem that had a periodic Green's function.

In Section 3.3, which deals with the problem of individual structures, the periodic function $G^P(x)$ is formed by replaying at regular intervals a windowed Green's function $G^L(x)$. This latter function is obtained by limiting the extension of the original Green's function $G(x)$ in order to avoid aliasing in the spatial domain, as is indicated in Appendix 3.A. Figures 3.5 and 3.8 show graphically the way $G^L(x)$ and $G^P(x)$ are formed.

In the case of PPs, the periodic Green's function $G^F(x)$ is formed by replaying at regular intervals the original Green's function $G(x)$, in such a way that we go from a problem with a periodic source to an equivalent problem with a space-limited source, as is indicated in expressions (3.75) through (3.79) or in Figure 3.30. Since $G(x)$ is not windowed in this case, we obtain a $G^F(x)$ that is formed by overlapping an infinite series of shifted Green's functions $G(x - mT)$, as is indicated in (3.79). This overlapping is equivalent to a spatial domain aliasing effect, which is not pernicious in this case; on the contrary, the overlapping is quite advantageous because it provides a direct way to consider the effect of a periodic source.

Thus, it appears that if we have a problem with individual structures, it is best to perform a discretization in the spatial domain; whereas if we are addressing a PP, we should choose a discretization in the spectral domain. However, this may not hold true for a problem with individual structures for which no analytical expressions are available to evaluate the Green's function easily in the spatial domain. In this case, if the Green's function can be evaluated easily in the spectral domain, we can discretize the Green's function in this domain by computing $\tilde{g}^P[m]$ directly by means of (3.71). In this way, we can avoid evaluating the Green's function in the spatial domain, as a direct evaluation of $g^P[i]$ in (3.46) would require. Thus, working in the spectral domain, we compute $\tilde{G}^L(k_x)$ from the convolution

$$\tilde{G}(k_x) = \frac{1}{2\pi}\tilde{G}(k_x) * \tilde{W}(k_x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{G}(k_x')\,\tilde{W}(k_x - k_x')\,\mathrm{d}k_x' \tag{3.90}$$

where $\tilde{W}(k_x)$ is the CFT of the window function considered in the spatial domain to limit $G(x)$.

Once $G^L(k_x)$ has been computed by means of (3.90), the rest of the terms in (3.71) can easily be evaluated to complete the calculation of $\tilde{g}^P[m]$. It can be seen that with this procedure we discretize the problem in the spectral domain by using the EPP defined to study problems with individual structures.

The convolution in (3.90) is not easy to compute because the integral extends over the whole spectral domain $(-\infty,\infty)$. An interesting alternative that has been proposed in [24] is to use window functions that are more suitable than simple pulses. In Chapter 11, the reader will find more details about such windows and approximate procedures for evaluating (3.90) that are much faster. Pioneering works using the windows procedure in the area of antenna array can be found in [25,26].

3.6 CONCLUDING REMARKS

In this chapter we have shown how to go from continuous to discrete convolutions. In particular, we have replaced a convolution integral between continuous Green's and source functions with a cyclic convolution between discrete Green's and source functions. It is usual in CG applications to perform the convolution many times using the same Green's function but a succession of different source functions. In order to increase the efficiency of the CG-FFT method, it is quite useful to obtain the discrete Green's function only once in the spectral domain and to store this function. With this technique, it is only necessary to compute a pair of FFTs each time a convolution is evaluated: a direct FFT to obtain the DFT of the discrete source function and the inverse FFT of the product of the DFT of the source with the DFT of the discrete Green's function.

We have considered two main ways of obtaining the DFT of the discrete Green's function. In the first one, which was explained in detail in Section 3.3.1, the Green's function is discretized in the spatial domain, and then the DFT of the discretized function is computed. This technique is applicable to the analysis of problems with individual structures. These problems require a zero padding with a length equal to the length of the source in order to avoid the aliasing in the spatial domain that would appear otherwise.

The other way is to obtain the DFT of the Green's function directly by using the spectral domain discretization procedure of (3.89) for PPs or that of (3.71) for problems with individual structures. In the PP case, no zero padding is required.

In this chapter we have stated on several occasions that, in principle, any kind of function can be considered for the basis and testing functions, $P(x)$ and $t(x)$, in the mathematical development. However, for the sake of simplicity in the drawings, we have used quite simple functions like pulses or delta functions. Now is the time for us to say something more about the choice of the basis and testing functions.

First, examination of the spectral series of (3.71) and (3.89) indicates that for fast convergence of the series, it is desirable to employ basis and testing functions for which $\tilde{P}(k_x)$ and $\tilde{t}(k_x)$ decrease quickly for $|k_x| \geq T_k/2$. This can help to reduce the number of terms of the series of (3.71) or (3.89) that must be considered in the generating cell of $\tilde{g}^P[m]$ or $\tilde{g}_g^P[m]$. But it would also be desirable for $\tilde{P}(k_x)$ and $\tilde{t}(k_x)$ to be as constant as possible in the spectral band $(-T_k/2, T_k/2)$ in order to avoid smoothing the solution. Therefore, $\tilde{P}(k_x)$ and $\tilde{t}(k_x)$ should decrease as quickly as possible for $|k_x| > T_k$ and be constant for $|k_x| < T_k$.

Obviously, it is hard to fulfill these requirements in a strict sense. Trade-off solutions must be considered. There are several functions that are candidates for $P(x)$ and $z(x)$, such as pulses, rooftops, piecewise sine, and piecewise Welch [27]. Of course, all of them must have a length of about Δx to satisfy the spectral band requirements as well as possible. No general criteria can be given for the choice of the best basis or testing function. However, it can be said that several of them, such as the piecewise sine or the piecewise Welch, provide the best performance for most cases.

The pair of functions, rooftop and razor blade, have been used throughout this book for the analysis of electrodynamic problems for one major reason and several minor ones. The minor reasons are that they are easy to draw and to treat mathematically, they provide the book with a uniform presentation, and, as mentioned previously, they can easily be changed if necessary.

The most important reason, as will be seen in the rest of the book, is related to the solution of electrodynamic problems by using the electric field integral equation in its mixed potential formulation, where two source functions must be handled: the current and its divergence (the charge). Theoretically, this would involve computing two potential functions: the magnetic vector potential A and the scalar electric potential ϕ. The major advantage of using rooftops and razor blades is that they

allow these problems to be solved by computing the convolution of only one potential function. This reduces memory storage requirements and CPU time by nearly half.

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Appendix 3A
Aliasing in the Spatial Domain

In order to avoid aliasing and other errors, the period of the EPP should be at least twice the length of the interval w over which the potential function $\phi(x)$ must be computed, where $\phi(x)$ is given by the following convolution:

$$\phi(x) = \int_{-\infty}^{\infty} \rho'(x')\, G(x')\, dx' \tag{3A.1}$$

Figure 3A.1 shows $\rho'(x)$, which is drawn as a pulse function in order to simplify the figures. The Green's function $G(x)$ is pictured in Figure 3.1. The space-limited Green's function $G'^{L}(x)$ can be forced to extend over an interval smaller than w:

$$G'^{L}(x) = P_{T'/2}(x)\, G(x) \tag{3A.2}$$

where $T'/2$ is smaller than w, as is shown in Figure 3A.1(b). The function $G'^{P}(x)$, shown in Figure 3A.1(c),

$$G'^{P}(x) = \sum_{m=-\infty}^{\infty} G'^{L}(x - mT') \tag{3A.3}$$

is defined by taking the minimum value for the period that avoids overlapping between consecutive replicas of the generating function $G'^{L}(x)$. Now we define the periodic source function $\rho'^{P}(x)$,

$$\rho'^{P}(x) = \sum_{m=-\infty}^{\infty} \rho(x - mT') \tag{3A.4}$$

with the same spatial cadence (Figure 3A.1(d)) as $G'^{P}(x)$.

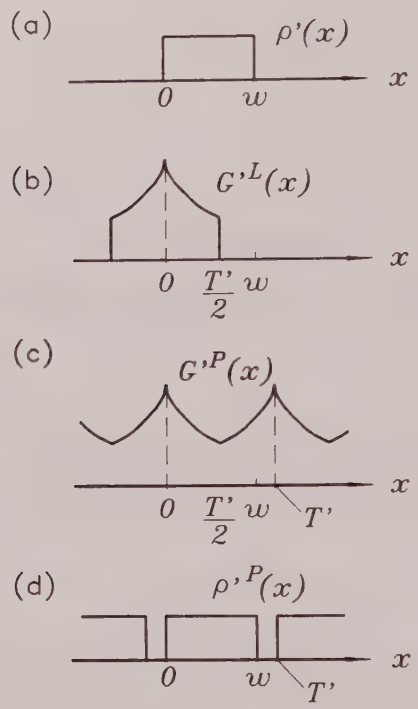


Figure 3A.1 Elementary functions to compute the convolution.

Once these functions have been defined, we are ready to compare the product function $h(x,x')$ defined in (3.6) with the new one:

$$h'^{L}(x,x') = \rho(x')\, G'^{L}(x,x') \tag{3A.5}$$

Figures 3A.2(a,b), for example, show the striking differences between $h^{L}(0,x')$ and $h'^{L}(0,x')$ due to aliasing. By considering the shape of both product functions $h^{L}(x,x')$ and $h'^{L}(x,x')$ for several values of x, we can imagine the form of $\phi(x)$ and $\phi'(x)$:

$$\phi'(x) = \int_0^T \rho'^{P}(x')\, G'^{P}(x - x')\, dx' \tag{3A.6}$$

Figures 3A.2(c,d) illustrate the accurate and aliased potential functions. Although the difference between $\phi(x)$ and $\phi'(x)$ has been highlighted in the figures,

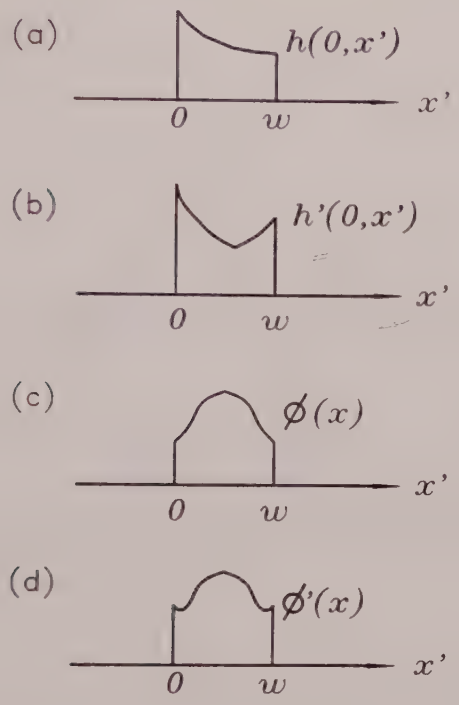


Figure 3A.2 Accurate and aliased product and potential functions.

it can be seen that aliasing introduces significant differences between the two potential functions for small and large values of x in the interval $(0,w)$. It is likely that a rapid decrease in the Green's function as the distance from the source point increases would attenuate the aliasing effect. In any case, in static or electrically small problems, this aliasing effect is not at all negligible and must definitely be avoided. If we are dealing with electrically large problems in time-varying cases, the oscillatory nature of the Green's function at large distances from the source may result in a negligible aliasing effect. However, as occurs in physical optics when computing in or near a direct, reflected, or diffracted direction, the phase of the source and the phase of the Green's function may occasionally result in the product function $h(x,x')$ taking on large values when x and x' are in the interval $(0,w)$ but are relatively far between x and x'. This can happen in the case of a wave propagating through the source region. In order to ensure that spatial domain aliasing is avoided, it is best to define the periodic functions in the EPP in such a way that the period is at least twice the length of the source.

Appendix 3B
Computation of the DFT_{2N} by Parts

Computation of the Direct DFT_{2N}

Our aim is to show how to compute the DFT_{2N} of a discrete signal like the source function $s^P[n]$, which has a length of $2N$, by means of two DFT_N's, assuming that $s^P[n]$ has been obtained by padding with zeros a function of length N, $s^D[n]$, as expression (3.49) indicates.

The DFT is defined as follows.

$$\bar{s}^P[i] = \mathrm{DFT}_{2N}(s^P[n]) = \sum_{n=0}^{2N-1} s^P[n] \exp\left(-j\frac{2\pi}{2N} in\right) \tag{3B.1}$$

$\bar{s}^P[i]$ can be computed separately for the even and odd values of i. For the even values we have

$$\bar{s}^P[i] = \bar{s}^P[2u] = \sum_{n=0}^{N-1} s^P[n] \exp\left(-j\frac{2\pi}{2N} 2un\right) \tag{3B.2}$$

By introducing a new periodic function $b[n]$ whose period is N, half that of $s^P[n]$,

$$b[n] = s^P[n] \qquad 0 \le n \le N-1 \tag{3B.3}$$

we can assimilate the DFT_{2N} of $\bar{s}^P[i]$ to $\tilde{b}[u]$ for the even values of the integer variable,

$$\bar{s}^P[2u] = \tilde{b}[u] \tag{3B.4}$$

because

$$\tilde{s}^P[2u] = \tilde{b}[u] = \mathrm{DFT}_N(b[n]) = \sum_{n=0}^{N-1} b[n] \exp\left(-j\frac{2\pi}{N}un\right) \tag{3B.5}$$

For the odd values of the integer spectral variable, we have

$$\begin{aligned} \tilde{s}^P[2u+1] &= \sum_{n=0}^{N-1} s^P[n] \exp\left(-j\frac{2\pi}{2N}(2u+1)n\right) \\ &= \sum_{n=0}^{N-1} b[n] \exp\left(-j\frac{\pi}{N}un\right) \exp\left(-j\frac{2\pi}{N}un\right) \end{aligned} \tag{3B.6}$$

which can be written in a more compact form as

$$s^P[2u+1] = \mathrm{DFT}_N\left(b[n] \exp\left(-j\frac{\pi}{N}n\right)\right) \tag{3B.7}$$

Thus, we have decomposed the DFT of length $2N$ (3B.1) into two DFTs of length N ((3B.5) and (3B.7)).

Computation of the Inverse DFT_{2N}

When computing the inverse DFT_{2N} of the potential function $\tilde{f}^P[i]$, we are only interested in the values of $f^P[n]$ for $0 \le n < N$, the interval over which $f^P[n]$ is representative of the continuous potential function.

From the definition of the inverse DFT, we have

$$f^P[n] = \frac{1}{2N}\sum_{i=1}^{2N} \tilde{f}^P[i] \exp\left(-j\frac{2\pi}{2N}ni\right) = \frac{1}{2}f_e^P[n] + \frac{1}{2}f_o^P[n] \tag{3B.8}$$

where $f_e^P[n]$ ($f_o^P[n]$) vanishes for n odd (even), and

$$f_e^P[n] = \frac{1}{N}\sum_{v=0}^{N-1} \tilde{f}^P[2v] \exp\left(-j\frac{2\pi}{2N}n2v\right) = \mathrm{DFT}_N^{-1}(\tilde{c}_e[i]) = c_e[n] \tag{3B.9}$$

where $c_e[n]$ is a new function with period N:

$$\tilde{c}_e[i] = \tilde{f}^P[i] \quad i = 0, 2, \ldots, N-1 \tag{3B.10}$$

For $f_o^P[n]$, we have

$$f_o^P[n] = \frac{1}{N}\sum_{v=0}^{N-1} \tilde{f}^P[2v+1]\exp\left(-j\frac{\pi}{N}n\right)\exp\left(-j\frac{2\pi}{N}vn\right)$$
$$= \exp\left(-j\frac{\pi}{Nn}\right)\mathrm{DFT}_N^{-1}(\tilde{c}_o[i]) = \exp\left(-j\frac{\pi}{N}n\right)c_o[n] \qquad \text{(3B.11)}$$

where $\tilde{c}_o[i]$ is a new periodic function with a period of length N defined as

$$\tilde{c}_o[i] = \tilde{f}^P[2i+1] \quad i = 0, 1, \ldots, N-1 \qquad \text{(3B.12)}$$

In conclusion, we can compute the direct DFT_{2N} of $s^P[n]$ and the inverse DFT_{2N} of $\tilde{f}^P[i]$ (for $0 \leq n < N$) using only a pair of matrices of length N, in which $b[n]$, $\tilde{c}_e[i]$, $\tilde{c}_o[i]$, and $f^P[n]$ are stored successively. This allows us to reduce the requirements for memory storage. In 3D problems this can become a serious concern due to the fact that the padding with zeros must be done in three directions, resulting in padded matrices eight times larger than the original ones. The authors have used this technique in the codes for 3D problems. Their experience indicates that the memory storage can be reduced by 75%, but the CPU time is increased by 300%.

Chapter 4
Conjugate Gradient Algorithms

4.1 INTRODUCTION

The purpose of this chapter is to present several different versions of the CG algorithm and to relate the CG technique with other iterative schemes. The classical direct solution to the scattering problem, the moment method, is also considered in some detail, because it is an obligatory reference when comparing different methods.

The CG algorithm is presented as a method of solving general linear equations in which a continuous operator may be involved, such as an integrodifferential operator; the linear algebraic system of equations is considered to be a particular case. Although from a practical point of view many cases are reduced to algebraic problems by discretization of the continuous operators, the authors think that this general point of view is more productive.

4.2 INTEGRAL EQUATION FORMULATION OF ELECTROMAGNETIC PROBLEMS

A large number of electromagnetic problems can be formulated as an integral equation of the general form

$$\int_D G(x,x')\,J(x')\,dx = E(x) \quad x,x' \quad \text{on} \quad D \tag{4.1}$$

In the electromagnetic context, $E(x)$ represents a known incident or impressed field, $J(x')$ is the response of the system, usually the induced current, and $G(x,x')$ is the Green's function that relates the field at a point x with the sources at x'.

We will use operational notation, so (4.1) is rewritten in abstract form as

$$LJ = E \tag{4.2}$$

The problem consists of finding the response J when the linear operator L and the excitation E are known. In most cases, the problem cannot be solved analytically and a numerical approach must be employed. If the numerical method provides us with an approximation J_N of the exact solution J, we naturally wonder how good this approximate solution is. In order to answer this question quantitatively, it is necessary to define a measure of the error. Bearing this goal in mind, we will first define the inner product between two integrable functions f and g defined on D:

$$\langle f,g\rangle = \int_D f(x)\, g^*(x)\, dx \tag{4.3}$$

where * denotes the complex conjugate. Using this operation, we can define the norm of a function f as

$$\|f\| = \langle f,f\rangle^{1/2} \tag{4.4}$$

It is interesting to note that from definitions (4.3) and (4.4), we can obtain

$$\|f\| \geq 0 \tag{4.5}$$

$$\|f\| = 0 \leftrightarrow f = 0 \quad \text{on} \quad D \tag{4.6}$$

On the other hand, it should be noted that the definition of the inner product in (4.3) is not unique. Different definitions of the inner product give rise to different norms and hence to different measures of the functions on the domain D. An interesting new inner product is defined as follows:

$$\langle f, g\rangle_2 = \langle Lf, Lg\rangle \tag{4.7}$$

and its associated norm is

$$\|f\|_2 = \langle f,f\rangle_2^{1/2} \tag{4.8}$$

The original product defined in (4.3) will be denoted by $\langle\ ,\ \rangle_1$. This will allow us to handle different norms with a common notation.

Once the inner product has been defined, it is possible to define an operator associated with L, which is called the adjoint operator L^A:

$$\langle Lf,g\rangle = \langle f,L^A g\rangle \tag{4.9}$$

When $L = L^A$, the operator L is said to be self-adjoint.

Another inner product can be defined if L is self-adjoint and positive (L is positive if $\langle f,Lf\rangle \geq 0$ for all $f \neq 0$ defined on D):

$$\langle f,g\rangle_3 = \langle f,Lg\rangle = \langle Lf,g\rangle \tag{4.10}$$

And the associated norm is defined as follows:

$$\|f\| = \langle f,f\rangle_3^{1/2} \tag{4.11}$$

Returning now to the task of quantifying how good the approximate solution J_N is, we define the residual in the following way.

$$r_N = E - LJ_N \tag{4.12}$$

It is intuitively clear that the norm of the residual is a good way of measuring the error of the solution J_N. Indeed, the root-mean-square (rms) error is usually defined as the norm of r_N:

$$e_r = \|r_N\| = \langle r_N,r_N\rangle^{1/2} \tag{4.13}$$

According to (4.5) and (4.6), $e_r \geq 0$, $e_r = 0$ if and only if $J_N = J$. Often rms errord is normalized; hence, we redefine the rms error as

$$\hat{e} = \frac{\|r_N\|}{\|E\|} \tag{4.14}$$

The normalized rms error has the properties that $\hat{e}_r = 0$, if and only if $J_N = J$, and $\hat{e}_r = 1$ if $J_N = 0$.

4.3 THE METHOD OF MOMENTS SOLUTION OF THE INTEGRAL EQUATION

One of the seminal ideas in the development of electromagnetism during the last three decades has been the application of the MM. The method was developed in the mid-1960s [1–5] and has now reached a high level of sophistication and is used for a broad range of applications. In this section, we will discuss the basic notions of the MM, because it is used so often in computational electromagnetics and is closely related to the CG-FFT method.

The MM starts off with the selection of a finite set of functions $\{\phi_i, i = 1, \ldots, N\}$, called expansion or basis functions, which are used to build an approximation to J:

$$J \simeq J_N(x') = \sum_{i=1}^{N} \alpha_i \phi_i(x') \quad x' \text{ on } D \tag{4.15}$$

In (4.15), the coefficients $\{\alpha_i\}$ are unknowns to be determined.

Substituting (4.15) into (4.2) and taking into account the linearity of the operator L, we obtain the following equation:

$$LJ \simeq LJ_N = \sum_{i=1}^{N} \alpha_i L \phi_i(x') \simeq E(x) \quad x,x' \text{ on } D \tag{4.16}$$

Equation (4.16) is an approximation of (4.2), but it must be discretized so that it can be solved by a computer. To do this, a new set of functions $\{W_j, j = 1, \ldots, N\}$, called the testing or weighting functions, is chosen. Then, taking the inner product of each weighting function with both terms of (4.16) results in a system of linear algebraic equations:

$$\sum_{i=1}^{N} \alpha_i \langle L\phi_i, W_j \rangle = \langle E, W_j \rangle \quad j = 1, \ldots, N \tag{4.17}$$

Equation (4.17) is usually written in the form

$$[Z][I] = [V] \tag{4.18}$$

where

$$[Z]_{ij} = \langle L\phi_i, W_j \rangle \tag{4.19a}$$

$$[I]_i = \alpha_i \tag{4.19b}$$

$$[V]_j = \langle E, W_j \rangle \tag{4.19c}$$

The approximate solution of the operator equation is then obtained by calculating the coefficients α_i by solving (4.17).

Equation (4.17) can also be obtained in the following way. First, the residual function is formed:

$$r_N = E - \sum_{i=1}^{N} \alpha_i L \phi_i \tag{4.20}$$

Next, the constraint that the inner product of the residual with each of the testing functions should be zero is enforced:

$$\langle r_N, W_j \rangle = 0 \quad j = 1, \ldots, N \tag{4.21}$$

Then, substituting (4.20) into (4.21), we again obtain the system of (4.17). From (4.21) it can be inferred that the residual function r_N converges to zero in a distributional sense. In fact, the approximation to J provided by the system of (4.17) may not become better as the number of basis and testing functions N increases. That is, $\hat{e}_N$ does not converge monotonically to zero [6].

When using an MM scheme, we can ensure that the error $\hat{e}_r$ will converge monotonically to zero by choosing the following set of testing functions $\{L^A \phi_j, j = 1, \ldots, N\}$, where L^A is the adjoint operator to L, and ϕ_j are the basis functions selected previously. In this case, the MM gives rise to the classical least-square scheme [6].

Even if the least-square scheme is used, the MM has an important disadvantage: the error cannot be determined until the problem is completely solved. Thus, we may expend a great deal of computational effort to solve the problem with unnecessary accuracy; or, on the other hand, the final error may be too large for our objectives. In the latter case, the only way to reduce the error is to increase the number of basis and testing functions and redo all the calculations.

In addition to the impossibility of controlling the error a priori, the MM is greatly limited with respect to the electrical size of the problems that can be solved. It is generally accepted, as a practical rule of thumb, that about 10 basis functions per wavelength must be used. Therefore, the size of the impedance matrix in the MM is about $(10D/\lambda)^{2m}$ in an m-dimensional problem. It is obvious that the electrical size of the scatterer is very limited for $2D$ and volumetric problems.

Since the 1970s, the basic ideas of the MM have enjoyed a period of rapid development, and very sophisticated versions of the MM scheme have been designed. The research effort has been directed towards improving basis and testing functions, exploiting symmetries and sparsity in the impedance matrix, using the method in conjunction with other approaches and techniques. In the search for numerically efficient solutions of scattering problems, several iterative schemes have been proposed, of which the CG method appears to be one of the most productive.

4.4 ITERATIVE SOLUTIONS OF THE INTEGRAL EQUATION

All iterative algorithms for the solution of the general equation

$$LJ = E \tag{4.22}$$

usually start by choosing an initial arbitrary approximation J_0 to the solution, and then generate successive estimates of the solution of the form

$$J_n = J_{n-1} + a_n P_n, \quad n \geq 1 \tag{4.23}$$

where J_{n-1} is the previous estimate of the solution, P_n is the searching direction, and a_n is the magnitude of movement in the direction P_n to correct the estimate.

At each step, we can determine the value of the residual:

$$r_n = E - LJ_n, \quad n \geq 0 \tag{4.24}$$

By using (4.23) in conjunction with (4.24), we obtain an iterative relationship for the residual:

$$r_n = r_{n-1} - a_n L P_n \tag{4.25}$$

In order to highlight the relation between the searching functions P_n and the successive estimates of the solution, we can use (4.23) to write the following series:

$$J_n = J_0 + \sum_{m=1}^{N} a_m P_m \tag{4.26}$$

A similar expression can be obtained for the residual:

$$r_n = r_0 - \sum_{m=1}^{n} a_m L P_m \tag{4.27}$$

Many iterative schemes exist, and the differences between them lie in the different criteria used to choose a_n and P_n. In the context of electromagnetics applications, a number of authors have attempted either to rationalize existing schemes or to develop new iterative methods [6–15]. For the sake of illustration, we shall now review some representative iterative algorithms following the classification of Kleinman and van den Berg [13].

The simplest iterative algorithm is called the Neumann or Picard-Poincaré-Neumann method. This can be obtained by choosing

$$a_n = 1, \quad P_n = r_{n-1} \tag{4.28}$$

in the iterative process (4.23) and (4.25). The result is

$$J_n = E + (I - L) J_{n-1} \tag{4.29}$$

$$r_n = (I - L)r_{n-1} \tag{4.30}$$

where I is the identity operator. The following expressions can then be derived from (4.29) and (4.30).

$$J_n = \sum_{m=0}^{n=1} (I - L)^m E + (I - L)^n J_0 \tag{4.31}$$

$$r_n = (I - L)^n r_0 \tag{4.32}$$

From the expressions above, it can be seen that while the process is very simple, the condition for convergence is very restrictive. In fact, for most operators of interest, the Neumann series does not converge [13,14].

The general procedure expressed in (4.23) is the basis for constructing any iterative scheme. The values of a_n and P_n must be chosen carefully in order to obtain a good iterative scheme, one that converges rapidly for a broad range of operators. Traditionally, the choice of a_n and P_n has been made following one of two different guidelines: either some error criterion has been minimized, or the choice has been based on some orthogonal expansion of the solution. These two points of view have now been unified, since it is known that the minimization of any functional can be associated with the orthogonalization properties of some functions. The following fundamental theorem establishes this relationship.

Let H be the Hilbert space in which we are working and M a subspace of it, then we have the condition that J_n minimizes $\|J - J_n\|_i$ on M if and only if $\langle J - J_n, W\rangle_i = 0$ for every $W \in M$.

The underlying idea enabling us to construct different schemes starting from (4.23) consists of minimizing different measures of the error $\|J - J_n\|_i$ on some subspaces M.

The next example that we consider is called the stationary over-relaxation method [7,14]. The method can be considered as a relaxation of the Neumann method by introducing two degrees of freedom in the choice of a_n and P_n. Thus, we have

$$a_n = \alpha \tag{4.33}$$

$$P_n = T\, r_{n-1} \tag{4.34}$$

where α is a constant and T is an arbitrary linear operator, both to be chosen.

Substituting (4.33) and (4.34) into the general expressions (4.23) and (4.25), we have

$$J_n = \alpha TE + (I - \alpha TL)J_{n-1} \tag{4.35}$$

$$r_n = (I - \alpha LT) r_{n-1} \tag{4.36}$$

From the above relationships, we can write

$$J_n = \sum_{m=0}^{n-1} (I - \alpha LT)^m \alpha TE + (I - \alpha TL)^n J_0 \tag{4.37}$$

$$r_n = (I - \alpha LT)^n r_0 \tag{4.38}$$

The convergence of the series in (4.37) depends on the operator $(I - \alpha LT)$, as has been discussed in [14,15]; thus, convergence depends on the choice of α and T. One proposed way of determining the values of α and T [13] is the following. The functional

$$F(J_1) = \|J - J_1\|_i \tag{4.39}$$

should be minimized on the subspace spanned by the function Tr_0. Taking the fundamental theorem into account, this implies that

$$\langle J - J_1, Tr_0 \rangle_i = 0 \tag{4.40}$$

Substituting in (4.35) and (4.36), expression (4.40) can be written

$$\langle J - J_0 - \alpha Tr_0, Tr_0 \rangle_i = 0 \tag{4.41}$$

Finally, from (4.41) the following expression is obtained

$$\alpha = \frac{\langle J - J_0, Tr_0 \rangle_i}{\|Tr_0\|_i^2} \tag{4.42}$$

It is interesting to note that (4.42) represents different versions of the method depending on the scalar product chosen. For instance, if $i = 2$ is chosen, we have

$$F(J_1) = \|J - J_1\|_2 = \langle L(J - J_1), L(J - J_1) \rangle^{1/2} = \|r_1\| \tag{4.43}$$

The parameter α in (4.42) depends explicitly on J, the unknown, and therefore can be calculated only for certain cases. In [13], Kleinman and van den Berg present several versions of this method. One of them, for instance, consists of choosing $i = 2$ and $T = I$. Thus, the functional in (4.43) is minimized and the expression for α is

$$\alpha = \frac{\langle r_0, Lr_0 \rangle}{\|Lr_0\|^2} \tag{4.44}$$

With the parameter above, we have all that is needed to perform the iterations in (4.35).

In the stationary over-relaxation method, the imposed condition is $a_n = \alpha$; that is, a_n has a fixed value for all the iterations. A little more complexity is introduced if a different value for a_n is allowed at each iteration. This method is called successive over-relaxation, and different versions have been presented [13]. A possible criterion for choosing a_n is the minimization at each iteration of the functional

$$F(J_n) = \|J - J_n\|_i \tag{4.45}$$

on the subspace spanned by the function Tr_{n-1}.

4.5 CONJUGATE GRADIENT METHODS

In this section we will present several versions of the CG method, some of which are derivable as particular cases of a general iterative method called the Krylov method [13,16]. The Krylov and CG methods can be considered generalizations of the over-relaxation methods.

4.5.1 Krylov Method

The Krylov method seeks iterative solutions as follows:

$$J_0 \text{ is arbitrary}$$

$$J_n = J_{n-1} + a_n P_n \quad n \geq 1 \tag{4.46}$$

$$r_n = r_{n-1} - a_n L P_n \quad n \geq 1 \tag{4.47}$$

$$P_1 = Tr_0 \tag{4.48}$$

$$P_n = Tr_{n-1} + \sum_{m=1}^{n-1} \gamma_{nm} P_m \quad n > 1 \tag{4.49}$$

The difference between this method and the over-relaxation method is that the searching direction is generated in a more sophisticated way. As can be observed in (4.49), it is a linear combination of all the previous directions. This new degree of

freedom, represented by introducing the new parameters $\gamma_{n,m}$, improves the convergence of the method. As a criterion for determining the values of a_n and $\gamma_{m,n}$, the functional $\|J - J_n\|_i$ is minimized at each iteration on the following subspace:

$$H_n = \text{sp}\{P_j\}_{j=1}^n = \text{sp}\{Tr_{j-1}\}_{j=1}^n = \text{sp}\{T(LT)^{j-1} r_0\}_{j=1}^n \tag{4.50}$$

where sp denotes spanned by. The equalities in (4.50) are demonstrated in [13].

Again, the fundamental theorem allows us to obtain equations from which the values of a_n and $\gamma_{n,m}$ can be calculated:

$$\langle J - J_n, P_n \rangle_i = 0; \quad j = 1, \ldots, n \tag{4.51}$$

The following expressions can be obtained from (4.51) without difficulty by using (4.46) through (4.49).

$$a_n = \frac{\langle J - J_{n-1}, P_n \rangle_i}{\|P_n\|_i^2} \tag{4.52}$$

$$\gamma_{nj} = -\frac{\langle Tr_{n-1}, P_j \rangle_i}{\|P_j\|_i^2}, \quad j = 1, \ldots, n-1 \tag{4.53}$$

Equations (4.52) and (4.53) are all that is needed to start the iterative process on (4.46) to (4.49). Nevertheless, the solution that is being sought, J, appears in (4.52) for a_n; consequently, a_n can be explicitly calculated only in a few cases. Table 4.1 summarizes the different versions of the Krylov method following the classification in [13].

4.5.2 Conjugate Gradient Methods

The most commonly used versions of the CG method are obtained as particularizations of the Krylov methods in those cases where operator TL is self-adjoint with respect to some of the scalar products defined above. When TL is self-adjoint, it can be easily demonstrated that

$$\gamma_{nj} = 0, \quad j = 1, \ldots, n-2 \tag{4.54}$$

$$\gamma_{n,n-1} = -\frac{\langle Tr_{n-1}, P_{n-1} \rangle_i}{\|P_{n-1}\|_i^2} = \frac{\langle Tr_{n-1}, L^{-1} r_{n-1} \rangle_i}{\langle Tr_{n-2}, L^{-1} r_{n-2} \rangle_i} \tag{4.55}$$

Table 4.1
Different Versions of the Krylov Method

i	T	*Functional Minimized*	a_n	$\gamma_{n,m}$
1	L^A	$\|J - J_n\|$	$\dfrac{\|r_{n-1}\|^2}{\|P_n\|^2}$	$-\dfrac{\langle L^A r_{n-1}, P_m\rangle}{\|P_m\|^2}$
2	I	$\|r_n\|$	$\dfrac{\langle r_{n-1}, Lr_{n-1}\rangle}{\|LP_n\|^2}$	$-\dfrac{\langle Lr_{n-1}, LP_m\rangle}{\|LP_m\|^2}$
2	L^A	$\|r_n\|$	$\dfrac{\|L^A r_{n-1}\|^2}{\|LP_n\|^2}$	$-\dfrac{\langle LL^A r_{n-1}, LP_m\rangle}{\|LP_m\|^2}$
3	I	$\langle J - J_n, L(J - J_n)\rangle$	$\dfrac{\|r_{n-1}\|^2}{\langle P_n, LP_n\rangle}$	$-\dfrac{\langle r_{n-1}, LP_m\rangle}{\langle P_m, LP_m\rangle}$

The case $i = 2$, $T = I$ is valid only if L is self-adjoint, and the case $i = 3$ is valid only if L is self-adjoint and positive.
Source: [13].

In this case, the functions P_n depend only on P_{n-1}, but the error is minimized on the subspace spanned by all the previous functions. This fact has an important effect from a computational point of view, because it can be avoided to save the functions P_n and r_n.

Table 4.2 shows the different versions of the CG method obtained from the corresponding versions of the Krylov method in Table 4.1. The methods in Table 4.2 are well known; in fact, some of them are identical to some of the CG versions presented by Sarkar, one of the authors that have done much to extend the CG method to solve electromagnetic problems. The method $i = 1$, $T = L^A$ is identical to the method called case B by Sarkar in [11]. The version obtained by considering $i = 2$ and $T = L^A$ is the same as case A in [11]. And finally, the method $i = 3$, $T = I$, valid only in the cases where the operator L is self-adjoint and positive, can be found in [17,18].

4.5.3 Preconditioned CG Algorithms

It is interesting to observe that if T is chosen equal to L^{-1} and $J_0 = 0$ in the general scheme of (4.46) to (4.49), then we have

$$r_0 = E - LJ_0 = E \tag{4.56}$$

Table 4.2
Different Versions of the CG Method

i	T	*Functional Minimized*	a_n	$\gamma_{n,n-1}$
1	L^A	$\|J - J_n\|$	$\dfrac{\|r_{n-1}\|^2}{\|P_n\|^2}$	$\dfrac{\|r_{n-1}\|^2}{\|r_{n-2}\|^2}$
2	I	$\|r_n\|$	$\dfrac{\langle r_{n-1}, Lr_{n-1}\rangle}{\|LP_n\|^2}$	$\dfrac{\langle r_{n-1}, Lr_{n-1}\rangle}{\langle r_{n-2}, Lr_{n-2}\rangle}$
2	L^A	$\|r_n\|$	$\dfrac{\|L^A r_{n-1}\|^2}{\|LP_n\|^2}$	$\dfrac{\|L^A r_{n-1}\|^2}{\|L^A r_{n-2}\|^2}$
3	I	$\langle J - J_n, L(J - J_n)\rangle$	$\dfrac{\|r_{n-1}\|^2}{\langle P_n, LP_n\rangle^2}$	$\dfrac{\|r_{n-1}\|^2}{\|r_{n-2}\|^2}$

The case $i = 2$, $T = I$ is valid only if L is self-adjoint, and the case $i = 3$ is valid only if L is self-adjoint and positive.
Source: [13].

$$P_1 = Tr_0 = L^{-1}E = J \tag{4.57}$$

That is to say, the exact solution is obtained in only one iteration. This may seem trivial, since it is obvious that if we knew the inverse operator, the problem would be solved. However, from a computational point of view, this observation is useful because the following rule holds: if we choose $T = L^{ap}$, where L^{ap} is a preconditioner such that LL^{ap} is close to the identity operator, then the CG scheme converges more quickly than L itself does. Nevertheless, this method depends on finding an approximate inverse operator, the preconditioner, which is not easy and is based on an analysis of each problem. As an example, case C, one of the two preconditioned CG schemes presented by Sarkar in [11], is rewritten here using the notation established above:

$$i = 2; \quad T = ((L^{ap})^A L^{ap})^{-1} L \tag{4.58}$$

$$\gamma_{n,n-1} = \frac{\|(L^{ap})^A L r_{n-1}\|^2}{\|(L^{ap})^A L r_{n-2}\|^2} \tag{4.59}$$

$$a_n = \frac{\|(L^{ap})^{-1} L_{n-1}\|^2}{\|LP_n\|^2} \tag{4.60}$$

4.6 THE GENERALIZED BICONJUGATE GRADIENT METHOD

For a non-self-adjoint operator, the CG method, instead of solving the equation $LJ = E$ directly, solves the modified equation $L^ALJ = L^AE$, where L^AL is now a self-adjoint operator [13,17,19]. This fact has consequences for the convergence rate of the CG method.

A parameter, called the condition number, can be associated with each operator L defined in a finite dimensional space. The condition number is defined as

$$\kappa = \frac{\lambda_{max}}{\lambda_{min}} \tag{4.61}$$

where λ_{max}, and λ_{min} are the maximum and minimum eigenvalues of the operator L.

The convergence of the CG method becomes slower as the condition number increases [6,19]. The problem is that the condition number of the modified operator, L^AL, is squared and, consequently, the convergence will be very slow if the condition number is very large (i.e., if the problem is ill-conditioned). Preconditioning is one way of dealing with this problem, but this is a difficult task, as has already been pointed out. Another possibility is to try to solve the original equation directly. This is the approach taken in the biconjugate gradient method.

To briefly trace the historical development of this method, outlined by Sarkar in [17], we should point out that the concept of biconjugate direction vectors was introduced by Lanczos [20] for the solution of nonsymmetric real matrix equations, and later extended by Fletcher and Reeves [21] to real indefinite systems of equations. Finally, the method was extended by Jacobs [22] and Wong [23] to the solution of complex nonsymmetric matrix equations. These last forms of the biconjugate gradient agree with the equations presented by Sarkar in [17], which are summarized here.

At each iteration, the following functional is minimized:

$$F(J_n,I_n) = \langle L(J - J_n);(I - I_n)\rangle + \langle (I - I_n);L(J - J_n)\rangle \tag{4.62}$$

where J is the solution of the equation

$$LJ = E \tag{4.63a}$$

and I is the solution of the adjoint equation

$$L^AI = Z \tag{4.63b}$$

In this method, it is not necessary to determine Z; the functions J_n and I_n are obtained iteratively by means of

$$J_{n+1} = J_n + a_n P_n \tag{4.64}$$

$$I_{n+1} = I_n + b_n W_n \tag{4.65}$$

where the search functions P_n and W_n satisfy the generalized L-conjugate relationship:

$$\langle LP_i, W_j \rangle = 0 = \langle P_i, L^A W_j \rangle \quad \text{for} \quad i \neq j \tag{4.66}$$

The minimization of the functional (4.62) in conjunction with condition (4.66) allows us to obtain the following iterative procedure for solving $LJ = E$:

$$J_{n+1} = J_n + a_n P_n \tag{4.67}$$

$$r_{n+1} = r_n - a_n L P_n \tag{4.68}$$

$$Q_{n+1} = Q_n - a_n^* L^A W_n \tag{4.69}$$

$$a_n = \frac{\langle r_n, Q_n \rangle}{\langle LP_n, W_n \rangle} \tag{4.70}$$

$$P_{n+1} = r_{n+1} + c_n P_n \tag{4.71}$$

$$W_{n+1} = Q_{n+1} + c_n^* W_n \tag{4.72}$$

$$c_n = \frac{\langle r_{n+1}, Q_{n+1} \rangle}{\langle r_n, Q_n \rangle} \tag{4.73}$$

The initial values can be chosen as follows:

$$J_1 = 0 \tag{4.74}$$

$$r_1 = E = P_1 \tag{4.75}$$

$$W_1 = Q_1 = E^* \tag{4.76}$$

It should be noted that the generalized biconjugate gradient method does not minimize either $\|J - J_n\|$ or $\|LJ_n - E\|$ at each iteration. Nevertheless, the iterations can be stopped when the residual is sufficiently small.

4.7 EXAMPLES OF CONVERGENCE RATES

The first example we will consider is the scattering of a plane wave normally incident to a slab of finite width. This case has been carefully analyzed in [13] in order to compare the convergence of different iterative schemes. In all cases, the rms error (4.14) was plotted versus the number of iterations. Each iterative scheme was used to analyze slabs of the same width, $l = 0.5\lambda$, but with different contrasts.

For the case of plane wave scattering by a slab, the operational equation $LJ = E$ takes the following form:

$$J(x) - \frac{jk}{2}\int_0^l \exp(jk|x - x'|)\,\chi(x)J(x)\,dx' = \exp(jkx) \tag{4.77}$$

where $\exp(jkx)$ represents the incident field in the slab, the unknown, $J(x)$, is the current in the slab, k is the wave number of the surrounding medium, l is the width of the slab, and $\chi(x')$ is the contrast function, defined as follows:

$$\chi(x') = \frac{k_s^2(x')}{k^2} - 1 \tag{4.78}$$

where $k_s(x')$ is the wave number of the slab.

Figure 4.1 shows the results of the Neumann method. It can be seen that the method diverges when the contrast increases. In Figure 4.2, the results are plotted for the stationary over-relaxation method, with $i = 2$ and $T = L^A$. With this method, an improvement is observed in the convergence. Finally, Figure 4.3 shows the results obtained by using the successive over-relaxation method, also with $i = 2$ and $T = L^A$. There appears to be no significant improvement with respect to the stationary scheme. A common characteristic of these three methods is that they do not ensure convergence, which is an obvious disadvantage.

It should be pointed out that almost all the versions of the methods we have discussed are thoroughly compared in [13]. Here we simply aim to provide a general picture of the convergence properties of each method. Figure 4.4 plots the convergence rates of the Krylov method in the version where $i = 2$ and $T = L^A$. The convergence of the Krylov method shows a great improvement over the three previous methods. An additional advantage of the Krylov and CG methods is that convergence is guaranteed in all cases, except for the version of the Krylov method in which $i = 2$ and $T = L$. The case where $i = 2$ and $T = L^A$ is theoretically equivalent to the CG method because LL^A is self-adjoint, but the values of Figure 4.4 have been calculated without taking into account the simplifications of (4.54) and (4.55), which characterize the CG method. Figure 4.5 shows the same results, but makes use of these simplifications. The results should be identical, but this is not true in practice,

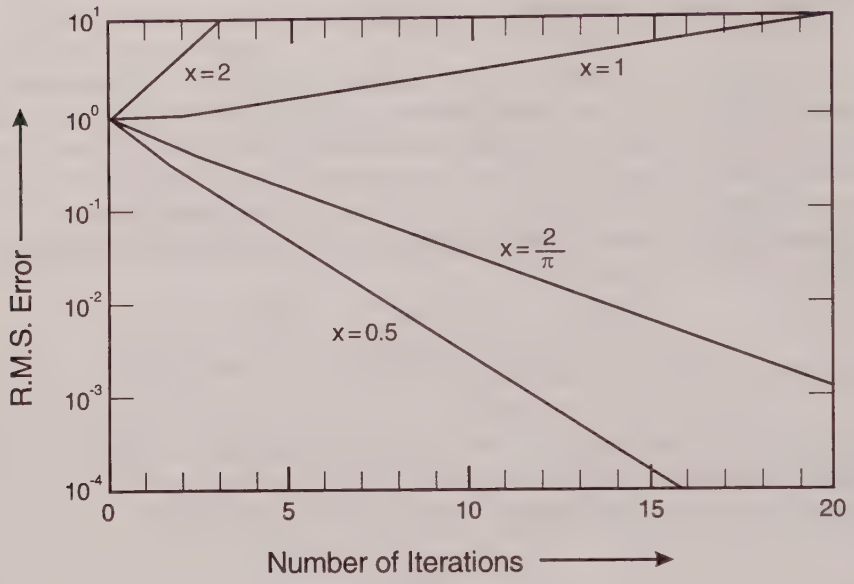


Figure 4.1 Results of the Neumann iterative method. (Source: [13].)

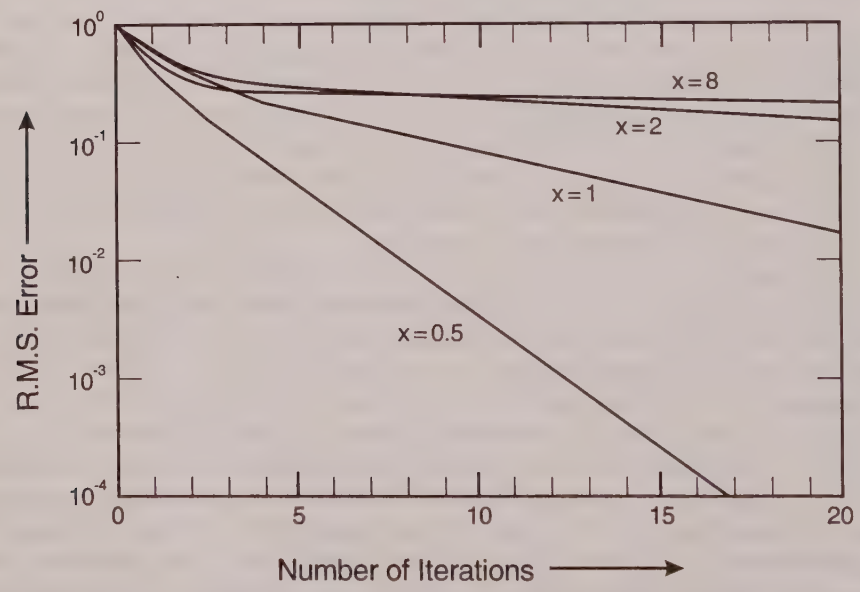


Figure 4.2 Results of the stationary over-relaxation method with $i = 2$ and $T = L_A$. (Source: [13].)

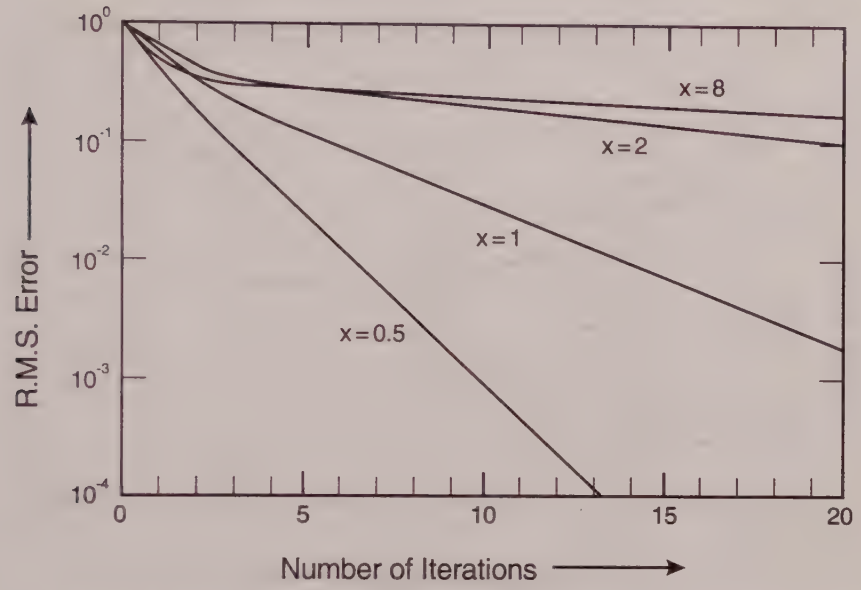


Figure 4.3 Results of the successive over-relaxation method with $i = 2$ and $T = L_A$. (Source: [13].)

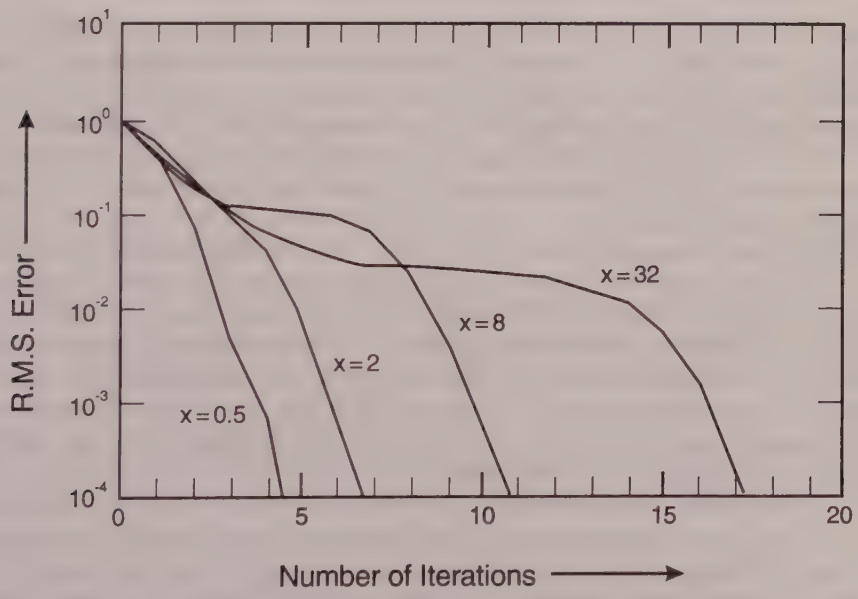


Figure 4.4 Convergence rates for the Krylov method in the version where $i = 2$ and $T = L^{\wedge}$. (Source: [13].)

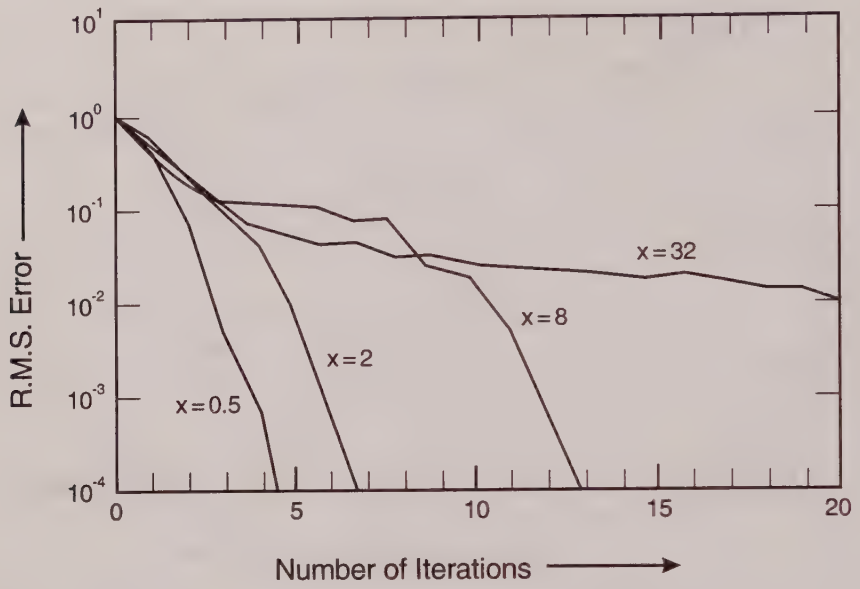


Figure 4.5 Results of the CG method in the version where $i = 2$ and $T = L^A$. (Source: [13].)

as the figures show. The explanation of this fact, reinforced by numerical experiments, is provided in [13]. It is found that the orthogonality properties are lost due to the round-off error involved in the numerical implementation of the CG scheme, whereas in the full Krylov method this condition is enforced at each iteration.

The next example presents the results obtained by Sarkar [11] when two different versions of the CG algorithm are used to solve an antenna problem. The two versions are called case A ($i = 2$, $T = L^A$) and case B ($i = 1$, $T = L^A$). Case B results in monotonic convergence in the space of the solution; that is, at each step, J_n converges to J. This is, in fact, a disadvantage, because it is difficult to evaluate the accuracy of the solution. Table 4.3 reproduces the results presented in [11] when methods A and B are applied to the scattering from a 0.8λ-long wire antenna with a radius of 0.001λ illuminated by a plane wave. It is noteworthy that the error (4.14) decreases monotonically for case A. For method B, the error measured in the range of the operator actually increases, although the error must finally decrease, since the estimate J_n converges to the exact solution.

Finally, we will discuss a comparison between the CG method and the generalized biconjugate gradient method. The comparison involves the CPU time versus the number of unknowns for each method when solving the scattering from a 2.5λ-long wire antenna of radius 0.001λ illuminated by a plane wave. Figures 4.6 and 4.7 show the results presented by Sarkar in [17]. It can be seen that, for this case,

Table 4.3
Results of When Methods A and B Are Applied to an Antenna Problem

Number of Iterations	*Error Case A*	*Error Case B*
1	1.000	1.000
2	0.984	61.558
3	0.972	77.731
4	0.961	88.222
5	0.951	94.835
6	0.942	98.433
7	0.933	99.625
8	0.925	98.909
9	0.916	96.716
10	0.907	93.419
11	0.898	89.361
12	0.889	84.002
13	0.881	100.192
14	0.849	24.200
15	0.590	1.9308
16	1.045 E-17	1.013 E-10

Source: [11].

the biconjugate gradient method is faster than the CG method (the particular version of the CG method is not specified in [17]). Nevertheless, the authors' experience in comparing the biconjugate scheme to the CG method (case A) for different electromagnetic problems indicates that there are no significant differences in the convergence rate and CPU time between the two methods.

4.8 FINAL REMARKS

Although there are many classes of iterative algorithms for the solution of linear equations, the CG method for non-self-adjoint operators (i.e., case A ($i = 2$, $T = L^A$)) appears to be one of the most commonly used algorithms in electromagnetic applications. As Sarkar pointed out, ". . . it may be better to use method A, as it provides some psychological satisfaction to the user, to observe that the magnitude of the residuals decreases monotonically at each iteration." Indeed, when we speak generically of the CG method in the rest of the book, we are referring to the case A version.

In some cases, when the problem is ill-conditioned, the convergence of the CG method may be degraded. For such cases, or to improve convergence in general, the full Krylov method can be used, but this solution has a great computational cost.

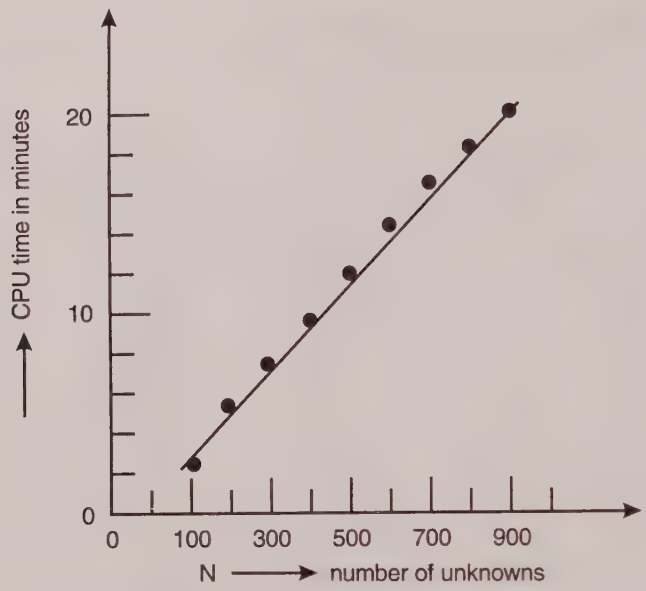


Figure 4.6 CPU time versus the number of unknowns for the generalized biconjugate gradient method. (Source: [17]).

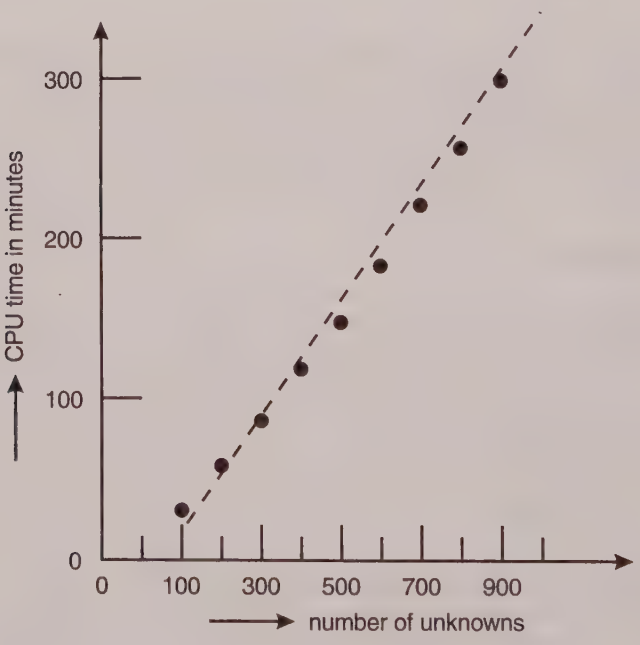


Figure 4.7 CPU time versus the number of unknowns for the CG method. (Source: [17].)

Preconditioning is another interesting option, but further research effort is needed in order to develop systematic methods of obtaining preconditioners.

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Appendix 4A

In this appendix, we present the case A version of the CG algorithm in a detailed form:

$$J_0 \text{ arbitrary (usually } J_0 = 0) \tag{4A.1}$$

$$r_0 = E - LJ_0 \tag{4A.2}$$

$$P_1 = L^A r_0 \tag{4A.3}$$

$$a_1 = \frac{\|L^A r_0\|^2}{\|LP_1\|^2} \tag{4A.4}$$

$$J_1 = J_0 + a_1 P_1 \tag{4A.5}$$

$$r_1 = E - LJ_1 \tag{4A.6}$$

for $n = 2, 3, 4, \ldots$

$$\gamma_{n,n-1} = \frac{\|L^A r_{n-1}\|}{\|L^A r_{n-2}\|} \tag{4A.7}$$

$$P_n = L^A r_{n-1} + \gamma_{n,n-1} P_{n-1} \tag{4A.8}$$

$$a_n = \frac{\|L^A r_{n-1}\|^2}{\|LP_n\|^2} \tag{4A.9}$$

$$J_n = J_{n-1} + a_n P_n \tag{4A.10}$$

$$r_n = r_{n-1} - a_n LP_n \tag{4A.11}$$

$$\text{if} \quad \frac{\|E - LJ\|}{\|E\|} \leq \varepsilon \quad \text{then} \quad \text{STOP} \tag{4A.12}$$

Chapter 5
Arbitrary Flat Conducting Plates

5.1 FORMULATION OF THE PROBLEM

Field equations aim to represent the space- and time-dependent responses of a linear field due to known excitations and subject to boundary conditions. The boundary conditions are generally imposed by the presence of arbitrary structures affecting the behavior of the fields, and the vector field solutions will be different depending on the geometry and physical properties of the structures. A general electromagnetic problem may be represented by a source located anywhere in space, which generates a known excitation. This excitation is applied to an environment and the modified fields will be intercepted by a receiver or scatterer. The direct problem consists of finding the modified fields that will be intercepted by the receiver when both the sources and environment are known. The inverse problem, on the other hand, involves determining the geometry or physical characteristics (properties) of the environment when the sources and modified fields are both known.

This chapter deals with the direct problem in which the environment consists of a flat (2D) conducting plate of arbitrary shape located anywhere in 3D free space. In this case, the physical properties are well known: a perfectly conducting body and free space. One of the most powerful approaches for solving electromagnetic problems is to find an integral equation that makes use of the simplest Green's function. It is well known that the solution to Maxwell's equations for a linear medium may be represented as the fields created by unit sources (dyadic Green's function). The original field can then be represented in terms of the overall excitation (an infinite sum of unit sources), thus giving rise to integral representations of the fields. Frequently, the general dyadic Green's function solution can be represented in terms of a scalar Green's function, which is much easier to obtain. Moreover, it is advantageous to reduce the original vector-field equations to scalar-wave equations and the corresponding solution to unit sources (scalar Green's function). A general theoretical development of these concepts can be found in [1] and specific applications in [2,3].

The complexity of the original problem and the final integral equation representation are determined by the solutions to the Green's function problem. These solutions also determine whether we will be dealing with the spatial or spectral domain and hence the process to be used in the discretization of the Green's function. These concepts are extremely important when trying to solve an electromagnetic problem by means of numerical procedures. In this chapter, time-harmonic electromagnetic fields with a time dependence of the form $\exp(j\omega_0 t)$ will be considered.

5.1.1 General Formulation

The general problem to be solved is shown in Figure 5.1. A perfectly conducting plate of arbitrary geometry is illuminated in the direction of the local angles (θ^i,ϕ^i) by an incident field $\mathbf{E}^i$ created by a source current $\mathbf{J}^i$ anywhere in space. This field will induce currents on the plate $\mathbf{J}$, which will generate a scattered field $\mathbf{E}^s$. At any point in space, the total field $\mathbf{E}^T$ will be the sum of the incident and scattered fields, assuming that no other sources exist. The scattered field will depend on the particular currents induced on the plate. Thus, the first step consists of obtaining these induced currents.

Let us begin by considering the plate of thickness d (i.e., the physical real case). Figure 5.2(a) shows the physical parameters related to the problem: L_x and L_y denote the maximum dimensions of the plate in the x- and y-directions, and PEC stands for perfect electric conducting body. Initially, we must consider a volumetric formulation. Since it is a PEC, the boundary conditions are as follows: (1) the tangential electric field vanishes on S, and (2) the tangential magnetic field is related to the surface current density on S.

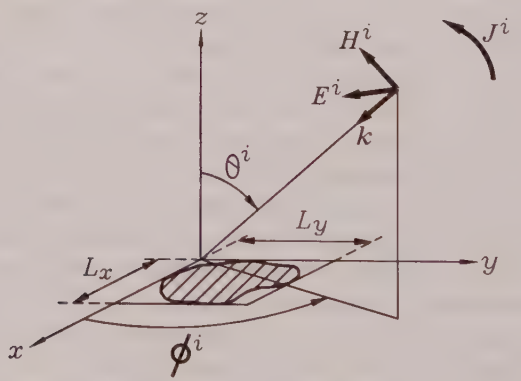


Figure 5.1 Schematic representation of the finite metallic plate problem.

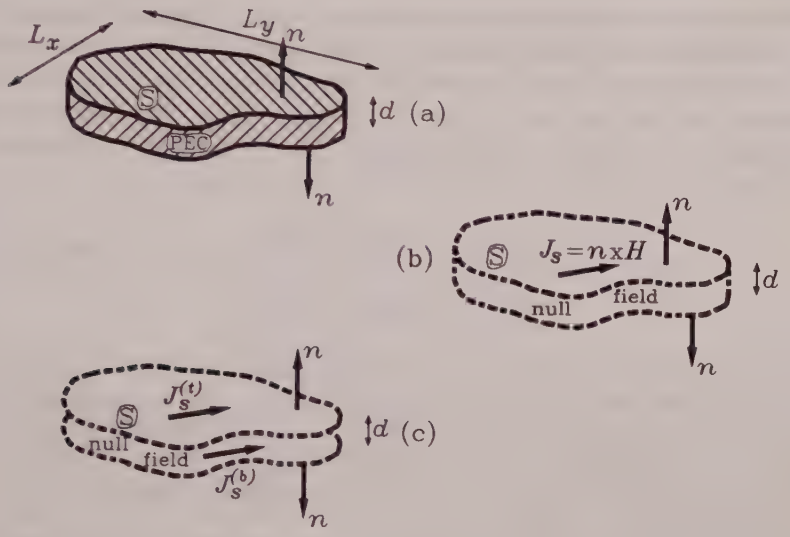


Figure 5.2 (a) Physical perfectly conducting plate problem (3D representation); (b) equivalent problem without any condition for the thickness; (c) equivalent problem obtained by imposing conditions on the thickness in (5.2).

$$\mathbf{E}_t = 0 \tag{5.1a}$$

$$\mathbf{n} \times \mathbf{H} = \mathbf{J}_s \tag{5.1b}$$

Consider now the equivalent structure in Figure 5.2(b), where the PEC has been eliminated. In this case, the surface currents will be those that make the field outside the plate equal to those of the original problem. The total field at any point in space outside the plate will be generated by $\mathbf{J}^i$ and the surface currents on the plate. By making the thickness d go to zero, we are able to assume that the surface currents on the vertical walls of the plate can be neglected. This will be valid when two conditions are fulfilled: (1) the wavelength is much greater than d, and (2) the minimum dimension of the plate is much greater than d. This will ensure that $\mathbf{J}_s$ on the top and bottom surfaces is much more relevant than the currents on the sides of the plate. Mathematically,

$$\begin{aligned} d &<< \lambda_0 \\ d &<< \min(L_x, L_y) \end{aligned} \tag{5.2}$$

where λ_0 is the wavelength corresponding to the time-harmonic frequency $f_o(\omega_o)$.

Under these conditions, the equivalent problem becomes that shown in Figure 5.2(c), where only the top and bottom surfaces are considered. The second boundary condition applied to the equivalent problem implies that the surface currents on the top or bottom of the plate are equal to the $+/-$ tangential component of the magnetic field at these surfaces (due to the opposite orientation of the normal vectors of the two surfaces).

$$\mathbf{H}_t^{(t)} = \mathbf{n} \times \mathbf{H}^{(t)} = \mathbf{J}_s^{(t)} \tag{5.3a}$$

$$-\mathbf{H}_t^{(b)} = -\mathbf{n} \times \mathbf{H}^{(b)} = \mathbf{J}_s^{(b)} \tag{5.3b}$$

By adding these two currents, we obtain the equivalent induced current we are searching for. Thus, we have gone from an initial volumetric formulation to a surface one in which the original boundary conditions in (5.1) are modified to become the following: (1) the tangential electric field on the surface is equal to zero, (2) the change in the tangential magnetic field from the top to the bottom surfaces is equal to the equivalent surface (induced) current.

$$\mathbf{n} \times \mathbf{E}^{(t)} = \mathbf{n} \times \mathbf{E}^{(b)} = 0 \tag{5.4a}$$

$$\mathbf{n} \times [\mathbf{H}^{(t)} - \mathbf{H}^{(b)}] = \mathbf{J} \tag{5.4b}$$

Using these concepts, we will develop a surface integral formulation to solve the plate problem. This can be done in either the spatial or spectral domains. The choice of an operative domain for this formulation is made according to the solution of the Green's function problem. In any case, it should be noted that the boundary conditions are always known in physical space, and if a spectral-spatial-based representation is chosen, the boundary conditions must be satisfied in the original domain (in which they are defined).

In this chapter, we will be concerned with a formulation in the spatial domain because the expression of the 3D Green's function for free space is well known and easy to evaluate. The spectral representation of this function will be used extensively when formulating the problem of an infinite periodic array of metallic patches in Chapter 9. In some cases, such as patches embedded in multilayer media (finite or as part of an infinite periodic array), it is very difficult to express the Green's function in the space domain. Here, the spectral domain is much easier to use, and it allows us to carry out a systematic study of any kind of multilayered structure (Chapters 9 to 11). Finally, there are some special cases in which it is useful to express the Green's function in the spectral domain for certain coordinates and in the spatial domain for the other coordinates, due to the special periodic characteristics of the structure (refer, for instance, to Chapter 12).

For a 3D free-space environment, which is our concern in this chapter, the scalar Green's function problem (Figure 5.3(a)) can be formulated in terms of the scalar wave equation,

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+k_0^2\right) G(\mathbf{r},\mathbf{r}') = -\delta(\mathbf{r},\mathbf{r}') \tag{5.5a}$$

The solution to (5.5a) is the well-known expression

$$G(\mathbf{r},\mathbf{r}') = \frac{\exp[-jk_0|\mathbf{r}-\mathbf{r}'|]}{4\pi\,|\mathbf{r}-\mathbf{r}'|} \tag{5.5b}$$

where $k_o = 2\pi/\lambda_0$ represents the wave number of the vacuum, $\mathbf{r}$ represents the observation point vector, and $\mathbf{r}'$ the source point vector, in which a unit delta function is located producing a field $G(\mathbf{r},\mathbf{r}')$ with $\mathbf{r}$ and $\mathbf{r}'$ dependence. Different methods of solving (5.5a) (Green's problem) may be found in [1–4]. Once the Green's function has been formulated, the solution to the scattered field will be given in terms of the overall response to a set of unit delta functions weighted by the current amplitude, giving rise to an integral equation (as an infinite sum of unit sources). These concepts can be strictly proved by using the theory of distributions (refer, for instance, to [5]

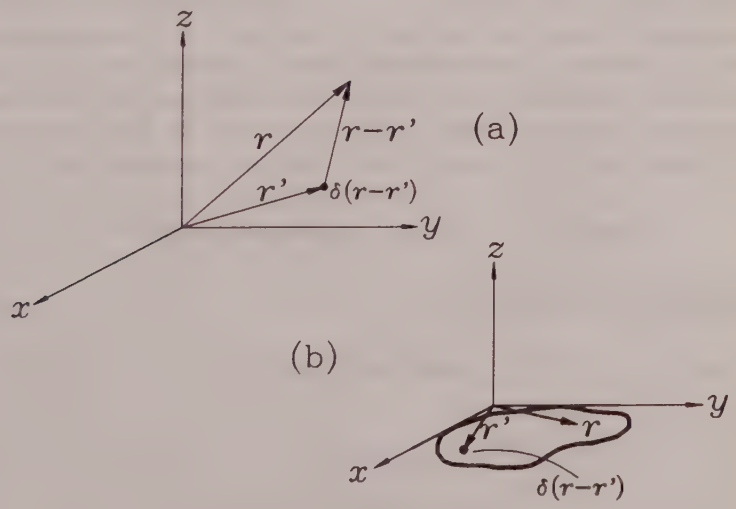


Figure 5.3 3D Green's function problem: (a) in free space, a unit impulse function is located at r' generating a field response $G(r,r')$; (b) particularized to a source located in the xy-plane.

as an introduction to the general problem). The electric field integral equation expressed in terms of the sources (scattered fields in terms of induced currents) is given by

$$\mathbf{E}^s(\mathbf{J}) = -j\omega\mathbf{A}(\mathbf{J}) - \nabla\phi(\mathbf{J}) \tag{5.6}$$

where $\mathbf{A}$ represents the electric vector potential and ϕ represents the electric scalar potential. These can be expressed in terms of the field response to a delta function $G(\mathbf{r},\mathbf{r}')$ weighted by the overall contribution from all the sources $\mathbf{J}$; that is,

$$\mathbf{A}(\mathbf{r}) = \mu_0 \int_V \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' \tag{5.7a}$$

$$\phi(\mathbf{r}) = \frac{1}{\varepsilon_0} \int_V \rho(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' \tag{5.7b}$$

$$\nabla\mathbf{J}(\mathbf{r}) = -j\omega\rho(\mathbf{r}) \tag{5.7c}$$

where $\mathbf{J}$ represents the induced currents over a volumetric source distribution and ρ is the induced charge density in the volume, which is related to the currents by the continuity equation (5.7c). Equations (5.7a–c) can be reduced to surface integrals by assuming that all sources are confined to a flat plate. $\mathbf{J}$ will then represent surface distributions of currents in the equivalent problem described previously. Recalling that the tangential component of the total electric field vanishes anywhere on the surface S, we can apply the boundary condition of (5.4). The total electric field anywhere in space will be the sum of the incident and scattered fields,

$$\mathbf{E}^T = \mathbf{E}^i(\mathbf{J}^i) + \mathbf{E}^s(\mathbf{J}) \quad \text{anywhere in space} \tag{5.8a}$$

and, in particular, on the surface,

$$\mathbf{n} \times \mathbf{E}^T = 0 \quad \text{on} \quad S \tag{5.8b}$$

If (5.8a,b) are combined, we obtain

$$-\mathbf{n} \times \mathbf{E}^i = \mathbf{n} \times \mathbf{E}^s \quad \text{on} \quad S \tag{5.9}$$

Considering a surface formulation of (5.6) and (5.7a–c), we can write (5.9) as

$$\mathbf{n} \times \mathbf{E}^i(\mathbf{r}) = \mathbf{n} \times \left[j\omega\mu_0 \int_S \mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' - \frac{1}{j\omega\varepsilon_0} \nabla \int_S \nabla \mathbf{J}(\mathbf{r}')\mathbf{G}(\mathbf{r},\mathbf{r}')\, ds' \right] \quad (5.10)$$

This integral equation, which is usually called the mixed potential integral equation (MPIE), explicitly considers the potentials due to both currents and charges (the first and second members on the right-hand side of the equality). This is the integral equation we are going to solve on the surface patch $z = 0$ (Figure 5.3(b)), where the position vectors take the form

$$\mathbf{r} = x\mathbf{x} + y\mathbf{y}, \quad z = 0 \quad (5.11a)$$

$$\mathbf{r}' = x'\mathbf{x} + y'\mathbf{y}, \quad z' = 0 \quad (5.11b)$$

Replacing ∇ with its expression in a cartesian coordinate system and noticing that the tangential components to the surface are x- and y-components in that system, we can rewrite (5.10) in a general form as

$$E^i_\alpha(\mathbf{r}) = k^c \int_S J_\alpha(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' + k^q \frac{\partial}{\partial\alpha} \int_S \left(\frac{\partial J_\alpha}{\partial\alpha'} + \frac{\partial J_\beta}{\partial\beta'} \right) G(\mathbf{r},\mathbf{r}')\, ds' \quad (5.12)$$

where α and β represent x or y, and the values of the constants are

$$k^c = j\omega\mu_0 \quad (5.13a)$$

$$k^q = -\frac{1}{j\omega\varepsilon_0} \quad (5.13b)$$

Equation (5.10) or (5.12) can be written in matrix form as

$$\begin{bmatrix} E^i_x \\ E^i_y \end{bmatrix} = k^c \begin{bmatrix} L^c_{xx} J_x \\ L^c_{yy} J_y \end{bmatrix} + k^q \begin{bmatrix} L^q_{xx} & L^q_{xy} \\ L^q_{yx} & L^q_{yy} \end{bmatrix} \begin{bmatrix} J_x \\ J_y \end{bmatrix} \quad (5.14)$$

where the operators in (5.14) applied to the general function A are given by

$$L^c_{\alpha\alpha} A = \int_S A(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, ds' \quad (5.15a)$$

$$L^q_{\alpha\beta} A = \frac{\partial}{\partial\alpha} \int_S \frac{\partial A(\mathbf{r}')}{\partial\beta'} G(\mathbf{r},\mathbf{r}')\, ds' \quad (5.15b)$$

Equations (5.12), (5.14), and (5.15) constitute the integrodifferential operator equation to be solved in order to obtain the induced currents on the structure. These currents can then be used to obtain any other parameters of interest such as radiated (scattered) fields and RCS.

5.2 DISCRETIZATION PROCESS

This section deals with the basic aspects involved in the discretization process. It includes a brief study of basis and testing functions and a discussion on the use of these functions. These ideas will be applied to the flat metallic plate problem in the final sections of this chapter and to other structures in the following chapters.

5.2.1 Expansion (Basis) Functions

Equations (5.10), (5.12), (5.14), and (5.15a,b) can be written in a general form as

$$LI = Y \tag{5.16}$$

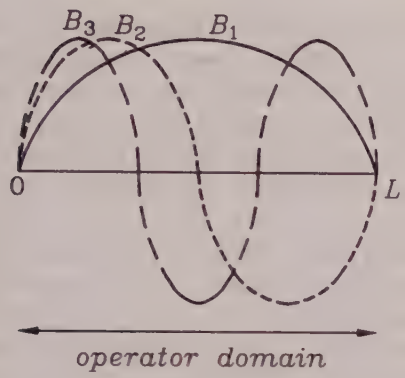
where I represents the unknown function (**J**), Y is the excitation function (E^i), and L is the integrodifferential operator applied to I. The unknown function will be expanded in terms of a set of expansion functions as

$$I = \sum_{n=1}^{N} \alpha_n B_n \tag{5.17}$$

where B_n represents a set of basis functions and α_n is a set of unknown coefficients that will determine the contribution of each basis function to the overall solution. Two kinds of expansion functions can be used to represent the approximate equation (5.17): (1) Entire domain functions, in which any B_n covers the whole domain of the operator. The shape of the functions changes with the index n. See Figure 5.4(a). (2) Subsectional functions, in which any B_n is defined over a subsection of the domain of the operator (see Figure 5.4(b)).

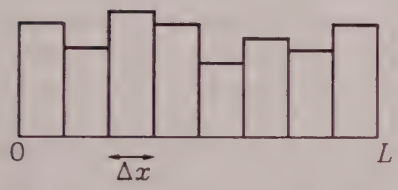
Entire domain functions are especially appropriate for those problems in which the geometry is limited by coordinate lines or surfaces. Consider, for example, a problem involving a 2D infinite (in the y-direction) strip in which the fields are not y-dependent (Figure 5.5).

In this case, the conducting surface is limited by two cartesian coordinate lines and the induced currents on the strip can be well represented by a set of entire domain basis functions.



$$B_n = \sin\left(\frac{n\pi x}{L}\right)$$

(a)



$$B_n = \begin{cases} 1 & (n-1)\,\Delta x < x < n\Delta x \\ 0 & \textit{elsewhere} \end{cases}$$

(b)

Figure 5.4 (a) Example of entire domain; (b) example of subsectional basis functions.

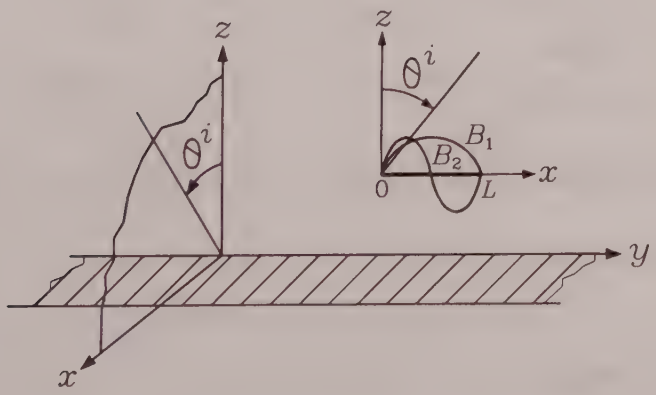


Figure 5.5 Representation of a 2D problem using entire domain basis functions.

Consider now the problem being analyzed in this chapter. The geometry of the plate cannot be defined by coordinate lines, and it is not suitable to represent the plate area by using entire domain basis functions. Subsectional basis functions, on the other hand, allow us to represent the plate area in a way that approaches the original geometry when the number of subsections N is increased. This example suggests that the expansion functions will have to satisfy certain conditions in order to make the representation of the problem valid.

Now consider the operator equation given in (5.12). Some derivatives of the expansion functions must be computed when applying the operator to the approximate expression of the currents in (5.17). In this particular case, second-order derivatives must be calculated to obtain the solution of the equation. Suppose the set of basis functions presented in Figure 5.4(b) is chosen. It can be seen that not only the second-order derivatives of B_n but also the first-order ones will be singular. It seems apparent that some restrictions must be applied when choosing basis functions. In general, these restrictions will depend on the boundary conditions of the problem (e.g., edge conditions) and the nature and form of the operator L (e.g., differentiability). These fundamental restrictions have been summarized by Sarkar [6] as follows: (1) expansion functions should be defined in the domain of the operator and should satisfy boundary and differentiability conditions of the problem (classical solution); (b) expansion functions must form a complete set with respect to the operator. This second condition means that the set of functions $\{B_n\}$ may not be complete in itself, but the result of applying the operator to this set $\{LB_n\}$ should form a complete set.

The most commonly used 1D basis functions are the following.

1. Entire domain: Trigonometric functions. See Figure 5.4(a).

$$B_n = \sin\left(\frac{(n-1)\pi}{L}\left(x + \frac{L}{2}\right)\right) \quad x \in \left(-\frac{L}{2}, \frac{L}{2}\right) \tag{5.18a}$$

$$B_n = \cos\left(\frac{(n-1)\pi}{L}\left(x + \frac{L}{2}\right) - \frac{\pi}{2}\right) \quad x \in \left(-\frac{L}{2}, \frac{L}{2}\right) \tag{5.18b}$$

2. Subsectional: Delta, pulse, triangle, and trigonometric functions. See Figure 5.6.

   a. Delta functions:

$$B_n = \delta(x - (n-1)\Delta x) \quad x \in (0, L) \tag{5.19a}$$

   b. Pulse functions:

$$B_n = \begin{cases} 1 & (n-1)\Delta x < x < n\Delta x \\ 0 & \text{elsewhere} \end{cases} \tag{5.19b}$$

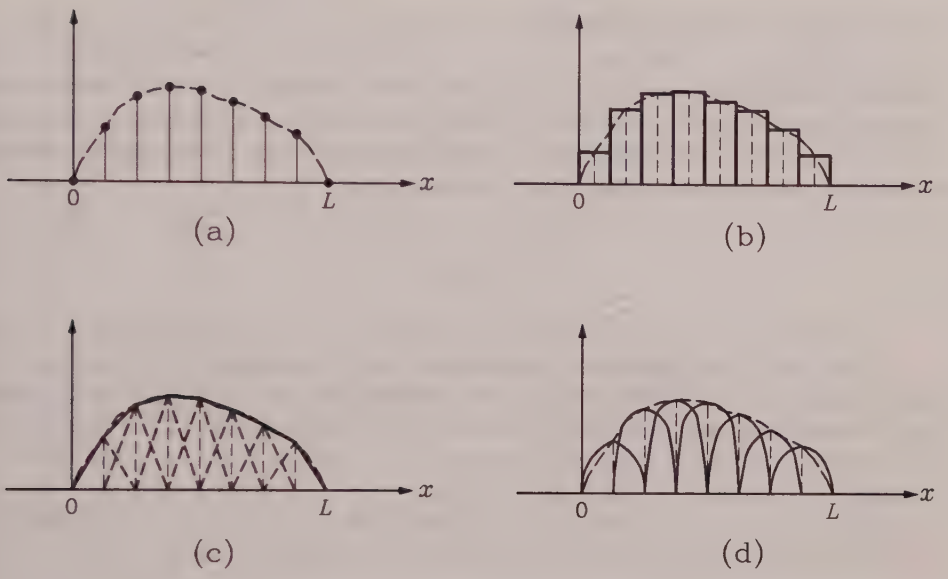


Figure 5.6 Arbitrary function represented by (a) delta, (b) pulse, (c) triangle, and (d) trigonometric functions.

c. Triangle functions:

$$B_n = \begin{cases} 1 - \dfrac{|x - n\Delta x|}{\Delta x} & (n-1)\Delta x < x < (n+1)\Delta x \\ 0 & \text{elsewhere} \end{cases} \tag{5.19c}$$

d. Trigonometric functions:

$$B_n = \begin{cases} \cos\left(\dfrac{\pi}{2\Delta x}|x - n\Delta x|\right) & (n-1)\Delta x < x < (n+1)\Delta x \\ 0 & \text{elsewhere} \end{cases} \tag{5.19d}$$

where

$$\Delta x = \frac{L}{N_x + 1} \quad \text{and} \quad N = N_x + 1$$

N = number of subsections, N_x = number of subdivisions

5.2.2 Testing (Weighting) Functions

Weighting functions appear in mathematics when a function $f(x)$ is expanded as a series of functions $\{\phi_n\}$. They were applied to computational electromagnetics by Harrington as part of the MM scheme [7,8]. If we try to solve (5.16) by using (5.17), we obtain a new equation that can be written as

$$\alpha_n \sum_{n=1}^{N} LB_n \simeq Y_N \tag{5.20}$$

Note that (5.20) represents one equation with N unknowns (α_n). One way to solve this problem is by defining a set of weighting functions $\{W_m\}$ in the range of the operator L and taking the inner product of (5.20) with each W_m:

$$\alpha_n \sum_{n=1}^{N} \langle LB_n, W_m \rangle = \langle Y, W_m \rangle \tag{5.21}$$

where the inner product of two arbitrary complex functions $f(x)$ and $g(x)$ is defined as

$$\langle f,g \rangle = \int_D f(x) g^*(x)\, dx \tag{5.22}$$

D denotes the domain of application of L and * the complex conjugate. It can be seen that (5.21) represents a system of N equations with N unknowns:

$$\begin{aligned}
&\alpha_1 \langle LB_1, W_1 \rangle + \alpha_2 \langle LB_2, W_1 \rangle + \ldots + \alpha_N \langle LB_N, W_1 \rangle = \langle Y, W_1 \rangle \\
&\alpha_1 \langle LB_1, W_2 \rangle + \alpha_2 \langle LB_2, W_2 \rangle + \ldots + \alpha_N \langle LB_N, W_2 \rangle = \langle Y, W_2 \rangle \\
&\qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\
&\alpha_1 \langle LB_1, W_N \rangle + \alpha_2 \langle LB_2, W_N \rangle + \ldots + \alpha_N \langle LB_N, W_N \rangle = \langle Y, W_N \rangle
\end{aligned} \tag{5.23}$$

or in matrix form,

$$[Z_{mn}][\alpha_n] = [y_m] \tag{5.24}$$

If $[Z_{mn}]$ in (5.24) is nonsingular, the solution for the coefficients $[\alpha_n]$ will be

$$[\alpha_n] = [Z_{mn}]^{-1}[y_m] \tag{5.25}$$

From (5.21) it can be seen that weighting functions can be viewed as functions that make the error, defined as $E = Y_N - Y$, go to zero. Equation (5.21) can then be written as $\langle E, W_m \rangle = 0$, which means that the error and weighting functions are orthogonal. Thus, weighting functions must be in the range of the operator and in the limit (infinite series with N going to infinity) should be able to represent Y [9,10]. The functions most commonly used as testing functions are the same ones employed as expansion functions; refer to Figures 5.4 and 5.6, as well as to (5.18a,b) and (5.19a–d).

5.2.3 Two-Dimensional Basis Functions—Representation of the Problem

Based on the preceding concepts, we will now try to discretize and represent the currents induced on a flat plate with arbitrary geometry (i.e., J_x and J_y in (5.14)), which are the unknown functions in the problem being considered. The geometry of the plate can be inscribed in a parallelogram of dimensions L_x and L_y (maximum dimensions of the plate in the x- and y-directions, respectively), which is subdivided into $N_x + 1$ by $N_y + 1$ small parallelograms, with N_x and N_y being the number of subdivisions (Figure 5.7).

The basis and testing functions will be defined over these subdomains (note that subsectional basis functions will be used to represent the currents). A 2D function can be represented in terms of expansion functions by a 2D version of (5.17):

$$I(x,y) = \sum_{m=1}^{M} \sum_{n=1}^{N} \alpha_{mn} B_{mn}(x,y) \tag{5.26}$$

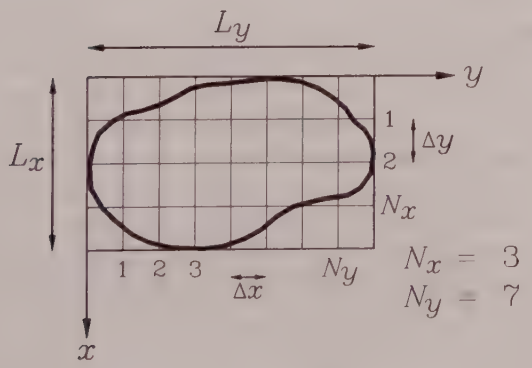


Figure 5.7 Discretization process of an arbitrary finite 2D geometry.

where B_{mn} denotes the 2D basis functions. Any combination in the x- and y-directions of the functions presented in Section 5.2.1 may be used as basis functions. First we will take pulses in both directions, as is shown in Figure 5.8.

$$B_{mn}(x,y) = P_{mn}(x,y) = \begin{cases} 1 & \text{if } \begin{cases} (m-1)\Delta x < x < m\Delta x \\ (n-1)\Delta y < y < n\Delta y \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{5.27}$$

Let us consider the representation of the induced currents by these functions. Direct observation of (5.15a,b) indicates that there is no problem when applying $L^c_{\alpha\alpha}$ to B_{mn} (for each one of the current components). However, problems may appear when trying to apply $L^q_{\alpha\beta}$ to B_{mn}. The first derivative of B_{mn} will be a train of delta functions. Thus, possible errors are to be expected if we try to represent the operator equation with pulse functions. Moreover, the boundary conditions (i.e., that any current component must vanish when normal to an edge of the plate) are not well satisfied, because the value of the pulse does not go to zero unless the entire pulse vanishes.

These boundary conditions are illustrated in Figure 5.9(a), where the normal component to the edge of a square plate will be J_x along a cut AA′ parallel to the x-direction and J_y along a cut BB′ parallel to the y-direction. As we have said, these two components must vanish at the respective edges. This behavior can be well

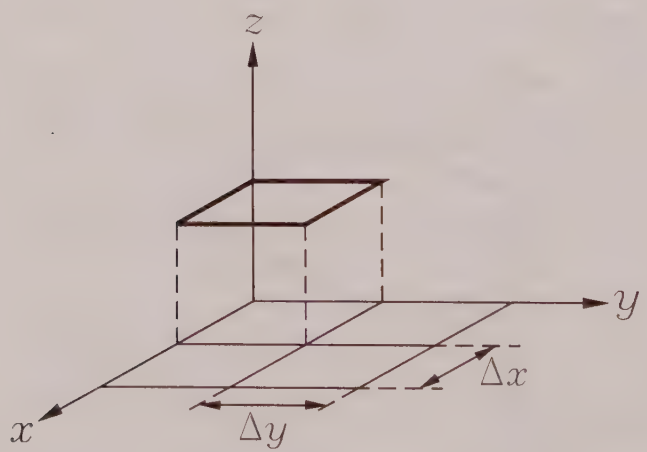


Figure 5.8 Representation of a 2D pulse function.

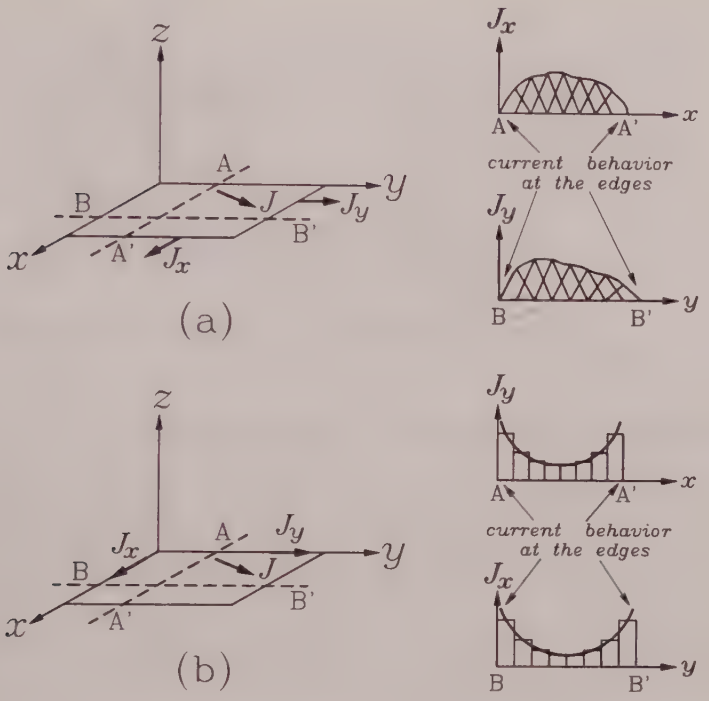


Figure 5.9 Edge condition along (a) a parallel cut and (b) a transversal cut with respect to the direction of the current components.

represented by triangle functions in the AA′ and BB′ directions for each one of the components of the current, as is shown in Figure 5.9(a), where expansion of $J_x(x,y)$ and $J_y(x,y)$ has been made with respect to the x- and y-coordinates, respectively. The asymptotic behavior of the currents J_y and J_x along the AA′ and BB′ cuts, respectively, can be represented by using pulse functions. Note that the use of triangle functions for these cuts would not allow these components to go to infinity while this behavior is well represented by means of pulse functions (see Figure 5.9(b)).

When these two representations are combined, we obtain as basis functions the well-known rooftop functions. The shape and position of these rooftops with respect to the discretization cell is shown in Figure 5.10, and the mathematical form of the currents will be

$$J_x(x,y) = \sum_{m=1}^{N_x} \sum_{n=1}^{N_y+1} J_x^D[m,n]\, T_{mn}^x(x,y) \tag{5.28a}$$

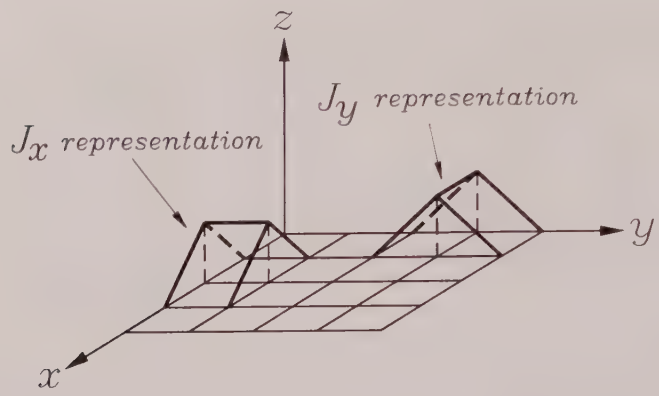


Figure 5.10 Basis functions used to represent the components of the current.

$$J_y(x,y) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y} J_y^D[m,n]\, T_{mn}^y(x,y) \tag{5.28b}$$

with the rooftops given by

$$T_{mn}^x(x,y) = \begin{cases} 1 - \dfrac{|x - \Delta x|}{\Delta x} & \text{for} \quad (m-1)\Delta x < x < (m+1)\Delta x \\ & \text{and} \quad (n-1)\Delta y < y < n\Delta y \\ 0 & \text{elsewhere} \end{cases} \tag{5.28c}$$

$$T_{mn}^y(x,y) = \begin{cases} 1 - \dfrac{|y - \Delta y|}{\Delta y} & \text{for} \quad (n-1)\Delta y < y < (n+1)\Delta y \\ & \text{and} \quad (m-1)\,\Delta x < x < m\Delta x \\ 0 & \text{elsewhere} \end{cases} \tag{5.28d}$$

The series limits have been adjusted to the discretization cell being used here. Note that the geometry is approximated in two different ways, depending on the current component, in order to satisfy the edge conditions. These two approximations are shown in Figure 5.11 with r_x^{mn} and r_y^{mn} being the centers of the rooftops for any current component:

$$r_x^{mn} = m\Delta x\mathbf{x} + (n - 0.5)\Delta y\mathbf{y} \tag{5.29a}$$

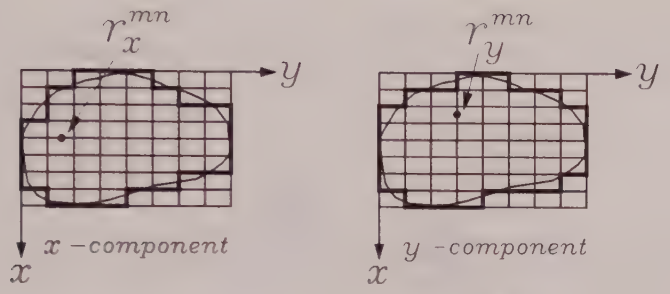


Figure 5.11 Geometrical approximation for both components of the current.

$$r_y^{mn} = (m - 0.5)\,\Delta x\mathbf{x} + n\Delta y\mathbf{y} \tag{5.29b}$$

Similar considerations must be taken into account when choosing the weighting functions. A combination of testing functions for each component may be used. One of the main reasons for using testing functions is to smooth the fields (i.e., take the average value of the fields around the point) in the direction in which a second derivative occurs ($L_{\alpha\alpha}^{q}$ in (5.15b) making $\beta = \alpha$); therefore, the simplest 2D function should be considered. This may be a 1D pulse function in a 2D space. Such functions are usually called razor blade functions and are expressed for the x- and y-components as

$$W_{mn}^{x}(x,y) = \begin{cases} 1 & \text{if } \begin{cases} (m-0.5)\Delta x < x < (m+0.5)\Delta x \\ y = 0.5n\Delta y \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{5.30a}$$

$$W_{mn}^{y}(x,y) = \begin{cases} 1 & \text{if } \begin{cases} x = 0.5m\Delta x \\ (n-0.5)\,\Delta y < y < (n+0.5)\Delta y \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{5.30b}$$

The centers of these functions are the same as those in (5.29a,b), respectively, and their form is shown in Figure 5.12.

The testing procedure consists of computing the integral of the field component (incident and scattered) multiplied by the corresponding weighting functions. As will

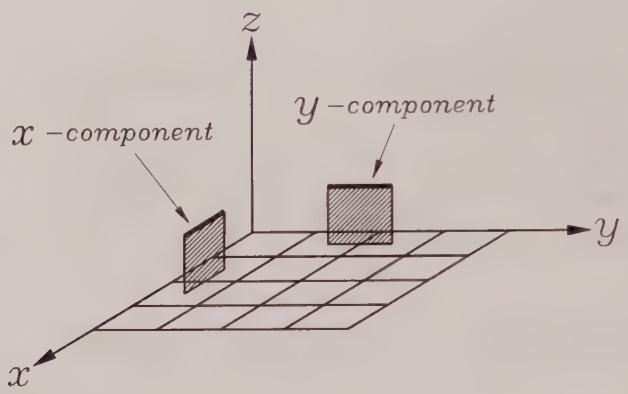


Figure 5.12 2D razor blade functions for both components of the field.

be seen later, only the first-order derivatives have to be computed, because this integral is canceled out by the partial derivative in $L^q_{\alpha\beta}$ in the same direction as the weighting function

$$V_\alpha(x,y) = \int_S W^\alpha_{mn}(x,y) E_\alpha(x,y)\, ds \tag{5.31}$$

where α represents x or y, depending on the field component under consideration.

5.3 DISCRETIZATION OF THE INTEGRAL EQUATION

This section deals with the discretization of the operator equation and subsequent discretization of the Green's function. The author's experience indicates that, using his approach, the best choice for the basis and testing functions for this problem are bidimensional rooftop and razor blade functions, respectively. Bearing this in mind, we carry out the discretization of the operator in the following way.

5.3.1 Inductive Term Due to the x-Component of the Current

This term is written as $L^c_{xx}J_x$ in (5.14). By making $\alpha = x$ and $A = J_x$ in (5.15a), we can write

$$E^c_{xx}(\mathbf{r}) = L^c_{xx}J_x = \int_S J_x(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.32}$$

This term is quite simple, but it is especially important when trying to understand the discretization process of the Green's function. Now let us apply the testing procedure of (5.31) to (5.32):

$$V_{xx}^{Dc}[m,n] = \int_S W_{mn}^{x}(x,y)\, E_{xx}^{c}(x,y)\, dx\, dy \tag{5.33}$$

where $W_{mn}^{x}(x,y)$ is defined as indicated in (5.30a). Equation (5.33) can be written as

$$V_{xx}^{Dc}[m,n] = \int_{(m-0.5)\Delta x}^{(m+0.5)\Delta x} E_{xx}^{c}(x,(n-0.5)\Delta y)\, dx \tag{5.34}$$

where the superscript D denotes a discrete function that depends on m- and n-variables. If Δx is considered small, the integral in (5.34) can be approximated by the value of the function at the midpoint ($x = m\,\Delta x$) multiplied by the value of the integration interval (Δx):

$$V_{xx}^{Dc}[m,n] \approx \Delta x\, E_{xx}^{c}(m\Delta x,(n-0.5)\Delta y) \tag{5.35}$$

The discrete parameters $[m,n]$ describe the behavior of the fields at the points given by (5.29a). By renaming

$$\begin{aligned} x_x^D &= m\Delta x \\ y_x^D &= (n-0.5)\Delta y \end{aligned} \tag{5.36}$$

we can write (5.35) in a simpler way as

$$V_{xx}^{Dc}[m,n] \approx \Delta x\, E_{xx}^{c}(x_x^D, y_x^D) \tag{5.37}$$

Let us concentrate now on the discretization of the operator in (5.37). By using the expansion expression for J_x in (5.28a) and substituting this into (5.32), we obtain

$$E_{xx}^{c}(\mathbf{r}) \approx \int_S \sum_{m'=1}^{N_x} \sum_{n'=1}^{N_y+1} J_x^D[m',n']\, T_{m'n'}^{x}(x',y') G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.38}$$

where $J_x^D[m',n']$ represents the set of unknown coefficients, $\{\alpha_n\}$ in (5.17), to be obtained (amplitude of each rooftop function). Note that J_x^D constitutes a set of discrete values only dependent on m' and n'. Equation (5.38) can be rewritten as

$$E^c_{xx}(x,y) \approx \sum_{m'=1}^{N_x} \sum_{n'=1}^{N_y+1} J^D_x[m',n'] \int_S T^x_{m'n'}(x',y')G(\mathbf{r},\mathbf{r}')\,ds' \tag{5.39}$$

where the integral in (5.39) may be expressed as

$$\int_S T^x_{m'n'}(x',y')G(\mathbf{r},\mathbf{r}')\,ds' = \int_S T^x_{m'n'}(x',y')G(x,y;x',y')\,dx'\,dy' \tag{5.40}$$

To compute (5.39), we must evaluate the integral in (5.40). A very useful technique for computing this integral is to approximate the rooftop with a pulse function with the same volume [11,12]. The new integral will be

$$\int_{D^x_{m'n'}} T^x_{m'n'}(x',y')G(\mathbf{r};x',y')\,dx'\,dy' \simeq \int_{D^x_{m'n'}} P^x_{m'n'}(x',y')G(\mathbf{r};x',y')\,dx'dy' \tag{5.41}$$

where $P^x_{m'n'}$ represents a 2D pulse function centered at $(x^D_{x'},y^D_{x'})$, and $D^x_{m'n'}$ and $D'^x_{m'n'}$ are the integration domains

$$D^x_{m'n'} = \begin{cases} (m'-1)\Delta x < x' < (m'+1)\Delta x \\ (n'-1)\Delta y < y' < n'\Delta y \end{cases} \tag{5.42a}$$

$$D'^x_{m'n'} = \begin{cases} (m'-0.5)\Delta x < x' < (m'+0.5)\Delta x \\ (n'-1)\Delta y < y' < n'\Delta y \end{cases} \tag{5.42b}$$

This transformation and both domains are illustrated in Figure 5.13. Using the result in (5.41), (5.39) can be written as

$$E^c_{xx}(\mathbf{r}) \approx \sum_{m'=1}^{N_x} \sum_{n'=1}^{N_y+1} J^D_x[m',n'] \int_{D'^x_{m'n'}} G(\mathbf{r};x',y')\,dx'\,dy' \tag{5.43}$$

where $P^x_{m'n'}$ in (5.41) has been replaced by its unit value on $D'^x_{m'n'}$. Introducing this result into (5.37), we obtain

$$V^{Dc}_{xx}[m,n] \approx \Delta x \sum_{m'=1}^{N_x} \sum_{n'=1}^{N_y+1} J^D_x[m',n'] \int_{D'^x_{m'n'}} G(x^D_x,y^D_x;x',y')\,dx'\,dy' \tag{5.44}$$

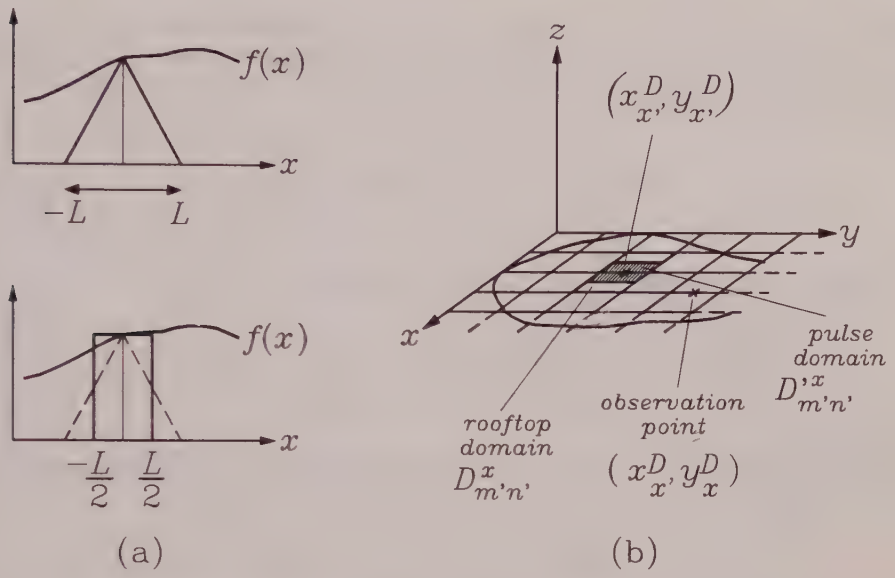


Figure 5.13 (a) Transformation of the integral of $f(x)$ times triangle to $f(x)$ times pulse; (b) 2D domains $D^{x}_{m'n'}$, and $D'^{x}_{m'n'}$ and observation point for the x-component inductive term.

5.3.2 Capacitive Term Due to the x-Component of the Current

This term is given by $L^{q}_{xx}J_x$, and by making $\alpha = \beta = x$ and $A = J_x$ in (5.15b), we obtain

$$E^{q}_{xx}(\mathbf{r}) = \frac{\partial}{\partial x}\int_S \frac{\partial J_x(\mathbf{r}')}{\partial x'} G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.45}$$

or

$$E^{q}_{xx}(\mathbf{r}) = \frac{\partial}{\partial x} I_x(\mathbf{r}) \tag{5.46a}$$

with

$$I_x(\mathbf{r}) = \int_S \frac{\partial J_x(\mathbf{r}')}{\partial x'} G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.46b}$$

where I_α is the general form of the integral of the Green's function multiplied by the partial derivative of the currents with respect to the direction α. As in the case of the inductive term, the testing procedure of (5.31) must be applied to (5.45):

$$V_{xx}^{Dq}[m,n] = \int_S W_{mn}^{x}(x,y)E_{xx}^{q}(x,y)\,dx\,dy \tag{5.47}$$

Taking into account the form of W_{mn}^{x}, we can write (5.47) as

$$V_{xx}^{Dq}[m,n] = \int_{(m-0.5)\Delta x}^{(m+0.5)\Delta x} E_{xx}^{q}(x,y_x^D)\,dx = \int_{(m-0.5)\Delta x}^{(m+0.5)\Delta x} \frac{\partial}{\partial x} I_x(x,y_x^D)\,dx \tag{5.48}$$

with y_x^D being defined as in (5.36). Equation (5.48) can be computed by making use of the property

$$\int_{x_1}^{x_2} \frac{\partial f(x,y)}{\partial x}\,dx = f(x_2,y) - f(x_1,y) \tag{5.49}$$

With this property, the partial derivative disappears and (5.48) can be rewritten as the difference between the values of the integral I_x particularized at the upper and lower limits of the integral

$$V_{xx}^{Dq}[m,n] = I_x(x^{D+},y_x^D) - I_x(x^{D-},y_x^D) \tag{5.50}$$

and

$$\begin{aligned} x^{D+} &= (m + 0.5)\Delta x \\ x^{D-} &= (m - 0.5)\Delta x \end{aligned} \tag{5.51}$$

Each one of these integrals can be represented in a general form as

$$I_x(x^{Ds},y_x^D) = \int_S \frac{\partial J_x(x',y')}{\partial x'} G(x^{Ds},y_x^D;x',y')\,dx'\,dy' \tag{5.52}$$

where the superscript s indicates either of the limits + or −. Before discretizing (5.52) by replacing J_x with the rooftop expansion, we must evaluate the partial derivative of (5.28a). This operation is easy to perform, since the derivative of each rooftop is a pair of pulse functions (see Figure 5.14). Mathematically,

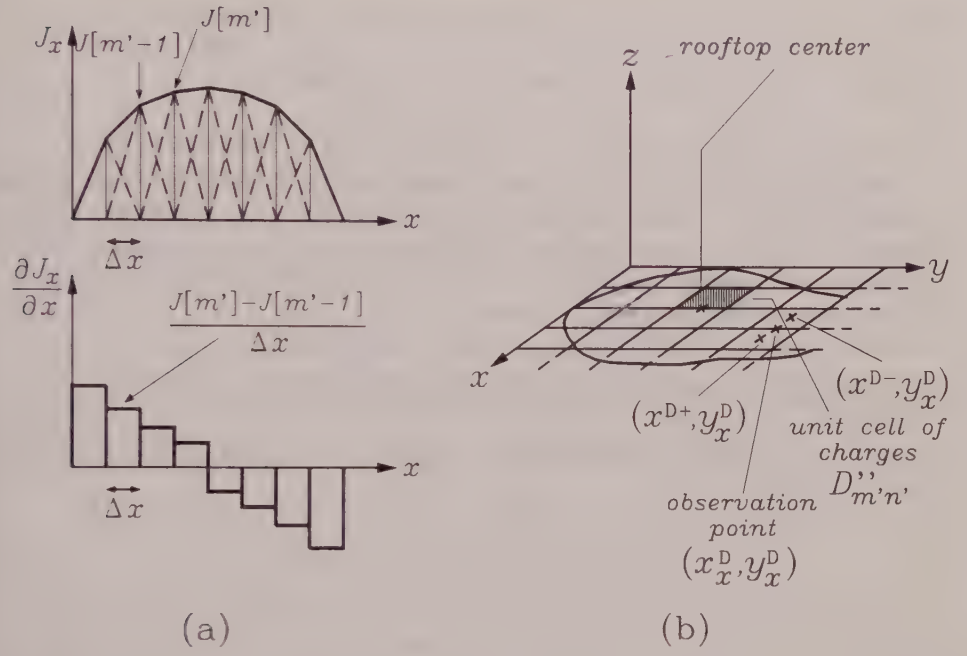


Figure 5.14 For the x-component of the current: (a) derivative of a train of triangle functions: (b) domains of definition for the pulses of charge.

$$\frac{\partial J_x}{\partial x} = \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_x^D[m',n'] - J_x^D[m'-1,n']}{\Delta x} P_{m'n'}(x',y') \tag{5.53}$$

where $P_{m'n'}$ are the 2D pulse functions already described in (5.27). Substituting (5.53) into (5.52), we have

$$I_x(x^{Ds}, y_x^D) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_x^D[m',n'] - J_x^D[m'-1,n']}{\Delta x} \int_S P_{m'n'}^x(\mathbf{r}') G(x^{Ds}, y_x^D; \mathbf{r}')\, ds' \tag{5.54}$$

The definition domains of the pulses of charge, which are shown in Figure 5.14(b), are denoted by $D''_{m'n'}$. Equation (5.54) can then be written as

$$I_x(x^{Ds}, y_x^D) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_x^D[m',n'] - J_x^D[m'-1,n']}{\Delta x} \int_{D''_{m'n'}} G(x^{Ds}, y_x^D; \mathbf{r}')\, ds' \tag{5.55}$$

where

$$D''_{m'n'} = \begin{cases} (m' - 1)\Delta x < x < m'\Delta x \\ (n' - 1)\Delta y < y < n'\Delta y \end{cases} \tag{5.56}$$

The relative positions of observation points and the source region are shown in Figure 5.14(b). Equations (5.55) and (5.56), together with (5.50) and (5.51), constitute the discrete expression of L^q_{xx}. The integral in (5.55) is very similar to that of (5.44) for the inductive term, and it will be seen that both can be represented by using the same discretized Green's function.

5.3.3 Capacitive Term Due to the *y*-Component of the Current

From (5.14), this term is given by $L^q_{xy}J_y$. Making $\alpha = x$, $\beta = y$, and $A = J_y$ in (5.15b), we obtain

$$E^q_{xy}(\mathbf{r}) = \frac{\partial}{\partial x}\int_S \frac{\partial J_y(\mathbf{r}')}{\partial y'} G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.57}$$

or

$$E^q_{xy}(\mathbf{r}) = \frac{\partial}{\partial x} I_y(\mathbf{r}) \tag{5.58a}$$

$$I_y(\mathbf{r}) = \int_S \frac{\partial J_y(\mathbf{r}')}{\partial y'} G(\mathbf{r},\mathbf{r}')\, ds' \tag{5.58b}$$

where I_y indicates the integral already defined in (5.46b), but with $\alpha = y$. As was done for the other terms, we apply the testing procedure of (5.31) to (5.57):

$$V^{Dq}_{xy}[m,n] = \int_S W^x_{mn}(x,y)\, E^q_{xy}(x,y)\, dx\, dy \tag{5.59}$$

Equation (5.59) can then be written as

$$V^{Dq}_{xy}[m,n] = \int_{(m-0.5)\Delta x}^{(m+0.5)\Delta x} E^q_{xy}\,(x, y^D_x)\, dx = \int_{(m-0.5)\Delta x}^{(m+0.5)\Delta x} \frac{\partial}{\partial x} I_y(x, y^D_x)\, dx \tag{5.60}$$

with y^D_x defined as in (5.36). Proceeding as before, the derivative with respect to the direction of the field component (i.e., the x-component) disappears, and making

$f(x,y) = I_y(x,y)$ in (5.49), we can write (5.60) as

$$V_{xy}^{Dq}[m,n] = I_y(x^{D+}, y_x^D) - I_y(x^{D-}, y_x^D) \tag{5.61}$$

Taking (5.58b) into account, we can easily express each of these integrals in general form (replacing + or − with s):

$$I_y(x^{Ds}, y_x^D) = \int_S \frac{\partial J_y(x',y')}{\partial y'} G(x^{Ds}, y_x^D; x', y')\, dx'\, dy' \tag{5.62}$$

Discretization of (5.62) involves the computation of the partial derivatives of J_y with respect to y. This computation, illustrated in Figure 5.15(a), is completely analogous to that described in (5.53). In this case, the dependence of the currents is on the y-direction, and the descriptive index n plays the role of m:

$$\frac{\partial J_y}{\partial y} = \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_y^D[m',n'] - J_y^D[m',n'-1]}{\Delta y} P_{m'n'}(x',y') \tag{5.63}$$

with $P_{m'n'}$ representing 2D pulse functions like those described in (5.27) and (5.53). Substituting (5.63) into (5.62), we have

$$I_y(x^{Ds}, y_x^D) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_y^D[m',n'] - J_y^D[m',n'-1]}{\Delta y} \int_S P_{m'n'}(\mathbf{r}') G(x^{Ds}, y_x^D; \mathbf{r}')\, ds' \tag{5.64}$$

The definition domains of the pulses of charge are shown in Figure 5.15(b), and (5.64) can be rewritten accordingly as

$$I_y(x^{Ds}, y_x^D) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_y^D[m',n'] - J_y^D[m',n'-1]}{\Delta y} \int_{D_{m'n'}^*} G(x^{Ds}, y_x^D; \mathbf{r}')\, ds' \tag{5.65}$$

Equation (5.65), together with (5.61), forms a discrete expression for L_{xy}^q. The integral in (5.65) is completely analogous to those in (5.44) and (5.55), and all of them can be computed in terms of the same Green's function integral.

5.3.4 y-Component of the Operator Equation

In Sections 5.3.1 to 5.3.3, we analyzed the discretization of the x-component of the operator equation (inductive and capacitive terms). In a similar way, we can discretize the y-component of the operator (L_{yy}^c, L_{yx}^q, and L_{yy}^q).

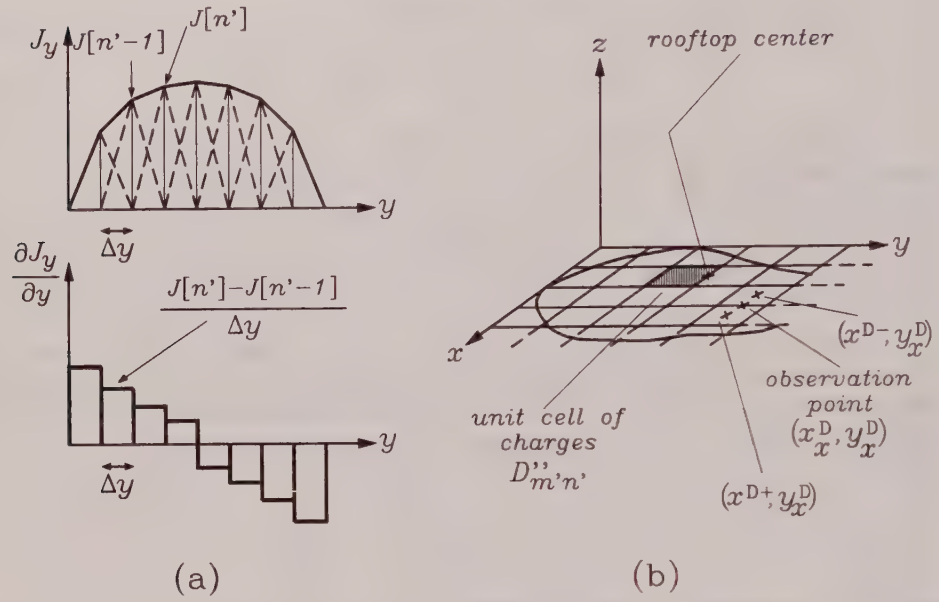


Figure 5.15 For the y-component capacitive term: (a) derivative of a train of triangle functions; (b) domains of definition for the pulses of charge.

1. $L^c_{yy}J_y$. The mathematical development is analogous to that presented in Section 5.3.1, but in this case the weighting functions will be y-oriented razor blade functions (W^y_{mn}), resulting in a new discretization of the observation point as follows:

$$(x_y^D, y_y^D) = [(m - 0.5)\Delta x, n\Delta y] \tag{5.66}$$

Similarly, the approximation in (5.35) will be affected by a Δy. The approximation of the currents is given by (5.28b) in terms of the y-oriented rooftops $T^y_{m'n'}$. These basis functions are approximated by 2D pulses $P^y_{m'n'}$ that extend over the domains $D'_{m'n'y}$:

$$D'_{m'n'y} = \begin{cases} (m' - 1)\Delta x < x < m'\Delta x \\ (n' - 0.5)\Delta y < y < (n' + 0.5)\Delta y \end{cases} \tag{5.67}$$

With these simple considerations, we can write the final discrete expression of the operator:

$$V_{yy}^{Dc}[m,n] \approx \Delta y \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y} J_y^D[m',n'] \int_{D'_{m'n'y}} G(x_y^D, y_y^D; x', y')\, dx'\, dy' \qquad (5.68)$$

2. $L_{yx}^q J_x$. The numerical treatment is analogous to that presented in Section 5.3.3, but now the weighting functions will be y-oriented razor blades, the partial derivative with respect to the y-coordinate disappears, and the approximation used in (5.60) and (5.61) will now be

$$V_{yx}^{Dq}[m,n] = I_x(x_y^D, y^{D+}) - I_x(x_y^D, y^{D-}) \qquad (5.69)$$

where I_x is the integral defined in (5.46b) and y^{D+} and y^{D-} are given by (5.70).

$$\begin{aligned} y^{D+} &= (n + 0.5)\Delta y \\ y^{D-} &= (n - 0.5)\Delta y \end{aligned} \qquad (5.70)$$

The approximate expression of the currents will be the same as in (5.28a), and the derivative will be the one presented in (5.53). The pulses of charge will be given by $P_{m'n'}$, and their definition domain will be the same as in (5.56), giving rise to an expression for I_x that is completely analogous to that in (5.55):

$$I_x(x_y^D, y^{Ds}) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_x^D[m',n'] - J_x^D[m'-1,n']}{\Delta x} \int_{D''_{m'n'}} G(x_y^D, y^{Ds}; \mathbf{r}')\, ds' \qquad (5.71)$$

3. $L_{yy}^q J_y$. Now the partial derivative with respect to y disappears ($\alpha = \beta = y$ and $A = J_y$ in (5.15b)), and we have

$$V_{yy}^{Dq}[m,n] = I_y(x_y^D, y^{D+}) - I_y(x_y^D, y^{D-}) \qquad (5.72)$$

The expression for I_y will be similar to that in (5.62):

$$I_y(x_y^D, y^{Ds}) \approx \sum_{m'=1}^{N_x+1} \sum_{n'=1}^{N_y+1} \frac{J_y^D[m',n'] - J_y^D[m',n'-1]}{\Delta y} \int_{D''_{m'n'}} G(x_y^D, y^{Ds}; \mathbf{r}')\, ds' \qquad (5.73)$$

5.3.5 Computation of Operator Convolutions

In previous sections, each one of the single operators in (5.14) has been written in terms of Green's function integrals, which can be seen in expressions (5.44), (5.55), (5.65), (5.68), (5.71), and (5.73).

The general form of these integrals is

$$G(x^D[m],y^D[n],m',n') = \int_{D_{m'n'}} G(x^D,y^D,x',y')\,dx'\,dy' \tag{5.74}$$

where $(x^D[m], y^D[n])$ is any one of the sampling points defined in (5.35), (5.51), and (5.70), and $D_{m'n'}$ are the corresponding integration domains given by (5.42b), (5.56), and (5.67).

Although these integrals have different sampling points and integration domains, it can be seen that all of them can be written in terms of a single integral

$$G(x^D[m],y^D[n],m',n') = G^D[m - m', n - n'] \tag{5.75}$$

where G^D is the discrete function

$$G^D[m,n] = \int_0^{\Delta x}\int_0^{\Delta y} G(x_m,y_n,x',y')\,dx'\,dy' \tag{5.76}$$

$$\begin{aligned} x_m &= (m + 0.5)\Delta x \\ y_n &= (n + 0.5)\Delta y \end{aligned} \tag{5.77}$$

Figure 5.16 shows the integration domain of (5.76) and the positions of the points where the integral must be evaluated. To compute this type of integral, a Gauss quadrature rule has been used throughout the book. The number of quadrature points considered has been 2×2, 8×8, or 32×32 as a function of the relative positions of source and observation domains (far, near, and coincident, respectively). This technique has provided very good results in all cases.

By using the discrete Green's function, we can write all the operators in (5.14) as discrete linear convolutions. For instance, the integral of (5.44) becomes

$$V_{xx}^{Dc}[m,n] = \Delta x \sum_{m'=1}^{N_x}\sum_{n'=1}^{N_y+1} J_x^D[m',n']G^D[m - m',n - n'] \tag{5.78}$$

This discrete convolution can be expressed in a more compact form as

$$V_{xx}^{Dc}[m,n] = \Delta x\, J_x^D[m',n'] * G^D[m,n] \tag{5.79}$$

In order to take advantage of FFTs to perform convolution, we must convert the linear convolutions like the one in (5.79) into cyclic convolutions. To do this, we can use the techniques shown in Chapter 3. In particular, the convolution of a pair of space-limited functions can be expressed in terms of the cyclic convolution

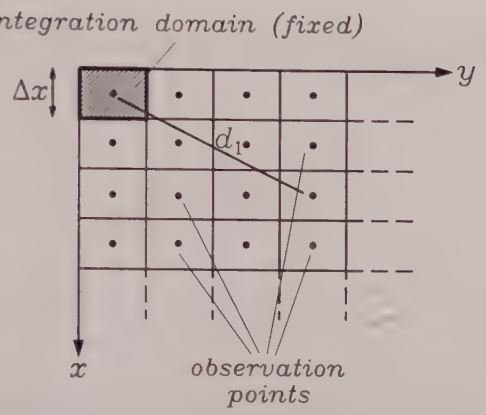


Figure 5.16 Geometry for the discrete Green's function.

of a pair of periodic functions if we are only interested in the result of this convolution in an interval of the independent variable. Following all the steps indicated in Chapter 3, we will introduce for the source function in (5.79) a new periodic function obtained by completing a period of length $(2N_x + 2, 2N_y + 2)$ by padding with zeros. For $0 < m \leq 2N_x + 2$ and $0 < n \leq 2N_y + 2$, the new source function is defined as

$$J^P[m,n] = \begin{cases} J^D[m,n] & \text{for} \quad 0 < m \leq N_x \quad \text{and} \quad 0 < n \leq N_y + 1 \\ 0 & \text{elsewhere} \end{cases} \tag{5.80}$$

For the Green's function, we will use a periodic function G^P defined as

$$G^P[m,n] = G^D[m,n] \quad \text{for} \quad |m| \leq N_x + 1 \quad \text{and} \quad |n| \leq N_y + 1 \tag{5.81}$$

with the period being $(2N_x + 2, 2N_y + 2)$, as in the previous case. With this formulation, the inductive term of (5.79) can be computed by means of

$$V_{xx}^{Dc}[m,n] = V_{xx}^{Pc}[m,n] \quad \text{for} \quad 0 < m < N_x + 1 \quad \text{and} \quad 0 < n < N_y + 1 \tag{5.82}$$

where

$$V_{xx}^{Pc}[m,n] = \Delta x J_x^P[m,n] \otimes G^P[m,n] \tag{5.83}$$

and the symbol $\otimes$ indicates a cyclic convolution, which can be computed efficiently with the help of FFT by using

$$V_{xx}^{Pc}[m,n] = \frac{\Delta x}{N_x^T N_y^T} \text{FFT}^{-1} \{\text{FFT}\{J^P[m,n]\} \text{FFT}\{G^P[m,n]\}\} \tag{5.84}$$

where

$$N_x^T = 2(N_x + 1) \tag{5.85}$$

and

$$N_y^T = 2(N_y + 1) \tag{5.86}$$

represent the size of the discrete periodic lattice that defines each one of the three functions in (5.84).

By carrying out the same process with (5.55), (5.65), (5.68), (5.71), and (5.73), it is possible to obtain the following expressions for the discrete periodic operators:

$$V_{xx}^{Pq}[m,n] \approx \frac{1}{\Delta x N_x^T N_y^T} \text{FFT}^{-1} \{(F_x[p] - 1)(1 - F_x^*[p]) \tilde{J}_x^P[p,q] \tilde{G}^P[p,q]\} \tag{5.87}$$

$$V_{xy}^{Pq}[m,n] \approx \frac{1}{\Delta y N_x^T N_y^T} \text{FFT}^{-1} \{(F_x[p] - 1)(1 - F_y^*[q]) \tilde{J}_y^P[p,q] \tilde{G}^P[p,q]\} \tag{5.88}$$

$$V_{yy}^{Pc}[m,n] \approx \frac{\Delta y}{N_x^T N_y^T} \text{FFT}^{-1} \{\tilde{J}_y^P[p,q] \tilde{G}^P[p,q]\} \tag{5.89}$$

$$V_{yx}^{Pq}[m,n] \approx \frac{1}{\Delta x N_x^T N_y^T} \text{FFT}^{-1} \{(F_y[q] - 1)(1 - F_x^*[p]) \tilde{J}_x^P[p,q] \tilde{G}^P[p,q]\} \tag{5.90}$$

$$V_{yy}^{Pq}[m,n] \approx \frac{1}{\Delta y N_x^T N_y^T} \text{FFT}^{-1} \{(F_y[q] - 1)(1 - F_y^*[q]) \tilde{J}_y^P[p,q] \tilde{G}^P[p,q]\} \tag{5.91}$$

where

$$V_{\alpha\beta}^{P\gamma}[m,n] = V_{\alpha\beta}^{D\gamma}[m,n] \quad \text{for} \quad 0 < m \leq N_x + 1 \quad \text{and} \quad 0 < n \leq N_y + 1 \tag{5.92}$$

In (5.87) and (5.88) and (5.90) and (5.91), the shift property of the DFT has been employed to compute the finite differences of the capacitive terms. The exponentials due to this property have been denoted by

$$F_x[p] = \exp\left(j\frac{2\pi}{N_x^T}p\right) \tag{5.93}$$

and

$$F_y[q] = \exp\left(j\frac{2\pi}{N_y^T}q\right) \tag{5.94}$$

Equations (5.84) and (5.87) through (5.91) constitute the basic expressions used to compute the direct operator convolutions that appear in the CG iterations.

5.3.6 Computation With the Adjoint Operator

The application of a CG scheme requires the use of the adjoint operator defined in terms of the inner product [13]

$$\langle L\mathbf{J}_a, \mathbf{J}_b\rangle = \langle \mathbf{J}_a, L^A\mathbf{J}_b\rangle \tag{5.95}$$

where L^A denotes the adjoint operator of L. The inner product is now defined over the 2D vector functions J_a and J_b, with L representing the matrix operator equation in (5.14), rewritten as

$$\mathbf{E} = L\mathbf{J} \tag{5.96a}$$

$$\begin{pmatrix} E_x^i \\ E_y^i \end{pmatrix} = \begin{pmatrix} L_{xx} & L_{xy} \\ L_{yx} & L_{yy} \end{pmatrix}\begin{pmatrix} J_x \\ J_y \end{pmatrix} \tag{5.96b}$$

By direct comparison with (5.14), it is simple to identify

$$\begin{aligned} L_{xx} &= k^c L_{xx}^c + k^q L_{xx}^q \\ L_{xy} &= k^q L_{xy}^q \\ L_{yx} &= k^q L_{yx}^q \\ L_{yy} &= k^c L_{yy}^c + k^q L_{yy}^q \end{aligned} \tag{5.97}$$

and by imposing condition (5.95) on two arbitrary vectors J_a and J_b in the range of the operator (definition domain), we can obtain the well-known result,

$$L^A = (L^T)^* = \text{Hermitian of } L \tag{5.98}$$

in terms of the complex conjugate (*) of the transpose (superscript T), or in matrix form,

$$L^A = \begin{pmatrix} L_{xx}^* & L_{yx}^* \\ L_{xy}^* & L_{yy}^* \end{pmatrix} \tag{5.99}$$

5.3.7 Final Discrete Expressions

In summary, the discrete operator can be computed for the direct case as follows:

$$\begin{aligned} V_{xx}^D[m,n] &= k^c V_x^{Pc}[m,n] + k^q V_{xx}^{Pq}[m,n] \\ &= \frac{k^c \Delta x}{N_x^T N_y^T} \text{FFT}^{-1}\{\tilde{J}_x^P[p,q]\tilde{G}^P[p,q]\} \\ &\quad + \frac{k^q}{\Delta x N_x^T N_y^T} \text{FFT}^{-1}\{(F_x[p]-1)(1-F_x^*[p])\tilde{J}_x^P[p,q]\tilde{G}^P[p,q]\} \end{aligned} \tag{5.100a}$$

$$\begin{aligned} V_{xy}^D[m,n] &= k^q V_{xy}^{Pq}[m,n] \\ &= \frac{k^q}{\Delta y N_x^T N_y^T} \text{FFT}^{-1}\{(F_x[p]-1)(1-F_y^*[q])\tilde{J}_y^P[p,q]\tilde{G}^P[p,q]\} \end{aligned} \tag{5.100b}$$

$$\begin{aligned} V_{yx}^D[m,n] &= k^q V_{yx}^{Pq}[m,n] \\ &= \frac{k^q}{\Delta x N_x^T N_y^T} \text{FFT}^{-1}\{(F_y[q]-1)(1-F_x^*[p])\tilde{J}_x^P[p,q]\tilde{G}^P[p,q]\} \end{aligned} \tag{5.100c}$$

$$\begin{aligned} V_{yy}^D[m,n] &= k^c V_y^{Pc}[m,n] + k^q V_{yy}^{Pq}[m,n] \\ &= \frac{k^c \Delta y}{N_x^T N_y^T} \text{FFT}^{-1}\{\tilde{J}_y^P[p,q]\tilde{G}^P[p,q]\} \\ &\quad + \frac{k^q}{\Delta y N_x^T N_y^T} \text{FFT}^{-1}\{(F_y[q]-1)(1-F_y^*[q])\tilde{J}_y^P[p,q]\tilde{G}^P[p,q]\} \end{aligned} \tag{5.100d}$$

Using the same equations and the result in (5.99), we obtain the final expression for the adjoint operator:

$$V_{xx}^{DA}[m,n] = \frac{k^{c*}\Delta x}{N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{\tilde{J}_x^p[p,q]\tilde{G}^{p*}[p,q]\} + \frac{k^{q*}}{\Delta y N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{(F_x^*[p]-1)(1-F_x[p])\tilde{J}_x^p[p,q]\tilde{G}^{p*}[p,q]\} \tag{5.101a}$$

$$V_{xy}^{DA}[m,n] = \frac{k^{q*}}{\Delta x N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{(F_y^*[q]-1)(1-F_x[p])\tilde{J}_y^p[p,q]\tilde{G}^{p*}[p,q]\} \tag{5.101b}$$

$$V_{yx}^{DA}[m,n] = \frac{k^{q*}}{\Delta y N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{(F_x^*[p]-1)(1-F_y[q])\tilde{J}_x^p[p,q]\tilde{G}^{p*}[p,q]\} \tag{5.101c}$$

$$V_{yy}^{DA}[m,n] = \frac{k^{c*}\Delta y}{N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{\tilde{J}_y^p[p,q]\tilde{G}^{p*}[p,q]\} + \frac{k^{q*}}{\Delta y N_x^T N_y^T}\,\mathrm{FFT}^{-1}\{(F_y^*[q]-1)(1-F_y[q])\tilde{J}_y^p[p,q]\tilde{G}^{p*}[p,q]\} \tag{5.101d}$$

5.4 RESULTS FOR INDUCED CURRENTS

By using (5.100) and (5.101) for the direct and adjoint operators, respectively, in the CG algorithm, we are able to obtain the induced currents on arbitrary structures. This section presents some of the results obtained for different flat-plate geometries. Most of the results are not compared with other solutions because values of this kind are not available in the literature.

5.4.1 Square Plate

Consider the square plate shown in Figure 5.17. A plane wave traveling in the k-direction (determined by θ^i and ϕ^i) is incident upon the plate. If $\theta^i = 0$ deg and $\phi^i = -90$ deg, the incident field will be x-polarized.

For this polarization and considering the case of $L = 1\lambda$, the CG-FFT scheme gives the results presented in Figure 5.18 for the two current components. The number of samples used to represent the current was $N_x = N_y = 15$.

Figures 5.19, 5.20, and 5.21 present a convergence study for the same structure in terms of the number of samples. The first two figures show the magnitude of the x-component of the current for two different cuts (AA′ and BB′) in the x- and y-directions, respectively. Figure 5.21 plots the relative error versus the number of iterations. The curves are representative of typical convergence rates obtained by

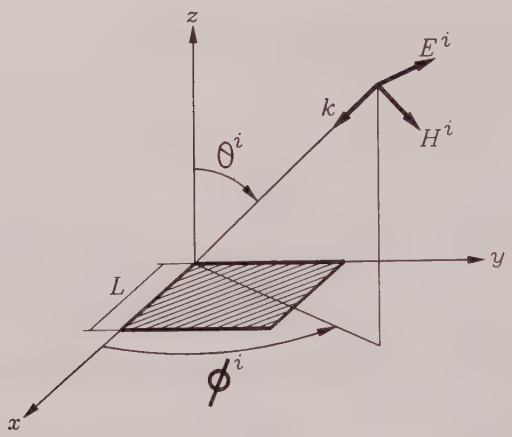


Figure 5.17 Schematic representation of the square-plate problem.

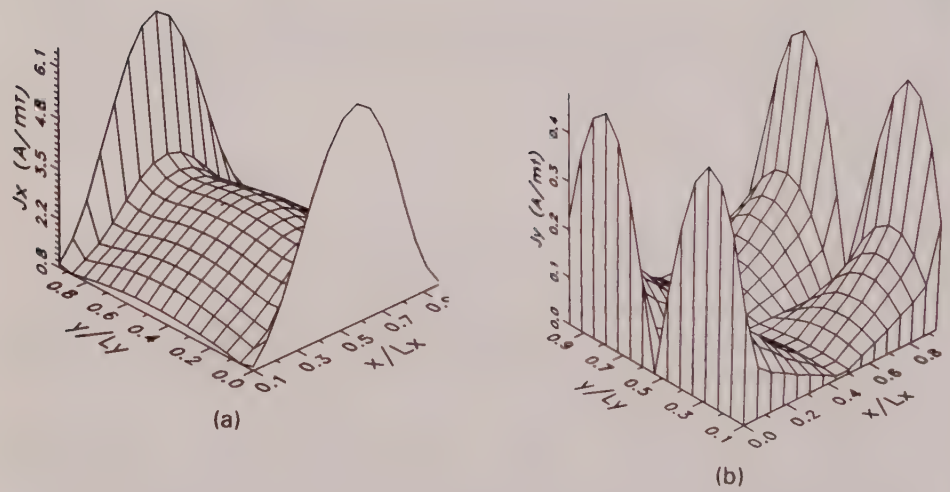


Figure 5.18 (a) 2D representation of the magnitude of the x-component of the current; (b) 2D representation of the magnitude of the y-component of the current. $L_x = L_y = L = 1\lambda$.

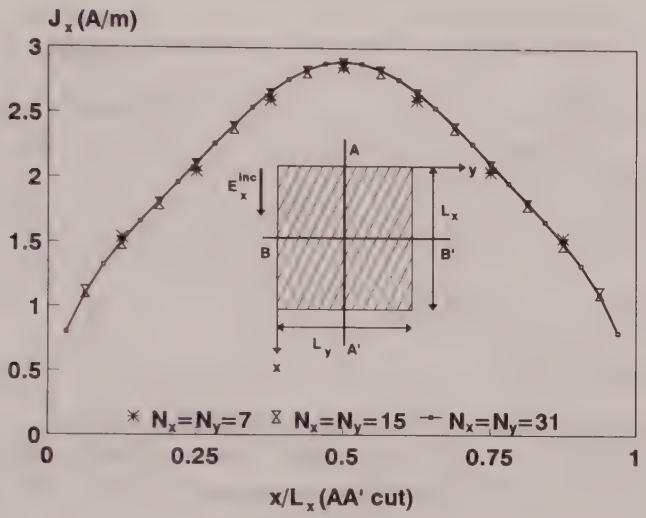


Figure 5.19 Convergence study for the x-component of the current along the AA' cut.

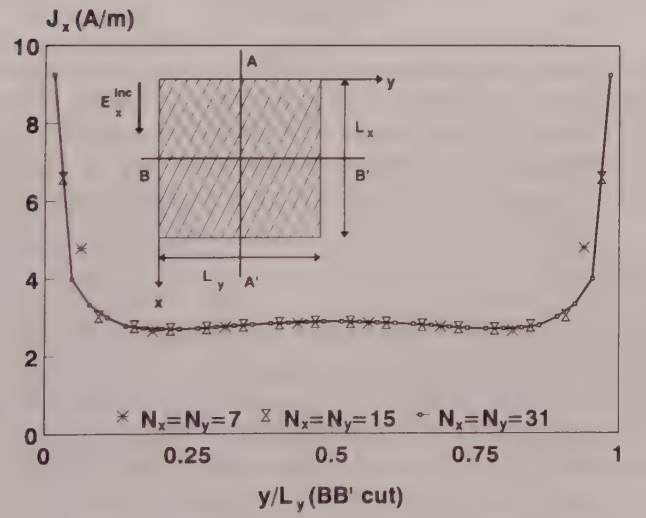


Figure 5.20 Convergence study for the x-component of the current along the BB' cut.

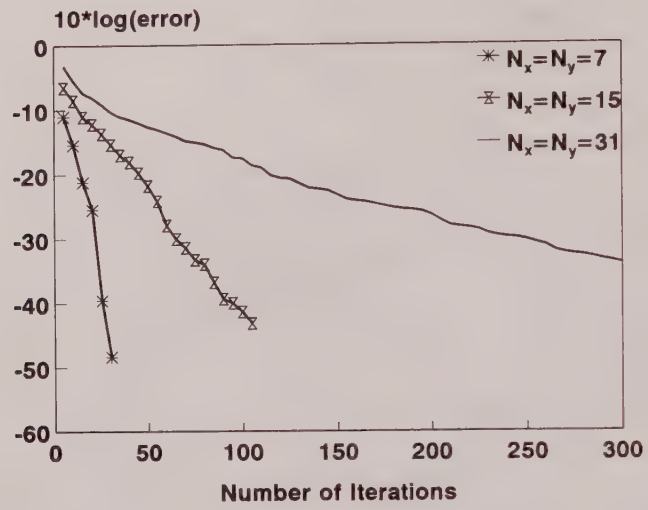


Figure 5.21 Relative error versus number of iterations in terms of the number of samples.

using a CG scheme combined with the FFT in the way previously described. The relative error is defined as

$$\varepsilon_r = 10 \log_{10} \frac{|LI - Y|}{|Y|} \text{ dB} \tag{5.102}$$

Figure 5.22 shows the x-component of the current compared with the results obtained by Glisson and Wilton [12] using an MM procedure. Cuts AA′ and BB′ are indicated in Figure 5.19. In this case, the number of samples was $N_x = N_y = 7$. The CPU time for this case was about 8 to 10 seconds using a PC-486 (33 MHz, 4 MB), which gives a good idea of the power of the scheme under consideration.

5.4.2 Circular Disk

Another important structure is the circular metallic disk. The problem is illustrated in Figure 5.23(a), where an x-polarized plane wave is normal incident upon the structure. The x- and y-components of the current are represented in Figures 5.23(b,c), respectively. The radius of the disk is denoted by a. This is a good test of the convergence of the method when the structure becomes large in terms of $\lambda(k_o a = 10)$. The number of samples used to represent the currents was $N_x = N_y = 31$, and the time required to get a relative error of 5.E-3 was about 3 minutes on a PC-486. In Section 5.5, we will compare the RCS obtained by using these currents with other results available in the literature.

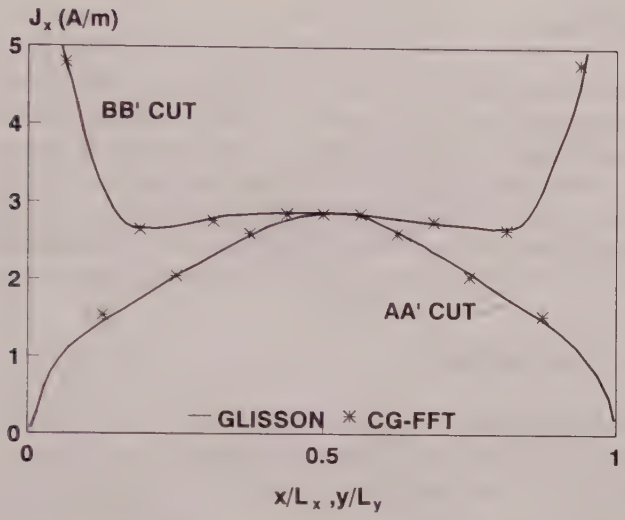


Figure 5.22 Comparison of the CG-FFT and MM solutions.

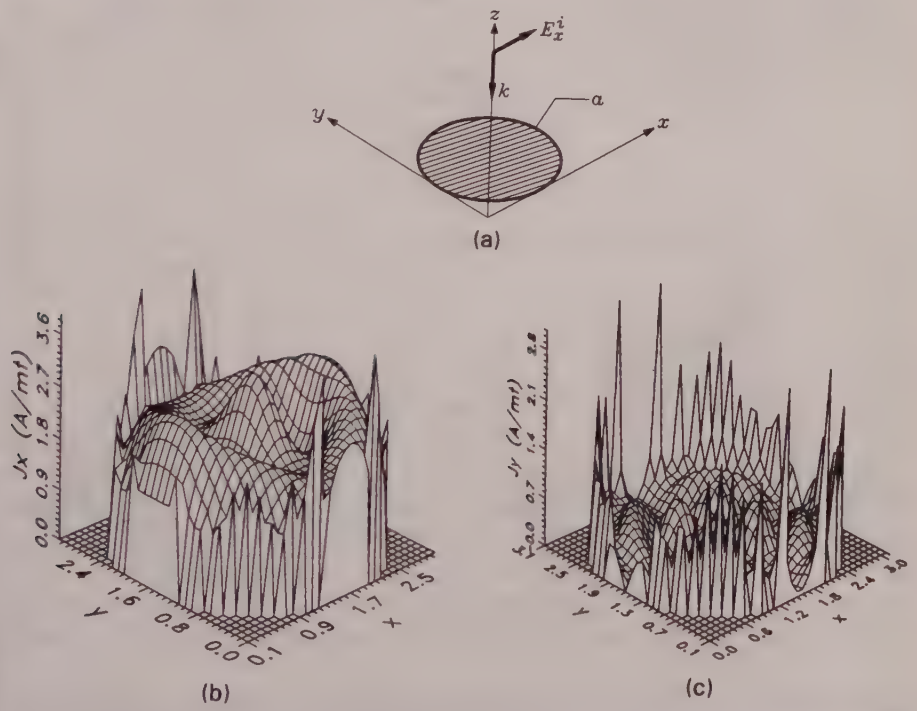


Figure 5.23 (a) Circular disk problem; (b) 2D representation of the x-component of the current; (c) 2D representation of the y-component of the current.

5.4.3 Circular Ring

In Figure 5.24(a), an x-polarized plane wave is normal incident upon a circular ring with outer and inner radii $r_1 = 0.5\lambda$ and $r_2 = 0.25\lambda$, respectively. The current components are represented in Figures 5.24(b,c). The number of samples used to approximate the currents was $N_x = N_y = 31$, and the time required to get a relative error of 5.E-3 was about 15 minutes on a PC-486. In Section 5.5, the bistatic cross section (BCS) obtained by using these currents will be compared with the results found by using the BOR formulation, which will be described in Chapter 8.

5.4.4 Irregular-Edge Surfaces

The first structure is presented in Figure 5.25(a), where an x-polarized plane wave is normal incident upon the plate. The radius of the curved edge is $r = 11.43$ cm, the total dimensions in both directions are $L_x = 25.3238$ cm and $L_y = 22.86$ cm,

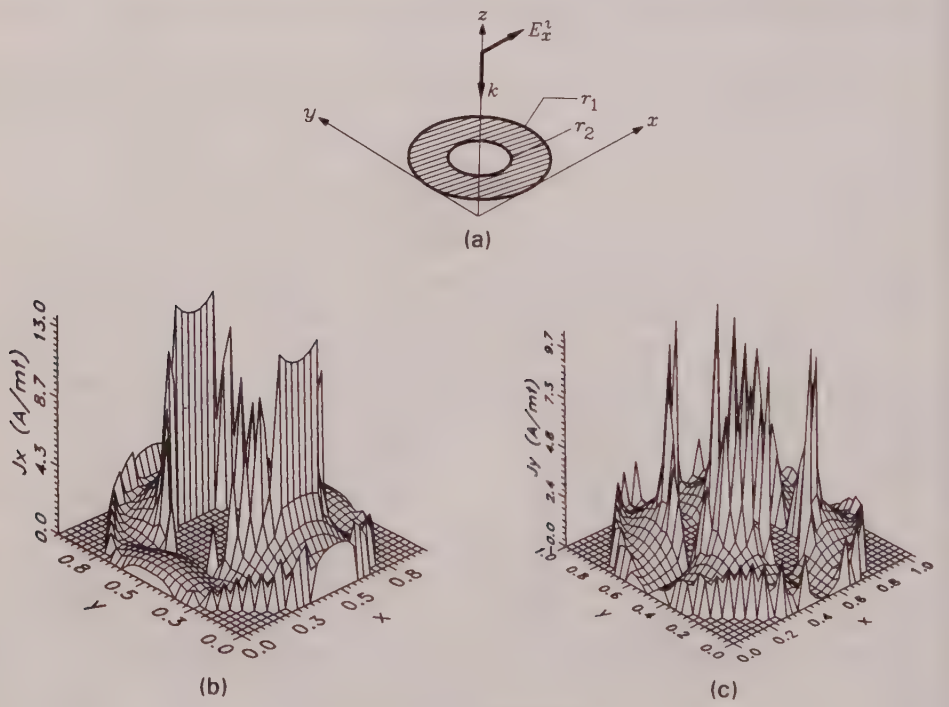


Figure 5.24 (a) Sketch of the circular ring problem and incident wave; (b) 2D representation of the x-component of the current; (c) 2D representation of the y-component of the current.

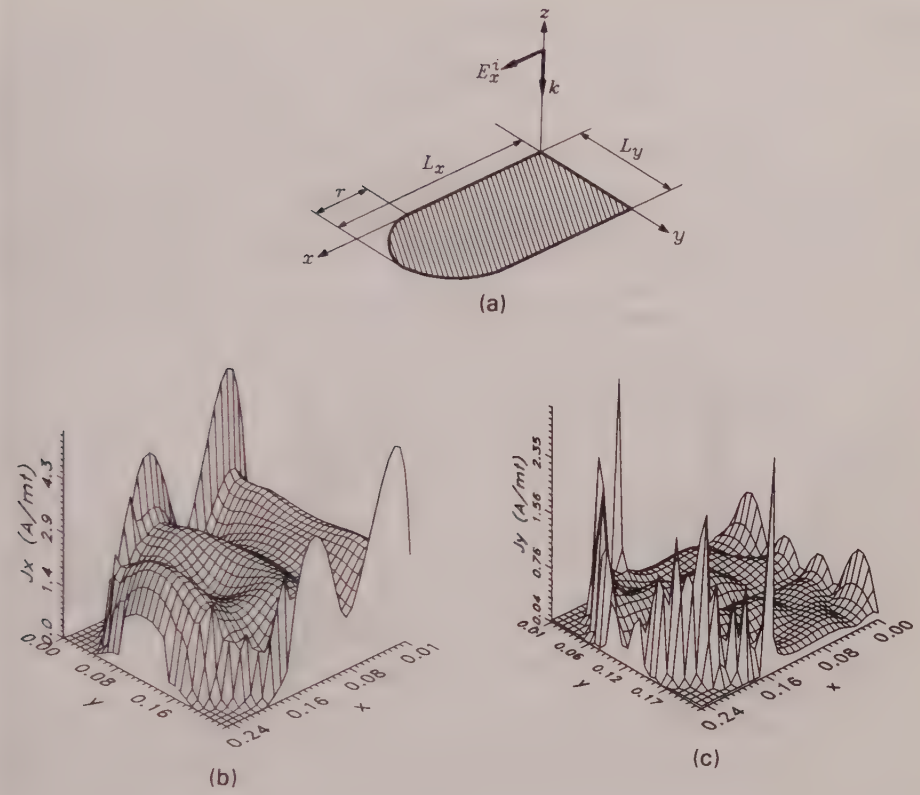


Figure 5.25 (a) Geometry of a square-disk plate; (b) x-component of the current; (c) y-component of the current.

and the frequency $f_0 = 2.9$ GHz. The x- and y-components of the current for $N_x = N_y = 31$ samples are represented in Figures 5.25(b) and 5.25(c), respectively. The time required to obtain a relative error of 5.E-3 was about 5 minutes on a PC-486.

The second structure under analysis is presented in Figure 5.26(a), where, as in previous cases, an x-polarized plane wave is normal incident upon the structure. The radius of the curved edge is $r = 11.43$ cm, the total dimensions in both directions are $L_x = 31.8262$ cm and $L_y = 22.86$ cm, and the frequency is $f_0 = 2.9$ GHz. Figures 5.26(b) and (c) represent the current components with $N_x = N_y = 31$ samples. The time required to obtain a relative error of 5.E-3 was about 6 minutes on a PC-486.

In the next section, the BCS obtained by using these currents for both structures will be compared with results from the literature.

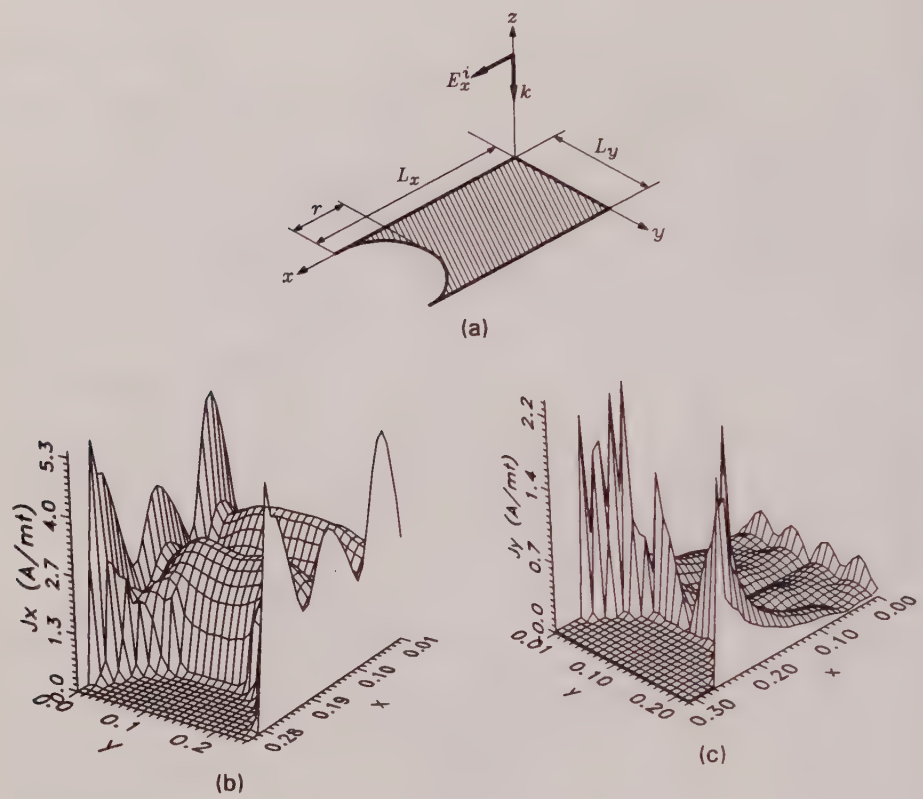


Figure 5.26 (a) Geometry of an irregular-edge surface problem; (b) representation of the x-component of the current; (c) representation of the y-component of the current.

5.5 APPLICATIONS TO RADIATION AND SCATTERING PROBLEMS

Once the equivalent currents on the plate have been found, the radiated fields can be obtained from

$$\mathbf{E} = -j\omega\mathbf{r} \times (\mathbf{A} \times \mathbf{r}) \tag{5.103}$$

where

$$\mathbf{A} = \frac{\mu_0}{4\pi} \frac{\exp[-jk_0 r]}{r} \int_S J(\mathbf{r}') \exp[jk_0 \mathbf{r}\mathbf{r}'] \, ds' \tag{5.104}$$

$$\mathbf{r} = x\mathbf{x} + y\mathbf{y} + z\mathbf{z} \tag{5.105a}$$

$$\mathbf{r}' = x'\mathbf{x} + y'\mathbf{y} \tag{5.105b}$$

and the unit position vector can be described by θ and ϕ as

$$\mathbf{r} = \sin\theta\cos\phi\mathbf{x} + \sin\theta\sin\phi\mathbf{y} + \cos\theta\mathbf{z} \tag{5.106}$$

The geometrical parameters of the problem are depicted in Figure 5.27, with θ and ϕ denoting the angles that define the position vector $\mathbf{r}$ in a spherical coordinate system.

Substituting (5.105b) and (5.106) into (5.104), we have

$$\mathbf{A} = \frac{\mu_0}{4\pi}\frac{\exp[-jk_0r]}{r}\int_0^{L_x}\int_0^{L_y}\mathbf{J}(\mathbf{r}')\exp[jk_0x'\sin\theta\cos\phi] \cdot \exp[jk_0y'\sin\theta\sin\phi]\,dx'\,dy' \tag{5.107}$$

The integral in (5.107) is the CFT of the currents considering the spectral variables

$$k_x = k_0\sin\theta\cos\phi \tag{5.108a}$$

$$k_y = k_0\sin\theta\sin\phi \tag{5.108b}$$

The local plane wave behavior of the field at the observation point $\mathbf{r}$ is described by E_θ and E_ϕ components. By using the corresponding transformations, E_θ

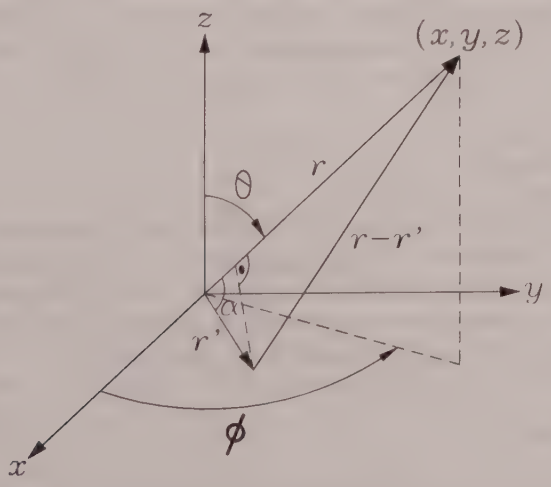


Figure 5.27 Parameters considered in the radiation problem.

and E_ϕ can be written in terms of the CFTs of J_x and J_y:

$$\mathbf{E} = E_\theta \hat{\theta} + E_\phi \hat{\phi} \tag{5.109a}$$

$$E_\theta = E_0[\tilde{J}_x(k_x,k_y) \cos\theta \cos\phi + \tilde{J}_y(k_x,k_y) \cos\theta \sin\phi] \tag{5.109b}$$

$$E_\phi = E_0[\tilde{J}_y(k_x,k_y) \cos\phi - \tilde{J}_x(k_x,k_y) \sin\phi] \tag{5.109c}$$

$$E_0 = -\frac{j\omega\mu_0}{4\pi} \frac{\exp(-jk_0 r)}{r} \tag{5.109d}$$

The RCS is defined in terms of the scattered fields by

$$\sigma = 10 \log\left(4\pi r^2 \frac{|E_\theta|^2 + |E_\phi|^2}{|E^i|^2}\right) \tag{5.110}$$

where E_θ and E_ϕ have already been computed in (5.109b,c) and E^i is the amplitude of the incident plane wave.

We will now present some RCS and BCS results of different structures calculated by using the currents obtained with the CG-FFT scheme previously described.

5.5.1 Square Plate

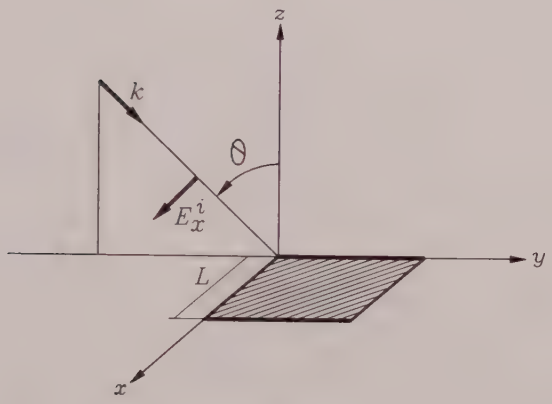
This structure is shown in Figure 5.28(a). An x-polarized plane wave is assumed incident. The results shown in Figure 5.28(b) correspond to the RCS as a function of the angle θ. The number of samples used is $N_x = N_y = 63$ and the numerical results are compared with measurements obtained from [14].

5.5.2 Circular Disk

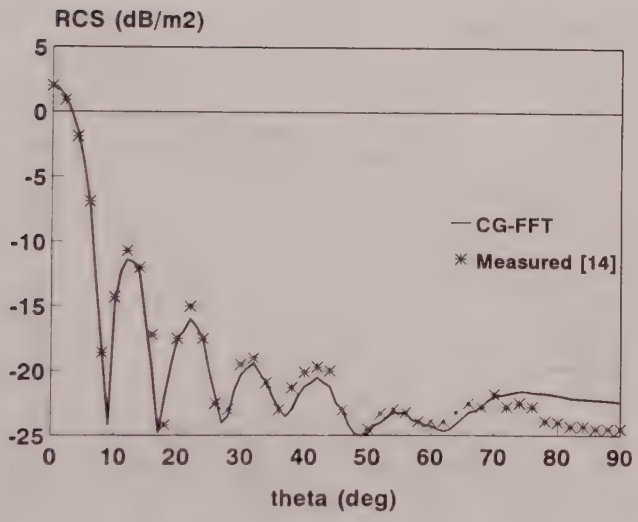
An x-polarized plane wave with normal incidence on a circular disk is depicted in Figure 5.29(a). The radius of the disk is denoted by a. RCS results ($\theta^i = \theta = 0$ deg) as a function of disk size are shown in Figure 5.29(b). The number of samples used to represent the currents varies depending on the size of the disk. The numerical results are compared with those presented by Hodge in [15].

5.5.3 Circular Ring

Figure 5.30 shows the results obtained for the BCS of a circular ring as a function of the scattering angle θ. Results for the E-plane ($\phi = 0$ deg) and H-plane ($\theta = 90$

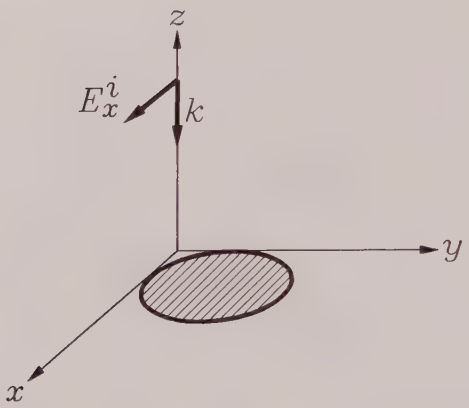


(a)

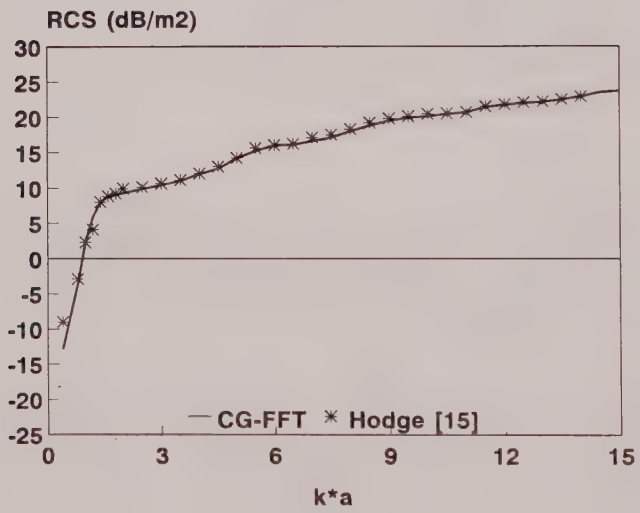


(b)

Figure 5.28 (a) Sketch of the problem; (b) RCS versus aspect angle of a square plate ($L = 3\lambda$, $f_0 =$ 10 GHz).

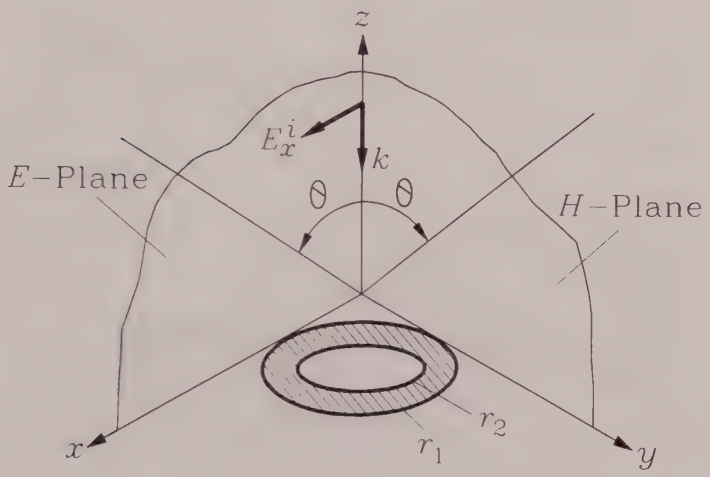


(a)

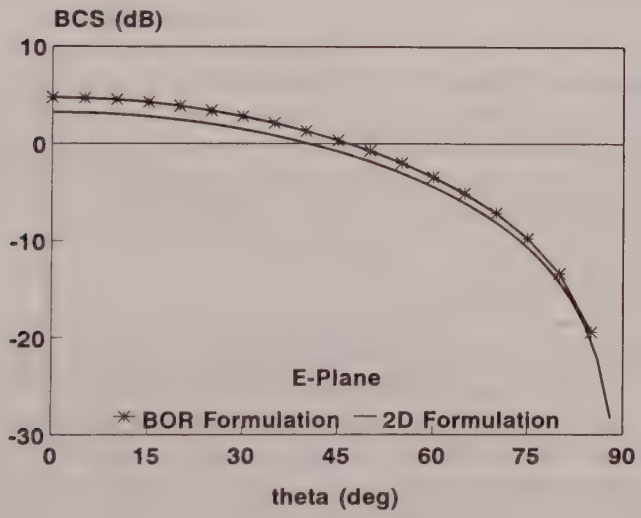


(b)

Figure 5.29 (a) Incident wave; (b) RCS versus size of a disk of radius a.

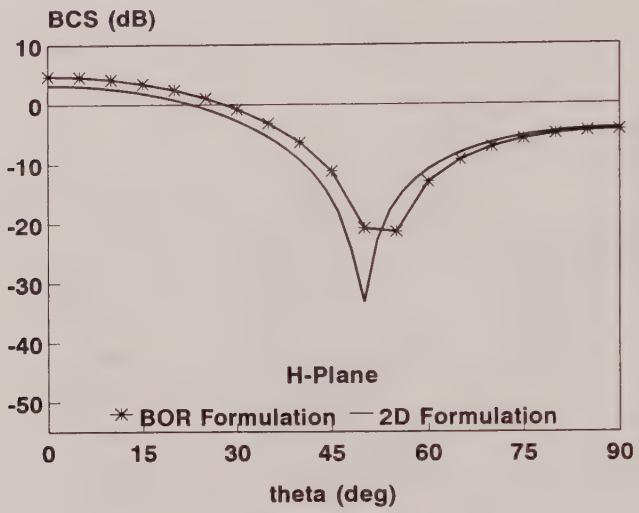


(a)



(b)

Figure 5.30 (a) Parameters considered for the BCS of a circular ring ($r_1 = 0.5\lambda$, $r_2 = 0.25\lambda$); (b) results for the E-plane; (c) results for the H-plane.



(c)

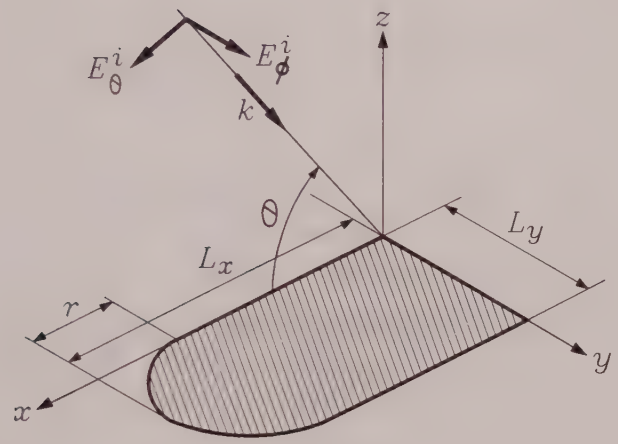
Figure 5.30 (Continued)

deg) are represented in Figures 5.30(b,c), respectively. The radii of the ring are $r_1 = 0.5\lambda$ and $r_2 = 0.25\lambda$. The excitation is once again an x-polarized plane wave normal to the ring, and the number of samples used to represent the current components is $N_x = N_y = 31$. The numerical results are compared with those obtained by using the BOR formulation to be described in Chapter 8.

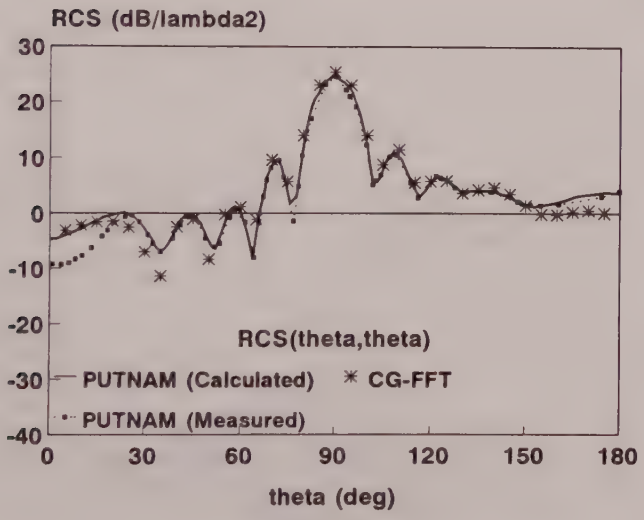
5.5.4 Irregular-Edge Surfaces

The first irregular-edge structure is presented in Figure 5.31(a). The radius of the curved edge is $r = 11.43$ cm, and the rectangular dimensions are $L_x = 25.3238$ cm and $L_y = 22.86$ cm, with a frequency of $f_0 = 2.9$ GHz. The results shown in Figures 5.31(b,c) correspond to the RCS ($\sigma_{\theta\theta}$ and $\sigma_{\phi\phi}$) as a function of θ. The number of samples used to represent the currents is $N_x = N_y = 63$. These numerical results are compared with other numerical data and measurements found in [16].

Figure 5.32(a) shows the second irregular-edge structure analyzed. The dimensions are $L_x = 31.8262$ cm and $L_y = 22.86$ cm, and the radius of the curved edge is $r = 11.43$ cm. The results shown in Figures 5.32(b,c) correspond to the RCS ($\sigma_{\theta\theta}$ and $\sigma_{\phi\phi}$) as a function of θ at a frequency $f_0 = 2.9$ GHz. $N_x = N_y = 63$ samples have been used to represent the currents. The numerical results are compared with measurements from [16].

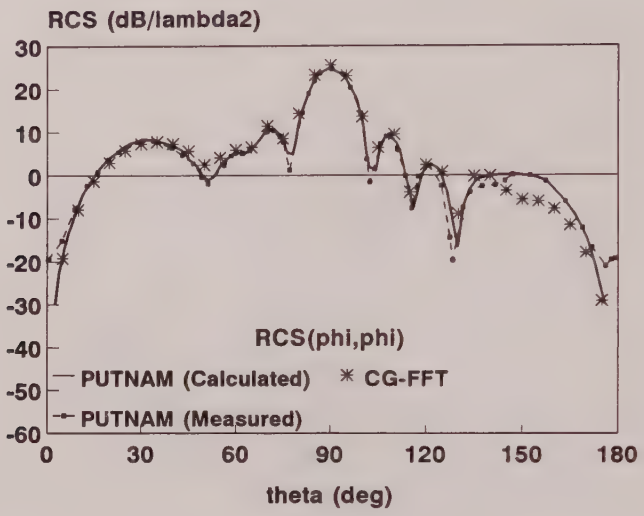


(a)



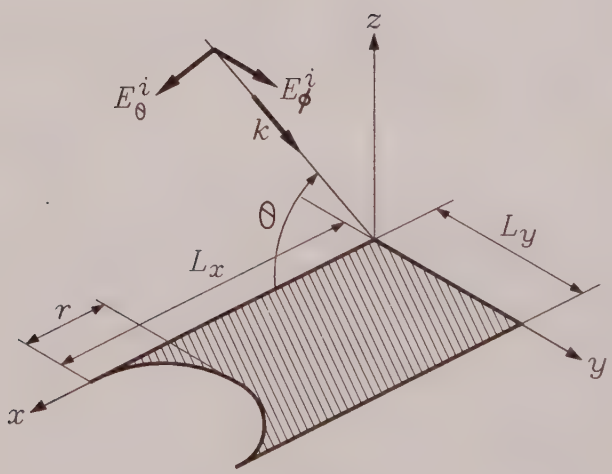
(b)

Figure 5.31 (a) Incident wave on the structure. RCS of the plate: (b) $\sigma_{\theta\theta}$; (c) $\sigma_{\phi\phi}$.



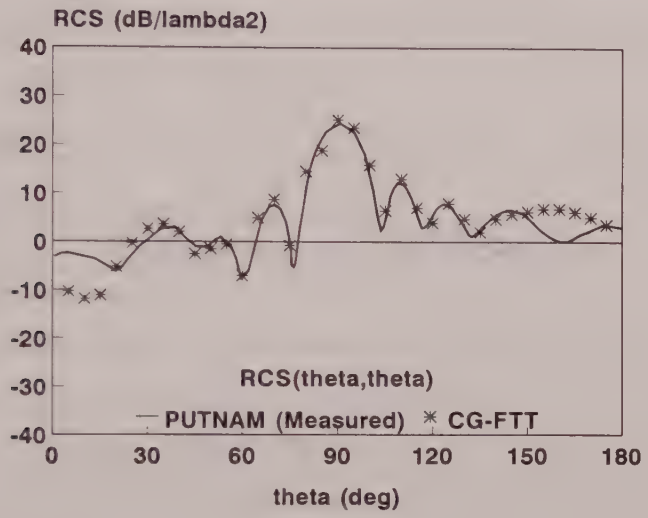
(c)

Figure 5.31 (Continued)

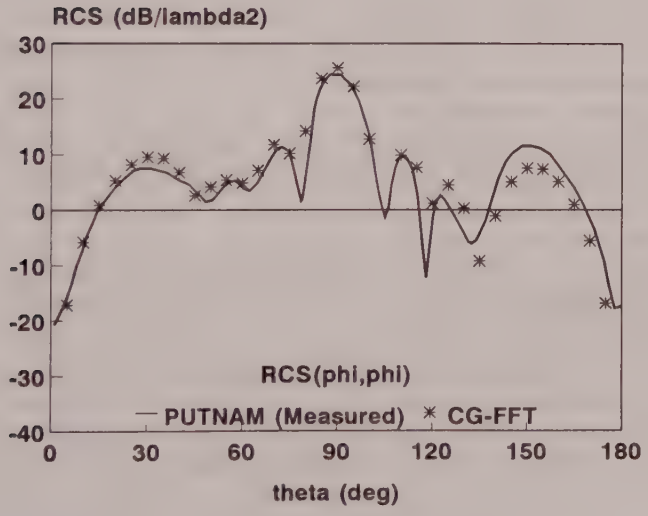


(a)

Figure 5.32 (a) Sketch of the problem. RCS of the plate: (b) $\sigma_{\theta\theta}$; (c) $\sigma_{\phi\phi}$.



(b)



(c)

Figure 5.32 (Continued)

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Chapter 6
Three-Dimensional Bodies

6.1 INTRODUCTION

This chapter deals with the analysis of 3D bodies of arbitrary geometry and material composition. Let us begin by considering the case in which the 3D body is a perfect conductor. Since the fields inside the body are null, we could use an approach based on a surface formulation. The domain of the operator (the surface S) will now be a 3D space instead of the 2D space used in the conducting-plate case. Thus, at first sight, it may appear that this problem is completely analogous to the one analyzed in Chapter 5, except that we are now dealing with a 3D structure. However, the aim of this chapter is broader in the sense that not only perfect conductors but also bodies with different material properties are to be analyzed. Consider, for instance, a homogeneous dielectric body (with or without losses). Although the fields inside do not vanish, we could try to obtain a 2D surface formulation by imposing adequate surface boundary conditions. However, boundary conditions are not easy to obtain when more arbitrary bodies are considered, such as nonhomogeneous structures like biological tissues.

Pioneering works treating 3D arbitrary bodies were based mainly on the MM [1–5]. Nearly all of them focused on the computation of the absorbed power density in human bodies and, due to the inherent limitations of this method, used quite simplified models for the geometry and the material composition of the structures analyzed. Moreover, simple basis functions were used to represent the equivalent volumetric current induced in the material bodies. A more accurate and efficient representation of the current was employed in [6], where a linear piecewise basis function defined over tetrahedrons provided an accurate evaluation of the second derivatives of the electric field integral equation (EFIE).

More recently, finite element and finite difference techniques have also been applied to the general 3D problem [7,8]. Although they lead to notable improvements in the computational efficiency, 3D electrodynamic radiation and scattering problems continue to be beyond the scope of these techniques when using affordable computational resources. Nevertheless, the FD-TD method [8] has significant advantages

for the analysis of problems with very broadband excitations, since the responses to such excitations are inherently provided by this method.

The CG method was introduced in the early 1980s as an efficient and accurate technique that made it possible to treat large problems with thousands or tens of thousands of unknowns by avoiding the storage of the MM matrices. This was accomplished by recomputing the elements of the MM matrix in each CG iteration (see [9] for example). This technique was improved [10] by means of a sophisticated treatment of the singularities in the source region of the Green's function in an MPIE formulation, which uses volumetric polarization currents [11]. However, none of these works took advantage of the properties of the Fourier transform and FFT algorithms to reduce the memory and CPU time required for the computation of the convolution integrals and derivatives in the EFIE.

The idea of employing an FFT algorithm in combination with iterative methods to solve integrodifferential equations in applied electromagnetics was actually introduced by Bojarski two decades ago in his K-space method [12]. This technique became the CG-FFT method when it was improved substantially by using a CG algorithm as the iterative method. The first applications of the CG-FFT method to volumetric material problems can be found in the analysis of the absorption of electromagnetic power by 2D or 3D models of human bodies [13]. Pulse functions defined over regular equidistant rectangular parallelepipeds have frequently been employed [13–15] to expand the current in terms of basis functions. Unfortunately, the use of pulse expansion functions yields poor results and slow convergence when dealing with a problem in which the field must be computed for components that are parallel to the directions of the current. This happens in the general 3D case or in the 2D volumetric problem that arises when a plane wave is incident upon an infinitely long cylinder, with the magnetic component of the incident field parallel to the cylinder axis (transverse electric (TE) case). This anomalous behavior in the numerical results when using pulses as basis functions was experimentally verified by Borup *et al.* [13] for the cylindrical case with TE polarization. A deeper understanding of this phenomenon was provided by Joachimowicz et al. [16], who related the inaccuracies in the results when using pulses to an incorrect treatment of the second derivatives in the EFIE. To mitigate the errors caused by the expansion of the current in pulses, they introduced an integrodifferential equation formulation that made use of generalized functions and included surface integrals to take into account the artificial surface charge densities originated by the pulse functions. Another way of alleviating the poor behavior of pulses in an EFIE formulation can be found in [17]. In this work, a careful treatment of the approximation of derivatives by finite differences and of the computation of the Green's function were presented in the context of a CG-FFT formulation, called in this work the DFT method, which takes advantage of the FFT as proposed by Bojarski in his K-space method.

The incorrect behavior of pulses as basis functions for the EFIE had actually been found earlier by researchers using the MM in the analysis of conducting wires and plates. To overcome this difficulty, Glisson and Wilton proposed the rooftop

function as a basis function in combination with the razor blade as a testing function for use in MM approaches when solving the EFIE [18] for wire or plate structures. This pair of functions provided quite an accurate treatment of the second derivatives appearing in this integral equation. Later, these functions were generalized for volumetric currents by using tetrahedrons, as was proposed in [6] for an MM scheme. In the CG-FFT scenario, volumetric generalizations of the rooftop functions have been used [19–23] for the analysis of bodies of arbitrary material composition, including conductors. In [19–21], the rooftops chosen as basis and testing functions were used in conjunction with the Galerkin procedure. In these works, the integrals of the EFIE were transformed after some analytical operations into an operator equation in which only convolutions with the magnetic potential vector appear. Good numerical predictions are obtained with this technique. However, this approach apparently requires the CG-FFT discrete mesh to have the same shift between samples in the three coordinate directions x, y, and z (i.e., $\Delta x = \Delta y = \Delta z$) and the same number of samples ($N_x = N_y = N_z$). In [22,23], rooftops were also used, but in this case razor blades were employed as testing functions in a line of work that can be considered as an extension to the volumetric case of the MM formulation developed in [18] or in Chapter 5 of this book to treat plate problems.

The purpose of this chapter is to illustrate how a 3D formulation can be applied together with the CG-FFT scheme to study arbitrary bodies. If CPU time and memory requirements are important in any numerical problem, they become serious concerns when dealing with 3D structures. For instance, in an MM formulation, the size of the matrices for a 3D problem will be on the order of $N^3 \times N^3$, where N is the number of basis functions in each direction of a 3D cartesian coordinate system. The MM formulation typically requires CPU time that is proportional to $(N^3)^3$; that is, N^9. Thus, the memory and CPU time required by an MM formulation are huge, which implies that only very small general 3D structures can be treated by such MM formulations. However, these kinds of very large problems can be studied efficiently when applying a 3D discrete formulation by using considerations similar to those employed in Chapter 5 for the flat metallic plate problem. It will be shown that with a CG-FFT formulation, the CPU time and memory requirements are reduced substantially, being proportional to $(N^3)^{3/2} \log N^3$ and N^3, respectively. This reduction makes it possible to tackle the analysis of 3D problems of relatively large size (e.g., several wavelengths). Once again we will be concerned with time-harmonic electromagnetic field problems at frequency f_0, and the spectral domain will now be a 3D discrete space related to a 3D discrete representation of the physical space by the direct or inverse DFT.

6.1.1 General Formulation

The general problem to be solved is shown in Figure 6.1. An arbitrary 3D body is illuminated by a field $\mathbf{E}^i$ incident in the direction (θ^i, ϕ^i). The field is created by an impressed source $\mathbf{J}^i$ anywhere in space. These fields induce a current $\mathbf{J}$ in the body.

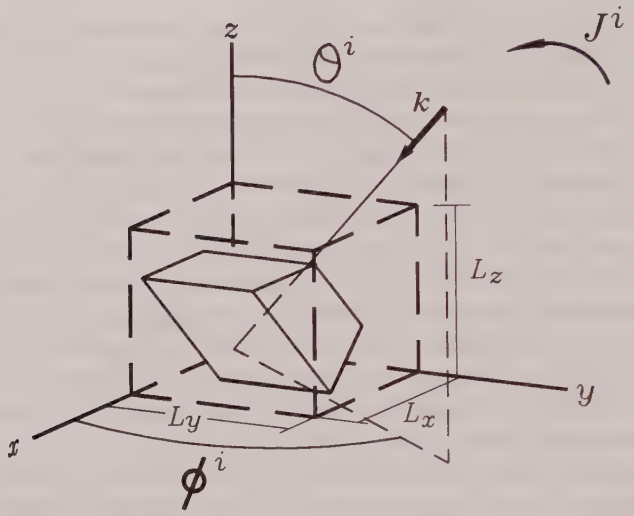


Figure 6.1 Schematic representation of the general 3D problem.

The *total field* (electric and magnetic) must satisfy Maxwell's equations anywhere in space, and in particular,

$$\nabla \times \mathbf{H} = \mathbf{J}^T + j\omega\varepsilon\mathbf{E} \tag{6.1}$$

where $\mathbf{E}$ and $\mathbf{H}$ denote the total electric and magnetic fields, respectively, and ε represents the dielectric constant. The term $\mathbf{J}^T$ represents the total physical currents; that is, the impressed current plus the current due to Ohm's law in the conducting medium. We can rewrite (6.1) as

$$\nabla \times \mathbf{H} = \mathbf{J}^i + \sigma\mathbf{E} + j\omega\varepsilon\mathbf{E} \tag{6.2}$$

where the total physical currents have been separated into impressed currents and those due to conduction phenomena (σ denotes the conductivity of the body). Writing (6.2) in terms of the relative dielectric constants, we obtain

$$\nabla \times \mathbf{H} = \mathbf{J}^i + j\omega\varepsilon_0\left(\varepsilon_r - j\frac{\sigma}{\omega\varepsilon_0}\right)\mathbf{E} \tag{6.3}$$

The expression in parentheses on the right-hand side represents the well-known complex dielectric constant,

$$\varepsilon_c = \varepsilon' - j\varepsilon''$$

$$\varepsilon' = \varepsilon_r \tag{6.4}$$

$$\varepsilon'' = \sigma/\omega\varepsilon_0$$

In equations (6.1) through (6.4), no restrictions have been imposed on ε and σ, which means that, in general, they will be point-dependent functions, $\varepsilon(\mathbf{r})$ and $\sigma(\mathbf{r})$.

If the equivalence theorem is applied [24], the fields can be obtained by replacing the original body by the vacuum plus the equivalent volumetric currents. These concepts are illustrated in Figure 6.2.

Rewriting (6.2) as

$$\nabla \times \mathbf{H} = \mathbf{J}^i + [j\omega(\varepsilon - \varepsilon_0) + \sigma]\mathbf{E} + j\omega\varepsilon_0\mathbf{E} \tag{6.5}$$

and enforcing

$$\mathbf{J}_{eq} = [j\omega(\varepsilon - \varepsilon_0) + \sigma]\mathbf{E} \quad \text{on} \quad V \tag{6.6}$$

we obtain the equation for the equivalent problem. Since $\mathbf{E}$ in (6.6) represents the total field in the volume (sum of incident and scattered fields), we have

$$\mathbf{J}_{eq} = [j\omega(\varepsilon - \varepsilon_0) + \sigma](\mathbf{E}^i + \mathbf{E}^s) \quad \text{on} \quad V \tag{6.7}$$

and

$$\mathbf{E}^i = \frac{\mathbf{J}_{eq}}{j\omega(\varepsilon - \varepsilon_0) + \sigma} - \mathbf{E}^s \quad \text{on} \quad V \tag{6.8}$$

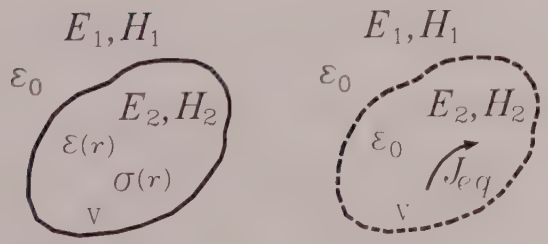


Figure 6.2 Equivalent problem for a 3D structure.

If $\varepsilon = \varepsilon_0$ and σ goes to infinity, (6.8) becomes the well-known equation for perfectly conducting bodies:

$$\mathbf{E}^i = -\mathbf{E}^s \quad \text{on} \quad V \tag{6.9}$$

Since the equivalent current must satisfy this equation, the current in a perfect conductor will be a surface current that can be treated as a particular case of volumetric currents. For the sake of simplicity, we will use $\mathbf{J}$ instead of $\mathbf{J}_{eq}$ to denote the equivalent current. We can then obtain an integral operator equation for the equivalent problem of the following form:

$$\mathbf{E}^i = \frac{\mathbf{J}}{j\omega(\varepsilon - \varepsilon_0) + \sigma} - \mathbf{E}^s(\mathbf{J}) \quad \text{on} \quad V \tag{6.10a}$$

$$\mathbf{E}^s(\mathbf{J}) = -j\omega\mu_0 \int_V \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' + \frac{1}{j\omega\varepsilon_0} \nabla \int_V \nabla \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' \quad \text{on} \quad V \tag{6.10b}$$

where

$$G(\mathbf{r},\mathbf{r}') = \frac{\exp(-jk_0|\mathbf{r} - \mathbf{r}'|)}{4\pi|\mathbf{r} - \mathbf{r}'|} \tag{6.11a}$$

$$\mathbf{r} = x\mathbf{x} + y\mathbf{y} + z\mathbf{z} \tag{6.11b}$$

$$\mathbf{r}' = x'\mathbf{x} + y'\mathbf{y} + z'\mathbf{z} \tag{6.11c}$$

Or, in matrix form,

$$\begin{bmatrix} E_x^i \\ E_y^i \\ E_z^i \end{bmatrix} = k^M \begin{bmatrix} J_x \\ J_y \\ J_z \end{bmatrix} + k^c \begin{bmatrix} L_{xx}^c J_x \\ L_{yy}^c J_y \\ L_{zz}^c J_z \end{bmatrix} + k^q \begin{bmatrix} L_{xx}^q & L_{xy}^q & L_{xz}^q \\ L_{yx}^q & L_{yy}^q & L_{yz}^q \\ L_{zx}^q & L_{zy}^q & L_{zz}^q \end{bmatrix} \begin{bmatrix} J_x \\ J_y \\ J_z \end{bmatrix} \tag{6.12}$$

where, as in Chapter 5 (5.13), k^c and k^q are defined as

$$k^c = j\omega\mu_0 \tag{6.13a}$$

$$k^q = -\frac{1}{j\omega\varepsilon_0} \tag{6.13b}$$

and

$$k^M(\mathbf{r}) = \frac{1}{j\omega[\varepsilon(\mathbf{r}) - \varepsilon_0] + \sigma(\mathbf{r})} \qquad \mathbf{r} \in V \tag{6.13c}$$

As in (5.15a,b), the single operators can now be represented as

$$L^c_{\alpha\alpha} A = \int_V A(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' \tag{6.14a}$$

$$L^q_{\alpha\beta} A = \frac{\partial}{\partial\alpha} \int_V \frac{\partial A(\mathbf{r}')}{\partial\beta'} G(\mathbf{r},\mathbf{r}')\, dv' \tag{6.14b}$$

with α and β now representing x, y, or z, depending on the particular term in (6.12). Once the currents are obtained, the problem is exactly the same as the one formulated in Chapter 5, except that volumetric sources (3D currents) are involved instead of surface currents.

6.2 DISCRETIZATION PROCESS

The mathematical formulation of the problem is completely described by (6.12). Using an approach similar to the one employed in Section 5.2 for the flat metallic plate problem, we will define a set of basis functions that provides a satisfactory approximation of the behavior of the volumetric currents.

6.2.1 Three-Dimensional Basis Functions—Representation of the Problem

First, let us inscribe the body within a parallelepiped of dimensions L_x, L_y, and L_z (maximum dimensions of the body in the x-, y-, and z-directions) and subdivide it into $(N_x + 1)$ by $(N_y + 1)$ by $(N_z + 1)$ small parallelepipeds, with N_x, N_y, and N_z being the number of subdivisions in each direction (Figure 6.3). In each one of the small parallelepipeds, the material is assumed to be homogeneous, and therefore k^M will be constant inside it. A nonhomogeneous body can be analyzed by approximating the material constant in each parallelepiped by its value at the parallelepiped center (piecewise constant approximation).

For the current, we will once again use a procedure that is similar to the one employed for a conducting plate in Chapter 5. Consider a cubic body like the one shown in Figure 6.4(a). The volumetric currents must be able to represent the behavior of the currents on each one of the surfaces of the cube. On $S1$, the normal component will be J_x, while the tangential components will be given by J_y and J_z.

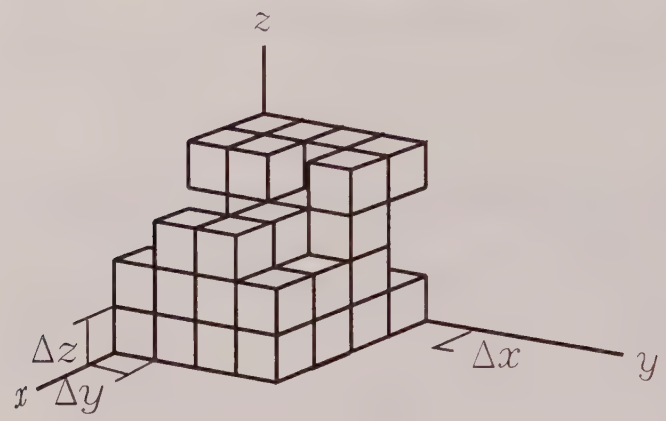


Figure 6.3 3D subsectional approximation of the original body.

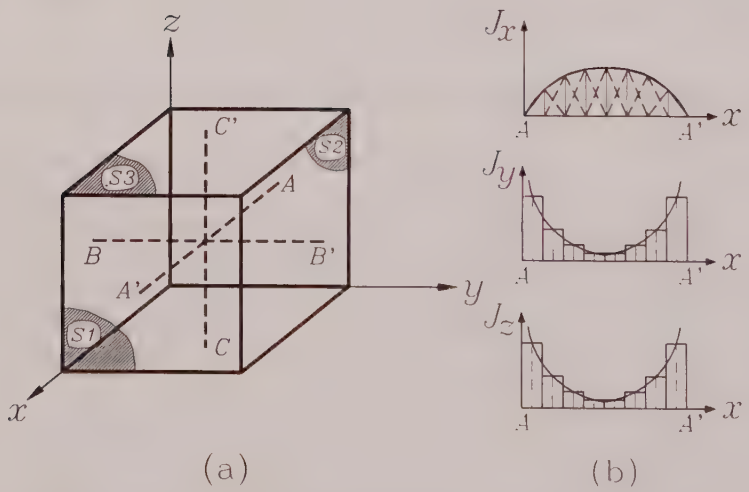


Figure 6.4 Expected behavior of the current components in a perfectly conducting cube.

This means that along the AA′ cut the current components must behave as is shown in Figure 6.4(b), allowing the normal component J_x to vanish at the surface. For a perfectly conducting cube, the tangential components of the current on *S1*, J_y and J_z, will vanish inside the cube (null currents inside a perfect conductor) and will approximate a delta function when approaching the surface (volumetric currents turning into surface ones).

As in the case of the plate problem, a good representation of this behavior can be obtained by using triangle functions for J_x and pulse functions for J_y and J_z in the x-direction, giving rise to the 3D version of rooftop functions. Similar considerations can be employed when analyzing the surfaces *S2* and *S3* and the cuts BB′ and CC′. The corresponding representations will be

$$J_x \begin{Bmatrix} \text{triangles in } x\text{-direction} \\ \text{pulses in } y\text{- and } z\text{-directions} \end{Bmatrix} x\text{-directed rooftops}$$

$$J_y \begin{Bmatrix} \text{triangles in } y\text{-direction} \\ \text{pulses in } x\text{- and } z\text{-directions} \end{Bmatrix} y\text{-directed rooftops}$$

$$J_z \begin{Bmatrix} \text{triangles in } z\text{-direction} \\ \text{pulses in } x\text{- and } y\text{-directions} \end{Bmatrix} z\text{-directed rooftops}$$

The 3D rooftop functions can then be defined as

$$T^x_{mnk}(x,y,z) = \begin{cases} 1 - \dfrac{|x - m\Delta x|}{\Delta x} & \text{over } D^x_{mnk} \\ 0 & \text{elsewhere} \end{cases} \tag{6.15a}$$

$$T^y_{mnk}(x,y,z) = \begin{cases} 1 - \dfrac{|y - m\Delta y|}{\Delta y} & \text{over } D^y_{mnk} \\ 0 & \text{elsewhere} \end{cases} \tag{6.15b}$$

$$T^z_{mnk}(x,y,z) = \begin{cases} 1 - \dfrac{|z - m\Delta z|}{\Delta z} & \text{over } D^z_{mnk} \\ 0 & \text{elsewhere} \end{cases} \tag{6.15c}$$

for the x-, y-, and z-components, respectively. The definition domains of these functions are depicted in Figure 6.5 and are given mathematically by

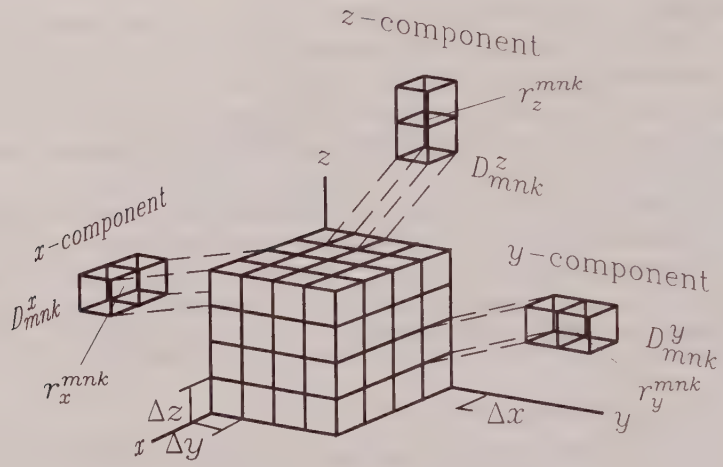


Figure 6.5 Basis and testing functions are defined in parallelepiped doublets.

$$D_{mnk}^x = \begin{cases} (m-1)\Delta x < x < (m+1)\Delta x \\ (n-1)\Delta y < y < n\Delta y \\ (k-1)\Delta z < z < k\Delta z \end{cases} \tag{6.16a}$$

$$D_{mnk}^y = \begin{cases} (m-1)\Delta x < x < m\Delta x \\ (n-1)\Delta y < y < (n+1)\Delta y \\ (k-1)\Delta z < z < k\Delta z \end{cases} \tag{6.16b}$$

$$D_{mnk}^z = \begin{cases} (m-1)\Delta x < x < m\Delta x \\ (n-1)\Delta y < y < n\Delta y \\ (k-1)\Delta z < z < (k+1)\Delta z \end{cases} \tag{6.16c}$$

The centers of the 3D rooftop functions will be

$$r_x^{mnk} = m\Delta x\mathbf{x} + (n-0.5)\Delta y\mathbf{y} + (k-0.5)\Delta z\mathbf{z} \tag{6.17a}$$

$$r_y^{mnk} = (m-0.5)\Delta x\mathbf{x} + n\Delta y\mathbf{y} + (k-0.5)\Delta z\mathbf{z} \tag{6.17b}$$

$$r_z^{mnk} = (m-0.5)\Delta x\mathbf{x} + (n-0.5)\Delta y\mathbf{y} + k\Delta z\mathbf{z} \tag{6.17c}$$

which constitutes an extension to three dimensions of the definition used in the 2D case (5.28). The currents can be written as

$$J_x(x,y,z) = \sum_{m=1}^{N_x} \sum_{n=1}^{N_y+1} \sum_{k=1}^{N_z+1} J_x^D[m,n,k]\, T_{mnk}^x(x,y,z) \tag{6.18a}$$

$$J_y(x,y,z) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y} \sum_{k=1}^{N_z+1} J_y^D[m,n,k]\, T_{mnk}^y(x,y,z) \tag{6.18b}$$

$$J_z(x,y,z) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} \sum_{k=1}^{N_z} J_z^D[m,n,k]\, T_{mnk}^z(x,y,z) \tag{6.18c}$$

where the summation limits have been adjusted in accordance with the current discretization process. The geometry is now approximated in three different ways, depending on the current component being considered, in order to represent the surface behavior described above.

The considerations taken into account when choosing the testing functions are very similar to those used in Chapter 5 (the fields around testing points are smoothed in the direction of the second derivative $L_{\alpha\alpha}^q$, making $\beta = \alpha$ in (6.14b)), and 3D razor blade functions are obtained, which are centered at r_x^{mnk}, r_y^{mnk}, and r_z^{mnk}, defined in (6.17a–c).

$$W_{mnk}^x(x,y,z) = \begin{cases} 1 & \text{if} \begin{cases} (m-0.5)\Delta x < x < (m+0.5)\Delta x \\ y = 0.5n\Delta y \\ z = 0.5k\Delta z \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{6.19a}$$

$$W_{mnk}^y(x,y,z) = \begin{cases} 1 & \text{if} \begin{cases} x = 0.5m\Delta x \\ (n-0.5)\Delta y < y < (n+0.5)\Delta y \\ z = 0.5k\Delta z \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{6.19b}$$

$$W_{mnk}^z(x,y,z) = \begin{cases} 1 & \text{if} \begin{cases} x = 0.5m\Delta x \\ y = 0.5n\Delta y \\ (k-0.5)\Delta z < z < (k+0.5)\Delta z \end{cases} \\ 0 & \text{elsewhere} \end{cases} \tag{6.19c}$$

These razor blades are similar in form to those presented in Figure 5.12, but they are extended into a 3D space (domains of definition in Figure 6.5), and the testing procedure is like (5.31) but extended now to the volume of the body:

$$V_\alpha(x,y,z) = \int_V W^\alpha_{mnk}(x,y,z)\, E_\alpha(x,y,z)\, dv \tag{6.20}$$

where α represents x, y, or z, depending on the field component under consideration (E_x, E_y, or E_z).

6.3 DISCRETIZATION OF THE INTEGRAL EQUATION, RESOLUTION OF THE OPERATOR, AND FINAL RESULTS

6.3.1 Discretization of the Integral Equation

The mathematical procedure for obtaining discrete expressions for equations (6.10–6.14) and the corresponding resolution of these expressions follow the scheme described in Section 5.3 for the 2D arbitrary flat-plate problem. In this case, there are three unknown functions (x-, y-, and z-components of the equivalent current) that have to be determined by using their approximate expressions in terms of 3D basis functions. Each field component will be smoothed by using the convolution procedure described in (6.20). These procedures (expansion and testing) will lead, after sampling with a train of delta functions, to a discrete expression of the original equation. As has already been described in the previous chapter, several approximations are involved in the process: (1) the integral of the testing functions times the inductive part of the operator is approximated by the value of this operator at the interval center times the value of the interval, (2) the integral of a triangle function times a general function f is approximated by the integral of a pulse function times f with the correspondingly adjusted integration interval, (3) the derivative of the current (the charge) is considered equal to the discrete value of the current at the present point minus the discrete value at the previous point divided by the value of the interval, and (4) the original integration domain V is transformed into discrete unit cell domains, and so on. Taking all these considerations into account, we obtain a final discrete operator in terms of a discrete Green's function integral, which is analogous to the one defined in (5.76), but is redefined for the 3D case:

$$G^D[m,n,k] = \int_0^{\Delta x}\int_0^{\Delta y}\int_0^{\Delta z} G(x_m,y_n,z_k;x',y',z')\, dx'\, dy'\, dz' \tag{6.21}$$

where the discrete observation points are now defined by

$$x_m = (m + 0.5)\Delta x$$

$$y_n = (n + 0.5)\Delta y \tag{6.22}$$

$$z_k = (k + 0.5)\Delta z$$

This last integral is evaluated using again a Gauss quadrature rule with a number of quadrature points $2 \times 2 \times 2$, $8 \times 8 \times 8$, or $32 \times 32 \times 32$, depending on the distance between source and observation domains: far, near, and coincident, respectively.

We obtain discrete linear convolutions that are completely equivalent to those in (5.79), except that they are extended to three dimensions.

$$V_{xx}^{Dc}[m,n,k] \approx \Delta x J_x^D[m,n,k] * G^D[m,n,k] \tag{6.23a}$$

$$V_{xx}^{Dq}[m,n,k] \approx \frac{1}{\Delta x}\left[J_x^D[m,n,k] - J_x^D[m-1,n,k]\right] * \left[G^D[m+1,n,k] - G^D[m,n,k]\right] \tag{6.23b}$$

$$V_{xy}^{Dq}[m,n,k] \approx \frac{1}{\Delta y}\left[J_y^D[m,n,k] - J_y^D[m,n-1,k]\right] * \left[G^D[m+1,n,k] - G^D[m,n,k]\right] \tag{6.23c}$$

$$V_{xz}^{Dq}[m,n,k] \approx \frac{1}{\Delta z}\left[J_z^D[m,n,k] - J_z^D[m,n,k-1]\right] * \left[G^D[m+1,n,k] - G^D[m,n,k]\right] \tag{6.23d}$$

$$V_{yy}^{Dc}[m,n,k] \approx \Delta y J_y^D[m,n,k] * G^D[m,n,k] \tag{6.24a}$$

$$V_{yx}^{Dq}[m,n,k] \approx \frac{1}{\Delta x}\left[J_x^D[m,n,k] - J_x^D[m-1,n,k]\right] * \left[G^D[m,n+1,k] - G^D[m,n,k]\right] \tag{6.24b}$$

$$V_{yy}^{Dq}[m,n,k] \approx \frac{1}{\Delta y}\left[J_y^D[m,n,k] - J_y^D[m,n-1,k]\right] * \left[G^D[m,n+1,k] - G^D[m,n,k]\right] \tag{6.24c}$$

$$V_{yz}^{Dq}[m,n,k] \approx \frac{1}{\Delta z}\left[J_z^D[m,n,k] - J_z^D[m,n,k-1]\right] * \left[G^D[m,n+1,k] - G^D[m,n,k]\right] \tag{6.24d}$$

$$V_{zz}^{Dq}[m,n,k] \approx \Delta z J_z^D[m,n,k] * G^D[m,n,k] \tag{6.25a}$$

$$V_{zx}^{Dq}[m,n,k] \approx \frac{1}{\Delta x}\left[J_x^D[m,n,k] - J_x^D[m-1,n,k]\right] * \left[G^D[m,n,k+1] - G^D[m,n,k]\right] \tag{6.25b}$$

$$V_{zy}^{Dq}[m,n,k] \approx \frac{1}{\Delta y}\left[J_y^D[m,n,k] - J_y^D[m,n-1,k]\right] * \left[G^D[m,n,k+1] - G^D[m,n,k]\right] \tag{6.25c}$$

$$V_{zz}^{Dq}[m,n,k] \approx \frac{1}{\Delta z}\left[J_z^D[m,n,k] - J_z^D[m,n,k-1]\right] * \left[G^D[m,n,k+1] - G^D[m,n,k]\right] \tag{6.25d}$$

6.3.2 Evaluation of the Operator

Each term is defined by a discrete linear convolution of the current components and G^D. The computations can be performed by turning these convolutions into cyclic ones, which involves a proper periodification of the current components and the Green's function integral. The periodification process is similar to the one presented in Section 5.3.5, but with the corresponding 3D extension to seven new cubes, as is shown in Figure 6.6. As mentioned in Chapter 3, the final period in each direction

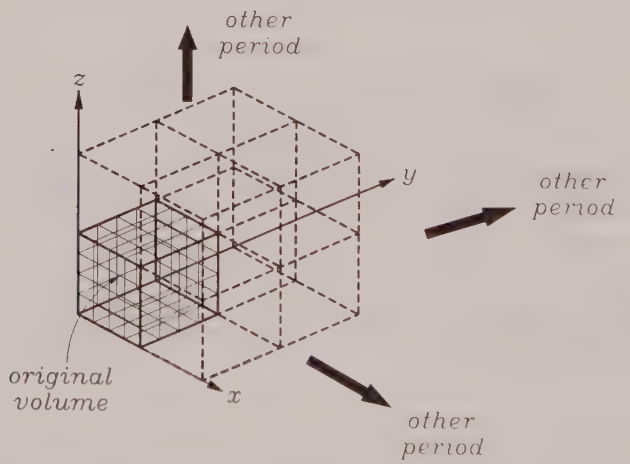


Figure 6.6 3D periodification process from the original volumetric definition domain.

must be at least twice the original problem size to avoid aliasing in the spatial domain and to obtain accurate results. Probably in some cases, mainly when electrically large problems are considered, this operation can be carried out without this large zero padding, assuming that some errors can result from the computational scheme, especially when particular wave propagation phenomena are associated with particular excitations. In any case, the increase in the memory storage requirements can be reduced by using two FFTs of length $N + 1$ instead of one of size $2N + 2$. These two FFTs relate the lower ($n < N + 1$) and upper bands ($n \geq N + 1$) of the sequence to the even and odd values of the discrete spectral variable. This trick reduces the storage requirements but increases the run time. For more details about this trick, the reader is referred to Chapter 3.

The new periodic functions, denoted by J_α^P and G^P, are defined in terms of the original ones as

$$J_\alpha^P[m,n,k] = \begin{cases} J_\alpha^D[m,n,k] & \text{for} \quad m < \dfrac{N_x^T}{2}, n < \dfrac{N_y^T}{2}, k < \dfrac{N_z^T}{2} \\ 0 & \text{for} \begin{cases} \dfrac{N_x^T}{2} < m < N_x^T \\ \dfrac{N_y^T}{2} < n < N_y^T \\ \dfrac{N_z^T}{2} < k < N_z^T \end{cases} \end{cases} \tag{6.26}$$

$$G^P[m,n,k] = \begin{cases} G^D[m,n,k] & \text{for} \quad m < \dfrac{N_x^T}{2}, n < \dfrac{N_y^T}{2}, k < \dfrac{N_z^T}{2} \\ G^D[N_x^T - m, N_y^T - n, N_z^T - k] & \text{for} \begin{cases} \dfrac{N_x^T}{2} < m < N_x^T \\ \dfrac{N_y^T}{2} < n < N_y^T \\ \dfrac{N_z^T}{2} < k < N_z^T \end{cases} \end{cases} \tag{6.27}$$

where

$$N_x^T = 2(N_x + 1) \quad N_y^T = 2(N_y + 1) \quad N_z^T = 2(N_z + 1) \tag{6.28}$$

As indicated in Chapter 3, these values represent the minimum periods necessary to obtain correct results for the convolution in the domain of interest.

6.3.3 Final Results

We are now ready to write the final results for the discrete operator (similar to those in (5.83) through (5.94)), where (p,q,r) denote the indexes in the spectral domain:

$$V_{xx}^{P}[m,n,k] = \frac{k^{c}\Delta x}{N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{\tilde{J}_x^P[p,q,r]\tilde{G}^P[p,q,r]\right\} + \frac{k^{q}}{\Delta x N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_x-1)(1-F_x^*)\tilde{J}_x^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29a}$$

$$V_{xy}^{P}[m,n,k] = \frac{k^{q}}{\Delta y N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_x-1)(1-F_y^*)\tilde{J}_y^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29b}$$

$$V_{xz}^{P}[m,n,k] = \frac{k^{q}}{\Delta z N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_x-1)(1-F_z^*)\tilde{J}_z^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29c}$$

$$V_{yx}^{P}[m,n,k] = \frac{k^{q}}{\Delta x N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_y-1)(1-F_x^*)\tilde{J}_x^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29d}$$

$$V_{yy}^{P}[m,n,k] = \frac{k^{c}\Delta y}{N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{\tilde{J}_y^P[p,q,r]\tilde{G}^P[p,q,r]\right\} + \frac{k^{q}}{\Delta y N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_y-1)(1-F_y^*)\tilde{J}_y^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29e}$$

$$V_{yz}^{P}[m,n,k] = \frac{k^{q}}{\Delta z N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_y-1)(1-F_z^*)\tilde{J}_z^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29f}$$

$$V_{zx}^{P}[m,n,k] = \frac{k^{q}}{\Delta x N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_z-1)(1-F_x^*)\tilde{J}_x^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29g}$$

$$V_{zy}^{P}[m,n,k] = \frac{k^{q}}{\Delta y N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_z-1)(1-F_y^*)\tilde{J}_y^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29h}$$

$$V_{zz}^{P}[m,n,k] = \frac{k^{c}\Delta z}{N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{\tilde{J}_z^P[p,q,r]\tilde{G}^P[p,q,r]\right\} + \frac{k^{q}}{\Delta z N_x^T N_y^T N_z^T}\,\mathrm{FFT}^{-1}\left\{(F_z-1)(1-F_z^*)\tilde{J}_z^P[p,q,r]\tilde{G}^P[p,q,r]\right\} \tag{6.29i}$$

where

$$V^D_{\alpha\beta}[m,n,k] = V^P_{\alpha\beta}[m,n,k]$$

$$\text{for} \quad 0 < m < \frac{N^T_x}{2},\ 0 < n < \frac{N^T_y}{2},\ 0 < k < \frac{N^T_z}{2} \tag{6.30}$$

The expressions in (6.23) to (6.25) and (6.29) correspond to the operator on the right-hand side of (6.12) without including the material composition term. The sampling points of the material constant k^M will be the same as for the charges (derivatives of the currents), as is shown in Figure 6.7 for a cut in a z-constant plane.

If two adjacent cells have different material constants, this should be taken into account by the operator equation. This can easily be accomplished by noticing that the centers of the rooftops are located halfway between the centers of the parallelepipeds in which the permittivity is considered constant. A simple and valid approximation is obtained by computing the average of the two constant values (in the adjacent unit cells) multiplied by the value of the interval ($\Delta\alpha$ in general) due to the convolution with the corresponding testing function. The first term on the right-hand side of (6.12) will then be approximated by

$$V^{MD}_x[m,n,k] = \frac{\Delta x}{2}\left[k^M[m+1,n,k] + k^M[m,n,k]\right] J^D_x[m,n,k] \tag{6.31a}$$

$$V^{MD}_y[m,n,k] = \frac{\Delta y}{2}\left[k^M[m,n+1,k] + k^M[m,n,k]\right] J^D_y[m,n,k] \tag{6.31b}$$

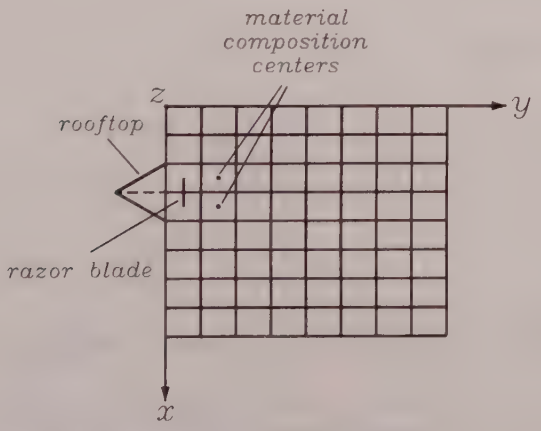


Figure 6.7 Centers of the material cells used to represent arbitrary bodies.

$$V_z^{MD}[m,n,k] = \frac{\Delta z}{2}\left[k^M[m,n,k+1] + k^M[m,n,k]\right] J_z^D[m,n,k] \tag{6.31c}$$

The total discrete version of (6.12) is given by the sum of (6.29a–c) plus (6.31a), (6.29d–f) plus (6.31b), and (6.29g–i) plus (6.31c) for the x-, y- and z-components of the fields, respectively.

Applying the definition of the adjoint operator in (5.95) to the present case, we obtain the following result, which is similar to (5.97) and (5.98):

$$L^A = (L^T)^* = \begin{pmatrix} L_{xx}^* & L_{yx}^* & L_{zx}^* \\ L_{xy}^* & L_{yy}^* & L_{zy}^* \\ L_{xz}^* & L_{yz}^* & L_{zz}^* \end{pmatrix} \tag{6.32}$$

By direct comparison with (6.12), each term in (6.32) can be identified:

$$\begin{aligned} L_{xx} &= k^M + k^c L_{xx}^c + k^q L_{xx}^q \\ L_{xy} &= k^M + k^q L_{xy}^q \\ L_{xz} &= k^M + k^q L_{xz}^q \end{aligned} \tag{6.33a}$$

$$\begin{aligned} L_{yx} &= k^M + k^q L_{yx}^q \\ L_{yy} &= k^M + k^c L_{yy}^c + k^q L_{yy}^q \\ L_{yz} &= k^M + k^q L_{yz}^q \end{aligned} \tag{6.33b}$$

$$\begin{aligned} L_{zx} &= k^M + k^q L_{zx}^q \\ L_{zy} &= k^M + k^q L_{zy}^q \\ L_{zz} &= k^M + k^c L_{zz}^c + k^q L_{zz}^q \end{aligned} \tag{6.33c}$$

6.4 RESULTS FOR INDUCED EQUIVALENT CURRENTS

By using (6.29a–i) and (6.31a–c), together with the CG algorithm selected in Chapter 4, we can obtain the induced currents in an arbitrary 3D structure. This section

presents and analyzes some representative results of induced currents, convergence rates, and CPU times.

6.4.1 Perfectly Conducting Cube

Consider a perfectly conducting cube with sides $L_x = L_y = L_z = L = 0.75\lambda$ illuminated by an x-polarized normal incident plane wave, as is shown in Figure 6.8.

Figures 6.9(a–c) show the most representative components of the volumetric current along three cuts, AA′, BB′, and CC′, passing through the center of the cube in the x-, y-, and z-directions, respectively. The number of samples used to represent the current was $N_x = N_y = N_z = 15$. Figure 6.9(a) shows the tangential component of the current J_z near the face, which presents the asymptotic behavior expected when approximating surface currents by volumetric ones. For this cut, J_x presents a vanishing behavior near the cube faces (J_x is normal to the cube faces at points A and A′). Its value at the center is very small compared with the value of the tangential component. This asymptotic behavior is also obtained along the BB′ cut, where both J_x and J_z are tangential to the corresponding faces (J_x is shown in Figure 6.9(b)). Along the CC′ cut, the behavior is similar (J_x tangential, J_z normal), but now the currents are not symmetric because of the phase delay of the incident plane wave along the z-direction (J_x is presented in Figure 6.9(c)). In all cases, J_y is very small (due to the x-polarization of the incident field) and is not represented. The value of the material constant, k^M in (6.13c), has been fixed by making $\varepsilon_r = 1$ and $\sigma = 10^{15}$

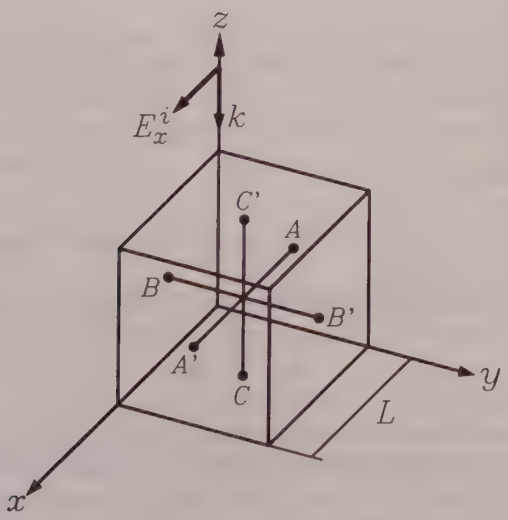
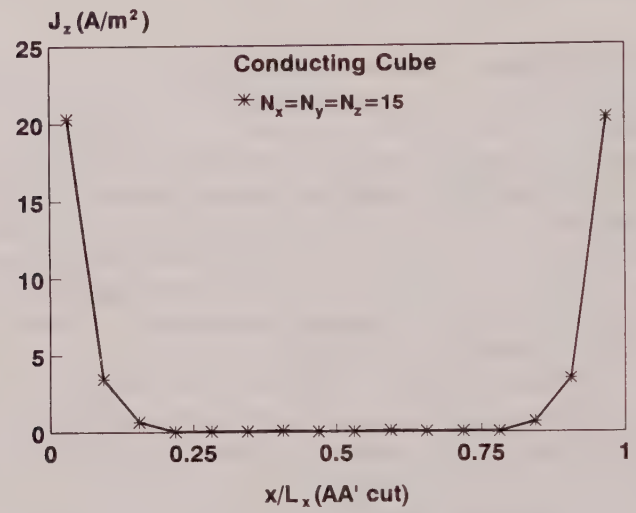


Figure 6.8 Schematic representation of the perfectly conducting cube.

(a)



(b)

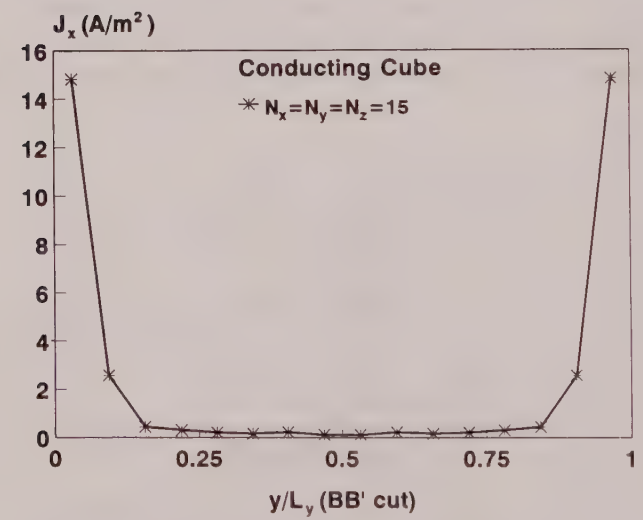


Figure 6.9 Representative components of the volumetric current in a perfectly conducting cube (L = 0.75λ): (a) z-component along AA′ cut; (b) x-component along BB′ cut; (c) x-component along CC′ cut.

(c)

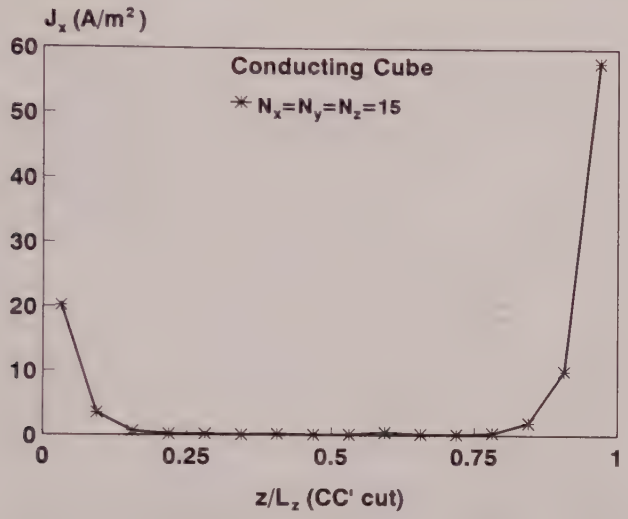


Figure 6.9 (Continued)

to represent the perfectly conducting properties. The CPU time to obtain a relative error of 5.10^{-3} (when the number of samples used was $N_x = N_y = N_z = 7$) was about 8 minutes in a PC-486, which gives an idea of the power of the scheme under consideration.

Figure 6.10 shows the convergence rates (relative error versus number of iterations) for two different numbers of subdivisions (basis functions). The convergence is slower than for a dielectric body (next section) because of the behavior of the currents in a conductor for which surface distributions must be approximated by volumetric current distributions.

6.4.2 Low-Permittivity Dielectric Sphere Without Losses

Next, consider a lossless dielectric sphere with low permittivity ($\varepsilon' = 4$, $\varepsilon'' = 0$) and radius a such that $k_0 a = 1$, which is illuminated by an x-polarized normal incident plane wave, as shown in Figure 6.11.

Figures 6.12(a–c) present some components of the volumetric current along the three cuts AA′, BB′, and CC′, passing through the center of the sphere in the x-, y-, and z-directions, respectively. Notice that the equivalent currents do not show the asymptotic behavior near the face surface previously described for conductors. Once again, the behavior of the currents is asymmetric along the CC′ cut due to the phase delay of the incident plane wave in the z-direction. In all cases, J_y is very

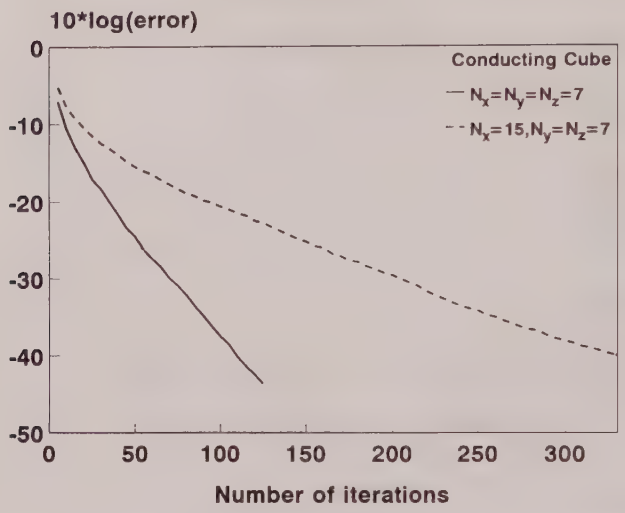


Figure 6.10 Relative error versus number of iterations for a perfectly conducting cube for two different numbers of samples.

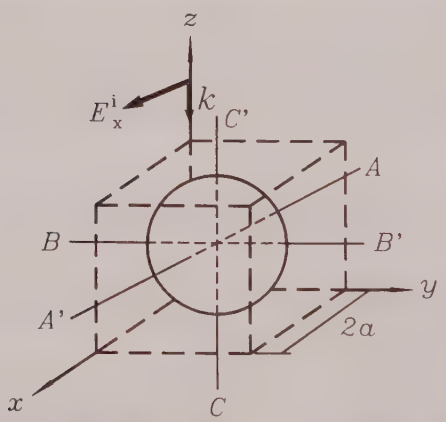
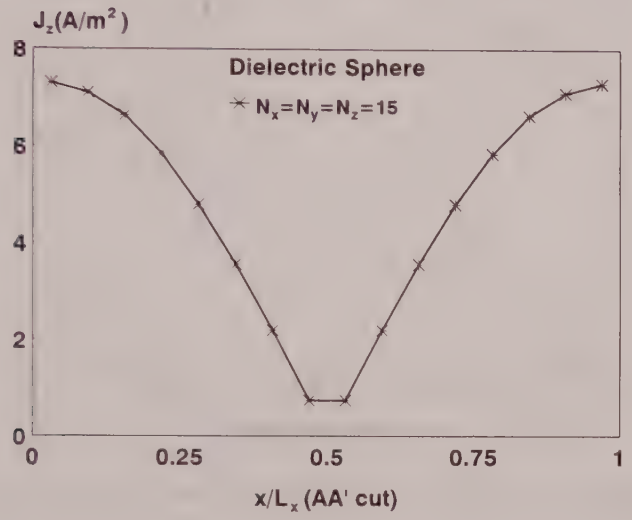


Figure 6.11 Schematic representation of the dielectric sphere ($k_0a = 1$, $\varepsilon_r = 4$, $\sigma = 0$).

(a)



(b)

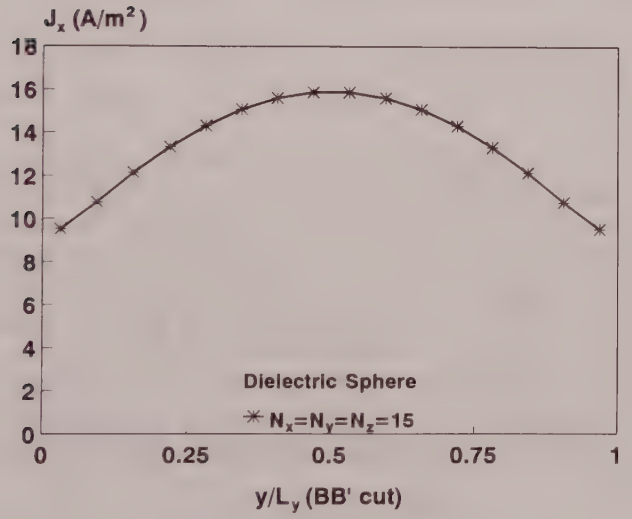


Figure 6.12 Some components of the volumetric current in a dielectric sphere ($k_0a = 1$, $\varepsilon_r = 4$, $\sigma = 0$): (a) z-component along AA′ cut; (b) x-component along BB′ cut; (c) x-component along CC′ cut.

(c)

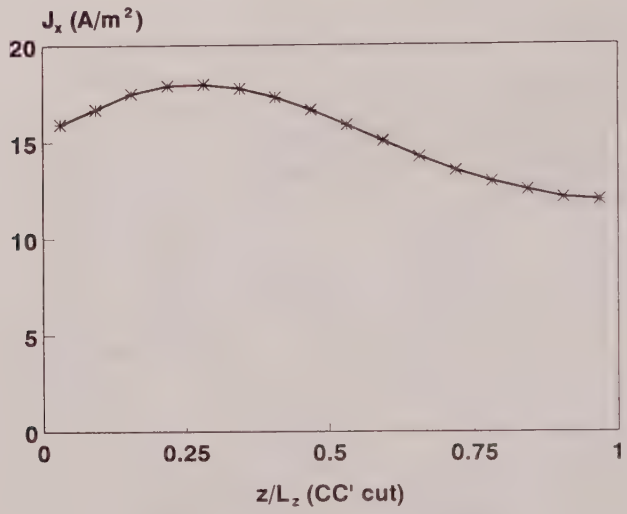


Figure 6.12 (Continued)

small (due to the x-polarized incident field) and is therefore not shown. These results were obtained by considering $N_x = N_y = N_z = 15$. The value of the material constant, k^M in (6.13c), has been fixed by making $\varepsilon_r = 4$ and $\sigma = 0$. The CPU time required to obtain a relative error of 10^{-4} when $N_x = N_y = N_z = 7$ was about 5 minutes in a PC-486.

Figure 6.13 shows the convergence rates (relative error versus number of iterations) for different numbers of subdivisions (basis functions). The convergence in this case is faster than for the perfectly conducting cube of the previous section because there is no asymptotic behavior of the currents near the face surfaces to be represented. Rapid convergence is always obtained for bodies with low dielectric constant. Even in the case of resonant-sized bodies, we have found that a small number of samples is sufficient to obtain good scattering values.

In the following sections it will be seen that the convergence is slower for dielectric bodies with high permittivity. Of course, the convergence will also depend on the complexity of the geometry being represented and on the presence or absence of physical losses.

6.4.3 High-Permittivity Dielectric Sphere Without Losses

Consider now a lossless dielectric sphere with a high permittivity ($\varepsilon' = 40$, $\varepsilon'' = 0$) and radius a such that $k_0 a = 1$, which is illuminated by an x-polarized normal incident plane wave, as shown in Figure 6.11. Figures 6.14(a–c) show examples of volu-

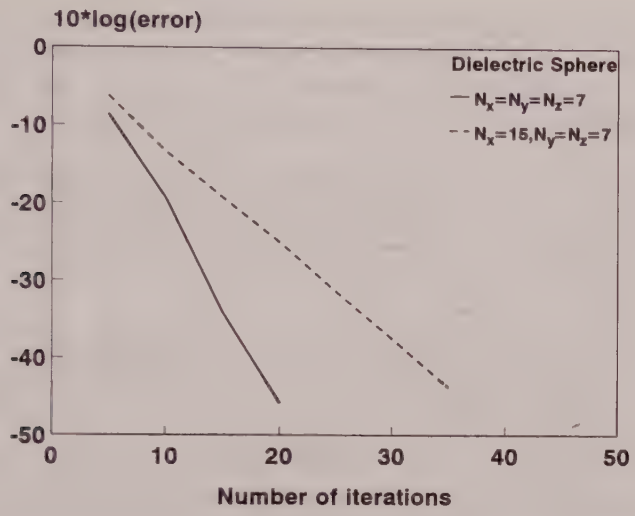


Figure 6.13 Relative error versus number of iterations for a low-permittivity, lossless dielectric sphere for two different numbers of samples.

(a)

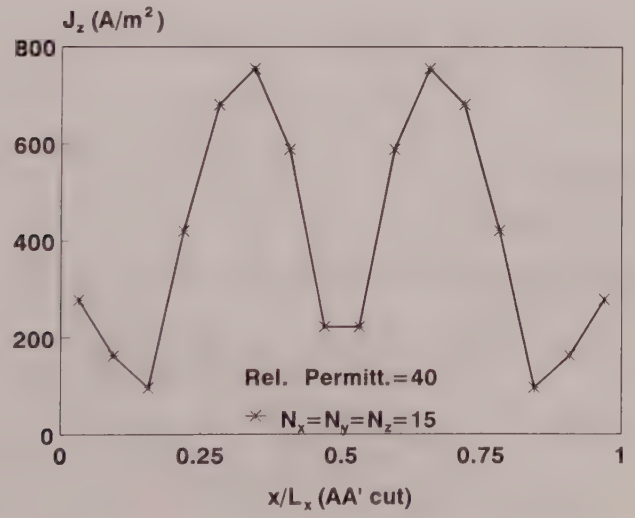
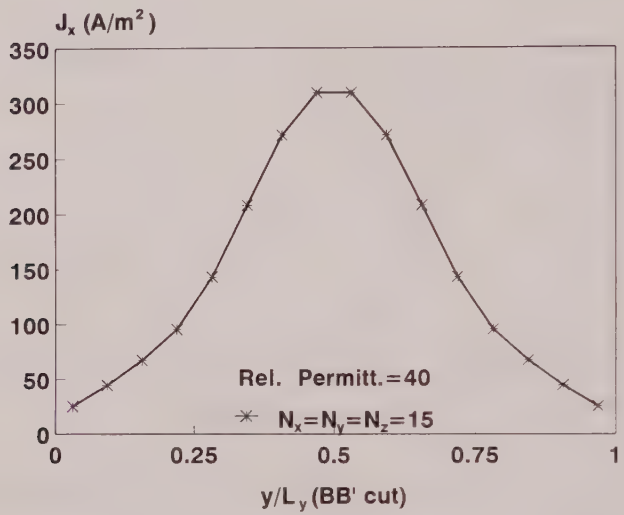


Figure 6.14 Examples of volumetric current components in a dielectric sphere ($k_0a = 1$, $\varepsilon_r = 40$, $\sigma = 0$): (a) z-component along the AA′ cut; (b) x-component along the BB′ cut; (c) x-component along the CC′ cut.

(b)



(c)

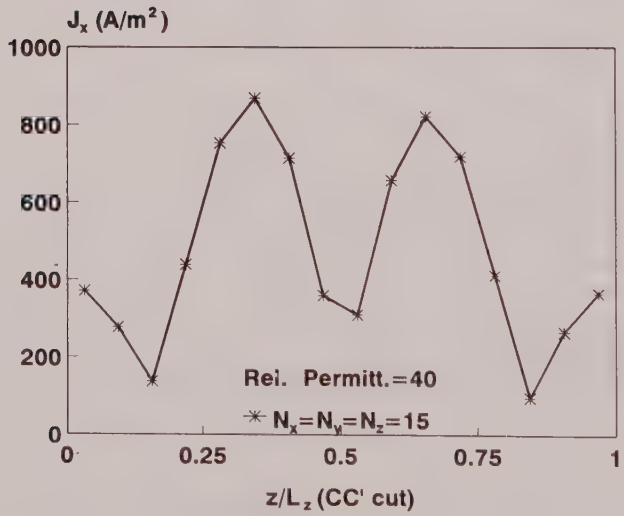


Figure 6.14 (Continued)

metric current components along the three cuts AA′, BB′, and CC′ of the previous examples for $N_x = N_y = N_z = 15$. The results are in part similar to the low-permittivity dielectric case, but now the equivalent current is significantly greater, as is the CPU time (110 minutes in a PC-486 to obtain a relative error of 10^{-3} when $N_x = N_y = N_z = 7$).

Figure 6.15 shows the convergence rates (relative error versus number of iterations) for the present case. Because of the high value of the dielectric constant, the convergence is now even slower than for the perfectly conducting cube.

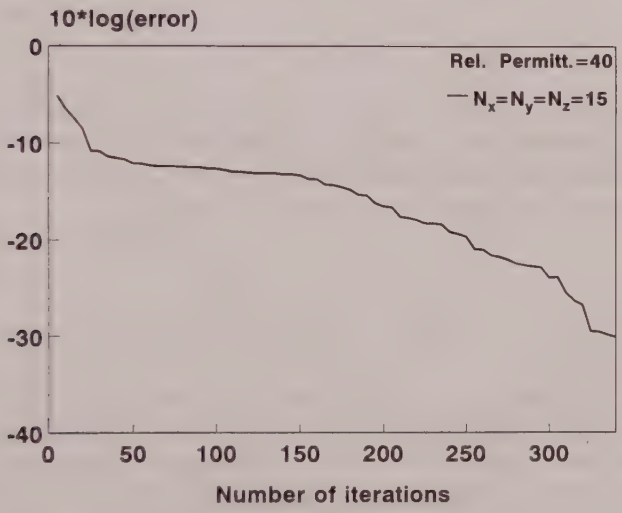


Figure 6.15 Relative error versus number of iterations for a high-permittivity dielectric sphere.

6.5 APPLICATION TO RADIATION AND SCATTERING PROBLEMS

Once the equivalent volumetric currents have been determined in the volume, the far fields can be obtained from the vector function **A**:

$$\mathbf{A}(\mathbf{r}) = \mu_0 \int_V \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, dv' \tag{6.34}$$

with $G(\mathbf{r},\mathbf{r}')$ being the function defined in (6.11a). The approximations used are completely equivalent to those in Section 5.5, and a similar result is obtained, although it is now in terms of the 3D Fourier transform of the volumetric currents:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\exp(-jk_0 r)}{r}\int_0^{L_x}\int_0^{L_y}\int_0^{L_z}\mathbf{J}(\mathbf{r}')\exp(jk_0 x'\sin\theta\cos\phi) \cdot \exp(jk_0 y'\sin\theta\sin\phi)\exp(jk_0 z'\cos\theta)\,dx'\,dy'\,dz' \tag{6.35}$$

The angular spectral values are defined by

$$k_x = k_0\sin\theta\cos\phi \tag{6.36a}$$

$$k_y = k_0\sin\theta\sin\phi \tag{6.36b}$$

$$k_z = k_0\cos\theta \tag{6.36c}$$

The plane wave behavior of the spherical wave (far-field approximation) at point $\mathbf{r}$ is again described by E_θ and E_ϕ. By using the corresponding transformations from the cartesian to the spherical coordinate systems, we obtain the following set of equations, which are similar to those in (5.109a–d):

$$\mathbf{E} = E_\theta\hat{\theta} + E_\phi\hat{\phi} \tag{6.37a}$$

$$E_\theta = E_0[\tilde{J}_x(k_x,k_y,k_z)\cos\theta\cos\phi + \tilde{J}_y(k_x,k_y,k_z)\cos\theta\sin\phi - \tilde{J}_z(k_x,k_y,k_z)\sin\theta] \tag{6.37b}$$

$$E_\phi = E_0[\tilde{J}_y(k_x,k_y,k_z)\cos\phi - \tilde{J}_x(k_x,k_y,k_z)\sin\phi] \tag{6.37c}$$

$$E_0 = -\frac{j\omega\mu_0}{4\pi}\frac{\exp(-jk_0 r)}{r} \tag{6.37d}$$

where k_x, k_y, and k_z are defined in (6.36a–c), and $\tilde{J}_x$, $\tilde{J}_y$, and $\tilde{J}_z$ are the CFT of the current components.

Finally, the expression for the RCS is identical to the one in (5.110).

$$\sigma = 10\log\left(4\pi r^2\frac{|E_\theta|^2 + |E_\phi|^2}{|E^i|^2}\right)$$

6.5.1 Perfectly Conducting Bodies

Figure 6.16 shows data for the RCS of a perfectly conducting cube as a function of the side length when illuminated by a y-polarized normal incident plane wave. The

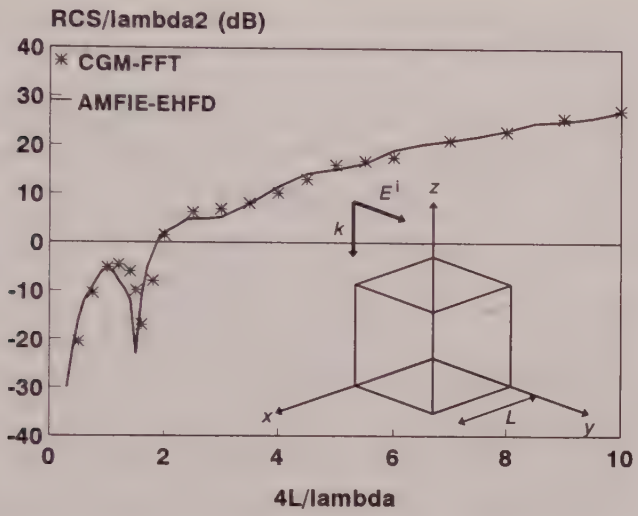


Figure 6.16 RCS of a cube of side L.

sizes being considered are near the resonance values. Our results are compared with those obtained by Yaghjian and McGahan [25] using an MM scheme combined with an augmented magnetic field integral equation (AMFIE) for L less than 1.5λ and an enhanced high-frequency diffraction (EHFD) technique for L greater than this value. The number of samples used to represent the currents was $N_x = N_y = N_z = 7$.

Figure 6.17 shows the BCS of a perfectly conducting cube of side $L = 0.75\lambda$, as a function of θ, when illuminated by a normal incident plane wave. The number of samples used to represent the currents was $N_x = N_y = N_z = 7$, as in the previous case. These results have been compared with measured data from [26]. The agreement was found to be good (within ± 1 dB), even though the number of samples used was small.

Finally, Figure 6.18 shows the results for the BCS of a perfectly conducting sphere of radius a such that $k_0 a = 1.1$. The structure is illuminated by an x-polarized plane wave incident in the z-direction. Data for both the E-plane ($x = 0$) and H-plane ($y = 0$) are presented.

The results are compared with analytical data from [27]. The number of samples used to represent the currents was again $N_x = N_y = N_z = 7$. The numerical values appear to be reasonably accurate in spite of the rough geometrical approximation of the sphere provided by such a small number of samples.

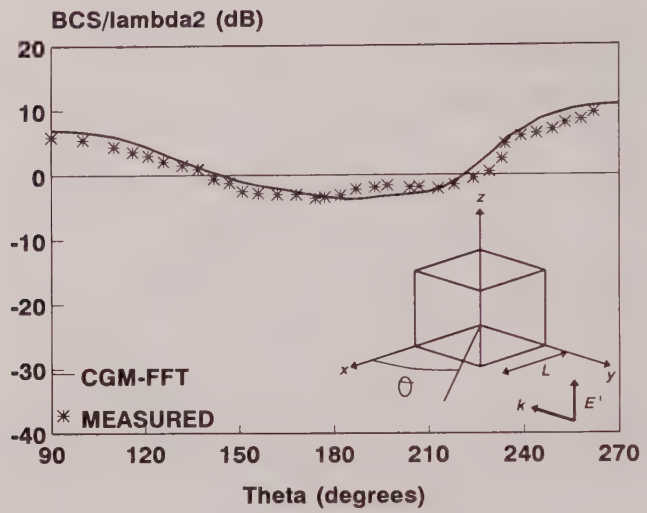


Figure 6.17 BCS of a perfectly conducting cube of side $L = 0.75\lambda$.

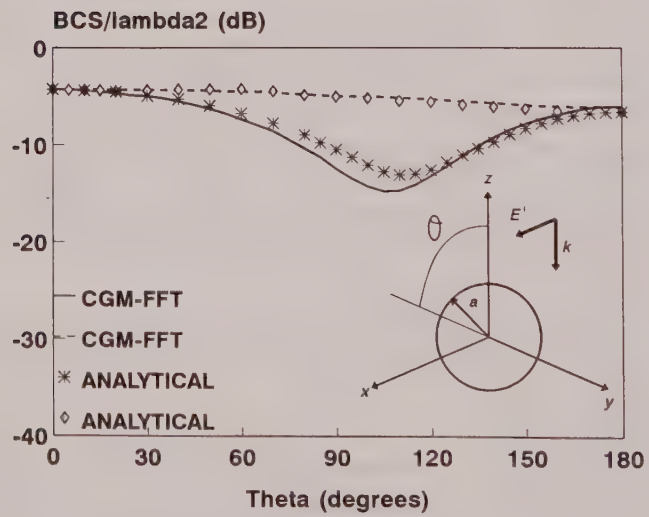


Figure 6.18 BCS of a perfectly conducting sphere with $k_0a = 1.1$.

6.5.2 Dielectric Bodies

Figure 6.19 shows data for the BCS of a homogeneous dielectric sphere with a low dielectric constant ($\varepsilon_r = 4.0$) and radius a, such that $k_0 a = 1$, illuminated by an x-polarized plane wave incident from the negative z-axis. The number of samples used to represent the currents was $N_x = N_y = N_z = 7$. The results are compared with those presented by Umashankar, Taflove and Rao [28] using an MM procedure. Although the geometrical approximation of the sphere is similar to that for the perfectly conducting sphere, the agreement in this case is excellent.

Figure 6.20(b) shows data for the RCS versus the aspect angle of the lossy oblate spheroid shown in Figure 6.20(a). The semiaxes are denoted by a and c, with $k_0 a = 3.824$ and $c/a = 0.2$. The complex relative dielectric constant is given by $\varepsilon_r = 3.14 - j0.036$. The number of samples used to represent the currents was $N_x = N_y = N_z = 7$. The results obtained are compared with the measurements made by Antar and Allan [29]. Notice that even for low levels the results are in good agreement with measured data. This can be considered an example of a dielectric body with losses.

Finally, Figure 6.21 shows some data for the BCS of two concentric, lossless dielectric spheres. The inner and outer radii of the spheres are denoted by r_1 and r_2,

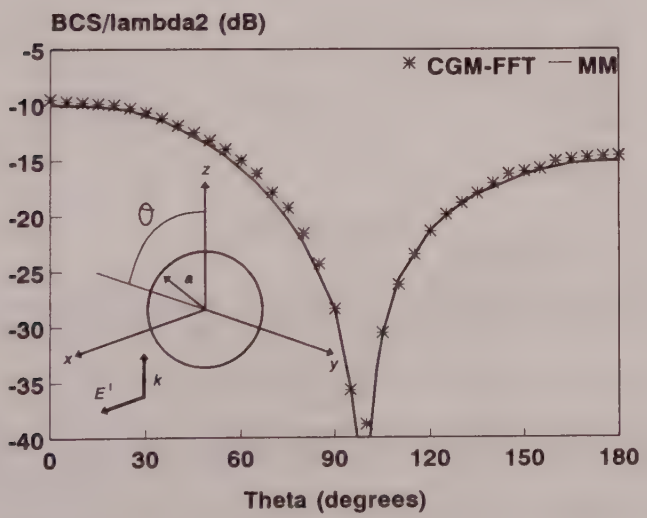
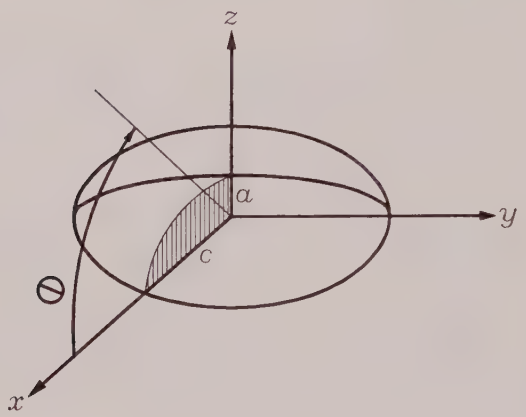


Figure 6.19 BCS of a dielectric sphere with $k_0 a = 1$ and $\varepsilon_r = 4$.

(a)



(b)

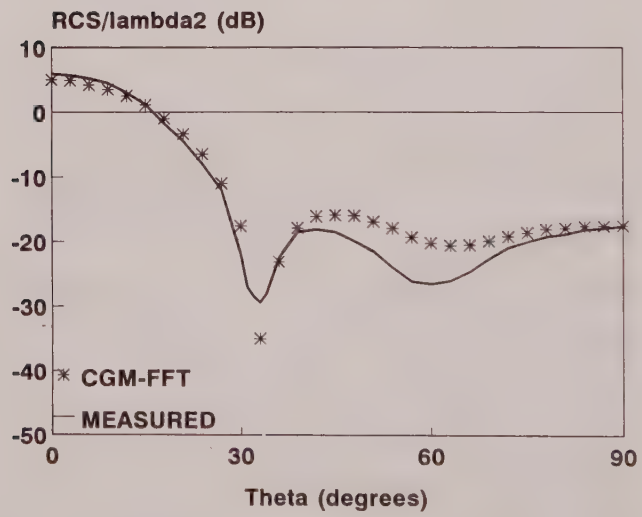


Figure 6.20 (a) Schematic representation and (b) RCS of a lossy oblate spheroid with $k_0a = 3.824$ and $c/a = 0.2$.

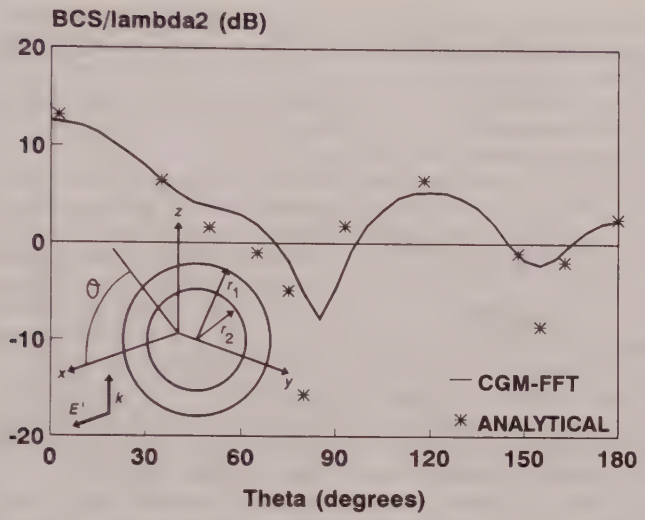


Figure 6.21 BCS of two concentric dielectric spheres ($k_0 r_2 = 1$, $r_1/r_2 = 3$).

respectively, with $k_0 r_2 = 1$ and $r_1/r_2 = 3$. The relative dielectric constants are $\varepsilon_{r1} = 4$ and $\varepsilon_{r2} = 2$. The number of samples used to represent the currents was $N_x = N_y = N_z = 15$. Our results are compared with analytical values obtained in [30]. For angles close to the forward and backward directions, the agreement is quite good, whereas for broadside angles, only a rough likeness of the shape can be recognized. Note that this number of samples provides a rather poor representation of the spheres, especially for the outer one.

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Chapter 7
Problems Formulated in Terms of Systems of Integral Equations

7.1 INTRODUCTION

So far, we have dealt with problems formulated by using the EFIE. As we have seen in Chapters 5 and 6, this equation is characterized by an integrodifferential operator that relates the unknown function (the electric current) to the known function (the impressed field). CG-FFT techniques similar to those in Chapters 5 and 6 can also be used to analyze problems characterized by the MFIE or other equations [1–3] in which a single unknown function is related to the known function by means of an operator.

In this chapter, we will address problems formulated in terms of systems of integral equations (SIE); that is, problems where several unknown and known functions are related by means of a matrix of integrodifferential operators. One example of this kind of problem arises when a body composed of several piecewise homogeneous dielectric materials is analyzed, considering electric and magnetic equivalent surface currents at the dielectric interfaces. In this case, the SIE is obtained by imposing continuity of the tangential components of the electric and magnetic fields at the dielectric surface boundaries. Another example of a problem that gives rise to an SIE is provided by the analysis of the magnetic material used for transformers, inductors, or other electric or electronic components. These materials, usually ferrites, are characterized by having very high values of both electric permittivity and magnetic permeability, ε and μ, respectively. For instance, a MnZn ferrite commonly used for the transformer in DC-DC converters can have an ε tens of thousands of times greater than the free-space permittivity ε_0, and a μ thousands of times greater than the free-space permeability μ_0. With these very large values for both material constants, it is necessary to consider electric and magnetic polarization currents if a volumetric formulation like the one in Chapter 6 is to be used. In this case, an SIE is obtained by relating the displacement current to the fields

$$\mathbf{J}(\mathbf{r}) = j\omega(\varepsilon - \varepsilon_0)[\mathbf{E}^s(\mathbf{r},\mathbf{J},\mathbf{M}) + \mathbf{E}^i(\mathbf{r})] \tag{7.1a}$$

$$\mathbf{M}(\mathbf{r}) = j\omega(\mu - \mu_0)[\mathbf{H}^s(\mathbf{r},\mathbf{J},\mathbf{M}) + \mathbf{H}^i(\mathbf{r})] \tag{7.1b}$$

where $\mathbf{J}$ and $\mathbf{M}$ are the electric and magnetic polarization currents, $\mathbf{E}^i(\mathbf{r})$ and $\mathbf{H}^i(\mathbf{r})$ are the impressed fields, and $\mathbf{E}^s(\mathbf{r},\mathbf{J},\mathbf{M})$ and $\mathbf{H}^s(\mathbf{r},\mathbf{J},\mathbf{M})$ are the induced fields, which are in turn related to the electric and magnetic polarization currents through integrodifferential operators [1–3]. Volumetric analysis by means of the CG-FFT method is quite appropriate for magnetic material with the geometries typically used in electronic devices. This is due to the fact that in such devices the parallelepiped volume that encloses the geometry of the problem to be analyzed by the CG-FFT method is filled almost completely by the magnetic material.

This chapter addresses the problem of radiation or scattering at the dielectric-to-dielectric or air-to-dielectric interfaces from dielectric sheets of finite size and with metallizations of arbitrary shape. The SIE will be obtained from the equivalence theorem [2,3], which allows the materials to be replaced by electric and magnetic surface currents in free space. Many microwave devices and antennas such as microstrip circuits or antennas, slot-line components, slot-line antennas (e.g., Vivaldi antennas), and radar-absorbing materials have this basic structure. Although the proposed scheme can be applied to multilayer geometries, for the sake of simplicity we will restrict our attention here to the single-layer planar problem illustrated in Figure 7.1. The scheme that will be used follows very closely the procedure presented in [4].

Most numerical formulations for treating single-layer or multilayer problems have assumed that the dielectric sheet and the lower metallization were infinite in size [5–9]. In Chapters 10 and 11 of this book, we take this as a premise. Formulations based on this assumption are quite efficient as long as the errors incurred by considering infinite structures instead of the actual size of the dielectric sheets can

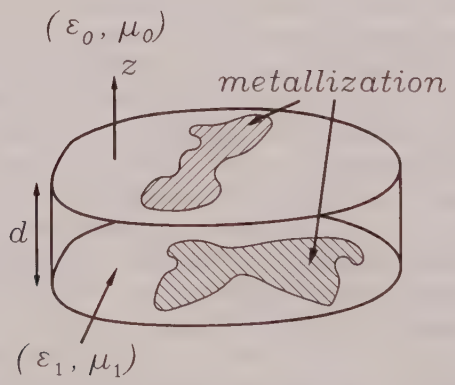


Figure 7.1 The problem to be analyzed consists of a dielectric sheet of arbitrary shape but with uniform thickness and finite size. Arbitrary metallizations can be considered on the upper (S^1) and lower (S^2) surfaces of the sheet. The lateral surface S^L is assumed to have no metallization.

be ignored. The efficiency of these formulations is due to the fact that they use Green's functions that implicitly take into account the presence of the dielectric layers and the infinite ground plane in the lower part of the structure. Thus, only the currents in the metallic patches of the device under study must be considered.

However, the finite size of the structure can have very significant effects, as in the case of linear tapered slot-line antennas (LTSA) or Vivaldi antennas, which have their main radiation lobe in the same plane as the planar structure. Another case where the finite real size of the structure is not negligible is in the analysis of microstrip antennas in directions close to the dielectric plane. In these cases, it is nearly impossible to find an analytical Green's function that implicitly takes into consideration the real size and shape of the dielectric structure. Therefore, equivalent currents must be used to take into account the dielectrics and the lower metallizations, as well as the currents in the metallic patches on the upper face of the structure. Considering these additional currents results in a huge increase in the computational cost of solving the problem. The total number of samples employed to represent the currents can easily be two or three orders of magnitude greater than when a Green's function is used.

Until very recently, few attempts to analyze these problems by considering the real size of the dielectric and ground metallizations had been reported [10,11], probably because of the numerical difficulties involved. In this chapter, we will describe in some detail a CG-FFT approach to these problems. As is pointed out in [4], this approach is quite useful for those cases where the discretization process leads to tens of thousands of samples (unknowns in an MM formulation).

7.2 FORMULATION OF THE CONTINUOUS SIE

An SIE can be derived by using the equivalence theorem [1] to analyze the problem indicated in Figure 7.1. According to this theorem, the electromagnetic fields in the region external to the structure can be computed by considering in a completely homogeneous space (ε_0,μ_0) the equivalent surface currents $\mathbf{J}^e(\mathbf{r})$ and $\mathbf{M}^e(\mathbf{r})$ on the dielectric surface, the current $\mathbf{J}_c^{ex}(\mathbf{r})$ on the surface of the metallic patches, and all the impressed currents $\mathbf{J}_i^{ex}(\mathbf{r})$ and $\mathbf{M}_i^{ex}(\mathbf{r})$ that exist in this external region. Figure 7.2(a) illustrates the equivalent problem for the external region. The fields in the dielectric region can be computed by considering the following currents in a new homogeneous region with material constants ε_1 and μ_1: the equivalent surface currents $-\mathbf{J}^e(\mathbf{r})$ and $-\mathbf{M}^e(\mathbf{r})$ on the dielectric faces, the current $\mathbf{J}_c^{in}(\mathbf{r})$ on the surface of the metallic patches, and all the impressed currents $\mathbf{J}_i^{in}(\mathbf{r})$ and $\mathbf{M}_i^{in}(\mathbf{r})$. Figure 7.2(b) shows the environment to be considered in the computation of the fields in the internal region (dielectric region). It can be seen that the equivalent problems of Figures 7.2(a) and 7.2(b) only share the surface currents $\mathbf{J}^e(\mathbf{r})$ and $\mathbf{M}^e(\mathbf{r})$ (with a change in the sign) at the parts of the interfaces between the external region and internal regions that are not metallized.

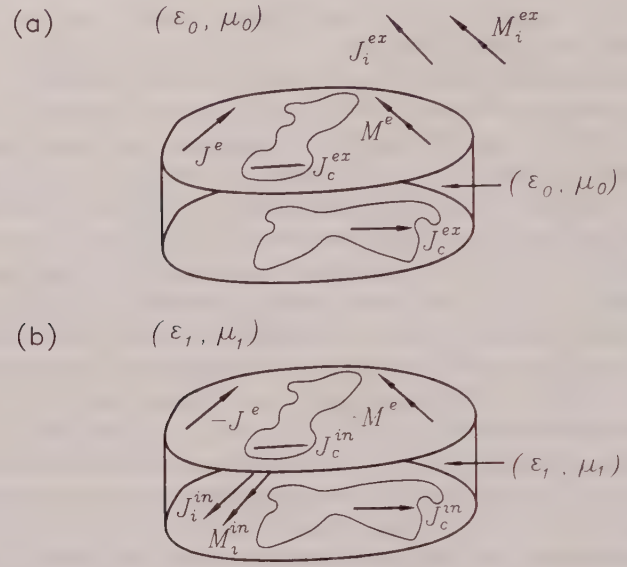


Figure 7.2 (a) Equivalent problem for computation of the electromagnetic fields in the region external to the dielectric sheet. All the space is considered homogeneous with material constants ε_0 and μ_0. (b) Equivalent problem for calculating the fields in the dielectric region. Now all the space is considered homogeneous with constants ε_1 and μ_1.

Now since the fields can be computed in both regions, we can formulate the SIE by imposing continuity of the tangential components of the fields at the dielectric interfaces, and the conducting boundary condition at the metallized surfaces.

The boundary conditions at the metallic surface S_c are

$$\mathbf{n} \times \mathbf{E}_i^{\text{ex}}(\mathbf{r}) = -L^{EJ}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{J}_c^{\text{ex}} + \mathbf{J}^e) - L^{EM}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{M}^e) \tag{7.2a}$$

for the face of the metallic patches in contact with the external region and

$$\mathbf{n} \times \mathbf{E}_i^{\text{in}}(\mathbf{r}) = -L^{EJ}(\mathbf{r},\varepsilon_1,\mu_1,\mathbf{J}_c^{\text{in}} - \mathbf{J}^e) - L^{EM}(\mathbf{r},\varepsilon_1,\mu_1,-\mathbf{M}^e) \tag{7.2b}$$

for the face of the metallic patches in contact with the dielectric.

The boundary conditions at the surface S_d between dielectrics are

$$\begin{aligned}\mathbf{n} \times (\mathbf{E}_i^{\text{ex}}(\mathbf{r}) - \mathbf{E}_i^{\text{in}}(\mathbf{r})) = {} & L^{EJ}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{J}_c^{\text{ex}} + \mathbf{J}^e) + L^{EM}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{M}^e) \\ & - L^{EJ}(\mathbf{r},\varepsilon_1,\mu_1,\mathbf{J}_c^{\text{in}} - \mathbf{J}^e) - L^{EM}(\mathbf{r},\varepsilon_1,\mu_1 - \mathbf{M}^e)\end{aligned} \tag{7.2c}$$

for the continuity of the tangential component of the electric field and

$$\mathbf{n} \times (\mathbf{H}_i^{\text{ex}}(\mathbf{r}) - \mathbf{H}_i^{\text{in}}(\mathbf{r})) = L^{HJ}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{J}_c^{\text{ex}} + \mathbf{J}^e) + L^{HM}(\mathbf{r},\varepsilon_0,\mu_0,\mathbf{M}^e) - L^{HJ}(\mathbf{r},\varepsilon_1,\mu_1,\mathbf{J}_c^{\text{in}} - \mathbf{J}^e) - L^{HM}(\mathbf{r},\varepsilon_1,\mu_1, -\mathbf{M}^e) \tag{7.2d}$$

for the continuity of the tangential component of the magnetic field. Here $\mathbf{n}$ is a unit vector normal to the dielectric surface and oriented in the outgoing direction, $(\mathbf{E}_i^{\text{ex}}(\mathbf{r}), \mathbf{H}_i^{\text{ex}}(\mathbf{r}))$ and $(\mathbf{E}_i^{\text{in}}(\mathbf{r}), \mathbf{H}_i^{\text{in}}(\mathbf{r}))$ are the impressed fields in the external and the dielectric regions, respectively (it is assumed that these fields are due to the impressed currents $\mathbf{J}_i^{\text{ex}}(\mathbf{r})$ and $\mathbf{M}_i^{\text{ex}}(\mathbf{r})$ for the external region, and $\mathbf{J}_i^{\text{in}}(\mathbf{r})$ and $\mathbf{M}_i^{\text{in}}(\mathbf{r})$ for the dielectric region). The integrodifferential operators L^{EJ}, L^{EM}, L^{HJ}, and L^{HM} are given by

$$L^{EJ}(\mathbf{r},\varepsilon,\mu,\mathbf{J}) = -\frac{j\omega\mu}{4\pi}\,\mathbf{n} \times \int_{S'} \phi(\mathbf{r} - \mathbf{r}',\varepsilon,\mu)\,\mathbf{J}(\mathbf{r}')\,ds' + \frac{1}{j\omega 4\pi\varepsilon}\,\mathbf{n} \times \nabla \int_{S'} \phi(\mathbf{r} - \mathbf{r}',\varepsilon,\mu)\,\nabla'\,\mathbf{J}(\mathbf{r}')\,ds' \tag{7.3a}$$

$$L^{EM}(\mathbf{r},\varepsilon,\mu,\mathbf{M}) = -\frac{\mathbf{n}}{4\pi} \times \oint_{S'} \mathbf{M}(\mathbf{r}') \times \nabla'\phi(\mathbf{r} - \mathbf{r}',\varepsilon,\mu)\,ds' \tag{7.3b}$$

$$L^{HJ}(\mathbf{r},\varepsilon,\mu,\mathbf{J}) = L^{EM}(\mathbf{r},\mu,\varepsilon,\mathbf{M}) \tag{7.3c}$$

$$L^{HM}(\mathbf{r},\varepsilon,\mu,\mathbf{M}) = L^{EJ}(\mathbf{r},\mu,\varepsilon,\mathbf{J}) \tag{7.3d}$$

where

$$\phi(\mathbf{r},\varepsilon,\mu) = \frac{\exp(-jk(\varepsilon,\mu)|\mathbf{r}|)}{|\mathbf{r}|} \tag{7.4}$$

and $k(\varepsilon,\mu)$ is the wave number for a homogeneous space with constants (ε,μ):

$$k(\varepsilon,\mu) = \omega(\varepsilon\mu)^{1/2} \tag{7.5}$$

The symbol ∇' indicates derivatives with respect to the variable $\mathbf{r}'$, and the symbol $\oint$ denotes a Cauchy principal value integral [1].

The equations of (7.2) constitute the SIE to be solved by the CG-FFT method. The electric and magnetic fields and currents in this SIE take values that are different by orders of magnitude. This circumstance is not ideal for a method like the CG which is based on minimizing a residual formed by the average error in the electric and magnetic fields. It would be better to normalize the electric and magnetic currents and fields so that they would have values of the same order of magnitude. The normalization procedure shown in Table 7.1 is suggested for the SIE of (7.2).

Table 7.1
Normalized Fields and Currents

$\mathbf{E}' = \frac{1}{\sqrt{\eta_0}}\mathbf{E}$	$\mathbf{H}' = \sqrt{\eta_0}\,\mathbf{H}$
$\mathbf{M}' = \frac{1}{\sqrt{\eta_0}}\mathbf{M}$	$\mathbf{J}' = \sqrt{\eta_0}\,\mathbf{J}$
$\mu' = \frac{1}{\sqrt{\eta_0}}\mu$	$\varepsilon' = \sqrt{\eta_0}\,\varepsilon$
$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}$	

7.3 DISCRETIZATION OF THE SIE

Once the SIE of (7.2) has been normalized, it will be discretized in the spatial domain following a scheme quite similar to the one presented in Chapter 5.

First, the horizontal surfaces of the structure are enclosed by the smallest rectangle possible, which will have sides of length L_x and L_y, as is shown in Figure 7.3. This rectangle is then covered by a dense mesh of very small rectangles of area $\Delta x \Delta y$, where

$$\Delta x = \frac{L_x}{N_x + 1} \tag{7.6a}$$

$$\Delta y = \frac{L_y}{N_y + 1} \tag{7.6b}$$

N_x and N_y are the number of subdivisions for directions x and y, as can be seen in Figure 7.3. The surfaces of the dielectric and the metallization are approximated by sets of these small rectangles, as is indicated in the figure.

The second step is to approximate the currents with a suitable set of basis functions. To maintain uniformity in the book, we have chosen rooftops as basis functions to represent the electric and magnetic surface currents. Using $S(r)$ to denote either the electric or magnetic currents in the SIE (7.2), we can write the following expansion in terms of rooftops:

$$S_x^v(\mathbf{r}) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y+1} S_x^{Dv}[i,j]\, T_{ij}^x \tag{7.7a}$$

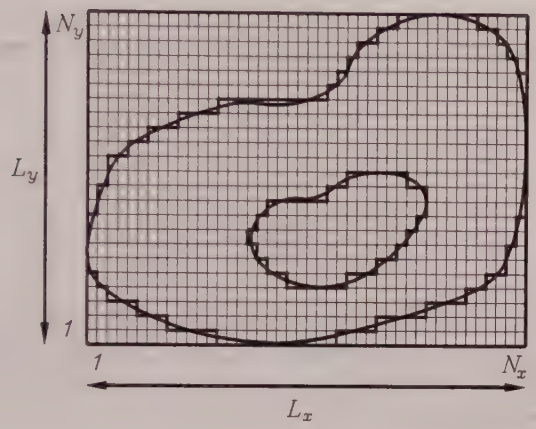


Figure 7.3 Rectangle with sides L_x and L_y and mesh of small rectangles of sides Δx by Δy covering one face of the dielectric structure.

$$S_y^v(\mathbf{r}) \approx \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y} S_y^{Dv}[i,j]\, T_{ij}^y \tag{7.7b}$$

where the superscript v indicates which dielectric surface is being considered; the top surface is denoted by S^1 and the bottom surface by S^2. The current along the lateral surface of the structure S^L has been considered negligible, although the formulation could easily be extended to include the lateral currents.

The mathematical expressions for the rooftops are given by (5.28c) and (5.28d) of Chapter 5. It should be recalled that the centers of the rooftops for the x- and y-components of the current r_x^{ij} and r_y^{ij}, which are defined by (5.29), do not coincide. This is illustrated clearly by Figure 5.10. It can be seen from this figure that we can express r_x^{ij} in terms of the four closest centers for the y-component as

$$r_x^{ij} = \frac{1}{4}\left(r_y^{ij} + r_y^{i+1j} + r_y^{ij-1} + r_y^{i+1j-1}\right) \tag{7.8a}$$

and by analogy r_y^{ij} can be written as

$$r_y^{ij} = \frac{1}{4}\left(r_x^{ij} + r_x^{ij+1} + r_x^{i-1j} + r_x^{i-1j+1}\right) \tag{7.8b}$$

As was indicated in Chapter 3, the following steps in the discretization scheme are to smooth with razor blade functions and then to sample the fields with a train of delta functions centered at r_x^{ij} and r_y^{ij} for the x- and y-components of the fields, respectively.

Using $Z(r)$ to represent any one of the fields (excitation or knowns of the SIE) on the left-hand side of (7.2), we proceed to make the approximations

$$Z_x^u(\mathbf{r}) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y+1} \left(W^x(\mathbf{r}) * Z_x^u(\mathbf{r})\right) \delta(\mathbf{r} - \mathbf{r}_x^{ij}) \tag{7.9a}$$

$$Z_y^u(\mathbf{r}) \approx \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y} \left(W^y(\mathbf{r}) * Z_y^u(\mathbf{r})\right) \delta(\mathbf{r} - \mathbf{r}_y^{ij}) \tag{7.9b}$$

where $W^x(\mathbf{r})$ and $W^y(\mathbf{r})$ are the razor blade functions defined in (5.30) and $*$ is the symbol for convolution. The superscript u indicates whether it is the top face ($u = 1$) or the bottom face ($u = 2$) of the structure. The equations of (7.9) can be written in terms of discrete functions as

$$Z_x^u(\mathbf{r}) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y+1} Z_x^{Du}[i,j]\, \delta(\mathbf{r} - \mathbf{r}_x^{ij}) \tag{7.10a}$$

$$Z_y^u(\mathbf{r}) \approx \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y} Z_y^{Du}[i,j]\, \delta(\mathbf{r} - \mathbf{r}_y^{ij}) \tag{7.10b}$$

Proceeding in a similar way with the functions on the right-hand side of the SIE of (7.2), we finally obtain the generic expression

$$Z_\alpha^{Du}[i,j] \approx \sum_{i'=-\infty}^{\infty} \sum_{j'=-\infty}^{\infty} \sum_{v=1}^{2} g_\alpha^{Duv}[i - i', j - j']\, S_\alpha^{Dv}[i',j'] \tag{7.11}$$

for any one of the equations in (7.2) after the discretization procedure. Expression (7.11) is a linear convolution that can be converted into a cyclic one by making the functions periodic. As in Chapter 5, the period will be $(2N_x + 2)(2N_y + 2)$.

The discrete Green's function $g_\alpha^{Duv}[i,j]$ takes into account the action on the surface u of a source located on surface v. The discrete function $g_\alpha^{Duv}[i,j]$ is formed either by the pair of operators L^{EJ}, L^{EM} or by L^{HJ}, L^{HM}, depending on which of the four equations in (7.2) is being considered. These pairs of operators are quite similar, as can be seen from (7.3c) and (7.3d); therefore, we will describe the discretization process only for the first pair.

Let us begin with L^{EJ}. In the discretization process, we must consider the cases L^{EJuv} for $u = 1, 2$ and $v = 1, 2$. The discretization process for L^{EJ11} and L^{EJ22} is exactly the same as the procedure used in Chapter 5 because the source and the observation points are in the same plane. To avoid unnecessary repetition, the reader can refer to Section 5.3 for details of the discretization process for these cases. For

the operators L^{EJ12} and L^{EJ21}, the discretization scheme is similar, except that the distance $\mathbf{r}$ in the potential function $\phi(\mathbf{r},\varepsilon,\mu)$ of (7.4) must be computed as

$$|\mathbf{r}| = (x^2 + y^2 + d^2)^{1/2} \tag{7.12}$$

instead of

$$|\mathbf{r}| = (x^2 + y^2)^{1/2} \tag{7.13}$$

which was used in Chapter 5. Therefore, it is also unnecessary in the cases of L^{EJ12} and L^{EJ21} to give more detail about the discretization.

The other operator defined in (7.3b), L^{EM}, will also be expanded into four terms L^{EMuv}, $u = 1, 2$ and $v = 1, 2$. The gradient of $\phi(\mathbf{r},\varepsilon,\mu)$ is given by

$$\nabla\phi(\mathbf{r},\varepsilon,\mu) = \mathbf{r}\frac{\mathrm{d}\phi(\mathbf{r},\varepsilon,\mu)}{\mathrm{d}r} = -\mathbf{r}\frac{jk(\varepsilon,\mu)r + 1}{r^3}\exp(-jk(\varepsilon,\mu)r) = -\mathbf{r}\Psi(r,\varepsilon,\mu) \tag{7.14}$$

Since the vectorial product $Mx\nabla\phi$ does not have any component in the xy-plane, L^{EM11} and L^{EM22} vanish. The operator L^{EM12}, on the other hand, can be written in terms of x- and y-components:

$$L^{EM12}(\mathbf{r},\varepsilon,\mu,\mathbf{M}) = E_x^{EM12}(\mathbf{r},\varepsilon,\mu)\mathbf{x} + E_y^{EM12}(\mathbf{r},\varepsilon,\mu)\mathbf{y} \tag{7.15}$$

which are given by

$$E_x^{EM12}(\mathbf{r},\varepsilon,\mu) = \frac{\mathrm{d}}{4\pi}\int_{S^2}\Psi(\mathbf{r}-\mathbf{r}',\varepsilon,\mu)\,M_y(\mathbf{r}')\,\mathrm{d}s' \tag{7.16a}$$

$$E_y^{EM12}(\mathbf{r},\varepsilon,\mu) = -\frac{\mathrm{d}}{4\pi}\int_{S^2}\Psi(\mathbf{r}-\mathbf{r}',\varepsilon,\mu)\,M_x(\mathbf{r}')\,\mathrm{d}s' \tag{7.16b}$$

Considering the discretization for the x-component, the next step is to smooth E_x^{EM12} with the razor blade $W^x(\mathbf{r})$ defined in (5.30) and to sample with a train of delta functions centered at the same points as the source $M_y(\mathbf{r}')$:

$$\begin{aligned} E_{xy}^{EM12}(\mathbf{r},\varepsilon,\mu) &= \sum_{i=-\infty}^{\infty}\sum_{j=-\infty}^{\infty}\left(W^x(\mathbf{r}) * E_x^{EM12}(\mathbf{r},\varepsilon,\mu)\right)\delta(\mathbf{r}-\mathbf{r}_y^{ij}) \\ &= \sum_{i=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} e_{xy}^{PEM12}[i,j]\,\delta(r_y^{ij}) \end{aligned} \tag{7.17}$$

As has been shown in Chapters 3 and 5, the discrete function in (7.17) can be computed by means of a 2D FFT with a period $(2N_x + 2)(2N_y + 2)$:

$$e_{xy}^{PEM12}[i,j] = \Delta x \, \mathrm{DFT}^{-1} \left(\tilde{g}^{PEM12}[m,n] \, \tilde{M}_y^P[m,n] \right) \tag{7.18}$$

where $\tilde{M}_y^P[m,n]$ is the DFT of the function $M_y^P[i,j]$. In a similar way, $\tilde{g}^{PEM12}[m,n]$ is the DFT of the periodic Green's function that can be obtained from

$$g^{DEM12}[i,j] = \frac{\mathrm{d}}{4\pi} \int_{-\Delta y/2}^{\Delta y/2} \int_{-\Delta x/2}^{\Delta x/2} \Psi(i\Delta x - x', j\Delta y - y', \varepsilon, \mu) \, \mathrm{d}x' \, \mathrm{d}y' \tag{7.19}$$

Note in (7.17) that $E_{xy}^{DEM12}(\mathbf{r},\varepsilon,\mu)$ is a series of delta functions located at r_y^{ij}, the central points of the rooftops considered to expand M_y. However, since all the other terms for the x-component in the SIE of (7.2) are given by a series of delta functions located at r_x^{ij}, the centers of the rooftops for the x-component of the currents, it is convenient to write a discrete function for $E_x^{EM12}(\mathbf{r},\varepsilon,\mu)$ which uses these centers:

$$E_x^{DEM12}(\mathbf{r},\varepsilon,\mu) = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} e_x^{PEM12}[i,j] \, \delta(\mathbf{r} - \mathbf{r}_x^{ij}) \tag{7.20}$$

where $e_x^{PEM12}[i,j]$ is related to $e_{xy}^{PEM12}[i,j]$ by the average

$$\begin{aligned} e_x^{PEM12}[i,j] = \frac{1}{4} \big(& e_{xy}^{PEM12}[i,j] + e_{xy}^{PEM12}[i+1,j] \\ & + e_{xy}^{PEM12}[i,j-1] + e_{xy}^{PEM12}[i+1,j-1] \big) \end{aligned} \tag{7.21}$$

which is the same average as that in (7.8a).

Using the shift property of the DFT, $e_x^{PEM12}[i,j]$ can be computed from

$$e_x^{PEM12}[i,j] = \Delta x \, \mathrm{DFT}^{-1} \left(D_x[m,n] \, \tilde{g}^{PEM12}[m,n] \, \tilde{M}_y^P[m,n] \right) \tag{7.22}$$

where

$$D_x[m,n] = \frac{1}{4} \left(1 + C_x[m] + C_y^*[n] + C_x[m] \, C_y^*[n] \right) \tag{7.23}$$

with

$$C_\alpha[l] = \exp\left(\frac{j2\pi(l-1)}{2N_\alpha + 2}\right) \qquad \alpha = x \quad \text{or} \quad y \tag{7.24}$$

and C_y^* is the conjugate of C_y.

Following a similar procedure, we can obtain a discrete function for the y-component of the fields

$$E_y^{DEM12}(\mathbf{r},\varepsilon,\mu) = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} e_y^{PEM12}[i,j]\, \delta(\mathbf{r} - \mathbf{r}_y^{ij}) \tag{7.25}$$

where

$$e_y^{PEM12}[i,j] = -\Delta y\, \mathrm{DFT}^{-1}(D_y[m,n]\, \tilde{g}^{PEM12}[m,n]\, \tilde{M}_x^P[m,n]) \tag{7.26}$$

and now

$$D_y[m,n] = \frac{1}{4}(1 + C_x^*[m] + C_y[n] + C_x^*[m]\, C_y[n]) \tag{7.27}$$

We have seen that to evaluate L^{EM12}, it is only necessary to compute and store one discrete Green's function $\tilde{g}^{DEM12}[m,n]$ for each equivalent problem defined in Figure 7.2 (external and internal problems, respectively). The evaluation of L^{EM21} is identical to that of L^{EM12} and can be performed with the help of the same pair of discrete Green's functions $\tilde{g}^{DEM12}[m,n]$.

7.4 RESULTS

The CG-FFT scheme described for the analysis of finite-sized planar dielectric structures has been checked considering two radiation problems. The first structure that will be discussed is the case of an electric dipole embedded in the center of a phantom dielectric layer, as shown in Figure 7.4. We refer to this structure as a phantom dielectric problem because, since the dielectric constant is equal to 1, the dipole is in effect radiating in free space, and therefore the analytical approach for finding the fields of a dipole in free space is entirely applicable to this problem. However, we can also use the CG-FFT scheme presented in this chapter to analyze this "dielectric" layer, compute the equivalent currents at the dielectric interface, and from these currents obtain the radiated fields. Therefore, in the CG-FFT analysis we have an electric impressed current (the dipole) in the equivalent problem (Figure 7.2(b)) for the internal region. This impressed current originates the equivalent currents to

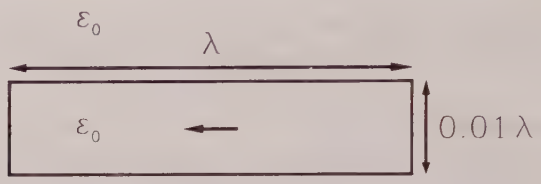


Figure 7.4 Electric dipole inserted in the center of a square phantom dielectric layer, $\varepsilon = \varepsilon_0$, having a side of one wavelength and a height of 0.01 wavelength. (Source: [4].)

be considered for the external equivalent problem (Figure 7.2(a)). In this case, we do not have any impressed field in the external equivalent problem; hence, the radiation field is computed only for the equivalent surface currents of the external problem. By comparing the radiation field computed in this way with the solution obtained by considering an electric dipole in free space, we can check the numerical scheme with an analytical solution. Figure 7.5 shows the results for the E-plane cut of the radiation pattern obtained by using both methods (continuous line: analytical solution; dots: CG-FFT). It is encouraging to see that there is very good agreement between the numerical and analytical values. The numerical results have been obtained considering 15 × 16 rooftops to represent each current component. As we have electric and magnetic currents on the upper and lower faces of the phantom structure and each current has x- and y-components, we are dealing with eight current

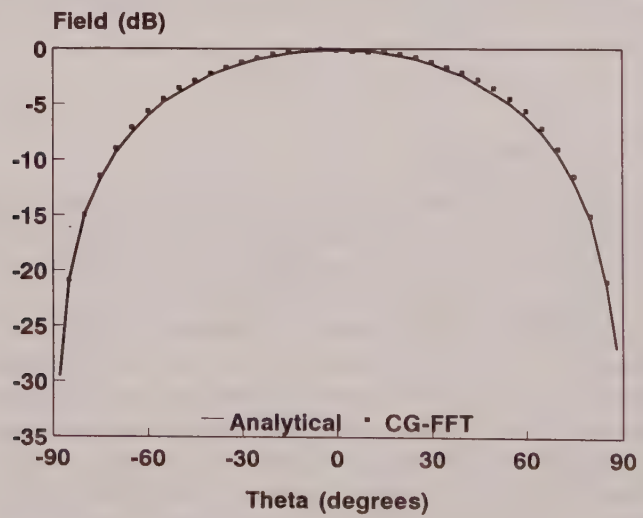


Figure 7.5 E-plane cut of the radiation pattern of the phantom problem depicted in Figure 7.4. (Source: [4].)

components ($2 \times 2 \times 2$). The total number of samples used in the numerical approach was therefore $15 \times 16 \times 8 = 1{,}920$.

The next case to be considered is the Vivaldi antenna shown in Figure 7.6. The antenna has been numerically fed considering a short electric dipole located as indicated in the figure. In the experimental setup, the antenna was fed by a strip line whose internal line crossed the slot line of the antenna approximately at the same point where the dipole is located in the numerically fed model. The strip line has not been considered in the CG-FFT simulation. It should also be pointed out that, in the simulation, the exponential line that defines the curve of the metallizations in the Vivaldi slot line has been approximated by a stair line, as is to be expected when using the CG-FFT mesh illustrated in Figure 7.3. In any case, the stair line approximation of the exponential line is good enough because a fine mesh has been considered: 31×32 rooftops for each current component, which represents a total of 7,936 samples for the eight current components of the problem. Figure 7.7 shows a comparison between computed and measured values for the E- and H-plane cuts. Again, there is good agreement between the values obtained with the CG-FFT method and the measurements.

Finally, in Figure 7.8 it can be seen that a typical curve is obtained for the convergence speed of the CG-FFT approach. The case indicated in Figure 7.4 was considered for this curve. A computer with a word length of 64 bits was used to obtain the convergence speed shown.

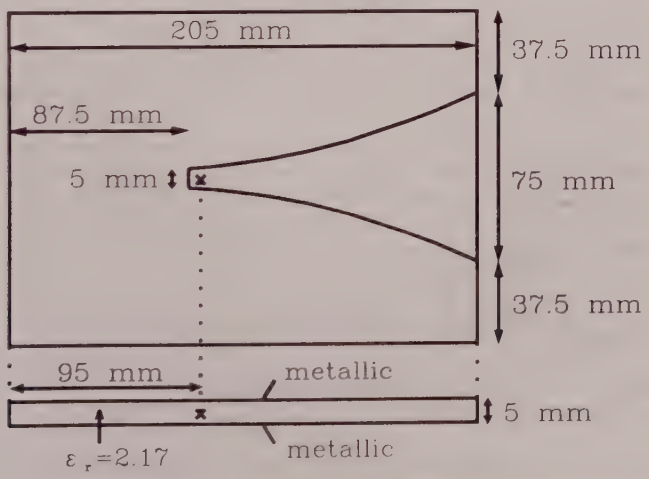
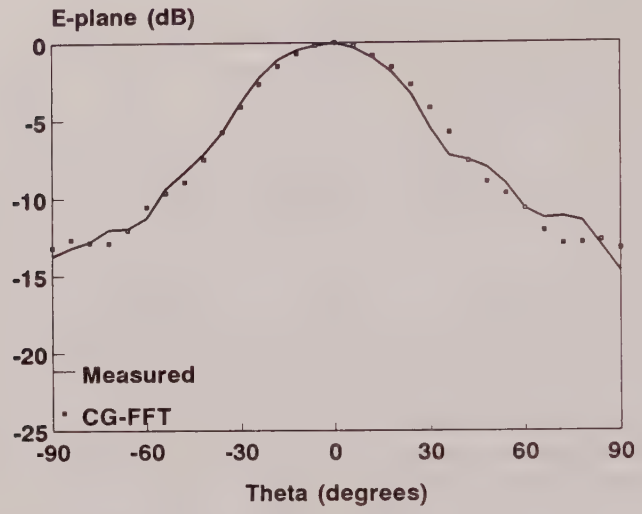
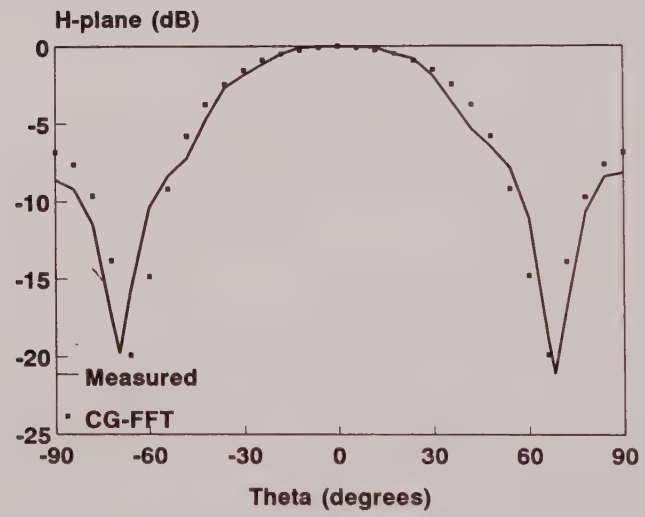


Figure 7.6 Vivaldi antenna for testing the CG-FFT approach. The working frequency is 2.5 GHz. The electrically short dipole used in the model of the antenna feeding is located at the cross. This dipole is parallel to the metallic plates and normal to the slot line of the antenna. (Source: [4].)



(a)



(b)

Figure 7.7 Measured and computed values for the (a) E-plane cut and (b) H-plane cut of the radiation pattern of the antenna sketched in Figure 7.6. (Source: [4].)

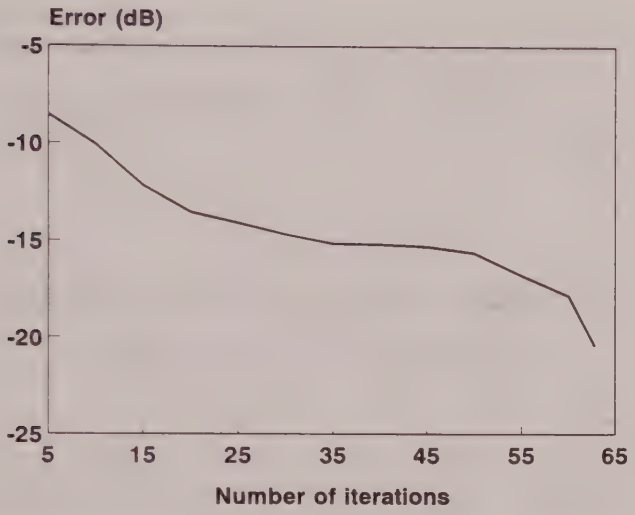


Figure 7.8 Number of iterations versus relative error for the problem of Figure 7.4. (Source: [4].)

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Chapter 8

Metallic Surfaces That Conform to Bodies of Revolution

8.1 INTRODUCTION

In this chapter, we undertake the electromagnetic analysis of metallic surfaces that conform to surfaces with symmetry of revolution (SSOR). An SSOR is formed by rotating a curve (called the generatrix) around an axis (the rotation axis). In many electromagnetic texts, SSORs are called BORs, and we, too, will use the latter term. If the rotation axis is taken as the z cartesian coordinate, a BOR can be studied easily in cylindrical coordinates because of the periodic behavior of the azimuthal coordinate (ϕ). BOR formulations can be applied to missiles, rockets, and satellites, as well as to antennas with symmetry of revolution such as conic horns and paraboloids. In these examples, we are referring to metallic surfaces that completely cover an SSOR. We will use the term complete BORs for such structures. Another interesting case arises when the conducting surface conforms perfectly to an SSOR but does not cover it completely. The term partial BOR will be used to distinguish such structures from the previous ones. Partial BORs are quite interesting and useful, because many problems can be represented by them. For example, a conical horn antenna with apertures in the lateral wall is a partial BOR. Many curved plates that must be analyzed for RCS purposes can also be assimilated to partial BORs.

Due to the generality of the problem being analyzed (i.e., any surface that conforms to any type of BOR), a numerical treatment is required to solve it. Several authors have analyzed the complete BOR problem by means of the MM. Mautz and Harrington, for instance, solved the EFIE with a combined-source method in order to avoid internal resonance problems [1,2]. Glisson and Wilton have presented techniques to improve MM formulations for BORs [3]. Although the MM method appears to be suitable when the metallic surface forms a complete BOR [4], it is not efficient for the treatment of partial BORs. To analyze such surfaces, the GTD in combination with the physical theory of diffraction have been used for electrically

large bodies [5]. Partial BORs of cylindrical shape can be analyzed with good numerical accuracy by using the MM and the finite bodies of translation (BOT) formulation of [6,7] in conjunction with quite specialized basis and testing functions. For this type of problem, a CG-FFT approach has been proposed [8] that provides the computational advantages of this method.

In this CG-FFT approach, the problem is discretized in the spatial domain. The 2D convolutions resulting from the discretization are then computed cyclically. As in Chapter 3, this computation will be performed by obtaining an equivalent periodic problem. The reference cell of this periodic problem must have sides that are at least twice as long as the sides of the minimum rectangle that completely envelopes the original problem. The length of the side associated with the variable ϕ need not be doubled, since the original problem is periodic for this coordinate.

In this chapter, we generalize this CG-FFT approach to the analysis of surfaces that conform to arbitrary BORs, showing its efficiency in terms of both CPU time and memory storage requirements [8]. In this case, the advantage of using the FFT to compute convolutions can be obtained only in the periodic coordinate (ϕ). The problem is not an invariant system for the other coordinates, and therefore the convolution property of the Fourier transform cannot be applied. Nevertheless, the CG-FFT approach is still advantageous, especially for partial BORs.

8.2 INTEGRAL EQUATION

Consider an arbitrary conducting surface S that conforms to a BOR (see Figure 8.1). Let $\mathbf{E}^{\text{inc}}$ denote the impressed field on this surface. This field induces a current distribution $\mathbf{J}^{\text{ind}}$, which generates a scattering field $\mathbf{E}(\mathbf{J}^{\text{ind}})$. Applying the superposition principle, we can write the total electric field as

$$\mathbf{E} = \mathbf{E}^{\text{inc}} + \mathbf{E}(\mathbf{J}^{\text{ind}}) \tag{8.1}$$

On the conducting surface,

$$\mathbf{n} \times \mathbf{E} = 0 \tag{8.2}$$

where $\mathbf{n}$ is an outgoing normal vector. Hence, we have

$$\mathbf{n} \times \mathbf{E}^{\text{inc}} = -\mathbf{n} \times \mathbf{E}(\mathbf{J}^{\text{ind}}) \tag{8.3}$$

On the other hand, from the Maxwell equations the field produced by a surface current distribution can be written

$$\mathbf{E}(\mathbf{J}) = -j\omega\mathbf{A} - \nabla\Phi \tag{8.4}$$

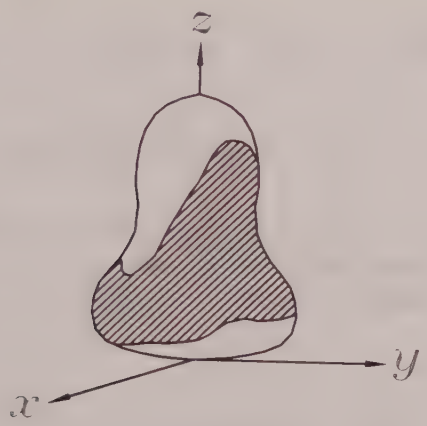


Figure 8.1 Arbitrary surface that conforms to an arbitrary BOR.

where $\nabla\Phi$ represents the gradient of the potential Φ, and $\mathbf{A}$ and Φ are given by

$$\mathbf{A} = \mu_0 \iint_{S'} \mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\,ds' \tag{8.5}$$

$$\Phi = \frac{1}{\varepsilon_0} \iint_{S'} \sigma(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\,ds' \tag{8.6}$$

In these expressions, G is the free-space Green's function:

$$G(\mathbf{r},\mathbf{r}') = \frac{\exp(-jk|\mathbf{r}-\mathbf{r}'|)}{4\pi|\mathbf{r}-\mathbf{r}'|} \tag{8.7}$$

From the continuity relation,

$$\nabla\mathbf{J} + j\omega\sigma = 0 \tag{8.8}$$

the charge distribution can be expressed as

$$\sigma = -\frac{\nabla\mathbf{J}}{j\omega} \tag{8.9}$$

Then the field produced by a current distribution can be written as

$$\mathbf{E}(\mathbf{J}) = -j\omega\mu_0 \iint_{S'} \mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\,ds' + \frac{1}{j\omega\varepsilon_0}\nabla \iint_{S'} \nabla'\mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\,ds' \tag{8.10}$$

where the first term on the right-hand side is called the inductive or current term and the second one the capacitive or charge term.

Thus, we have obtained an expression for (8.3) having the generic form

$$\mathbf{E} = L\mathbf{J} \tag{8.11}$$

The following sections develop the solution of this equation by means of the CG-FFT scheme.

8.3 SURFACES THAT CONFORM TO CYLINDERS

The EFIE presented in the previous section will now be particularized to the analysis of the simplest possible structure: a metallic surface that conforms to a cylinder of radius a and height h (see Figure 8.2). It seems evident that a cylindrical coordinate system (ρ,ϕ,z) should be used for the analysis. In this case, ρ is a constant (a) and its partial derivatives are null.

In this system, we can write the following expressions for the field, current distribution, relative distance, gradient, divergence, and surface differential element, respectively:

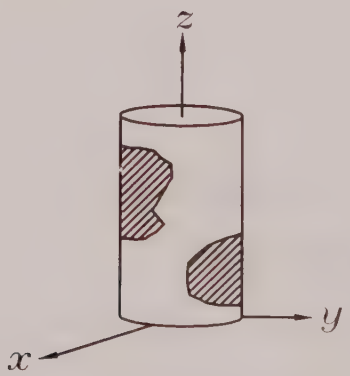


Figure 8.2 An arbitrary surface that conforms to a cylinder.

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}(\phi,z) = \boldsymbol{\phi} E_\phi(\phi,z) + \mathbf{z} E_z(\phi,z) \tag{8.12}$$

$$\mathbf{J}(\mathbf{r}) = \mathbf{J}(\phi,z) = \boldsymbol{\phi} J_\phi(\phi,z) + \mathbf{z} J_z(\phi,z) \tag{8.13}$$

$$|\mathbf{r} - \mathbf{r}'|^2 = 2a^2[1 - \cos(\phi - \phi')] + (z - z')^2 \tag{8.14}$$

$$\nabla \Phi = \boldsymbol{\phi} \frac{1}{a} \frac{\partial \Phi}{\partial \phi} + \mathbf{z} \frac{\partial \Phi}{\partial z} \tag{8.15}$$

$$\nabla \mathbf{J} = \frac{1}{a} \frac{\partial J_\phi}{\partial \phi} + \frac{\partial J_z}{\partial z} \tag{8.16}$$

$$ds = a \, d\phi \, dz \tag{8.17}$$

Using these expressions, (8.3) can be expressed in cylindrical component form as

$$\begin{aligned} E_\phi^{\text{inc}}(\mathbf{r}) = {} & k^c \int\int_{S'} J_\phi(\mathbf{r}') G(\mathbf{r},\mathbf{r}') \cos(\phi - \phi') \, ds' \\ & + k^q \frac{1}{a} \frac{\partial}{\partial \phi} \int\int_{S'} \left[\frac{1}{a} \frac{\partial J_\phi(\mathbf{r}')}{\partial \phi'} + \frac{\partial J_z(\mathbf{r}')}{\partial z'} \right] G(\mathbf{r},\mathbf{r}') \, ds' \end{aligned} \tag{8.18}$$

$$\begin{aligned} E_z^{\text{inc}}(\mathbf{r}) = {} & k^c \int\int_{S'} J_z(\mathbf{r}') G(\mathbf{r},\mathbf{r}') \, ds' \\ & + k^q \frac{\partial}{\partial z} \int\int_{S'} \left[\frac{1}{a} \frac{\partial J_\phi(\mathbf{r}')}{\partial \phi'} + \frac{\partial J_z(\mathbf{r}')}{\partial z'} \right] G(\mathbf{r},\mathbf{r}') \, ds' \end{aligned} \tag{8.19}$$

with $k^c = j\omega\mu_0$ and $k^q = -1/(j\omega\varepsilon_0)$.

The next step in the process is to expand the current distributions in terms of appropriate basis functions defined on the cylinder surface, which can be represented as a plane surface lying in the plane ϕz (see Figure 8.3). These expansions are of the form

$$J_\phi(\phi,z) = \sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_z+1} J_\phi[i,j] \, T_{ij}^{\phi}(\phi,z) \tag{8.20}$$

$$J_z(\phi,z) = \sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_z} J_z[i,j] \, T_{ij}^{z}(\phi,z) \tag{8.21}$$

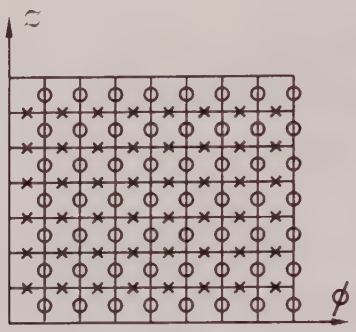


Figure 8.3 An ϕz-plane representation of the cylinder surface showing the centers of the domains of definition of the basis functions.

T_{ij}^{α} are rooftop functions fulfilling the shift property $T_{ij}^{\alpha} = T^{\alpha}(\phi - \phi_{\alpha}^{i}, z - z_{\alpha}^{j})$, centered on the points shown in Figure 8.3 (crosses for the z-component, ϕ_z^i and z_z^j; circles for the ϕ-component, ϕ_ϕ^i and z_z^j), which are given by

$$T^{\phi}(\phi,z) = \begin{cases} 1 - \dfrac{|\phi|}{\Delta\phi} & \text{if} \quad |\phi| \le \Delta\phi \quad \text{and} \quad |z| \le \Delta z/2 \\ 0 & \text{otherwise} \end{cases} \tag{8.22a}$$

$$T^{z}(\phi,z) = \begin{cases} 1 - \dfrac{|z|}{\Delta z} & \text{if} \quad |\phi| \le \Delta\phi/2 \quad \text{and} \quad |z| \le \Delta z \\ 0 & \text{otherwise} \end{cases} \tag{8.22b}$$

$$\begin{cases} \phi_{\phi}^{i} = i\Delta\phi & i = 1, 2, \ldots, N_{\phi} + 1 \\ z_{\phi}^{j} = (j - 0.5)\Delta z & j = 1, 2, \ldots, N_{z} + 1 \end{cases} \tag{8.22c}$$

$$\begin{cases} \phi_{z}^{i} = (i - 0.5)\Delta\phi & i = 1, 2, \ldots, N_{\phi} + 1 \\ z_{z}^{j} = j\Delta z & j = 1, 2, \ldots, N_{z} \end{cases} \tag{8.22d}$$

The next step is to test the field components ((8.18) and (8.19)) with razor blade functions in the corresponding directions. From this process we obtain

$$\int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} E_\phi^{\text{inc}}(\mathbf{r}) a \, d\phi = \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \left[k^c \iint_{S'} J_\phi(\mathbf{r}') G(\mathbf{r},\mathbf{r}') \cos(\phi - \phi') \, ds' + k^q \frac{1}{a} \frac{\partial}{\partial \phi} \iint_{S'} \left[\frac{1}{a} \frac{\partial J_\phi(\mathbf{r}')}{\partial \phi'} + \frac{\partial J_z(\mathbf{r}')}{\partial z'} \right] G(\mathbf{r},\mathbf{r}') \, ds' \right] a \, d\phi \tag{8.23}$$

$$\int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} E_z^{\text{inc}}(\mathbf{r}) \, dz = \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \left[k^c \iint_{S'} J_z(\mathbf{r}') G(\mathbf{r},\mathbf{r}') \, ds' + k^q \frac{\partial}{\partial z} \iint_{S'} \left[\frac{1}{a} \frac{\partial J_\phi(\mathbf{r}')}{\partial \phi'} + \frac{\partial J_z(\mathbf{r}')}{\partial z'} \right] G(\mathbf{r},\mathbf{r}') \, ds' \right] dz \tag{8.24}$$

These equations can be written in a more compact form as

$$V_\phi = V_\phi^J + V_\phi^C \tag{8.25a}$$

$$V_z = V_z^J + V_z^C \tag{8.25b}$$

where V_α represents the part of the operator related to the α-component of the impressed field. Considering this field constant along each testing function, we have

$$V_\phi[i,j] \approx a\Delta\phi E_\phi^{\text{inc}}(\phi_\phi^i, z_\phi^j) \tag{8.26}$$

$$V_z[i,j] \approx \Delta z E_z^{\text{inc}}(\phi_z^i, z_z^j) \tag{8.27}$$

The other terms in (8.25) will be developed in the next sections.

8.3.1 V_ϕ^J Term

This term corresponds to the ϕ field component due to the current distribution, which is the first term on the right-hand side of (8.23). Replacing J_ϕ by its expansion in terms of basis functions, we have

$$V_\phi^J[i,j] = k^c \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \left[\sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_z+1} J_\phi[i',j'] \cdot \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_{(j'-1)\Delta z}^{j'\Delta z} G(\mathbf{r},\mathbf{r}') \cos(\phi - \phi') \, ds' \right] a \, d\phi \tag{8.28}$$

In (8.28), we have used Glisson and Wilton's approximation [3] for the integrals in which the triangles in the ϕ-direction (basis functions) have been replaced by pulses that extend over half of the integration interval. Thus, the following discrete expression is obtained for the point $\phi = \phi^i_\phi$, $z = z^j_\phi$ before testing:

$$E^J_\phi[i,j] = k^c \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_z+1} J_\phi[i',j'] G_{dn}[i-i',j-j'] \tag{8.29}$$

where

$$G_{dn}[i-i',j-j'] = \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_{(j'-1)\Delta z}^{j'\Delta z} G(\phi^i_\phi - \phi', z^j_\phi - z') \cos(\phi^i_\phi - \phi')\, ds' \tag{8.30}$$

which represents the integral of the Green's function centered on the point (ϕ^i_ϕ, z^j_ϕ) over the domain depicted in Figure 8.4. Considering the cosine function constant over the integration interval, we have

$$\begin{aligned} G_{dn}[i-i',j-j'] &= \cos[(i-i')\Delta\phi] G_d[i-i',j-j'] \\ &= \cos[(i-i')\Delta\phi] \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_{(j'-1)\Delta z}^{j'\Delta z} G(\phi^i_\phi - \phi', z^j_\phi - z')\, ds' \end{aligned} \tag{8.31}$$

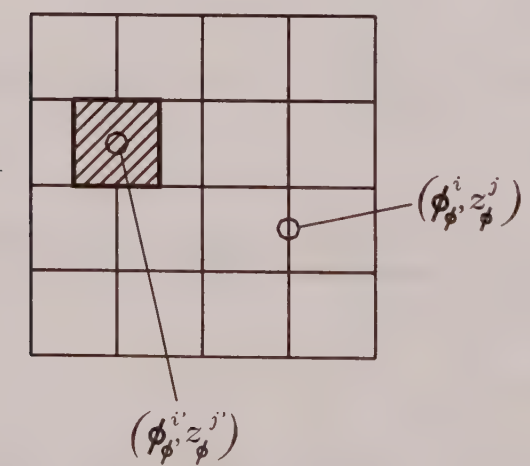


Figure 8.4 The dashed rectangle is the integration domain corresponding to (8.30). The Green's function is centered on (ϕ^i_ϕ, z^j_ϕ).

Expression (8.29) is a discrete convolution of two functions. Once again assuming the fields to be constant along the testing functions, we obtain

$$V_{\phi}^{J}[i,j] = k^{c} a \Delta\phi J_{\phi}[i,j] \otimes G_{dn}[i,j] \tag{8.32}$$

and by applying the DFT, we have

$$\tilde{V}_{\phi}^{J}[m,n] = k^{c} a \Delta\phi \tilde{J}_{\phi}[m,n] \frac{\tilde{G}_{d}[m+1,n] + \tilde{G}_{d}[m-1,n]}{2} \tag{8.33}$$

For these applications of the DFT, it is necessary to take into account the comments about the computation of operator convolutions in Chapter 3. In order to obtain the correct result by using the DFT, the sequences must be periodic. For the cylindrical case, we are dealing with structures that are intrinsically periodic in the ϕ-direction, but this does not hold for the z-direction. In order to compute the correct values of the convolutions in the region of interest, we have to pad the currents with zeros to obtain new functions with period $(N_{\phi} + 1, 2N_{z} + 2)$. Throughout this chapter, it will be assumed that this step is implicitly involved whenever cyclic convolutions are introduced for the cylindrical case. However, in the approach for arbitrary BORs, no padding is required.

8.3.2 V_{ϕ}^{C} Term

This term corresponds to the ϕ field component due to the charge distribution, which is the second term on the right-hand side of (8.23). This term is analyzed in two parts, the first one related to the charge due to the ϕ-component of the current ($V_{\phi}^{C\phi}$) and the second related to the charge due to the z-component of the current (V_{ϕ}^{Cz}).

8.3.2.1 $V_{\phi}^{C\phi}$ Term

This term corresponds to the expression

$$V_{\phi}^{C\phi}[i,j] = k^{q} \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \left[\frac{1}{a} \frac{\partial}{\partial\phi} \int\int_{S'} \frac{1}{a} \frac{\partial J_{\phi}(\mathbf{r}')}{\partial\phi'} G(\mathbf{r},\mathbf{r}') \, ds' \right] a \, d\phi \tag{8.34}$$

If the potential along ϕ is defined as

$$\Phi_{\phi}(\phi,z) = \int\int_{S'} \frac{1}{a} \frac{\partial J_{\phi}(\mathbf{r}')}{\partial\phi'} G(\mathbf{r},\mathbf{r}') \, ds' \tag{8.35}$$

then (8.34) becomes

$$V_{\phi}^{C\phi}[i,j] = k^q \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \frac{1}{a} \frac{\partial \Phi_{\phi}(\phi, z_{\phi}^{j})}{\partial \phi} a \, d\phi$$
$$= k^q [\Phi_{\phi}(\phi_z^{i+1}, z_{\phi}^{j}) - \Phi_{\phi}(\phi_z^{i}, z_{\phi}^{j})] \tag{8.36}$$

or, in discrete form,

$$V_{\phi}^{C\phi}[i,j] = k^q [\Phi_{\phi}[i+1,j] - \Phi_{\phi}[i,j]] \tag{8.37}$$

It is necessary to take into account the fact that the indexes (i,j) do not correspond to the same point for V and Φ. When dealing with V_{ϕ}, they represent the center of the domain of the basis function T_{ij}^{ϕ}. This center is indicated by a cross in Figure 8.5. In the same figure, the domain over which T_{ij}^{ϕ} extends is denoted by D_{ij}^{ϕ} and it is the area enclosed by the rectangle formed by the solid line.

$$(\phi_{\phi}^{i}, z_{\phi}^{j}) = (i\Delta\phi, (j-0.5)\Delta z) \tag{8.38}$$

For Φ, on the other hand, the indexes (i,j) refer to the center (solid point) of the domain D_{ij} indicated by a dashed line. This center point is given by

$$(\phi_z^{i}, z_{\phi}^{j}) = ((i-0.5)\Delta\phi, (j-0.5)\Delta z) \tag{8.39}$$

In Figure 8.6, the functions used to obtain $V_{\phi}(i,j)$ from $J_{\phi}(i,j)$ are shown in an intuitive way. Using $C_{\phi}(\phi,z)$ to denote

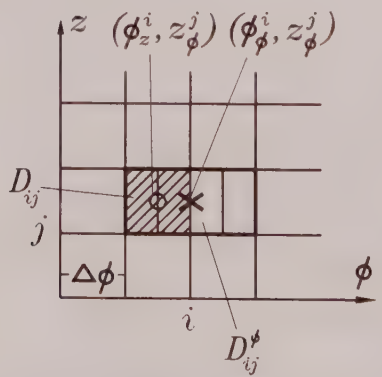


Figure 8.5 Definition of domains and their centers.

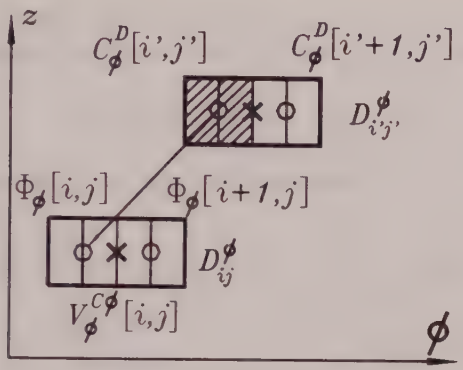


Figure 8.6 Schematic representation of the functions used to obtain $V_\phi^{C\phi}$ from J_ϕ.

$$C_\phi(\phi,z) = \frac{1}{a}\frac{\partial J_\phi}{\partial \phi'} = \frac{1}{a}\frac{\partial}{\partial \phi'}\left[\sum_{i=1}^{N_\phi+1}\sum_{j=1}^{N_z+1} J_\phi[i,j]\, T_{ij}^\phi(\phi,z)\right] \tag{8.40}$$

and taking into account the fact that the current is represented by triangle functions in the ϕ-direction so that its derivatives are pulses of amplitude

$$C_\phi^D[i,j] = \frac{J_\phi[i,j] - J_\phi[i-1,j]}{a\Delta\phi} \tag{8.41}$$

we can write (8.40) as

$$C_\phi(\phi,z) = \sum_{i=1}^{N_\phi+1}\sum_{j=1}^{N_z+1} C_\phi^D[i,j]\, P_{ij}(\phi,z) \tag{8.42}$$

The potential Φ_ϕ can be written

$$\Phi_\phi[i,j] = \sum_{i'=1}^{N_\phi+1}\sum_{j'=1}^{N_z+1} C_\phi^D[i',j']G_d[i-i',j-j'] \tag{8.43}$$

where the Green's function is given by

$$G_d[i-i',j-j'] = \int_{(i'-1)\Delta\phi}^{i'\Delta\phi}\int_{(j'-1)\Delta z}^{j'\Delta z} G(\phi_z^i - \phi', z_\phi^j - z')\, ds' \tag{8.44}$$

which is equal to $G_d[i - i', j - j']$ in (8.31) with the integrand and the integration domain shifted by $\Delta\phi/2$. Hence, the potential can be expressed as the convolution

$$\Phi_\phi[i,j] = C_\phi^D[i,j] \otimes G_d[i,j] \tag{8.45}$$

The transform of this expression (substituting its value for C_ϕ^D) is

$$\tilde{\Phi}_\phi[m,n] = \frac{1}{a\Delta\phi}\left[1 - F_\phi^*[m]\right]\tilde{J}_\phi[m,n]\tilde{G}_d[m,n] \tag{8.46}$$

and the transform of (8.37) is

$$\tilde{V}_\phi^{C\phi}[m,n] = k^q\left[F_\phi[m] - 1\right]\tilde{\Phi}_\phi[m,n] \tag{8.47}$$

where F_ϕ is a term that takes into account the relative shift

$$F_\phi[m] = \exp[jm2\pi/(N_\phi + 1)] \tag{8.48}$$

Thus, we can write the transform of the discrete operator as a function of the transform of the current distribution:

$$\tilde{V}_\phi^{C\phi}[m,n] = \frac{k^q}{a\Delta\phi}\left[F_\phi[m] - 1\right]\left[1 - F_\phi^*[m]\right]\tilde{J}_\phi[m,n]\tilde{G}_d[m,n] \tag{8.49}$$

8.3.2.2 V_ϕ^{Cz} Term

The V_ϕ^{Cz} term corresponds to the following expression:

$$V_\phi^{Cz}[i,j] = k^q \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \left[\frac{1}{a}\frac{\partial}{\partial\phi}\int\int_{S'} \frac{\partial J_z(\mathbf{r}')}{\partial z'} G(\mathbf{r},\mathbf{r}')\, ds'\right] dz \tag{8.50}$$

This term is analyzed in a similar way to the previous one (the main functions in the process are illustrated in Figure 8.7).

Substituting in the z-potential,

$$\Phi_z(\phi,z) = \int\int_{S'} \frac{\partial J_z(\mathbf{r}')}{\partial z'} G(\mathbf{r},\mathbf{r}')\, ds' \tag{8.51}$$

we can write (8.50) in the form

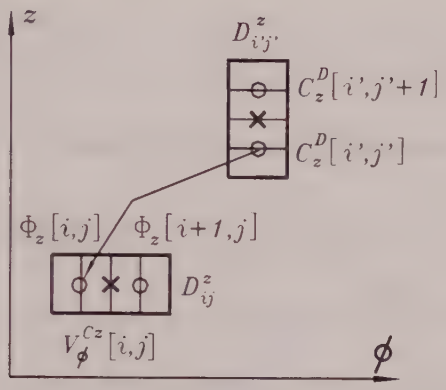


Figure 8.7 Schematic representation of the functions used to obtain V_ϕ^{Cz} from J_z.

$$\begin{aligned} V_\phi^{Cz}[i,j] &= k^q \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} \frac{1}{a} \frac{\partial \Phi_z(\phi, z_\phi^j)}{\partial \phi} a \, \mathrm{d}\phi \\ &= k^q[\Phi_z(\phi_z^{i+1}, z_\phi^j) - \Phi_z(\phi_z^i, z_\phi^j)] \\ &= k^q[\Phi_z[i+1,j] - \Phi_z[i,j]] \end{aligned} \tag{8.52}$$

The derivative of the current in this case is given by

$$C_z(\phi,z) = \frac{\partial J_z}{\partial z'} = \frac{\partial}{\partial z'} \left[\sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_z} J_z[i,j] T_{ij}^z(\phi,z) \right] \tag{8.53}$$

The pulse amplitude is

$$C_z^D[i,j] = \frac{J_z[i,j] - J_z[i,j-1]}{\Delta z} \tag{8.54}$$

so that (8.53) becomes

$$C_z(\phi,z) = \sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_z} C_z^D[i,j] P_{ij}(\phi,z) \tag{8.55}$$

The potential (8.51) can now be written as

$$\Phi_z(\phi,z) = \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_z} C_z^D[i',j'] \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{(j'-1)\Delta z}^{j'\Delta z} G(\phi - \phi', z - z')\, ds' \tag{8.56}$$

The Green's function again coincides with (8.44); hence,

$$\Phi_z[i,j] = \frac{J_z[i,j] - J_z[i,j-1]}{\Delta z} \otimes G_d[i,j] \tag{8.57}$$

which, in the transform domain, becomes

$$\tilde{\Phi}_z[m,n] = \frac{1}{\Delta z}\left[1 - F_z^*[n]\right]\tilde{J}_z[m,n]\tilde{G}_d[m,n] \tag{8.58}$$

where

$$F_z[n] = \exp[jn2\pi/(2N_z + 2)] \tag{8.59}$$

On the other hand, the transform of (8.52) is

$$\tilde{V}_\phi^{Cz}[m,n] = k^q\left[F_\phi[m] - 1\right]\tilde{\Phi}_z[m,n] \tag{8.60}$$

So that by substituting in (8.58), we obtain the following final expression:

$$\tilde{V}_\phi^{Cz}[m,n] = \frac{k^q}{\Delta z}\left[F_\phi[m] - 1\right]\left[1 - F_z^*[n]\right]\tilde{J}_\phi[m,n]\tilde{G}_d[m,n] \tag{8.61}$$

8.3.3 V_z^J Term

As can be seen in (8.18) and (8.19), the expressions for the two field components are similar except for the cosine term that appears in (8.18). Hence, the mathematical derivation for the z-component is similar and therefore will not be presented in detail.

Briefly,

$$V_z^J[i,j] = k^c \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \left[\iint_{S'} J_z(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds'\right] dz \tag{8.62}$$

Substituting in the J_z expansion and employing an approach similar to the one used for V_ϕ^J, we obtain

$$V_z^J[i,j] = k^c \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \left[\sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_z} J_z[i',j'] \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{(j'-0.5)\Delta z}^{(j'+0.5)\Delta z} G(\mathbf{r},\mathbf{r}')\, ds' \right] dz \quad (8.63)$$

Using the following notation for the Green's function,

$$G_d''[i-i',j-j'] = \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{(j'-0.5)\Delta z}^{(j'+0.5)\Delta z} G(\phi_z^i - \phi', z_z^j - z')\, ds' \quad (8.64)$$

which again coincides with (8.44), we have

$$V_z^J[i,j] = k^c J_z[i,j] \otimes G_d[i,j] \quad (8.65)$$

Thus, the operator in the transform domain can be expressed as

$$\tilde{V}_z^J[m,n] = k^c \Delta z \tilde{J}_z[m,n] \tilde{G}_d[m,n] \quad (8.66)$$

8.3.4 V_z^C Term

Once again we analyze the two parts of the term separately.

8.3.4.1 $V_z^{C\phi}$ Term

This term is given by

$$V_z^{C\phi}[i,j] = k^q \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \left[\frac{\partial}{\partial z} \iint_{S'} \frac{1}{a} \frac{\partial J_\phi(\mathbf{r}')}{\partial \phi'} G(\mathbf{r},\mathbf{r}')\, ds' \right] dz \quad (8.67)$$

Analogously with Section 8.2.2.1, we can write (8.67) as

$$\begin{aligned} V_z^{C\phi}[i,j] &= k^q \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \frac{\partial \Phi_\phi(\phi, z_z^j)}{\partial z} dz \\ &= k^q [\Phi_\phi(\phi_z^i, z_\phi^{j+1}) - \Phi_\phi(\phi_z^i, z_\phi^j)] \end{aligned} \quad (8.68)$$

or, in simplified form,

$$V_z^{C\phi}[i,j] = k^q [\Phi_\phi[i,j+1] - \Phi_\phi[i,j]] \quad (8.69)$$

where Φ_ϕ is the potential defined by (8.35).

The resulting final expression in the transform domain is

$$\tilde{V}_z^{C\phi}[m,n] = \frac{k^q}{a\Delta\phi}\,[F_z[n] - 1]\big[1 - F_\phi^*[m]\big]\tilde{J}_\phi[m,n]\tilde{G}_d[m,n] \tag{8.70}$$

8.3.4.2 V_z^{Cz} Term

This term corresponds to the expression

$$V_z^{Cz}[i,j] = k^q \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \left[\frac{\partial}{\partial z}\int\int_{S'} \frac{\partial J_z(\mathbf{r}')}{\partial z'}\, G(\mathbf{r},\mathbf{r}')\, ds'\right] dz \tag{8.71}$$

Substituting in the z-potential defined in (8.51) and (8.52), we have

$$\begin{aligned} V_z^{Cz}[i,j] &= k^q \int_{(j-0.5)\Delta z}^{(j+0.5)\Delta z} \frac{\partial \Phi_z(\phi, z_\phi^j)}{\partial z}\, dz \\ &= k^q\big[\Phi_z(\phi_z^i, z_\phi^{j+1}) - \Phi_z(\phi_z^i, z_\phi^j)\big] \end{aligned} \tag{8.72}$$

or, in the transform domain,

$$\tilde{V}_z^{Cz}[m,n] = \frac{k^q}{\Delta z}\,[F_z[n] - 1]\big[1 - F_z^*[n]\big]\tilde{J}_z[m,n]\tilde{G}_d[m,n] \tag{8.73}$$

8.3.5 Final Expressions for the Operators

The operators can be computed by using the expression

$$\begin{bmatrix} V_\phi^{\text{inc}} \\ V_z^{\text{inc}} \end{bmatrix} = \text{FFT}^{-1}\left(\begin{bmatrix} \tilde{L}_{\phi\phi} & \tilde{L}_{\phi z} \\ \tilde{L}_{z\phi} & \tilde{L}_{zz} \end{bmatrix}\begin{bmatrix} \tilde{J}_\phi \\ \tilde{J}_z \end{bmatrix}\right) \tag{8.74}$$

It should be noted that the FFT must have a period of $(N_\phi + 1, 2N_z + 2)$ due to the periodic nature in the ϕ-direction and the necessity of obtaining a periodic behavior in the z-direction by padding with zeros until a period is obtained with a length at least twice the original extension of the function for the z-direction. As mentioned previously and in Chapter 3, this padding is required to transform the original linear convolution in z into a cyclic convolution.

The final expressions for the operators appearing in (8.74) can be summarized as follows:

$$\tilde{J}_{\phi}[i,j] = \text{FFT}[J_{\phi}[p,q]] \tag{8.75}$$

$$\tilde{J}_{z}[i,j] = \text{FFT}[J_{z}[p,q]] \tag{8.76}$$

$$\tilde{L}_{\phi\phi} = k^{c} a \Delta\phi \tilde{G}_{dd} + \frac{k^{q}}{a\Delta\phi}(F_{\phi} - 1)(1 - F_{\phi}^{*})\tilde{G}_{d} \tag{8.77}$$

$$\tilde{L}_{\phi z} = \frac{k^{q}}{\Delta z}(F_{\phi} - 1)(1 - F_{z}^{*})\tilde{G}_{d} \tag{8.78}$$

$$\tilde{L}_{z\phi} = \frac{k^{q}}{a\Delta\phi}(F_{z} - 1)(1 - F_{\phi}^{*})\tilde{G}_{d} \tag{8.79}$$

$$\tilde{L}_{zz} = k^{c} \Delta z \tilde{G}_{dd} + \frac{k^{q}}{\Delta z}(F_{z} - 1)(1 - F_{z}^{*})\tilde{G}_{d} \tag{8.80}$$

$$F_{\phi}[n] = \exp[j2\pi n/(N_{\phi} + 1)] \tag{8.81}$$

$$F_{z}[n] = \exp[j2\pi n/(2N_{z} + 2)] \tag{8.82}$$

The discrete Green's function G_d is periodic with periods of $(N_\phi + 1)$ and $2(N_z + 1)$ for the ϕ- and z-coordinates, respectively:

$$G_{d}[i,j] = \int_{-\Delta\phi/2}^{\Delta\phi/2} \int_{-\Delta z/2}^{\Delta z/2} G(i\Delta\phi - \phi', j\Delta z - z')\, a\, d\phi'\, dz' \tag{8.83}$$

$$\tilde{G}_{dd}[i,j] = \frac{\tilde{G}_{d}[i+1,j] + \tilde{G}_{d}[i-1,j]}{2} \tag{8.84}$$

where the tilde denotes transform quantities.

The adjoint operator that appears in the CG scheme is the transpose and complex conjugate of the direct operator of (8.74).

8.4 SURFACES THAT CONFORM TO ARBITRARY BORs

This section deals with the analysis of surfaces that conforms to arbitrary BORs (Figure 8.1). At first sight, it may seem that the cylindrical coordinate system (ρ,ϕ,z) used in the previous section would be adequate for this analysis. However, a new

coordinate system (ϕ,t,n), defined by the vectors $\mathbf{t}$ and $\mathbf{n}$ that are tangent and normal, respectively, to the surface along the generatrix and the vector $\boldsymbol{\phi}$ in the azimuthal direction, proves to be better. In this new coordinate system, which is shown in Figure 8.8, the computations are reduced to two components (ϕ,t) (i.e., the tangential components to the surface). Note that $\mathbf{n}$ is always an outgoing vector and that $\mathbf{t}$ is given by the vectorial product $\mathbf{n} \times \boldsymbol{\phi}$. These vectors are related to the cartesian coordinates through the expressions:

$$\mathbf{n} = \cos\gamma\cos\phi\mathbf{x} + \cos\gamma\sin\phi\mathbf{y} - \sin\gamma\mathbf{z} \tag{8.85}$$

$$\boldsymbol{\phi} = -\sin\phi\mathbf{x} + \cos\phi\mathbf{y} \tag{8.86}$$

$$\mathbf{t} = \sin\gamma\cos\phi\mathbf{x} + \sin\gamma\sin\phi\mathbf{y} + \cos\gamma\mathbf{z} \tag{8.87}$$

where γ is the angle formed by $\mathbf{t}$ and the z-axis (see Figure 8.9).

A BOR is defined by its generatrix. In the case being considered, the generatrix is defined by a set of ordered points along the t-coordinate (see Figure 8.10). The generatrix is approximated by the straight segments between these points for which the angle γ remains constant.

The field and current can be expressed in this system as

$$\mathbf{E}(\mathbf{r}) = \boldsymbol{\phi}E_{\phi}(\phi,t) + \mathbf{t}E_{t}(\phi,t) \tag{8.88}$$

$$\mathbf{J}(\mathbf{r}) = \boldsymbol{\phi}J_{\phi}(\phi,t) + \mathbf{t}J_{t}(\phi,t) \tag{8.89}$$

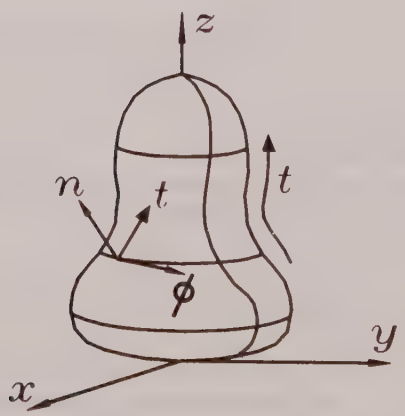


Figure 8.8 Coordinate system for the analysis.

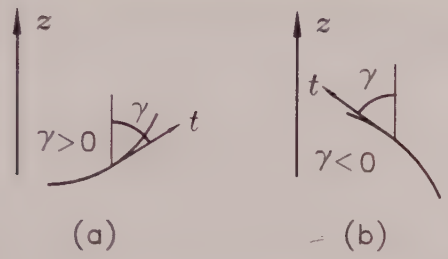


Figure 8.9 Definition of the angle γ: (a) if t is directed away from the z-axis, γ is positive; (b) γ is negative if t points towards the z-axis.

The free-space Green's function (8.7) is used in the development, with the relative distance between points given in cylindrical coordinates by

$$|\mathbf{r} - \mathbf{r}'|^2 = \rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi') + (z - z')^2 \tag{8.90}$$

In this coordinate system, the gradient, divergence, and surface differential element are given by

$$\nabla\Phi = \mathbf{t}\frac{\partial\Phi}{\partial t} + \boldsymbol{\phi}\frac{1}{\rho}\frac{\partial\Phi}{\partial\phi} \tag{8.91}$$

$$\nabla\mathbf{A} = \frac{1}{\rho}\frac{\partial}{\partial t}(\rho A_t) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi} \tag{8.92}$$

$$ds = \rho\, d\phi\, dt \tag{8.93}$$

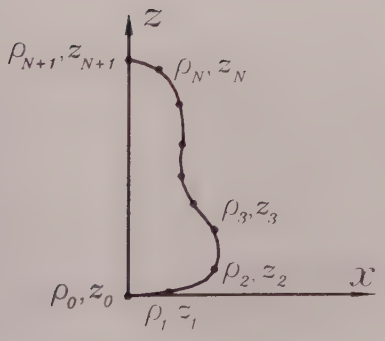


Figure 8.10 Definition of the generatrix.

Starting from (8.3) and taking into account Figure 8.11, we can write the EFIE in component form as

$$E_{\phi}^{\text{inc}} = \beta_{11}(J_{\phi}) + \beta_{12}(J_t) \tag{8.94}$$

$$E_{t}^{\text{inc}} = \beta_{21}(J_{\phi}) + \beta_{22}(J_t) \tag{8.95}$$

where

$$\begin{aligned}\beta_{11} = {} & k^c \iint_{S'} J_{\phi}(\mathbf{r}') \cos(\phi - \phi') G(\mathbf{r},\mathbf{r}')\, ds' \\ & + k^q \frac{1}{\rho}\frac{\partial}{\partial \phi} \iint_{S'} \frac{1}{\rho'} \frac{\partial J_{\phi}(\mathbf{r}')}{\partial \phi'} G(\mathbf{r},\mathbf{r}')\, ds'\end{aligned} \tag{8.96}$$

$$\begin{aligned}\beta_{12} = {} & -k^c \iint_{S'} J_t(\mathbf{r}') \sin\gamma' \sin(\phi - \phi') G(\mathbf{r},\mathbf{r}')\, ds' \\ & + k^q \frac{1}{\rho}\frac{\partial}{\partial \phi} \iint_{S'} \frac{1}{\rho'} \frac{\partial}{\partial \phi'} (\rho' J_t(\mathbf{r}')) G(\mathbf{r},\mathbf{r}')\, ds'\end{aligned} \tag{8.97}$$

$$\begin{aligned}\beta_{21} = {} & k^c \iint_{S'} J_{\phi}(\mathbf{r}') \sin\gamma \sin(\phi - \phi') G(\mathbf{r},\mathbf{r}')\, ds' \\ & + k^q \frac{\partial}{\partial t} \iint_{S'} \frac{1}{\rho'} \frac{\partial J_{\phi}(\mathbf{r}')}{\partial \phi'} G(\mathbf{r},\mathbf{r}')\, ds'\end{aligned} \tag{8.98}$$

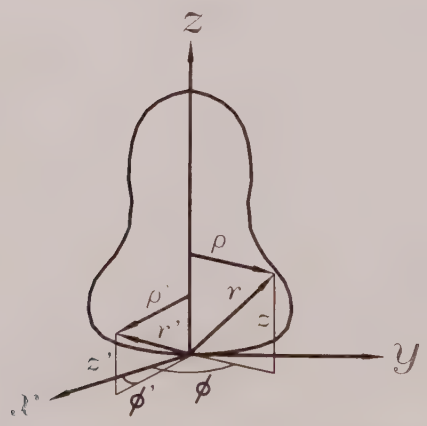


Figure 8.11 Points r and r' and related magnitudes.

$$\beta_{22} = -k^c \iint_{S'} J_t(\mathbf{r}')\,(\sin\gamma \sin\gamma' \cos(\phi - \phi') + \cos\gamma \cos\gamma')G(\mathbf{r},\mathbf{r}')\,ds' + k^q \frac{\partial}{\partial t} \iint_{S'} \frac{1}{\rho'} \frac{\partial}{\partial t'} (\rho' J_t(\mathbf{r}'))G(\mathbf{r},\mathbf{r}')\,ds' \tag{8.99}$$

The next step in the process is to define the points at which the sampling in the ϕ- and t-directions will be performed. Figure 8.12 shows both a 3D and a plane representation of these points (crosses for ϕ and circles for t) for a general structure. In the ϕ-direction, the samples must be equidistant, but not necessarily in the t-direction. This implies the loss of the convolutive character of the operators in the t-direction. The sampling points are defined as follows.

- Along ϕ (crosses):

$$\phi_\phi^i = i\Delta\phi \qquad i = 1, 2, \ldots, N_\phi + 1$$

$$t_\phi^j = \begin{cases} \rho_\phi^j = \dfrac{\rho_j + \rho_{j-1}}{2} \\ z_\phi^j = \dfrac{z_j + z_{j-1}}{2} \end{cases} \qquad j = 1, 2, \ldots, N_t + 1 \tag{8.100}$$

- Along t (circles):

$$\phi_t^i = (i - 0.5)\Delta\phi \quad i = 1, 2, \ldots, N_\phi + 1$$

$$t_t^j = \begin{cases} \rho_t^j = \rho_j \\ z_t^j = z_j \end{cases} \quad j = 1, 2, \ldots, N_t \tag{8.101}$$

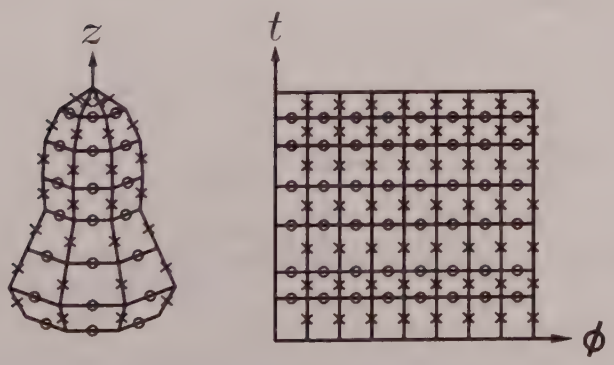


Figure 8.12 3D and ϕt-plane representations of the subdomain mesh and the centers of the basis functions (which coincide with the sampling points).

The current in the ϕ-direction is expanded in basis functions (Figure 8.13) in the following manner:

$$J_\phi(\phi,t) = \sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_t+1} J_\phi[i,j]\, T_{ij}^{\phi}(\phi,t) \tag{8.102}$$

with

$$T_j^{\phi}(\phi,t) = \begin{cases} 1 - \dfrac{|\phi|}{\Delta\phi} & |\phi| \le \Delta\phi,\ 0 \le t \le \Delta t_j \\ 0 & \text{otherwise} \end{cases} \tag{8.103}$$

$$T_{ij}^{\phi}(\phi,t) = T_j^{\phi}(\phi - \phi_\phi^i, t - t_\phi^j) \tag{8.104}$$

and $J_\phi[i,j]$ a discrete function.

To expand the t-component of the current, and in order to improve the computation of the radial divergence, a new function is introduced:

$$I_t(\phi,t) = 2\pi\rho J_t(\phi,t) \tag{8.105}$$

This function is expanded in terms of basis functions,

$$I_t(\phi,t) = \sum_{i=1}^{N_\phi+1} \sum_{j=1}^{N_t} I_t[i,j]\, T_{ij}^{t}(\phi,t) \tag{8.106}$$

with the rooftops (Figure 8.14) defined as follows:

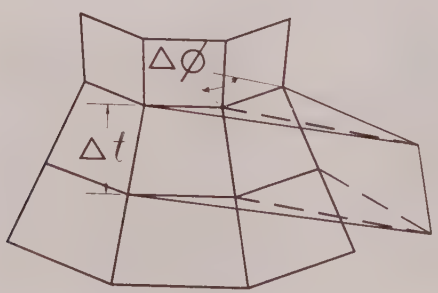


Figure 8.13 Basis function for the ϕ-component.

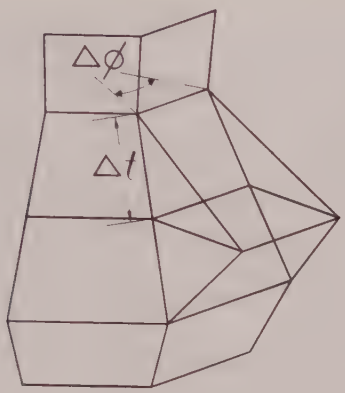


Figure 8.14 Basis function for the t-component.

$$T_j^t(\phi,t) = \begin{cases} 1 - \dfrac{t}{\Delta t_{j+1}} & |\phi| \le \Delta\phi/2,\ 0 \le t \le \Delta t_{j+1} \\ \dfrac{t}{\Delta t_j} & |\phi| \le \Delta\phi/2,\ 0 \le t \le \Delta t_j \\ 0 & \text{otherwise} \end{cases} \tag{8.107}$$

$$T_{ij}^t(\phi,t) = T_j^t(\phi - \phi_t^i, t - t_t^j) \tag{8.108}$$

Finally, the impressed electric field is tested using the modified razor blade functions depicted in Figure 8.15:

$$V_\phi^{iD}[i,j] = \int_{(i-0.5)\Delta\phi}^{(i+0.5)\Delta\phi} E_\phi^{\text{inc}}(\phi,t_\phi^j)\,\rho\,\mathrm{d}\phi \tag{8.109}$$

$$V_t^{iD}[i,j] = \int_{\Delta t_j/2}^{\Delta t_j} E_t^{\text{inc}}(\phi_t^i,t)\,\mathrm{d}t + \int_0^{\Delta t_{j+1}/2} E_t^{\text{inc}}(\phi_t^i,t)\,\mathrm{d}t \tag{8.110}$$

Once again considering the impressed field to be constant over the testing function domain, we obtain the potentials

$$V_\phi^{iD}[i,j] = \frac{\rho_j + \rho_{j-1}}{2}\,\Delta\phi E_\phi^{\text{inc}}(\phi_\phi^i,t_\phi^j) \tag{8.111}$$

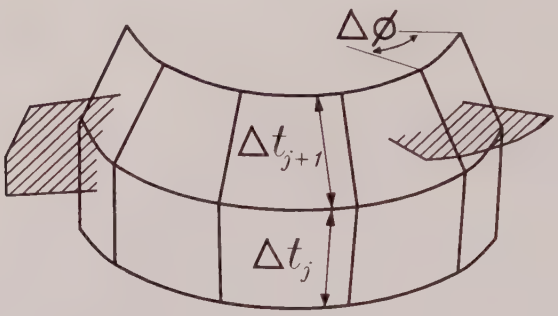


Figure 8.15 Testing functions in both the t- and ϕ-directions.

$$V_t^{iD}[i,j] = \frac{\Delta t_j + \Delta t_{j+1}}{2} E_t^{\text{inc}}(\phi_t^i, t_t^j) \tag{8.112}$$

8.4.1 β_{11} Term

This term can, as usual, be decomposed into current- and charge-related terms,

$$\beta_{11} = \beta_{11J} + \beta_{11C} \tag{8.113}$$

which will be analyzed independently.

8.4.1.1 β_{11J} Term

This term is given by

$$\beta_{11J}(\phi,t) = k^c \iint_{S'} J_\phi(\phi',t') \cos(\phi - \phi') G(\phi - \phi', t, t')\, ds' \tag{8.114}$$

Since it is similar to the cylindrical case, we can express this term, after testing, in the transform domain as

$$\tilde{V}_{11J}[m,j] = \frac{\rho_j + \rho_{j-1}}{2} \Delta\phi k^c \sum_{j'=1}^{N_t+1} \tilde{J}_\phi[m,j'] \tilde{G}_{11}[m,j,j'] \tag{8.115}$$

$\tilde{G}_{11}[m,j,j']$ is the Fourier transform with respect to the variable i of

$$G_{11}[i - i',j,j'] = \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_0^{\Delta t_{j'}} G_C(\phi_\phi^i - \phi',t_\phi^j,t')\,ds' \tag{8.116}$$

where

$$G_C(\phi - \phi',t,t') = \cos(\phi - \phi')G(\phi - \phi',t,t') \tag{8.117}$$

8.4.1.2 β_{11C} Term

This term corresponds to

$$\beta_{11C}(\phi,t) = k^q \frac{1}{\rho}\frac{\partial}{\partial\phi}\iint_{S'}\frac{1}{\rho'}\frac{\partial J_\phi(\phi',t')}{\partial\phi'}G(\phi - \phi',t,t')\,ds' \tag{8.118}$$

which again is similar to the cylindrical case. Proceeding in a similar way, we have

$$\tilde{V}_{11C}[m,j] = \frac{k^q}{\Delta\phi}(F_\phi - 1)(1 - F_\phi^*)\sum_{j'=1}^{N_t+1}\tilde{J}_\phi[m,j']\tilde{G}_d[m,j,j'] \tag{8.119}$$

where $\tilde{G}_d$ is the Fourier transform along i of the function:

$$G_d[i - i',j,j'] = \int_{(i'-1)\Delta\phi}^{i'\Delta\phi}\int_0^{\Delta t_{j'}} G(\phi_t^i - \phi',t_\phi^j,t')\,dt'\,d\phi' \tag{8.120}$$

and

$$F_\phi[m] = \exp[j2\pi m/(N_\phi + 1)] \tag{8.121}$$

with * denoting the complex conjugate.

8.4.2 β_{12} Term

This term can also be expressed as the sum of its current and charge parts:

$$\beta_{12} = \beta_{12J} + \beta_{12C} \tag{8.122}$$

which will be analyzed separately.

8.4.2.1 β_{12J} Term

This term is given by

$$\beta_{12J}(\phi,t) = k^c \iint_{S'} J_t(\phi',t') \sin\gamma' \sin(\phi - \phi') G(\phi - \phi',t,t')\, ds' \quad (8.123)$$

If J_t is replaced by its expression in (8.105) and its approximate value (8.106), then (8.123) becomes

$$\beta_{12J}(\phi,t) = -\frac{k^c}{2\pi} \iint_{S'} \left[\sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t} I_t[i',j'] T^t_{i'j'}(\phi',t') \right] \sin\gamma' G_S(\phi - \phi',t,t') \frac{ds'}{\rho'} \quad (8.124)$$

with

$$G_S(\phi - \phi',t,t') = \sin(\phi - \phi') G(\phi - \phi',t,t') \quad (8.125)$$

Interchanging sums and integrals and approximating the rooftops by pulses, as in the previous section, we obtain the following expression for the transform of the operator:

$$\tilde{V}_{12J}[m,j] = -\frac{k^c}{2\pi} \frac{\rho_j + \rho_{j+1}}{2} \Delta\phi \sum_{j'=1}^{N_t} \tilde{I}_t[m,j'] \tilde{G}_{12}[m,j,j'] \quad (8.126)$$

where

$$G_{12}[i - i',j,j'] = \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{0.5\Delta t_{j'}}^{\Delta t_{j'}} \sin\gamma'_j G_S(\phi^i_\phi - \phi', t^j_\phi, t') \frac{ds'}{\rho'} + \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{0}^{0.5\Delta t_{j'+1}} \sin\gamma'_{j+1} G_S(\phi^i_\phi - \phi', t^j_\phi, t') \frac{ds'}{\rho'} \quad (8.127)$$

8.4.2.2 β_{12C} Term

This term corresponds to

$$\beta_{12C}(\phi,t) = k^q \frac{1}{\rho} \frac{\partial}{\partial\phi} \iint_{S'} \frac{1}{\rho'} \frac{\partial}{\partial t'} [\rho' J_t(\phi',t')] G(\phi - \phi',t,t')\, ds' \quad (8.128)$$

Substituting in I_t and simplifying, we have

$$\beta_{12C}(\phi,t) = k^q \frac{1}{2\pi\rho} \frac{\partial}{\partial\phi} \iint_{S'} \frac{\partial I_t(\phi',t')}{\partial t'} G(\phi - \phi',t,t') \frac{ds'}{\rho'} \tag{8.129}$$

The following function is introduced:

$$C_t(\phi,t) = \frac{\partial I_t(\phi,t)}{\partial t} \tag{8.130}$$

which, using the derivative approach, becomes

$$C_t(\phi',t') = \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t+1} C_t^D[i',j'] P_{i'j'}(\phi',t') \tag{8.131}$$

where

$$C_t^D[i',j'] = \frac{I_t[i',j'] - I_t[i',j'-1]}{\Delta t_{j'}} \tag{8.132}$$

The domains of definition of the pulses in (8.131), C_t, are $D_{i'j'}$, whereas the domains of the rooftops of I_t are D_{ij}^t (see Figure 8.7 for similitude).

Introducing the potential,

$$\Phi_t(\phi,t) = \iint_{S'} \frac{\partial I_t(\phi',t')}{\partial t'} G(\phi - \phi',t,t')\, dt'\, d\phi' \tag{8.133}$$

and the approximation for the derivative of (8.131),

$$\Phi_t(\phi,t) = \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t+1} C_t^D[i',j'] \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_0^{\Delta t_{j'}} G(\phi - \phi',t,t')\, dt'\, d\phi' \tag{8.134}$$

we obtain the following expression for the discrete potential

$$\Phi_t[i,j] = \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t+1} C_t^D[i',j'] G_d[i - i',j,j'] \tag{8.135}$$

where G_d is defined in (8.120). The potential can also be expressed as

$$\Phi_t[i,j] = \sum_{j'=1}^{N_t+1} C_t^D\,[i',j'] \otimes G_d[i,j,j'] \tag{8.136}$$

Performing the testing of (8.129), we obtain

$$V_{12C}[i,j] = \frac{k^q}{2\pi} \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \frac{1}{\rho} \frac{\partial \Phi_t}{\partial \phi} \rho \, d\phi = \frac{k^q}{2\pi} \left[\Phi_t(\phi_t^{i+1}, t_\phi^j) - \Phi_t(\phi_t^i, t_\phi^j)\right] \tag{8.137}$$

which can be expressed as

$$V_{12C}[i,j] = \frac{k^q}{2\pi} \left[\Phi_t[i+1,j] - \Phi_t[i,j]\right] \tag{8.138}$$

Performing the Fourier transform on the variable i in the previous expression, we obtain

$$\tilde{V}_{12C}[m,j] = \frac{k^q}{2\pi} \left[F_\phi[m] - 1\right] \tilde{\Phi}_t[m,j] \tag{8.139}$$

where

$$\tilde{\Phi}_t[m,j] = \sum_{j'=1}^{N_t+1} \tilde{C}_t^D\,[m,j']\tilde{G}_d[m,j,j'] \tag{8.140}$$

and

$$\tilde{C}_t^D[m,j'] = \frac{\tilde{I}_t[m,j'] - \tilde{I}_t[m,j'-1]}{\Delta t_{j'}} \tag{8.141}$$

Thus, the final expression for the discrete operator is

$$\tilde{V}_{12C}[m,j] = \frac{k^q}{2\pi} \left[F_\phi[m] - 1\right] \sum_{j'=1}^{N_t+1} \left[\frac{\tilde{I}_t[m,j'] - \tilde{I}_t[m,j'-1]}{\Delta t_{j'}}\right] \tilde{G}_d[m,j,j'] \tag{8.142}$$

8.4.3 β_{21} Term

Once again, we write the term as the sum of its current and charge parts:

$$\beta_{21} = \beta_{21J} + \beta_{21C} \tag{8.143}$$

8.4.3.1 β_{21J} Term

This part is given by

$$\beta_{21J}(\phi,t) = k^c \iint_{S'} J_\phi(\phi',t') \sin\gamma \sin(\phi - \phi') G(\phi - \phi',t,t')\, ds' \quad (8.144)$$

Replacing the current by its approximate value and approximating the integral, we have

$$\beta_{21J}(\phi,t) = k^c \sin\gamma \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t+1} J_\phi[i',j'] \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_0^{\Delta t_{j'}} G_S(\phi - \phi',t,t')\, ds' \quad (8.145)$$

where

$$G_S(\phi - \phi',t,t') = \sin(\phi - \phi') G(\phi - \phi',t,t') \quad (8.146)$$

After testing and performing the Fourier transform in the ϕ-direction, we obtain the following expression for the discrete operator:

$$\tilde{V}_{21J}[m,j] = \frac{\Delta t_j + \Delta t_{j+1}}{2} k^c \sin\gamma_j' \sum_{j'=1}^{N_t+1} \tilde{J}_\phi[m,j'] \tilde{G}_{21}[m,j,j'] \quad (8.147)$$

with

$$\gamma_j' = \frac{\Delta t_j \gamma_j + \Delta t_{j+1}\gamma_{j+1}}{\Delta t_j + \Delta t_{j+1}} \quad (8.148)$$

and

$$G_{21}[i - i',j,j'] = \int_{(i'-0.5)\Delta\phi}^{(i'+0.5)\Delta\phi} \int_0^{\Delta t_{j'}} G_S(\phi_i^i - \phi',t_t^j,t')\, ds' \quad (8.149)$$

8.4.3.2 β_{21C} Term

This term is given by

$$\beta_{21C}(\phi,t) = k^q \frac{\partial}{\partial t} \iint_{S'} \frac{1}{\rho'} \frac{\partial J_\phi}{\partial \phi'} G(\phi - \phi',t,t')\, ds' \quad (8.150)$$

The process is similar to that used for β_{12C}, so that we can write

$$\beta_{21C}(\phi,t) = k^q \frac{\partial}{\partial t}[\Phi_\phi(\phi,t)] \tag{8.151}$$

where the potential Φ_ϕ, once the derivative has been approximated, is given by

$$\Phi_\phi(\phi,t) = \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t+1} C_\phi^D[i',j'] \iint_{D_{i'j'}} G(\phi - \phi',t,t')\, dt'\, d\phi' \tag{8.152}$$

After testing along t in (8.150), we have

$$V_{21C}[i,j] = k^q [\Phi_\phi(\phi_t^i, t_\phi^{j+1}) - \Phi_\phi(\phi_t^i, t_\phi^j)] \tag{8.153}$$

This last expression can be written in discrete form after performing the Fourier transform in the ϕ-direction as follows:

$$\tilde{V}_{21C}[m,j] = \frac{k^q}{\Delta\phi}[1 - F_\phi^*[m]] \sum_{j'=1}^{N_t+1} \tilde{J}_\phi[m,j'][\tilde{G}_d[m,j+1,j'] - \tilde{G}_d[m,j,j']] \tag{8.154}$$

8.4.4 β_{22} Term

This term can be written as

$$\beta_{22} = \beta_{22J} + \beta_{22C} \tag{8.155}$$

8.4.4.1 β_{22J} Term

For this term we can write

$$\begin{aligned}\beta_{22J}(\phi,t) = k^c \Bigg[&\sin\gamma \iint_{S'} J_t(\phi',t') \sin\gamma' \cos(\phi - \phi') G(\phi - \phi',t,t')\, ds' \\ &+ \cos\gamma \iint_{S'} J_t(\phi',t') \cos\gamma'\, G(\phi - \phi',t,t')\, ds' \Bigg]\end{aligned} \tag{8.156}$$

Using β_{22JS} and β_{22JC} to denote the first and second terms on the right-hand side of (8.156) and following an approach similar to those used in previous sections (i.e., replacing J_t with the approximate expressions and approximating the integrals), we can write

$$\beta_{22JS}(\phi,t) = \frac{k^c}{2\pi} \sin\gamma \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t} I_t[i',j'] \iint_{D^t_{i'j'}} \sin\gamma' \, G_C(\phi - \phi',t,t') \, dt' \, d\phi' \quad (8.157)$$

$$\beta_{22JC}(\phi,t) = \frac{k^c}{2\pi} \cos\gamma \sum_{i'=1}^{N_\phi+1} \sum_{j'=1}^{N_t} I_t[i',j'] \cos\gamma' \iint_{D^t_{i'j'}} G(\phi - \phi',t,t') \, dt' \, d\phi' \quad (8.158)$$

After discretizing both terms and performing the Fourier transform in the ϕ-direction, we obtain the following discrete operator:

$$\tilde{V}_{22J}[m,j] = \frac{k^c}{2\pi} \frac{\Delta t_j + \Delta t_{j+1}}{2} \sum_{j'=1}^{N_t} \tilde{I}_t[m,j'] \left[\sin\gamma'_j \, \tilde{G}_{22}[m,j,j'] + \cos\gamma'_j \, \tilde{G}_{05}[m,j,j']\right] \quad (8.159)$$

where

$$G_{22}[i - i',j,j'] = \sin\gamma'_{j+1} \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_0^{0.5\Delta t_{j'+1}} G_C(\phi^i_t - \phi',t^j_t,t') \, dt' \, d\phi' + \sin\gamma'_j \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{0.5\Delta t_{j'}}^{\Delta t_{j'}} G_C(\phi^i_t - \phi',t^j_t,t') \, dt' \, d\phi' \quad (8.160)$$

$$G_{05}[i - i',j,j'] = \cos\gamma'_{j+1} \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_0^{0.5\Delta t_{j'+1}} G(\phi^i_t - \phi',t^j_t,t') \, dt' \, d\phi' + \cos\gamma'_j \int_{(i'-1)\Delta\phi}^{i'\Delta\phi} \int_{0.5\Delta t_{j'}}^{\Delta t_{j'}} G(\phi^i_t - \phi',t^j_t,t') \, dt' \, d\phi' \quad (8.161)$$

8.4.4.2 β_{22C} Term

This term is given by

$$\beta_{22C}(\phi,t) = k^q \frac{\partial}{\partial t} \iint_{S'} \frac{1}{\rho'} \frac{\partial}{\partial t'} [\rho' J_t] G(\phi - \phi',t,t') \, ds' \quad (8.162)$$

Substituting in (8.162) the current approximations and introducing the potential Φ_t, we have

$$\beta_{22C}(\phi,t) = \frac{k^q}{2\pi} \frac{\partial}{\partial t} [\Phi_t(\phi,t)] \quad (8.163)$$

When this is discretized and expressed in the transform domain, it becomes

$$\tilde{V}_{22C}[m,j] = \frac{k^q}{2\pi}\left[\tilde{\Phi}_t[m,j+1] - \tilde{\Phi}_t[m,j]\right] \tag{8.164}$$

so that the final expression is

$$\tilde{V}_{22C}[m,j] = \frac{k^q}{2\pi}\sum_{j'=1}^{N_t+1}\frac{\tilde{I}_t[m,j'] - \tilde{I}_t[m,j'-1]}{\Delta t_{j'}}\left(\tilde{G}_d[m,j+1,j'] - \tilde{G}_d[m,j,j']\right) \tag{8.165}$$

8.4.5 Final Expressions for the Operators

The final operator expression can be written in the following form:

$$\begin{bmatrix} V_\phi^{\text{inc}} \\ V_t^{\text{inc}} \end{bmatrix} = \text{FFT}^{-1}\left(\begin{bmatrix} \tilde{L}_{\phi\phi} & \tilde{L}_{\phi t} \\ \tilde{L}_{t\phi} & \tilde{L}_{tt} \end{bmatrix}\begin{bmatrix} \tilde{J}_\phi \\ \tilde{I}_t \end{bmatrix}\right) \tag{8.166}$$

where the symbol FFT indicates a fast Fourier transform applied to the ϕ-direction (the index i in the development).

The final expressions for the transform operators appearing in (8.166) can be summarized as

$$\tilde{J}_\phi[m,j] = \text{FFT}[J_\phi[i,j]] \tag{8.167}$$

$$\tilde{I}_t[m,j] = \text{FFT}[I_t[i,j]] \tag{8.168}$$

$$\tilde{L}_{\phi\phi}[m,j,j'] = \left[\frac{\rho_j + \rho_{j-1}}{2}\Delta\phi k^c \tilde{G}_{11}[m,j,j'] + \frac{k^q}{\Delta\phi}(F_\phi - 1)(1 - F_\phi^*)\tilde{G}_d[m,j,j']\right] \tag{8.169}$$

$$\tilde{L}_{\phi t}[m,j,j'] = \left[-\frac{k^c}{2\pi}\frac{\rho_j + \rho_{j-1}}{2}\Delta\phi\tilde{G}_{12}[m,j,j'] + \frac{k^q}{2\pi}(F_\phi - 1)\left(\frac{\tilde{G}_d[m,j,j']}{\Delta t_{j'}} - \frac{\tilde{G}_d[m,j,j'+1]}{\Delta t_{j'+1}}\right)\right] \tag{8.170}$$

$$\tilde{L}_{t\phi}[m,j,j'] = \left[\frac{\Delta t_j + \Delta t_{j+1}}{2} k^c \sin \gamma_j^t \tilde{G}_{21}[m,j,j'] + \frac{k^q}{\Delta\phi}(1 - F_\phi^*)(\tilde{G}_d[m,j+1,j'] - \tilde{G}_d[m,j,j']) \right] \tag{8.171}$$

$$\tilde{L}_{\pi}[m,j,j'] = \frac{k^c}{2\pi} \frac{\Delta t_j + \Delta t_{j+1}}{2} (\sin \gamma_j^t \tilde{G}_{22}[m,j,j'] + \cos \gamma_j^t \tilde{G}_{05}[m,j,j']) + \frac{k^q}{2\pi} \left(\frac{\tilde{G}_d[m,j+1,j'] - \tilde{G}_d[m,j,j']}{\Delta t_{j'}} - \frac{\tilde{G}_d[m,j+1,j'+1] - \tilde{G}_d[m,j,j'+1]}{\Delta t_{j'+1}} \right) \tag{8.172}$$

where the tilde denotes transform quantities and the asterisk indicates the complex conjugate.

The adjoint operator used in the CG algorithm is obtained from the elements of $\tilde{L}_{\alpha\beta}$ by taking $(\tilde{L}_{\beta\alpha})^*$ (the transpose and complex conjugate) and interchanging the indexes j and j'.

8.5 RESULTS

In this final section of the chapter, we present some results obtained with both BOR formulations: the one for cylindrical surfaces and the more general formulation applicable to arbitrary BOR surfaces. Results of current distributions, radiation patterns, and RCS are shown. These results are compared with those obtained by other authors with the MM, when available, and with experimental data for the rest.

The first results presented correspond to the magnitude (Figure 8.16) and phase (Figure 8.17) of the t and ϕ current components along ϕ = 0-deg and ϕ = 90-deg cuts of a hollow open-ended cylinder with radius $a = \lambda/2\pi$ and length λ for an axially incident plane wave. The number of samples used is 16 × 16 so that the generatrix is defined by 17 points with a distance between them of 0.0625λ. It can be seen that the agreement with the results obtained by Davis and Mittra in [9] is very good. The 90-deg phase difference in the t-component is due to a different phase reference.

Figure 8.18 shows the modulus of the t and ϕ current components along ϕ = 0-deg and ϕ = 90-deg cuts, respectively, for a conesphere with a conic angle α = 20 deg and a hemisphere radius $a = 0.2\lambda$ (see sketch in the figure). The generatrix has been modeled by means of 17 points separated by a distance of 0.09λ in the conic part and 0.078λ in the hemisphere. The field incidence is axial and x-polarized.

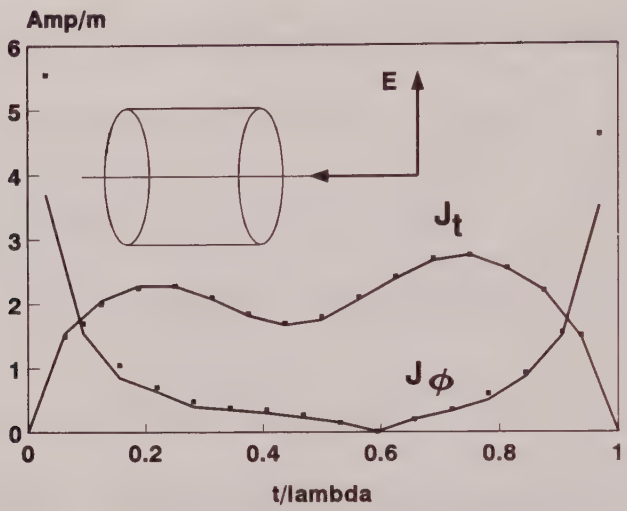


Figure 8.16 Current magnitude for the one-wavelength-long hollow open-ended cylinder, $ka = 1$. The axial incident field is x-polarized as shown in the sketch of Figure 8.17 and has an amplitude of 120π. The z-component is along a $\phi = 0$-deg cut, while the ϕ-component is along a $\phi = 90$-deg cut. Continuous lines correspond to results from [9] and crosses and asterisks to CG-FFT results.

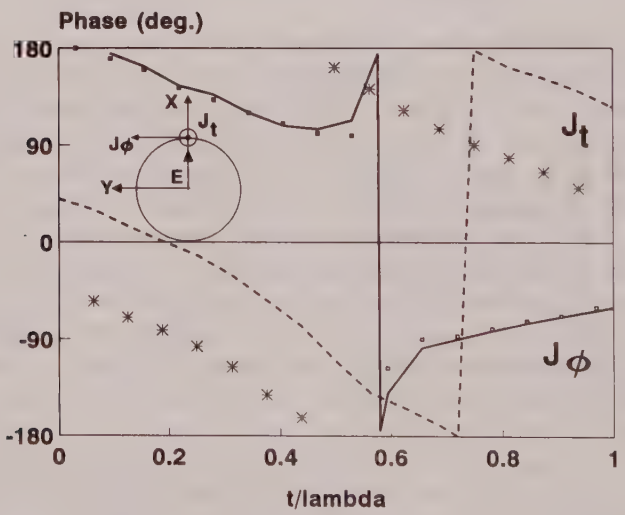


Figure 8.17 Current phase for the problem defined in Figure 8.16.

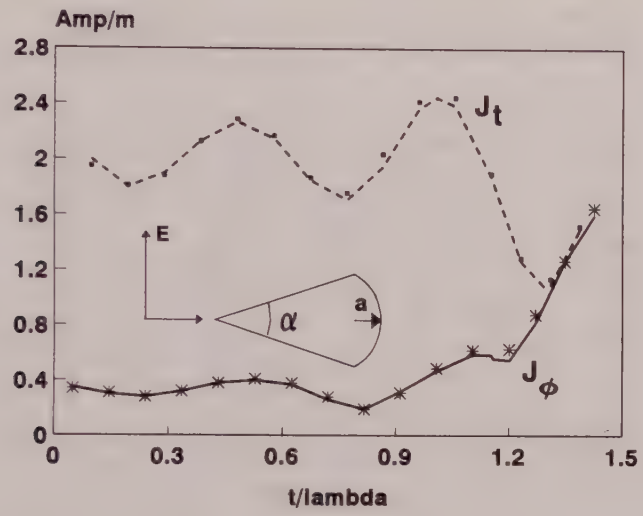


Figure 8.18 Modulus of ϕ and t current components for the conesphere described in the text.

Our results are compared with those of Glisson and Wilton [3]. The solid lines correspond to the results from [3], and the asterisks to those of the BOR formulation presented here.

Since many applications of BOR formulations lie in the field of RCS computation, we present here the analysis of two examples of this type. The first one considers the backscattering from the ogive shown in Figure 8.19. The body is formed by the revolution of a circumference arc forming an angle of α deg with the revolution axis. The backscattering is presented as a function of the ratio between the circumference diameter and the wavelength for an angle α of 60 deg. In all cases, the distance taken between samples is less than 0.1λ. The results are compared with experimental data from [10].

The second example considers the monostatic RCS for both the θ-θ and ϕ-ϕ polarizations (Figures 8.20 and 8.21, respectively) for the hollow, open-ended, truncated cylinder shown in the figures. The cylinder radius is $a = 0.216\lambda$ and its maximum height is $h = 1.216\lambda$. The cylinder is cut by a plane that forms an angle of 45 deg with the base. The generatrix has been modeled by taking a sample every 0.0784λ up to a height of 0.784λ (i.e., to the top of the complete cylindrical surface). From here on up, the samples are taken at intervals of 0.03927λ in order to achieve a better model in the cut region (i.e., incomplete cylindrical surface). The results are compared with experimental data from Putnam and Medgyesi [7]. Once again,

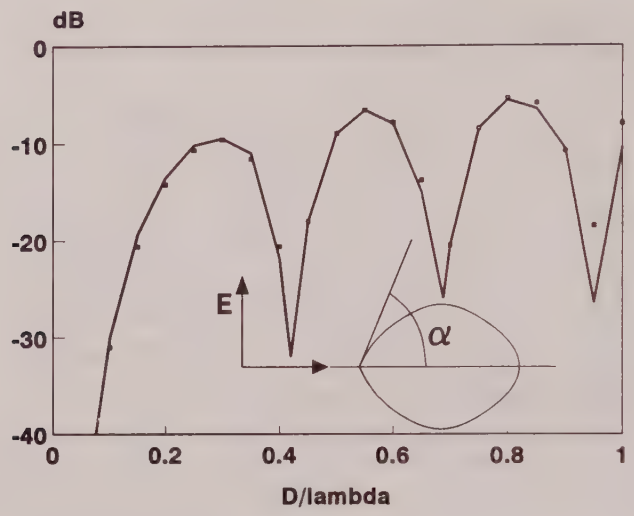


Figure 8.19 Backscattering for an ogive with axial incidence. Backscattering is represented as a function of the ratio between the diameter (D) and the wavelength (lambda).

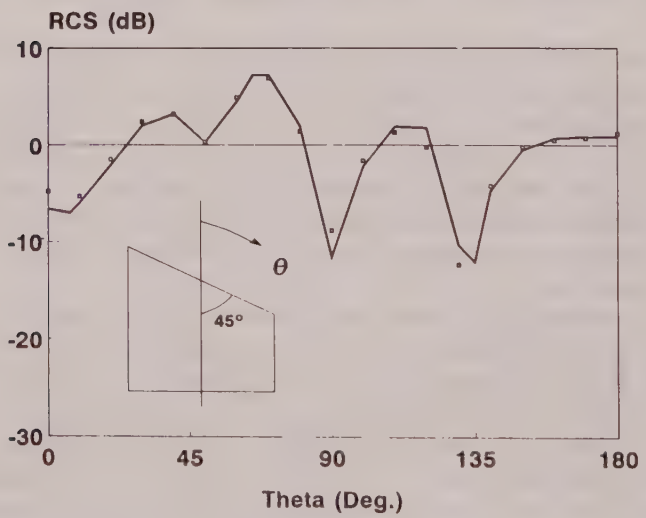


Figure 8.20 RCS (θ,θ) for a hollow, open-ended, truncated cylinder.

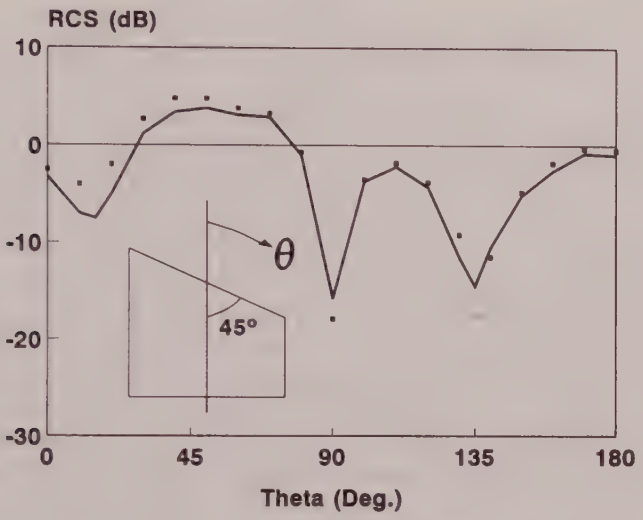


Figure 8.21 RCS (ϕ,ϕ) for a hollow, open-ended, truncated cylinder.

there is good agreement between the results obtained with this method and the experimental data.

Finally, we will present some results for the radiation patterns of antennas with symmetry of revolution. In particular, we will concentrate on antennas fed by cylindrical waveguides of the type shown in Figure 8.22.

Two perpendicular dipoles located in the xy-plane at an arbitrary position z are used to simulate the feeding of this antenna (see Figure 8.23). The second dipole can be multiplied by a factor (a $\exp(-j\phi)$) if some particular polarization is of interest. We are interested in circular polarization, so the values of the factor will be $a = 1$, $\phi = \pi/2$.

With this pair of dipoles, we can obtain the following expressions for the impressed fields in the antenna walls by using superposition and the general expressions for the fields produced by an infinitesimal dipole placed at the origin and oriented along the z-axis

$$E_\theta = -j\frac{Z_0 Idl}{4\pi k_0}\left(\frac{1}{r^3} + \frac{jk_0}{r^2} - \frac{k_0^2}{r}\right)\sin\theta\exp(-jk_0 r) \tag{8.173a}$$

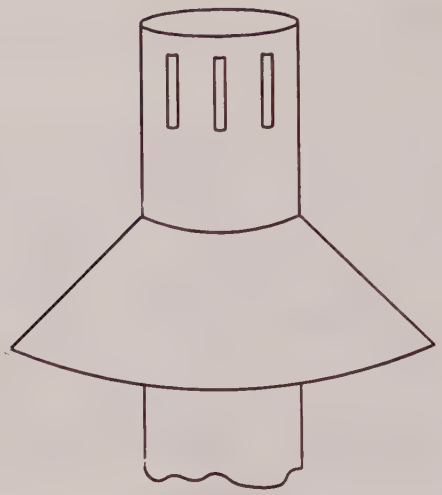


Figure 8.22 Basic structure of a telemetry, tracking, and control (TTC) antenna fed by a cylindrical waveguide.

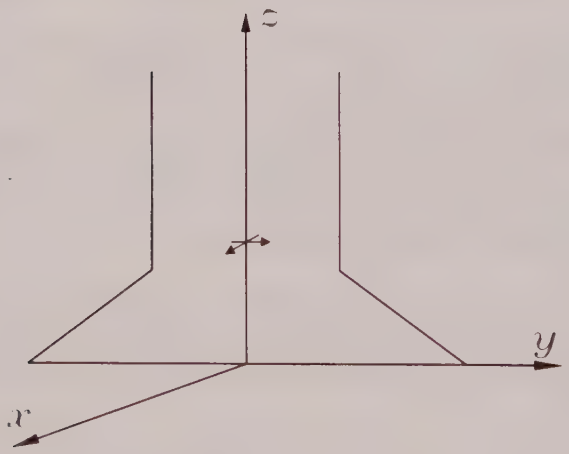


Figure 8.23 Location of the dipoles used to feed the antenna.

$$E_r = j\frac{Z_0 Idl}{2\pi k_0}\left(-\frac{1}{r^3}-\frac{jk_0}{r^2}\right)\cos\theta\exp(-jk_0 r) \qquad (8.173b)$$

bearing in mind that the fields in (8.173) must be expressed in the BOR coordinate system. From these impressed fields, we can obtain the scattering fields that, together with the direct fields produced by the dipoles, constitute the radiated field.

As an illustrative example, the radiation patterns of the structure shown in Figure 8.24 have been analyzed at a frequency of 10.7 GHz. The structure consists basically of a cylindrical waveguide 22 mm in diameter with eight equidistant apertures 1.08 mm wide by 5 mm long, ending in a section of waveguide that spreads out at an angle of approximately 85 deg with eight 188-mm-long apertures that are shifted 22.5 deg with respect to the first set. All dimensions are shown in Figure 8.24. The number of samples used in the ϕ-direction was 63, and 40 segments were used in the t-direction. Figure 8.25 shows the radiation patterns for this antenna. Copolar (dashed line), cross-polar (dotted line), and total (continuous line) patterns are plotted as a function of θ for the $\phi = 0$ cut.

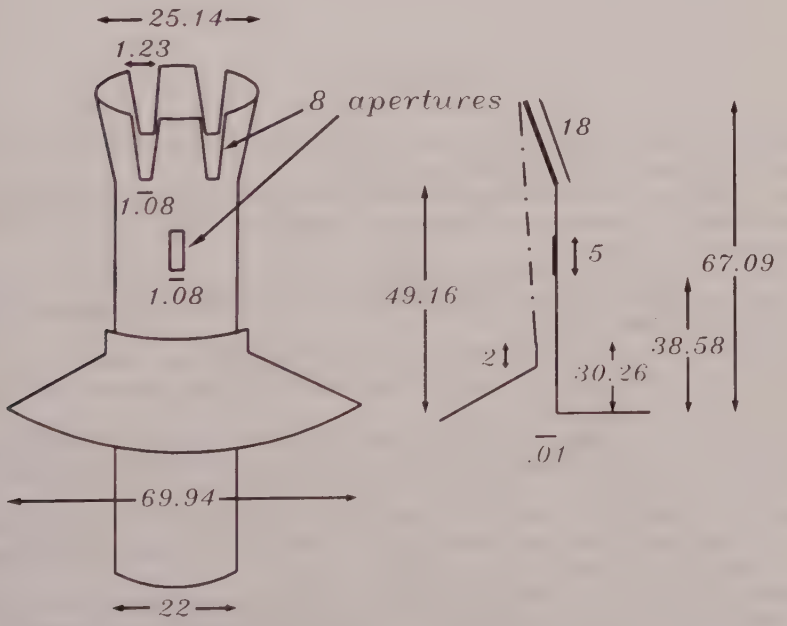


Figure 8.24 Sketch showing dimensions (in millimeters) of the TTC antenna analyzed.

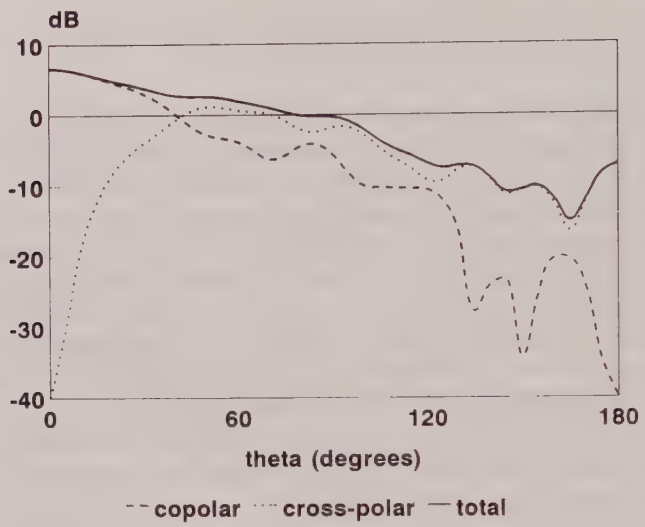


Figure 8.25 Radiation patterns of the antenna in Figure 8.24 for the $\phi = 0$ cut.

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Chapter 9
Flat Periodic Structures

9.1 INTRODUCTION

Periodic structures have been a focus of study in a number of disciplines, including optics, crystallography, solid-state physics, and quantum mechanics, among others. This widespread interest is due to the special interaction between these structures and all kinds of waves. In electromagnetism and optics, the study of periodic structures has been important since the early days. In fact, the study of diffraction by gratings was fundamental to the understanding and acceptance of the wave nature of light. At present, diffraction gratings and related structures are used in many applications at optical and infrared frequencies, such as beam splitting, shaping and sampling, coherent light generation, convolution, correlation, and processing of optical signals, holographic optical elements, and so on [1].

At millimeter and microwave frequencies, there are also many devices that are based on periodic structures: antenna radomes and polarizers, spatial filters and diplexers, lens antennas, artificial dielectrics, dichroic subreflectors, and so on [2,3]. A common characteristic of all these devices is that they are built using periodic screens (i.e., 2D periodic structures). There are two kinds of periodic screens. One kind, which consists of a periodic arrangement of metallic patches, has a low-pass frequency response. Screens of this type are usually called capacitive frequency-selective surfaces (FSS). The other kind consists of a metallic screen with holes that are periodically distributed. Screens of this type present a dual-frequency behavior and are called inductive FSSs. Both kinds can be either isolated structures or sandwiched between dielectric slabs. All such structures are spatial filters, since they have passband and stopband characteristics in the frequency spectrum. It is because of this important property that the term frequency-selective surface is commonly used. It should be noted that these periodic structures are also selective with respect to the polarization and incident direction of electromagnetic waves. In light of this fact, we will refer to all of the above-mentioned screens as flat periodic structures (FPS).

When a plane wave illuminates an FPS, the scattered field can be described in terms of a discrete spectrum of plane waves. If the incident field is not a plane wave, the response can be analyzed by using the plane wave spectrum representation of the incident field. The directions and wave numbers of the scattered spectrum are determined by the periodicity of the FPS, but the relative power of each mode depends on the geometrical and electrical characteristics of the structure. When the incident wavelength is close to the size of the cells of the FPS, the scattered modes depend strongly on these characteristics. Therefore, a full-wave analysis must be carried out by solving a boundary value problem.

Solutions to the boundary value problem have been found by several authors who have considered the problem from different points of view. The mode-matching technique has been used to analyze scattering from conducting screens perforated periodically [4–7]. The problem of scattering from a periodic array of conducting patches has been solved by using the MM [8–10]. The major problem with these techniques is that they use basis functions that are specific to the geometry under analysis. Therefore, the number of geometries that can be analyzed is limited to a few canonical forms: patches or holes having the form of rectangles, circles, or crosses. Solutions can be obtained for arbitrary geometries by using subdomain basis functions in conjunction with the MM [11]. The problem with this method is that the number of unknowns grows very large and the method becomes inefficient.

A number of alternatives that use iterative techniques to solve the integral equations have also been proposed [12–18]. In this chapter we develop a scheme for calculating the scattered field from isolated (free-standing) FPSs, which is based on the CG-FFT method. The integral equation is discretized in the spatial domain for the most part, although the discretization procedure is completed in the spectral domain. FPSs embedded in dielectric multilayered media will be dealt with in the next chapter. In this case, we will present a different scheme that uses, from the beginning, a spectral domain expression for the integral equation.

9.2 DIRECT AND RECIPROCAL LATTICES

The general idea of a periodic structure is a very intuitive notion with which most readers will be familiar. Nevertheless, we will present some concepts that may make it easier to understand the electromagnetic properties of periodic structures.

A FPS is shown in Figure 9.1. The vectors $\mathbf{T}_1$ and $\mathbf{T}_2$ define a parallelogram that is often called a unit cell or generating cell, because the entire structure can be reproduced by translating the generating cell, which results in a periodic structure. The translation operation is defined as

$$\mathbf{T} = n\mathbf{T}_1 + m\mathbf{T}_2 \quad n,m = \pm 1, \pm 2, \ldots, \pm\infty \tag{9.1}$$

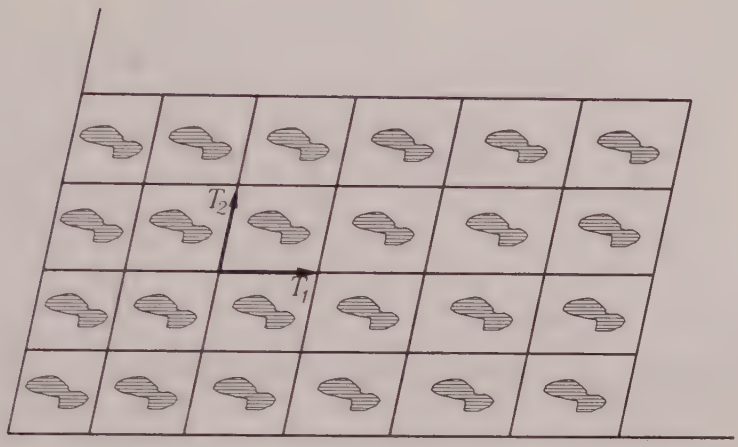


Figure 9.1 A flat periodic structure with arbitrary generating cell.

Consider a function $f(\mathbf{r})$ which represents any characteristic of the structure (e.g., geometrical, physical). If we know the value of this function at any point of the unit cell, a suitable definition of periodicity is provided by the following relationship:

$$f(\mathbf{r}') = f(\mathbf{r}), \quad \mathbf{r} \text{ on } S_0, \quad \text{if } \mathbf{r}' = \mathbf{r} + m\mathbf{T}_1 + n\mathbf{T}_2 \tag{9.2}$$

where S_0 represents the unit cell surface, and m,n are integers. From the preceding relationships, we can see that the periodic structure looks the same whether it is viewed from a point $\mathbf{r}$ in the unit cell or from a point $\mathbf{r}'$ as defined in (9.2).

The set of points

$$\mathbf{r}_{mn} = n\mathbf{T}_1 + m\mathbf{T}_2 \quad n,m = \pm 1, \pm 2, \ldots, \pm\infty \tag{9.3}$$

defines the direct or spatial lattice, which is a regular periodic arrangement of points in space. An infinite number of different FPSs can be associated with the same lattice, but, as we will see later, the lattice structure determines certain properties of the FPS independently of the particular characteristics of the unit cell. Figure 9.2 depicts the lattice associated with the FPS of Figure 9.1. Lattices have several interesting properties related to symmetry, which are described in any book dealing with crystallography. See, for instance, [19].

Associated with any spatial lattice, there is a periodic arrangement of points in the spectral domain, which is usually called the reciprocal lattice. As we will show, the reciprocal lattice allows us to obtain at first glance some important aspects of the scattered field. The periodicity of the reciprocal lattice, which is defined by vectors $\mathbf{K}_1$ and $\mathbf{K}_2$, is obtained from the following relationship:

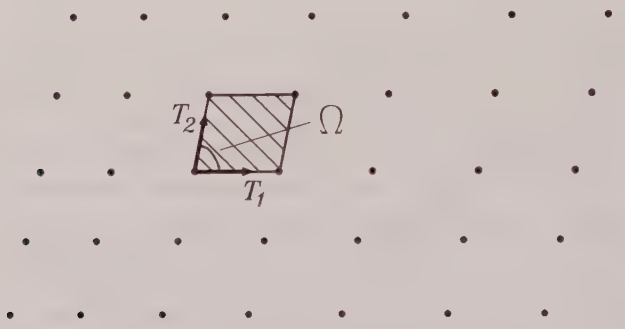


Figure 9.2 Spatial lattice associated with the FPS in Figure 9.1. The periodicity is clearly defined by vectors $\mathbf{T}_1$ and $\mathbf{T}_2$.

$$\mathbf{T}_i \cdot \mathbf{K}_i = 2\pi\delta_{ij} \tag{9.4}$$

where $\mathbf{T}_i$ are the periodicity vectors of the spatial lattice, and δ_{ij} is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if} \quad i = j \\ 0 & \text{if} \quad i \neq j \end{cases} \tag{9.5}$$

Figure 9.3 shows the reciprocal lattice corresponding to the spatial lattice of Figure 9.2. It can be observed that the lines of points of the reciprocal lattice are perpendicular to the lines in the spatial one.

As usual, we express the periodicity vectors in terms of cartesian components:

$$\mathbf{T}_i = (t_{ix}, t_{iy}) \tag{9.6}$$

$$\mathbf{K}_j = (k_{jx}, k_{jy}) \tag{9.7}$$

Now the periodicity of the spatial or direct lattice is defined by the matrix

$$[T] = \begin{bmatrix} t_{1x} & t_{2x} \\ t_{1y} & t_{2y} \end{bmatrix} \tag{9.8}$$

and the periodicity of the reciprocal lattice is also expressed in matrix form as follows:

$$[K] = \begin{bmatrix} k_{1x} & k_{2x} \\ k_{1y} & k_{2y} \end{bmatrix} \tag{9.9}$$

Taking (9.4) into account, it is easy to prove that

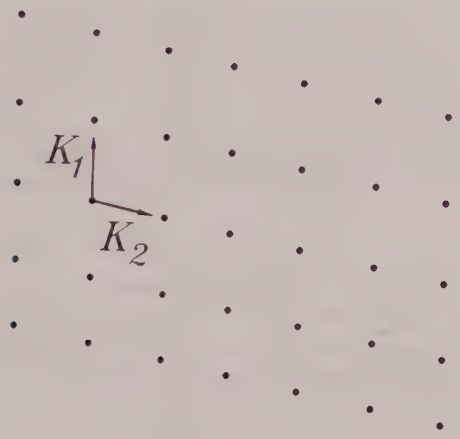


Figure 9.3 Reciprocal lattice corresponding to the spatial lattice in Figure 9.2. The periodicity of the lattice is defined by the vectors $\mathbf{K}_1$ and $\mathbf{K}_2$.

$$[T][K] = 2\pi[I] \tag{9.10}$$

where $[I]$ is the 2D unit matrix.

In the case of a spatial cartesian lattice, we have

$$\mathbf{T}_1 = T_x\mathbf{x}; \quad \mathbf{T}_2 = T_y\mathbf{y} \tag{9.11}$$

and

$$[T] = \begin{bmatrix} T_x & 0 \\ 0 & T_y \end{bmatrix} \tag{9.12}$$

From (9.10), we can obtain the periodicity matrix of the reciprocal lattice:

$$[K] = \begin{bmatrix} \dfrac{2\pi}{T_x} & 0 \\ 0 & \dfrac{2\pi}{T_y} \end{bmatrix} \tag{9.13}$$

By observing (9.13), we can see that the reciprocal lattice of a cartesian direct lattice is also cartesian. The set of points that forms this lattice can be obtained from the relationship

$$\mathbf{K}_{mn} = [K][m,n]^t = \frac{2\pi m}{T_x}\mathbf{x} + \frac{2\pi n}{T_y}\mathbf{y} \tag{9.14}$$

In the case of an arbitrary direct lattice, like the one in Figure 9.2, the cartesian components of the periodicity vectors can always be written as

$$\mathbf{T}_1 = a\mathbf{x} \tag{9.15a}$$

$$\mathbf{T}_2 = b\cos\Omega\mathbf{x} + b\sin\Omega\mathbf{y} \tag{9.15b}$$

The cartesian axes have been chosen so that $\mathbf{T}_1$ is oriented in the x-direction. Then the periodicity matrix has the following expression:

$$[T] = \begin{bmatrix} a & b\cos\Omega \\ 0 & b\sin\Omega \end{bmatrix} \tag{9.16}$$

The periodicity of the reciprocal lattice is obtained by using definition (9.10):

$$[K] = \begin{bmatrix} \dfrac{2\pi}{a} & 0 \\ -\dfrac{2\pi}{a}\operatorname{cotg}\Omega & \dfrac{2\pi}{b}\sin\Omega \end{bmatrix} \tag{9.17}$$

In conclusion, note that the points (m,n) in the reciprocal lattice associated with any 2D periodic structure can be obtained from the following relationship:

$$\mathbf{K}_{mn} = [U][m,n]^t = \frac{2\pi m}{a}\mathbf{x} + \left(\frac{2\pi n}{b\sin\Omega} - \frac{2\pi m}{a}\operatorname{cotg}\Omega\right)\mathbf{y} \tag{9.18}$$

9.3 FLOQUET'S THEOREM

Floquet's theorem is very important because it determines the general behavior of the scattered field of a flat periodic structure. By using this theorem, we can express the scattered field as a discrete spectrum of plane waves whose wave vectors depend only on the periodicity of the structure and the wave vector of the incident plane wave.

The original work of Floquet dealt with the problem of wave propagation into periodic continuous media solved by means of differential equations with periodic

coefficients [20]. In this chapter, we are interested in the case of periodic boundary conditions, which can be considered as an extension of Floquet's work. A very clear formulation of Floquet's theorem from this point of view was developed by Amitay, Galindo, and Wu [21].

We shall assume an FPS with cartesian periodicity (T_x,T_y) and a plane wave incident on the structure. This incident field has the form

$$\mathbf{E}^i(x,y,z) = \mathbf{E}_0^i \exp(-j(k_{0x}x + k_{0y}y + k_{0z}z)) \tag{9.19}$$

If the FPS is in the $z = 0$ plane of the reference system, as shown in Figure 9.4, and a reference cell is defined as

$$0 \le x \le T_x \tag{9.20a}$$

$$0 \le y \le T_y \tag{9.20b}$$

Then the excitation field over this cell is given by

$$\mathbf{E}^i(x,y,0) = \mathbf{E}_0^i \exp(-j(k_{0x}x + k_{0y}y)) \tag{9.21}$$

The excitation field over an arbitrary cell (m,n) of the periodic structure is related to the field in the reference cell by

$$\begin{aligned}\mathbf{E}^i(x',y',0) &= \mathbf{E}_0^i \exp(-j(k_{0x}(x + mT_x) + k_{0y}(y + nT_y))) \\ &= \mathbf{E}_0^i \exp(-j(k_{0x}x + k_{0y}y)) \exp(-j(k_{0x}mT_x + k_{0y}nT_y))\end{aligned} \tag{9.22}$$

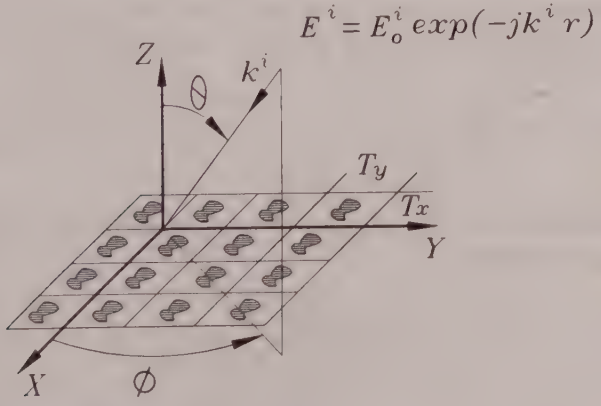


Figure 9.4 Arbitrary FPS with cartesian periodicity (T_x,T_y) in the $z = 0$ plane of the reference system. An incident plane wave is also represented.

where the (m,n) cell is defined by the relationships

$$mT_x \leq x' \leq (m+1)T_x \tag{9.23a}$$

$$nT_y \leq y' \leq (n+1)T_y \tag{9.23b}$$

A point (x',y') in this cell can be expressed as

$$x' = x + mT_x \tag{9.24a}$$

$$y' = y + nT_y \tag{9.24b}$$

where (x,y) is a point in the reference cell.

From (9.22) we can obtain

$$\mathbf{E}^i(x + mT_x, y + nT_y, 0) = \mathbf{E}^i(x,y,0) \exp(-j(k_{0x}mT_x + k_{0y}nT_y)) \tag{9.25}$$

This expression shows that the excitation field is periodic over the structure except for a phase factor. We say that the excitation field is also a quasiperiodic function.

The periodicity of the boundary conditions in our problem and the linearity of Maxwell's equations allow us to assume that the response to a quasiperiodic excitation is quasiperiodic. Therefore, we can write the scattered field over the structure as

$$\mathbf{E}^s(x + mT_x, y + nT_y, 0) = \mathbf{E}^s(x,y,0) \exp(-j(k_{0x}mT_x + k_{0y}nT_y)) \tag{9.26}$$

It has been shown [21] that the scattered field at any point in the space can be expressed in terms of a discrete plane wave spectrum as follows:

$$\mathbf{E}^s(x',y',z') = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \mathbf{E}^s(k_{mx},k_{ny}) \exp(j(k_{mx}x' + k_{ny}y' \pm k_{mnz}z')) \tag{9.27}$$

where the positive sign is for $z' > 0$ and the negative for $z' < 0$. The wave numbers k_{mx}, k_{ny}, and k_{mnz} are defined by

$$k_{mx} = \frac{2\pi m}{T_x} - k_{0x} \tag{9.28a}$$

$$k_{ny} = \frac{2\pi n}{T_y} - k_{0y} \tag{9.28b}$$

$$k_{mnz} = \pm \sqrt{\left(\frac{2\pi}{\lambda}\right)^2 - k_{mx}^2 - k_{ny}^2} \tag{9.28c}$$

and $\mathbf{E}^s(k_{mx},k_{ny})$ are the complex amplitudes of each plane wave in the spectrum. In (9.28c), the negative sign is for real values of the square root and the positive sign for imaginary values in order to obtain decreasing exponential functions that represent the evanescent modes. The fact that the spectrum is discrete is a consequence of the quasiperiodic nature of the excitation and scattered fields. In fact, the scattered field over the structure can be written from (9.27) as

$$\mathbf{E}^s(x',y',0) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \mathbf{E}^s(k_{mx},k_{ny}) \exp(j(k_{mx}x' + k_{ny}y')) \tag{9.29}$$

and (9.29) is the Fourier series expansion of a quasiperiodic function. The scattered field on the structure (9.24) could be obtained by generalizing the formulation of Section 3.4. This will be considered a formulation of the Floquet theorem from a signal processing point of view.

It is interesting to note that the discrete nature of the spectrum and the wave vectors of these plane waves depend only on the incident wave vector and the periodicity of the structure. On the other hand, the amplitude $\mathbf{E}^s(k_{mx},k_{ny})$ of each wave in the spectrum depends on the geometrical shape and electrical properties of the cells in the periodic structure.

One of the most important properties of the reciprocal lattice is that the wave vectors of the discrete spectrum scattered from the direct lattice structure are defined by the reciprocal lattice points. In fact, we can rewrite the wave numbers in (9.28) by substituting in the relationships of (9.14). Thus, Floquet's theorem can be generalized to any 2D periodic structure. The wave vectors of the discrete spectrum are obtained from the following relationships:

$$k_{mnx} = [1,0][K][m,n]^t - k_{0x} \tag{9.30a}$$

$$k_{mny} = [0,1][K][m,n]^t - k_{0y} \tag{9.30b}$$

$$k_{mnz} = \pm \sqrt{\left(\frac{2\pi}{\lambda}\right)^2 - k_{mnx}^2 - k_{mny}^2} \tag{9.30c}$$

The wave number in the direction transverse to the structure, k_{mnz}, changes from real values to imaginary ones when m and n are sufficiently large.

Therefore, the plane wave spectrum contains a finite number of propagating modes and an infinite number of evanescent ones which form the reactive field. These evanescent modes do not contribute to the far field, but they have very important effects on the near field; for instance, by coupling the structure to antennas or other close periodic structures.

Once the frequency has been fixed, the number of modes in the scattered field that are propagating depends on the direction of the incident plane wave (k_{0x},k_{0y}), as can be observed in (9.30). A graphical method of obtaining the number of propagating modes when the frequency and incident wave have been fixed is to represent in the spectral domain the reciprocal lattice and the circle whose equation is

$$k_x^2 + k_y^2 - k_{0x}^2 - k_{0y}^2 \leq \left(\frac{2\pi}{\lambda}\right)^2 \tag{9.31}$$

As is illustrated by the example of Figure 9.5, the center of this circle is at the point (k_{0x},k_{0y}) and its radius is k_0. All the points of the reciprocal lattice inside the circle represent propagating modes, and the points outside it evanescent ones. As the frequency rises, the radius becomes greater and the number of propagating modes increases. When the incident direction changes, the circle moves and a larger or smaller number of modes is included in the propagative area of the spectral space.

In most devices based on FPSs, it is important that there should be only one propagating mode. It is easy to obtain the following condition for this:

$$\max\{|T_1|,|T_2|\} \leq \frac{\lambda}{1 + |\sin \theta_i|} \tag{9.32}$$

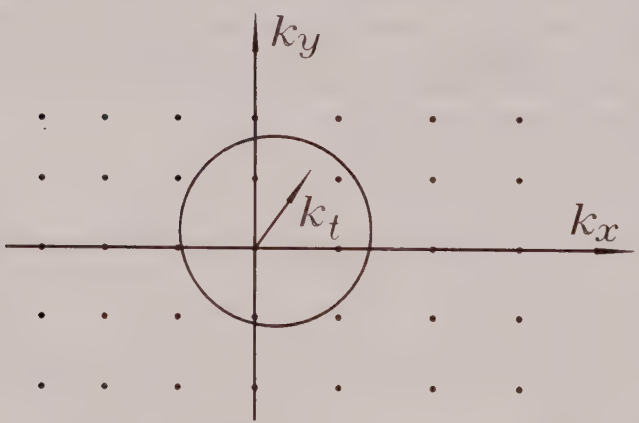


Figure 9.5 Reciprocal lattice of a cartesian FPS. The circle of (9.31) is shown. All the points of the lattice inside the circle represent propagating modes. This circle is centered at the point $k_t = (k_{0x},k_{0y})$.

9.4 MPIE FORMULATION FOR PERIODIC STRUCTURES

As we indicated in the introduction, the principal aim of this chapter is to introduce periodic structures and to study a class of them: the isolated flat periodic structures or free-standing periodic structures. These can be considered as a type of arbitrary flat conducting plates, so initially we can assume that the formulation of Section 5.1 holds. In fact, starting from the general MPIE (5.10), we will obtain a particular form to be applied to FPS.

Recall that the MPIE has the form

$$\mathbf{n} \times \mathbf{E}^i(\mathbf{r}) = \mathbf{n} \times \left[j\omega\mu_0 \int_S \mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' - \frac{1}{j\omega\varepsilon} \nabla \int_S \nabla' \mathbf{J}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' \right] \mathbf{r}' \text{ on } S \tag{9.33}$$

where $G(\mathbf{r},\mathbf{r}')$ represents the free-space Green's function:

$$G(\mathbf{r},\mathbf{r}') = \frac{\exp(-jk_0|\mathbf{r}-\mathbf{r}'|)}{4\pi|\mathbf{r}-\mathbf{r}'|} \tag{9.34}$$

and S is the conducting surface that we assume to be in the $z = 0$ plane.

Equation (9.33) holds for the case of periodic structures, but the integration domain S consists of the whole periodic structure. Taking this fact into consideration, we can rewrite (9.33) as

$$\mathbf{n} \times \mathbf{E}^i = \mathbf{n} \times \left[k^c \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \int_{S_{mn}} \mathbf{J}_{mn}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds + k^q \nabla \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \int_{S_{mn}} \nabla' \mathbf{J}_{mn}(\mathbf{r}')G(\mathbf{r},\mathbf{r}')\, ds' \right] \tag{9.35}$$

where S_{mn} is the metallic surface of the (m,n) cell, $\mathbf{J}_{mn}(\mathbf{r}')$ represents the induced current in this cell, and k^c and k^q are $j\omega\mu_0$ and $-1/j\omega\varepsilon_0$, respectively.

As a consequence of Floquet's theorem, the current in the (m,n) cell of the structure can be written as a function of the current on a reference cell (the (0,0) cell):

$$\mathbf{J}_{mn}(\mathbf{r}') = \mathbf{J}_{00}(\mathbf{r}') \exp(-j(k_{0x}mT_x + k_{0y}nT_y)) = \mathbf{J}_0(\mathbf{r}') \exp(-j\mathbf{k}^i \cdot \mathbf{T}_{mn}) \tag{9.36}$$

Substituting the above expression for the current into (9.35), we obtain

$$\mathbf{n} \times \mathbf{E}^i = \mathbf{n} \times \left[k^c \int_{S_0} \mathbf{J}_0(\mathbf{r}') G^F(\mathbf{r},\mathbf{r}')\, ds' + k^q \nabla \int_{S_0} \nabla' \mathbf{J}_0(\mathbf{r}') G^F(\mathbf{r},\mathbf{r}')\, ds' \right] \tag{9.37}$$

where S_0 is the metallic patch in the (0,0) cell and where

$$G^F(\mathbf{r},\mathbf{r}') = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} G(\mathbf{r},\mathbf{r}' + \mathbf{T}_{mn}) \exp(-j\mathbf{k}^i \cdot \mathbf{T}_{mn}) \tag{9.38}$$

Expression (9.38) defines a new Green's function which takes into account the periodic nature of the structure. It is interesting to note that the new MPIE holds only in one cell and that the unknown is the current $\mathbf{J}_0(\mathbf{r}')$ in this cell.

Equation (9.37) will be our initial equation that must be discretized so that it can be solved numerically. In order to understand completely the operations involved in (9.37) we will write it in a more explicit form:

$$\begin{aligned} E^i_\alpha(\mathbf{r}) = {} & k^c \int_{S_0} J_{0\alpha}(\mathbf{r}') G^F(\mathbf{r},\mathbf{r}')\, ds' \\ & + k^q \frac{\partial}{\partial \alpha} \int_{S_0} \left(\frac{\partial J_{0\alpha}(\mathbf{r}')}{\partial \alpha'} + \frac{\partial J_{0\beta}(\mathbf{r}')}{\partial \beta'} \right) G^F(\mathbf{r},\mathbf{r}')\, ds' \end{aligned} \tag{9.39}$$

where α and β can be x or y, and $J_{0\alpha}(\mathbf{r}')$ and $J_{0\beta}(\mathbf{r}')$ represent the cartesian components of $\mathbf{J}_0(\mathbf{r}')$.

From now on, we will write the current in the reference cell as

$$\mathbf{J}_0(\mathbf{r}') = (J_x(\mathbf{r}')\mathbf{x} + J_y(\mathbf{r}'))\, \Phi(\mathbf{r}') \tag{9.40}$$

where

$$\Phi(\mathbf{r}') = \exp(-j(k_{0x}x' + k_{0y}y')) \tag{9.41}$$

Taking into account the above expressions, we can express (9.37) in terms of a matrix operator:

$$\begin{bmatrix} E^i_x \\ E^i_y \end{bmatrix} = \begin{bmatrix} L^c_{xx} J_x \\ L^c_{yy} J_y \end{bmatrix} + \begin{bmatrix} L^q_{xx} & L^q_{xy} \\ L^q_{yx} & L^q_{yy} \end{bmatrix} \begin{bmatrix} J_x \\ J_y \end{bmatrix} \tag{9.42}$$

where

$$L^c_{\alpha\alpha} = k^c \int_{S_0} G^F(\mathbf{r},\mathbf{r}')\Phi(\mathbf{r}')\,ds' \tag{9.43a}$$

$$L^q_{\alpha\beta} = k^q \frac{\partial}{\partial\alpha} \int_{S_0} G^F(\mathbf{r},\mathbf{r}')\Phi(\mathbf{r}')\left(\frac{\partial}{\partial\beta'} - jk_{0\beta}\right) ds' \tag{9.43b}$$

9.5 DISCRETIZATION PROCESS

In this section, we will develop a discretization process that allows the MPIE to be solved iteratively by using the CG method. A scheme similar to that developed in Section 5.3 will be used. As was done in that section, rooftop and razor blade functions will be used as basis and testing functions, respectively.

While the process is similar to that of Section 5.3, there are some differences. In the case of finite-sized problems, the discretization of the MPIE is completely performed in the spatial domain, including the discretization of the Green's function. When the process is finished, the operator and function are completely discrete and the FFT is applied.

For periodic structures, we will see that the discretization process in the real domain involves the current and its derivatives, but not the Green's function. This is because the Green's function for periodic structures, G^F, is a discrete function in the spectral domain, which is a consequence of its quasiperiodic form in the spatial domain.

Regarding the location of the basis and testing functions, we observe that the metallic patch of the reference cell must be enclosed by a rectangular parallelogram, as was done in Chapter 5. But L_x and L_y of Figure 5.7 must now be equal to T_x and T_y. The current is approximated by $(N_x + 1) \times (N_y + 1)$ basis functions located similarly to those in Chapter 5. However, there can exist a new row of rooftop functions in each direction, as is illustrated in Figure 9.6. This new row of rooftops is centered on the boundary between the reference and the contiguous one. In this way, current is allowed to flow between adjacent cells of the structure. Therefore, the approximation for the current is

$$J_\alpha(\mathbf{r}) \cong \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} J^D_\alpha[m,n]\, T^\alpha_{mn}(x,y) \tag{9.44}$$

where α can be x or y, and T^α_{mn} are the rooftop functions [22] in the α-direction with centers at the points

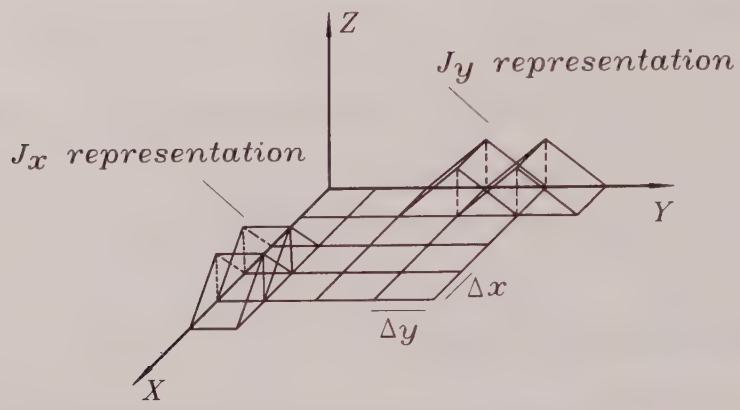


Figure 9.6 Rooftop function centered on the boundary of the reference cell.

$$\mathbf{r}^x_{mn} = x^x_{mn}\mathbf{x} + y^x_{mn}\mathbf{y} = m\Delta x\mathbf{x} + \left(n - \frac{1}{2}\right)\Delta y\mathbf{y} \quad \text{for} \quad T^x_{mn} \tag{9.45a}$$

$$\mathbf{r}^y_{mn} = x^y_{mn}\mathbf{x} + y^y_{mn}\mathbf{y} = \left(m - \frac{1}{2}\right)\Delta x\mathbf{x} + n\Delta y\mathbf{y} \quad \text{for} \quad T^y_{mn} \tag{9.45b}$$

9.5.1 Discretization Process in the Spatial Domain

We start the discretization process of the MPIE with the easier term, the impressed field. This allows us to introduce the notation and some basic ideas. The impressed field $E^i_\alpha(\mathbf{r})$ is discretized by using the testing or smoothing functions. In the context of the MM, the result is usually called the impressed voltage matrix.

$$V^\alpha[m,n] = \int_{S_0} E^i_\alpha(\mathbf{r})\, W^\alpha_{mn}(\mathbf{r})\, \mathrm{d}s \tag{9.46}$$

where W^α_{mn} are the razor blade functions for the α-direction (5.30a,b). The same result can be obtained if the testing procedure is expressed in an alternative form:

$$E^{iD}_\alpha(\mathbf{r}^\alpha_{mn}) = \mathit{III}\,(\mathbf{r}^\alpha_{mn}) \int_{S_0} E^i_\alpha(\mathbf{r}')\, W_\alpha(\mathbf{r} - \mathbf{r}')\, \mathrm{d}s' \tag{9.47}$$

where W_α can be W^x_{00} or W^y_{00}.

This discretization procedure, as it is expressed in (9.47), may appear to be rather strange. Nevertheless, the result is the same as with MM schemes, and this procedure proves to be advantageous because the sampling function is considered explicitly. Further advantages of expressing the testing procedure as a convolution will be seen in the next chapter.

If the impressed field is an incident plane wave,

$$E_\alpha^i = E_{0\alpha} \exp(-j\mathbf{k}^i \cdot \mathbf{r}) \tag{9.48a}$$

the following result is obtained from (9.47):

$$E_\alpha^{iD}(\mathbf{r}) = \Pi\Pi(\mathbf{r}_{mn}^\alpha)\left[\Phi(r)\,\Delta\alpha E_{0\alpha}\,\mathrm{sinc}\left(k_\alpha^i \frac{\Delta\alpha}{2}\right)\right] \tag{9.48b}$$

Thus, the first term of the MPIE equation has been discretized. The next term in the MPIE that we consider is the inductive one; that is, the part of the operator that takes into account the field scattered directly from the current:

$$E_\alpha^{SC}(\mathbf{r}) = L_{\alpha\alpha}^c J_\alpha \quad r \text{ on } S_0 \tag{9.49}$$

The continuous function $E_\alpha^{SC}(\mathbf{r})$ must be discretized and sampled at the same points $\mathbf{r}_{mn}^\alpha$ as the impressed field, with averaged boundary conditions being enforced. To carry out the discretization, we will proceed as in the previous section:

$$\begin{aligned} E_\alpha^{SCD}(\mathbf{r}) &= \Pi\Pi(\mathbf{r}_{mn}^\alpha) \int_{S_0} E_\alpha^{SC}(\mathbf{r}')\, W_\alpha(\mathbf{r} - \mathbf{r}')\, \mathrm{d}s' \\ &\simeq \Pi\Pi(\mathbf{r}_{mn}^\alpha)\, \Delta\alpha E_\alpha^{SC}(\mathbf{r}) \end{aligned} \tag{9.50}$$

The approach of (9.50) assumes that the function $E_\alpha^{SC}(\mathbf{r})$ changes very little in intervals of length $\Delta\alpha$.

In the next step, the current is expanded in terms of rooftop basis functions:

$$\begin{aligned} E_\alpha^{SCD}(\mathbf{r}) &= \Pi\Pi(\mathbf{r}_{mn}^\alpha)\, \Delta\alpha L_{\alpha\alpha}^c \left[\sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} J_\alpha^D[i,j]\, T_{ij}^\alpha(\mathbf{r}')\right] \\ &\simeq \Pi\Pi(\mathbf{r}_{mn}^\alpha)\, \Delta\alpha L_{\alpha\alpha}^c \left[\sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} J_\alpha^D[i,j]\, P_{ij}^\alpha(\mathbf{r}')\right] \end{aligned} \tag{9.51}$$

In the above expression, the rooftop T_{ij}^α has been approximated by the 2D pulse function P_{ij}^α, which has the same volume and center point as T_{ij}^α. This approach, which is justified in [22], allows us to simplify the formulation.

Finally, we write the explicit expression for the capacitive operator $L^c_{\alpha\alpha}$ to obtain

$$E^{SCD}_{\alpha}(\mathbf{r}) = \Pi(\mathbf{r}^{\alpha}_{mn})\,\Delta\alpha \int_{S_0} G^F(\mathbf{r}-\mathbf{r}')\,\Phi(\mathbf{r}') \sum_{i=1}^{N_x+1}\sum_{j=1}^{N_y+1} J^D_{\alpha}[i,j]P^{\alpha}_{ij}(\mathbf{r}')\,ds' \tag{9.52}$$

At this point, we will stop expanding the expressions because we are interested in the form of (9.52). However, it is interesting to note that we could continue by interchanging the sum and the integral in (9.52) to obtain an expression like

$$E^{SCD}_{\alpha}(\mathbf{r}^{\alpha}_{mn}) = \sum_{i=1}^{N_x+1}\sum_{j=1}^{N_y+1} J^D_{\alpha}[i,j]Z_{\alpha}[i,j;m,n] \tag{9.53}$$

where

$$Z_{\alpha}[i,j;m,n] = \Delta\alpha \int_{S_0} G^F(\mathbf{r}^{\alpha}_{mn}-\mathbf{r}')\,\Phi(\mathbf{r}')P^{\alpha}_{ij}(\mathbf{r}')\,ds' \tag{9.54}$$

Equation (9.53) is clearly identified as the algebraic equation of the MM scheme, in which $Z_{\alpha}[i,j;m,n]$ is usually referred to as the impedance matrix [11]. In this case, it would be the capacitive part of the impedance matrix.

The rest of the operator in (9.42), $L^q_{\alpha\beta}$, acts on the current components through their derivatives. Hence, these terms are referred to as capacitive terms. The contribution to the scattered field can be written as

$$E^{Sq}_{\alpha\beta}(\mathbf{r}) = L^q_{\alpha\beta}\,J_{\beta} \quad \mathbf{r} \text{ on } S_0 \tag{9.55}$$

The process used to discretize $E^{Sq}_{\alpha\beta}$ is similar to that followed for the inductive term

$$\begin{aligned}
E^{SqD}_{\alpha\beta}(\mathbf{r}) &= \Pi(\mathbf{r}^{\alpha}_{mn}) \int_{S_0} E^{Sq}_{\alpha\beta}(\mathbf{r}')\,W_{\alpha}(\mathbf{r}-\mathbf{r}')\,ds' \\
&= \Pi(\mathbf{r}^{\alpha}_{mn}) \int_{r^-_{\alpha}}^{r^+_{\alpha}} \frac{\partial}{\partial\alpha} V(\mathbf{r})\,\Phi(\mathbf{r}) \\
&= \Pi(\mathbf{r}^{\alpha}_{mn})[V(\mathbf{r}^+_{\alpha})\,\Phi(\mathbf{r}^+_{\alpha}) - V(\mathbf{r}^-_{\alpha})\,\Phi(\mathbf{r}^-_{\alpha})] \\
&\simeq \Pi(\mathbf{r}^{\alpha}_{mn})\Phi(\mathbf{r})\left[V(\mathbf{r}^+_{\alpha}) - V(\mathbf{r}^-_{\alpha}) - jk_{0\alpha}\Delta\alpha\,\frac{v(\mathbf{r}^+_{\alpha}) + V(\mathbf{r}^-_{\alpha})}{2}\right]
\end{aligned} \tag{9.56}$$

where

$$\mathbf{r}_{\alpha}^{\pm} = \mathbf{r} \pm \frac{\Delta\alpha}{2}\boldsymbol{\alpha} \tag{9.57}$$

$$\begin{aligned} V(\mathbf{r}) &= \Phi^*(\mathbf{r}) \int_{S_0} G^F(\mathbf{r}-\mathbf{r}')\Phi(\mathbf{r}')\left(\frac{\partial}{\partial\beta} - jk_{0\beta}\right) J_\beta(\mathbf{r}')\, ds' \\ &\simeq \frac{\Phi^*(\mathbf{r})}{\Delta\beta} \int_{S_0} G^F(\mathbf{r}-\mathbf{r}')\Phi(\mathbf{r}') \sum_{i=1}^{N_x+1}\sum_{j=1}^{N_y+1} P_{ij}^q \left(J_\beta^{D\partial}[i,j] \right. \\ &\quad \left. - jk_{0\beta}\frac{\Delta\beta}{2} J_\beta^{D\Delta}[i,j] \right) ds' \end{aligned} \tag{9.58}$$

and P_{ij}^q is a 2D pulse over the small rectangle of area $\Delta x \Delta y$ whose center is

$$\mathbf{r}_{i,j}^q = \left(i - \frac{1}{2}\right)\Delta x\mathbf{x} + \left(j - \frac{1}{2}\right)\Delta y\mathbf{y} \tag{9.59}$$

and

$$J_x^{D\Delta}[i,j] = (J_x^D[i,j] + J_x^D[i-1,j]) \tag{9.60a}$$

$$J_y^{D\Delta}[i,j] = (J_y^D[i,j] + J_y^D[i,j-1]) \tag{9.60b}$$

$$J_x^{D\delta}[i,j] = (J_x^D[i,j] - J_x^D[i-1,j]) \tag{9.60c}$$

$$J_y^{D\delta}[i,j] = (J_y^D[ij] - J_y^D[i,j-1]) \tag{9.60d}$$

The pulses P_{ij}^q as well as the above discrete functions (9.60) appear in a natural way when the derivatives of the current are carried out. In addition, it should be noted that the approach in (9.56) and (9.58) is valid if the electrical lengths of Δx and Δy are small. From experience, we can say that the electrical length must be no greater than $\lambda/8$.

9.6 COMPLETING THE DISCRETIZATION IN THE SPECTRAL DOMAIN

The inductive and capacitive contributions to the scattered field have been approximated in the spatial domain and finally reduced to the same mathematical form:

$$W^{\delta}(\mathbf{r}) = \mathit{\Pi\!I}(\mathbf{r}_{mn})\, W(\mathbf{r}) \tag{9.61}$$

where

$$W(\mathbf{r}) = \int_{S_0} G^{F}(\mathbf{r} - \mathbf{r}')S(\mathbf{r}')\, \mathrm{d}s' \tag{9.62a}$$

$$S(\mathbf{r}') = \Phi(\mathbf{r}') \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} S^{D}[i,j]P^{i,j}(\mathbf{r}') \tag{9.62b}$$

In these expressions, $\mathbf{r}_{mn}$ can be $\mathbf{r}^{\alpha}_{mn}$ (9.45) or $\mathbf{r}^{q}_{mn}$ (9.59), $P^{ij}(\mathbf{r}')$ can be P^{α}_{ij} (9.51) or P^{q}_{ij} (9.58), and $S^{D}[i,j]$ will be the discrete function $J^{D}[i,j]$ or its derivatives (9.60). Note that all the integrals required to compute the scattered field have been written in terms of a convolution integral like (9.62).

The discrete function $W^{\delta}(\mathbf{r})$ can be computed by using the following rule:

$$W^{\delta}(\mathbf{r}) = \mathrm{CFT}^{-1}\{\mathrm{CFT}[\mathit{\Pi\!I}(\mathbf{r}_{mn})] \otimes \mathrm{CFT}[W(\mathbf{r})]\} \tag{9.63}$$

The convolution of the spectral domain functions in (9.63) can be computed as

$$\begin{aligned}\tilde{W}^{S}(\mathbf{k}) &= \mathrm{CFT}\{\mathit{\Pi\!I}(\mathbf{r}_{mn}\} \otimes \mathrm{CFT}\{W(\mathbf{r})\} \\ &= \frac{1}{\Delta x \Delta y} \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} \tilde{W}(k_x - m'T^{K}_{x}; k_y - n'T^{K}_{y})\end{aligned} \tag{9.64}$$

where

$$\tilde{W}(\mathbf{k}) = \mathrm{CFT}\,\{W(\mathbf{r})\} \tag{9.65}$$

and

$$T^{K}_{\alpha} = \frac{2\pi}{\Delta\alpha} \quad \alpha = x,y \tag{9.66}$$

The function $W(\mathbf{r})$ is obtained in the real domain from a convolution integral, so we have

$$\tilde{W}(\mathbf{k}) = \tilde{G}^{F}(\mathbf{k})\tilde{S}(\mathbf{k}) \tag{9.67}$$

where G^F and $\tilde{S}$ are the CFTs of $G^F(\mathbf{r})$ and $S(\mathbf{r})$, respectively. As can be observed in (9.64), $\mathbf{W}^\delta(\mathbf{k})$ is a periodic function as a consequence of sampling in the spatial domain.

Since the CFT of $G(\mathbf{r})$ is known, the transform of $G^F(\mathbf{r})$ can be easily found:

$$\tilde{G}^F(\mathbf{k}) = \frac{(2\pi)^2}{T_x T_y} \mathit{\Pi}(\mathbf{k}^{mn} - \mathbf{k}^i) \left(\frac{2\pi}{(k_x^2 + k_y^2 - k_0^2)^{1/2}} \right) \tag{9.68}$$

where

$$\mathbf{k}^{mn} = k_x^m \mathbf{x} + k_y^n \mathbf{y} = \frac{2\pi m}{T_x} \mathbf{x} + \frac{2\pi n}{T_y} \mathbf{y} = m \left(\frac{T_x^K}{N_x} \right) \mathbf{x} + n \left(\frac{T_y^K}{N_y} \right) \mathbf{y} \tag{9.69}$$

As can be seen from (9.68) and (9.69), the Green's function in the spectral domain is discrete and, in accordance with Floquet's theorem, only has values at the spectral points $(\mathbf{k}^{mn} - \mathbf{k}^i)$. To compute $\tilde{W}(\mathbf{k})$ in (9.67), we only need to know $\tilde{P}(\mathbf{k}^{mn} - \mathbf{k}^i)$.

Taking into account the general expression for $S(\mathbf{r})$ in the spatial domain (9.62b), we can obtain in a straightforward way the values of the CFT of $S(\mathbf{r})$ at the points of interest:

$$\tilde{S}(\mathbf{k}^{mn} - \mathbf{k}^i) = \tilde{P}(\mathbf{k}^{mn}) \tilde{S}^D[m,n] \tag{9.70}$$

where

$$\tilde{P}(k_x,k_y) = \Delta x \Delta y \operatorname{sinc}\left(k_x \frac{\Delta x}{2} \right) \operatorname{sinc}\left(k_y \frac{\Delta y}{2} \right) \tag{9.71}$$

$$\tilde{S}^D[m,n] = N_x N_y \operatorname{DFT}(S^D[i,j]) \tag{9.72}$$

The function $\tilde{W}(\mathbf{k})$ can be written from (9.67) to (9.70) as follows:

$$\tilde{W}(\mathbf{k}) = \frac{(2\pi)^2 N_x N_y}{T_x T_y} \sum_{m,n=-\infty}^{\infty} S^D[m,n] P(\mathbf{k}) G(\mathbf{k}) \delta(k_x - (k_x^m - k_x^i)) \delta(k_y - (k_y^n - k_y^i)) \tag{9.73}$$

Our aim is to obtain an expression that will allow us to compute $\tilde{W}^\delta(\mathbf{k})$ as easily as possible. Bearing this in mind, we introduce (9.73) into (9.64):

$$\tilde{W}^{\delta}(\mathbf{k}) = \sum_{m',n'} \sum_{m,n} S^{D}[m,n]\tilde{W}_0(k_x + m'T_x^k, k_y + n'T_y^k)\delta[k_x - (k_x^m - k_x^i)]\delta[k_y - (k_y^n - k_y^i)] \tag{9.74}$$

where

$$\tilde{W}_0(\mathbf{k}) = (2\pi)^2 \tilde{P}(\mathbf{k})\tilde{G}(\mathbf{k}) \tag{9.75}$$

As we indicated previously, $\tilde{W}^{\delta}(\mathbf{k})$ is a periodic function in the spectral domain and $W(\mathbf{k})$ is a discrete function. Therefore, $W^{\delta}(\mathbf{k})$ is a discrete and periodic function in the spectral domain. This is not surprising if we remember that $W^{\delta}(\mathbf{k})$ is the CFT of a sampled periodic function in the spatial domain. However, (9.74) does not show periodicity clearly, so we will continue rewriting it. First, it should be noted that $\tilde{S}^{D}[m,n]$ is a discrete and periodic function because it is obtained as a DFT. To account for this, we can write

$$m = m_0 + tN_x \tag{9.76a}$$

$$n = n_0 + zN_y \tag{9.76b}$$

$$m_0, n_0 = 0, 1, 2, 3, \ldots, N_x, N_y$$

$$t, z = 0, \pm 1, \pm 2, \ldots, \pm\infty$$

so that

$$\tilde{S}^{D}[m,n] = \tilde{S}^{D}[m_0 + tN_x, n_0 + zN_y] = \tilde{S}^{D}[m_0,n_0] \tag{9.77}$$

By using the notation introduced above,

$$\tilde{W}^{\delta}(\mathbf{k}) = \sum_{m_0,n_0} \tilde{S}^{D}[m_0,n_0] \sum_{m',n'} \sum_{t,z} \tilde{W}_0\left[\frac{2\pi}{\Delta x}\left(m' + t + \frac{m_0}{N_x}\right), \frac{2\pi}{\Delta y}\left(n' + z + \frac{n_0}{N_y}\right)\right] \tag{9.78}$$
$$\delta[k_x - (k_x^{m_0+tN_x} - k_x^i)]\delta[k_y - (k_y^{n_0+zN_y} - k_y^i)]$$

Now we can define a periodic function as the sum of the series

$$\tilde{W}^{P}[m_0,n_0] = \sum_{h=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} \tilde{W}_0\left[\frac{2\pi}{\Delta x}\left(h + \frac{m_0}{N_x}\right), \frac{2\pi}{\Delta y}\left(k + \frac{n_0}{N_y}\right)\right] \tag{9.79}$$

where we have used the new variables $h = m' + t$ and $k = n' + z$ to simplify the double series in (9.78). In (9.79), it can be seen that $\tilde{W}^{P}$ has a period of (N_x,N_y).

Finally, we can write a compact expression for $\tilde{W}^{\delta}(\mathbf{k})$:

$$\tilde{W}^{\delta}(\mathbf{k}) = \sum_{m_0=0}^{N_x} \sum_{n_0=0}^{N_y} \tilde{S}^{D}[m_0,n_0]\tilde{W}^{P}[m_0,n_0]\delta(k_x - (k_x^{m_0} - k_x^{i}))\delta(k_y - (k_y^{n_0} - k_y^{i})) \quad (9.80)$$

In conclusion, it can be seen that $\tilde{W}^{\delta}(\mathbf{k})$ is a discrete and periodic function, whose period is (T_x^K, T_y^K), since we can obtain its inverse transform as

$$\begin{aligned} W^{\delta}(\mathbf{r}) &= \mathrm{CFT}^{-1}\{\tilde{W}^{\delta}(\mathbf{k})\} \\ &= \Phi(\mathbf{r}) \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \delta(x - x^{i})\delta(y - y^{j})\, \mathrm{DFT}^{-1}\{\tilde{W}^{P}[m_0,n_0]\tilde{S}^{D}[m_0,n_0]\} \end{aligned} \quad (9.81)$$

This is a very interesting result: the effect of approximating the current and the field by a finite number of subdomain basis functions is that the Floquet spectrum is also approximated by the same finite number of different harmonics. Nevertheless, to determine the exact value of the harmonics, we must compute the series in (9.79). Fortunately, this series converges quickly. This subject has been studied thoroughly in [18]; we will simply point out that a good approximation can be obtained by considering a few terms of the series; that is,

$$\tilde{W}^{P}[m_0,n_0] \simeq \sum_{h=-M}^{M} \sum_{k=-N}^{N} \tilde{W}_0\left(\frac{2\pi}{\Delta x}\left(h + \frac{m_0}{N_x}\right), \frac{2\pi}{\Delta y}\left(K + \frac{n_0}{N_y}\right)\right] \quad (9.82)$$

It has been shown that the differences between using $N = M = 3$ and computing the series exactly are negligible.

In summary, it is important to note that the different terms of the MPIE operator, which have been written in the general form in (9.61) and (9.62), can be computed by making use of the properties of the CFT. The most time-consuming computations are two DFTs, one direct (9.72) and the other inverse (9.81), which can be efficiently computed by using FFTs.

9.7 OPERATIONAL FORM OF THE MPIE

To highlight the advantages of the scheme developed in the preceding section, we will rewrite it in an operational form. First of all, we write the following expression for the discretized scattered field using our notation:

$$E_{\alpha}^{SD}(\mathbf{r}_{mn}^{\alpha}) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} \delta(x - x_{\alpha}^{m})\delta(y - y_{\alpha}^{n})E_{\alpha}^{\delta}[m,n] \quad (9.83)$$

The values of the discrete function $E_\alpha^\delta[m,n]$ can be obtained from

$$\begin{bmatrix} E_x^\delta[m,n] \\ E_y^\delta[m,n] \end{bmatrix} = \text{FFT}^{-1}\left\{ \begin{bmatrix} Y_{xx}(i,i) & Y_{yx}(i,j) \\ Y_{xy}(j,i) & Y_{yy}(j,j) \end{bmatrix} \begin{bmatrix} \tilde{W}^p(i,j)\,\text{FFT}(J_x^D[m,n]) \\ \tilde{W}^p(i,j)\,\text{FFT}(J_y^D[m,n]) \end{bmatrix} \right\} \tag{9.84}$$

where

$$Y_{\alpha\alpha}[i,j] = C_\alpha^C + C_\alpha^q(1 - F_\alpha^*[i])(F_\alpha[i] - 1) \tag{9.85a}$$

$$Y_{\alpha\beta}[i,j] = C_\alpha^q(1 - F_\alpha^*[j])(F_\beta[i] - 1) \tag{9.85b}$$

$$C_\alpha^C = \left(\frac{N_x N_y}{(2\pi)^2}\right)\left(\frac{-j\mu\omega\Delta\alpha}{4\pi}\right) \tag{9.86a}$$

$$C_\alpha^q = \left(\frac{N_x N_y}{(2\pi)^2}\right)\left(\frac{-j}{4\pi\varepsilon\omega\Delta\alpha}\right) \tag{9.86b}$$

where $\tilde{W}^p[i,j]$ is obtained from (9.82), and

$$F_\alpha[i] = \exp\left(j\frac{2\pi i}{N_x}\right) - j\frac{k_\alpha^i \Delta\alpha}{2} \tag{9.87}$$

This last function is used to express concisely the relationships between the current and its derivatives, as well as to take into consideration the effect in the spectral domain of shifting the function $V(\mathbf{r})$ in (9.56).

We are interested in solving the integral equation by using the iterative CG method. To do this, we must compute iteratively the result of (9.84) by using successive approximations of J_x^D and J_y^D.

9.8 REFLECTION AND TRANSMISSION COEFFICIENTS

The reflection and transmission coefficients are the most important quantities when it comes to characterizing a FPS. They include all the information about the modifications occurring in the incident fields on these structures. They also have the advantage of being measurable quantities, which are comparable with the results of any analysis method. The definitions of these coefficients are similar to those used in the case of microwave circuits, since the techniques employed to design microwave devices can also be used in the context of designing devices based on FPSs.

These coefficients can be defined for each plane wave in the Floquet spectrum, including propagating and evanescent modes. The complex amplitudes of the Floquet modes on the structure can be obtained according to (9.83) and (9.84) as

$$\begin{bmatrix} \tilde{E}_x(k_{xi},k_{yj}) \\ \tilde{E}_y(k_{xi},k_{yj}) \end{bmatrix} = \begin{bmatrix} Y_{xx}[i,i] & Y_{yx}[i,j] \\ Y_{xy}[j,i] & Y_{yy}[j,j] \end{bmatrix} \begin{bmatrix} \tilde{W}^p[i,j] & \text{FFT}(J_x^D) \\ \tilde{W}^p[i,j] & \text{FFT}(J_y^D) \end{bmatrix} \tag{9.88}$$

All the functions involved in (9.88) have been defined in the previous section.

Since the scattered field is strongly dependent on polarization, both the incident and scattered field are usually decomposed into transverse electric (TE) and transverse magnetic (TM) with respect to z-direction. Once this is done, we can rewrite the scattered field as

$$\mathbf{E}^s(x,y,z) \simeq \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} [\tilde{\mathbf{E}}^s_{\text{TE}}(k_{xi},k_{yi}) + \tilde{\mathbf{E}}^s_{\text{TM}}(k_{xi},k_{yj})] e^{j(k_{xi}x + k_{yij}y \pm k_{zij}z)} \tag{9.89}$$

Following the work of Cwik and Mittra [23,24], we define the normalized Floquet voltage waves as

$$V_{ij}^{\text{TE}}(z) = \tilde{E}^s_{\text{TE}}(k_{xi},k_{yj})(\eta^*_{\text{TE},ij})^{-1/2}(k_{xi}^2 + k_{yj}^2) \exp(\pm jk_{zij}) \tag{9.90a}$$

$$V_{ij}^{\text{TM}}(z) = \tilde{E}^s_{\text{TM}}(k_{xi},k_{yj})(\eta_{\text{TM},ij})^{1/2}(k_{xi}^2 + k_{yj}^2) \exp(\pm jk_{zij}) \tag{9.90b}$$

where the η functions represent wave impedances:

$$\eta_{\text{TE},ij} = \frac{\omega\mu}{k_{zij}} \tag{9.91a}$$

$$\eta_{\text{TM},ij} = \frac{k_{zij}}{\omega\varepsilon} \tag{9.91b}$$

The normalized voltage waves describe the field in any plane perpendicular to **z**. The squares of these waves have units of power and are interpreted as propagating or evanescent power waves. Starting from the voltage waves, generalized scattering coefficients are defined as the ratios between incident and scattered waves. A detailed discussion is presented in [23]. Here, as an example, we will express the generalized reflection coefficient:

$$R_{pi'j'}^{qij}(z) = \frac{V_{ij}^q(z)}{V_{i'j'}^p(z)} \tag{9.92}$$

The symbols p and q can be TE or TM, and (i',j') and (i,j) are the orders of the incident and reflected modes, respectively. The coefficient may be interpreted as the reflected energy in the mode (i,j) due to the coupling of energy from the incident mode (i',j').

In many cases, propagating modes are of interest because they represent the radiation field. We rewrite here the reflection coefficient as a function of the tangential components of the incident plane wave and the scattered propagating mode for the particular case of a structure with only one propagating mode:

$$R_{\mathrm{TE}_{00}}^{\mathrm{TE}_{00}} = \frac{\tilde{E}_{t\mathrm{TE}}^{s}(k_{x0},k_{y0})}{\tilde{E}_{t\mathrm{TE}}^{i}(k_{x0},k_{y0})} \tag{9.93a}$$

$$R_{\mathrm{TE}_{00}}^{\mathrm{TM}_{00}} = \frac{\tilde{E}_{t\mathrm{TM}}^{s}(k_{x0},k_{y0})}{\tilde{E}_{t\mathrm{TE}}^{i}(k_{x0},k_{y0})\cos\theta^{i}} \tag{9.93b}$$

$$R_{\mathrm{TM}_{00}}^{\mathrm{TM}_{00}} = \frac{\tilde{E}_{t\mathrm{TM}}^{s}(k_{x0},k_{y0})}{\tilde{E}_{t\mathrm{TM}}^{i}(k_{x0},k_{y0})} \tag{9.93c}$$

$$R_{\mathrm{TM}_{00}}^{\mathrm{TE}_{00}} = \frac{\tilde{E}_{t\mathrm{TE}}^{s}(k_{x0},k_{y0})\cos\theta^{i}}{\tilde{E}_{t\mathrm{TM}}^{i}(k_{x0},k_{y0})} \tag{9.93d}$$

where θ^{i} is the azimuthal angle of the incident wave.

For illustrative purposes, some characteristic results of transmission and reflection coefficients are presented. Figure 9.7 shows the power reflection coefficient for normal incidence of an FPS composed of dipoles. The results obtained by using the method presented in this chapter are compared with those obtained by Chen and measured by Ott [8]. It can be observed that, at resonance frequency, when the length of the dipole is about $\lambda/2$, the structure is very reflective.

Another example, with practical interest in the design of frequency-selective subreflectors, is an array of cross-dipoles. In Figure 9.8, the power reflection coefficient is presented.

As an example of an inductive FPS, Figure 9.9 presents the reflection coefficient (in magnitude (a) and phase (b)) of a periodic structure of perforated crosses on a conducting plane. The results are compared with those obtained by Rubin and Bertoni using an MM scheme [11]. In the case of an inductive FPS, the structure presents high transmission at resonance frequency.

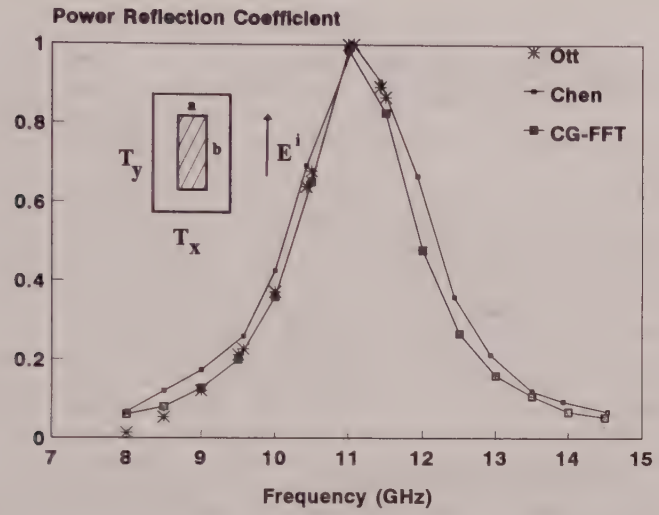


Figure 9.7 Power reflection coefficient of an FPS composed of dipoles. $T_x = T_y = 1.78$ cm, $a = 0.127$ cm, $b = 1.27$ cm.

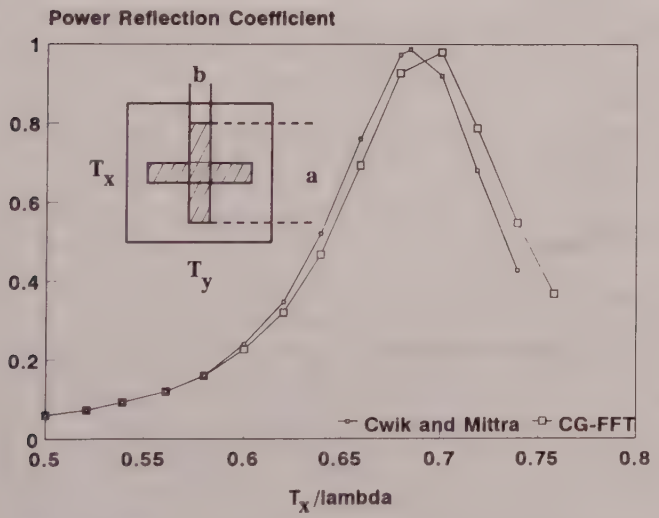
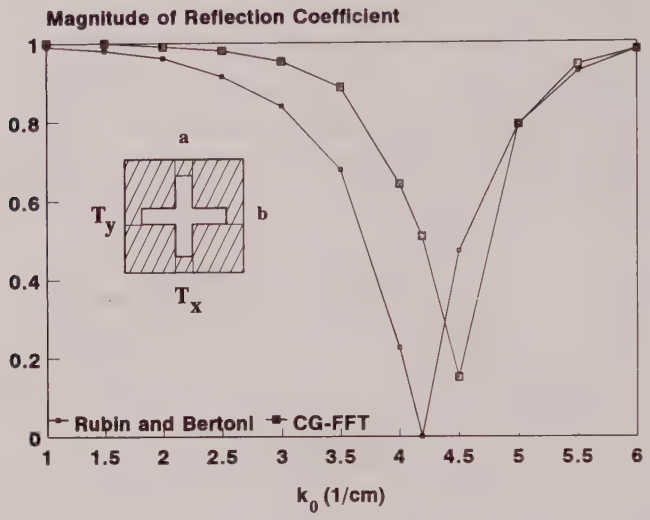
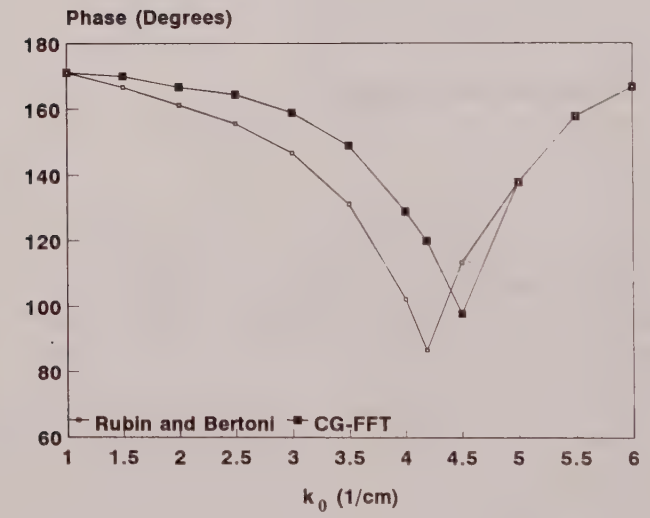


Figure 9.8 Power reflection coefficient of an array of cross-dipoles. $T_x = T_y = \lambda$, $a = 0.6875\lambda$, $b = 0.0625\lambda$.



(a)



(b)

Figure 9.9 (a) Magnitude of the reflection coefficient of an array of perforated crosses. $T_x = T_y = 1.0$ cm, $a = 0.14$ cm, $b = 0.70$ cm. (b) Phase.

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Chapter 10
Flat Periodic Structures in Multilayer Media

10.1 INTRODUCTION

This chapter deals with flat periodic structures (FPS), which are embedded in dielectric multilayer media. The dielectric materials that surround the metallic arrays serve to strengthen the structures mechanically and to protect them from corrosion. But also, and perhaps more importantly, the number of slabs, their thickness, and the dielectric constant are degrees of freedom that can be used to control the electromagnetic performance of periodic structures.

The early numerical methods for analyzing FPSs in multilayer media were developed for periodic arrays of rectangular holes in metallic screens sandwiched between dielectric slabs. These techniques were based on the MM. The work of C. C. Chen [1] and S. W. Lee [2] are illustrative examples. The complementary problem (i.e., the scattering from a periodic array of rectangular metallic patches on a dielectric substrate) was first solved by J.P. Montgomery [3]. One of the early studies that attempted to use dielectrics to control the response of metallic screens was done by R.J. Luebbers and B.A. Munk [4]. These analyses of the effect of the dielectric on screens were aimed at applying this knowledge to the improvement of antenna radomes [2] and to the design of phase-delay-type artificial dielectrics [5]. A major focus of current research is the design of frequency-selective subreflectors in multifrequency reflector antenna systems [6–8]. Some of the interesting applications in the microwave and millimeter-wave range are lobe suppression [9], circular polarizers and linear polarization rotators [10–12], and the possibility of making lenses based on the grating of slow-varying periodicity [13,14].

At the present time, there are many different methods of analyzing FPSs. These may be divided into three general classes: equivalent circuit methods, MMs, and iterative methods. In the equivalent circuit methods, the periodic structure is represented by a set of inductive and capacitive elements in a transmission line [15,16]. This method is easy and computationally efficient, but some problems arise when considering oblique incidence, cross-polarization effects and dielectric supports. The method is being improved by considering higher order modes in the model [17].

The MM has been used in conjunction with a set of expansion functions specific to the geometry of the generating cell in order to solve the integral equations obtained from a harmonic matching point of view [1–5]. The problem can also be formulated as a scattering problem where the induced current is the central unknown. Once the current has been found, the scattered field and the transmission and reflection coefficients are calculated. The integral equation that arises can be solved either by using basis functions specific to each geometry [18,19] or by using subsectional basis functions [20].

To improve the computational efficiency of the MM, iterative techniques have been applied to FPS problem. The combination of the CG method with the FFT appears to be one of the most efficient [21–24].

In the following sections, we will present a scheme based on the CG-FFT method to analyze FPSs in multilayer dielectric media. This method is different from the one developed in the previous chapter for the analysis of free-standing FPSs in that the problem is posed in the spectral domain from the beginning. In this new approach, no operations are performed with Green's functions in the real domain. The major reason for a new scheme is the fact that the Green's functions of multilayer problems are given in the real domain by numerical expressions that are quite cumbersome to evaluate, whereas the Green's functions in the spectral domain have relatively simple analytical expressions.

10.2 INTEGRAL EQUATION IN THE SPECTRAL DOMAIN

In this section, we will formulate the scattering problem in the spectral domain by performing the CFT of the operator equation. It is interesting to observe how the derivatives and convolutions that relate current and scattered fields in the real domain are transformed into simple products in the spectral domain. However, a certain price must be paid: in the real domain, the integral equations relate currents and fields that are finite-limited in the real space (for instance, the metallic patches of an FPS); in the spectral domain, both fields and currents are unlimited functions. In the case of periodic structures, whether free-standing or in multilayer media, this spectrum is discrete and the DFT is therefore well suited to use as a computational tool. In the case of finite structures, the spectrum is continuous and infinite; but, as we shall see in the next chapters, the use of the DFT can also be appropriate.

10.2.1 EFIE for Free-Space Cases

In this section, we present an integral equation for the electric field (EFIE) which relates the scattered field with an induced current in planar surfaces. This equation is very similar to the MPIE presented in Chapter 5, but it is easier to transform to

the spectral domain. First of all, we will consider once again the case of a plane metallic patch of finite size and negligible thickness in free space.

Let $\mathbf{J}$ be the induced current in the metallic patch; then the magnetic vector potential $\mathbf{A}$ can be obtained by means of the relationship

$$\mathbf{A}(\mathbf{r}) = \frac{\mu}{4\pi} \int_S \frac{\exp(-jk_0|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} \mathbf{J}(\mathbf{r}')\, ds' \tag{10.1}$$

where $k_0 = 2\pi/\lambda$ with λ being the wavelength in free space. Here free space means a vacuum or any unbounded, homogeneous, isotropic dielectric medium.

The scattered field can be expressed as a function of the magnetic vector $\mathbf{A}$ and the electric scalar potential Φ:

$$\mathbf{E}^S(\mathbf{r}) = -j\omega\mathbf{A}(\mathbf{r}) - \nabla\Phi(\mathbf{r}) \tag{10.2}$$

where

$$\Phi(\mathbf{r}) = \frac{\varepsilon}{4\pi} \int \frac{\exp(-jk_0|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} \sigma(\mathbf{r}')\, ds' \tag{10.3}$$

with $\sigma(\mathbf{r}')$ being the electric charge density on the surface.

The scattered field can be expressed in terms of current alone if we relate it with charge by means of the continuity equation. In Chapter 5 (Section 5.1), this was done and the MPIE was obtained. Now we will use a different relationship, the Lorentz condition, which relates the two potentials $\mathbf{A}$ and Φ directly:

$$\Phi(\mathbf{r}) = \frac{-1}{j\omega\varepsilon\mu} \nabla \cdot \mathbf{A} \tag{10.4}$$

By using (10.4), the scattered field in (10.2) can be rewritten as follows:

$$\mathbf{E}^S(\mathbf{r}) = -j\omega\mathbf{A}(\mathbf{r}) + \frac{1}{j\omega\varepsilon\mu} \nabla(\nabla \cdot \mathbf{A}(\mathbf{r})) \tag{10.5}$$

The operator (10.5) allows us to calculate the scattered field if we know the induced current, but the usual situation is to know the incident field and the patch geometry. Hence, the induced current becomes the initial unknown of the problem. In this case, we need more information to solve the problem, and it is given by the boundary condition of the tangential components of the fields over the metallic surface:

$$\mathbf{E}_t^S(\mathbf{r}) = -\mathbf{E}_t^i(\mathbf{r}) \qquad \text{if} \quad \mathbf{r} \text{ is on } S \tag{10.6}$$

Returning to (10.5), the tangential components of the field can be rewritten in a more explicit form as

$$\begin{bmatrix} E_x^i(\mathbf{r}) \\ E_y^i(\mathbf{r}) \end{bmatrix} = \frac{-1}{j\omega\varepsilon\mu} \begin{bmatrix} \dfrac{\partial^2}{\partial x^2} + k_0^2 & \dfrac{\partial^2}{\partial x \partial y} \\ \dfrac{\partial^2}{\partial y \partial x} & \dfrac{\partial^2}{\partial^2} + k_0^2 \end{bmatrix} \begin{bmatrix} A_x(\mathbf{r}) \\ A_y(\mathbf{r}) \end{bmatrix} \tag{10.7}$$

By performing the CFT with respect to all the functions and operations in both terms of (10.7), the EFIE in the spectral domain is obtained:

$$\begin{bmatrix} \tilde{E}_x^i(\mathbf{k}) \\ \tilde{E}_y^i(\mathbf{k}) \end{bmatrix} = \frac{-1}{j\omega\varepsilon\mu} \begin{bmatrix} k_0^2 - k_x^2 & -k_x k_y \\ -k_y k_x & k_0^2 k_y^2 \end{bmatrix} [F(\mathbf{k})] \begin{bmatrix} \tilde{J}_x(\mathbf{k}) \\ \tilde{J}_y(\mathbf{k}) \end{bmatrix} \tag{10.8}$$

where $\mathbf{k} = k_x\mathbf{x} + k_y\mathbf{y}$ is the spectral position vector, and

$$[F(\mathbf{k})] = \frac{-j}{2(k_0^2 - k_x^2 - k_y^2)^{1/2}} [I] = \frac{-j}{2k_z} [I] \tag{10.9}$$

where $[I]$ is the 2D identity matrix.

Expression (10.8) can be written as

$$\begin{bmatrix} \tilde{E}_x^i(\mathbf{k}) \\ \tilde{E}_y^i(\mathbf{k}) \end{bmatrix} = \begin{bmatrix} \tilde{G}_{xx}(\mathbf{k}) & \tilde{G}_{xy}(\mathbf{k}) \\ \tilde{G}_{yx}(\mathbf{k}) & \tilde{G}_{yy}(\mathbf{k}) \end{bmatrix} \begin{bmatrix} \tilde{J}_x(\mathbf{k}) \\ \tilde{J}_y(\mathbf{k}) \end{bmatrix} \tag{10.10}$$

where

$$\tilde{G}_{\alpha\alpha}(\mathbf{k}) = \frac{1}{2\omega\varepsilon\mu} \frac{k_0^2 - k_\alpha^2}{k_z} \tag{10.11}$$

$$\tilde{G}_{\alpha\beta}(\mathbf{k}) = \tilde{G}_{\beta\alpha}(\mathbf{k}) = -\frac{1}{2\omega\varepsilon\mu} \frac{k_\alpha k_\beta}{k_z} \tag{10.12}$$

In these expressions α and β can be x or y. The matrix $[G_{\alpha\beta}]$ in (10.10) is usually called Green's dyad. The simple relationship that exists between fields and currents in the spectral domain can be observed.

10.2.2 The Case of Dielectric Multilayer Media

In the case where the embedding medium of the conducting patch is a stratified region, as illustrated in Figure 10.1, the relationship between induced currents and fields is still a convolution-type operation

$$\mathbf{E}^i(\mathbf{r}) = \int_S G(\mathbf{r},\mathbf{r}')\mathbf{J}(\mathbf{r}')\, ds' \tag{10.13}$$

The Green's functions in the spatial domain have cumbersome expressions in terms of Sommerfeld-type integrals. Nevertheless, the general form of the Fourier transform of (10.13) is (10.10), and the Green's dyad for stratified media in the spectral domain is given by relatively simple expressions. These can be obtained easily by using the equivalent transmission line method. This method has been developed and used by many authors (see, for instance, [25–27]). Therefore, we will only sketch the basic outline of the solution.

For a general structure like the one shown in Figure 10.1, an analogy will be established with a transmission line circuit like the one depicted in Figure 10.2. However, each mode of the spectrum must first be decomposed into orthogonal TE and TM polarizations relative to z. Then a transmission line analogy is established for each polarization and mode: the length of the lines in the circuit represent the width of the dielectric slabs, and the characteristic impedances of the lines correspond to the impedances of the TE or TM modes in the corresponding medium:

$$Z_{0i}^{\mathrm{TE}}(\mathbf{k}) = \sqrt{\frac{\mu_i}{\epsilon_i}}\frac{k_{0i}}{\sqrt{k_{0i}^2 - k_x^2 - k_y^2}} \tag{10.14}$$

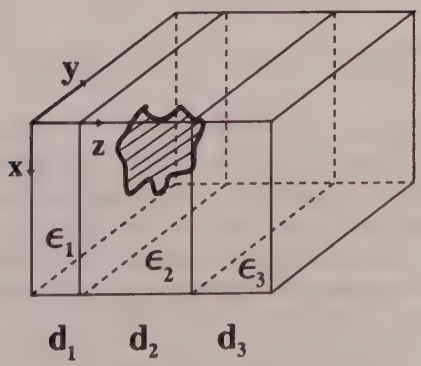


Figure 10.1 Finite metallic patch surrounded by a stratified medium.

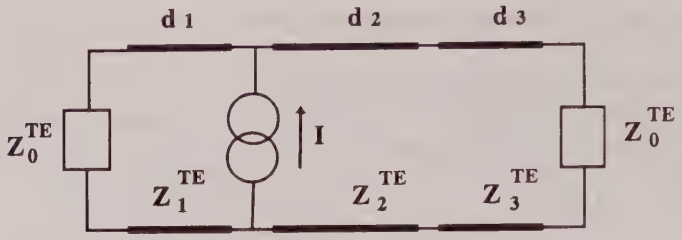


Figure 10.2 Transmission line analogy for the scatterer sketched in Figure 10.1.

$$Z_{0i}^{TM}(\mathbf{k}) = \sqrt{\frac{\mu_i}{\varepsilon_i}} \frac{\sqrt{k_{0i}^2 - k_x^2 - k_y^2}}{k_{0i}} \tag{10.15}$$

where

$$k_{0i} = \omega\sqrt{\varepsilon_i \mu_i} = \frac{2\pi}{\lambda_i}$$

The analogy is completed by the following correspondences for each mode and polarization:

$$I^{TE}(k_x,k_y) \rightleftarrows \tilde{J}^{TE}(k_x,k_y)$$

$$V^{TE}(k_x,k_y) \rightleftarrows \tilde{E}^{TE}(k_x,k_y)$$

$$Z^{TE}(k_x,k_y) \rightleftarrows \tilde{G}^{TE}(k_x,k_y)$$

$$V^{TE} = Z^{TE} I^{TE} \rightleftarrows \tilde{E}^{TE} = \tilde{G}^{TE} \tilde{J}^{TE}$$

The same correspondences are considered for the TM polarization.

The impedances Z^{TE} and Z^{TM}, which relate the voltages and currents in planes transverse to z, are obtained by using the conventional transmission line analysis. Once the impedances, and hence the Green's functions, have been found for both the TE and TM modes, it is easy to transform them into a cartesian dyad.

The equivalent transmission line analogy gives rise to the following expressions for the Green's dyad in the spectral domain:

$$\tilde{G}_{xx}(\mathbf{k}) = \frac{-1}{4\pi}\left[Z^{TM}\frac{k_x^2}{k_u^2} + Z^{TE}\frac{k_y^2}{k_u^2}\right] \tag{10.16a}$$

$$\tilde{G}_{xy}(\mathbf{k}) = \frac{1}{4\pi}\left[(Z^{\mathrm{TE}} - Z^{\mathrm{TM}})\frac{k_x k_y}{k_u^2}\right] \tag{10.16b}$$

where $k_u^2 = k_x^2 + k_y^2$.

The expressions for $\tilde{G}_{yy}(\mathbf{k})$ and $\tilde{G}_{yx}(\mathbf{k})$ can be obtained by interchanging the roles of x and y. Examination of (10.16) shows that Green's dyad in the spectral domain for multilayer media is not cumbersome and is easy to implement.

10.2.3 Periodic Structures

In the previous sections, we have illustrated how the scattering problem can be stated in the spectral domain. No restriction has been imposed on the geometry of the scatterer; we have only assumed that it is a perfect conductor with negligible thickness. Thus, the case of a periodic structure can be considered as a scatterer with a particular geometry. As a consequence of Floquet's theorem (Section 9.3), the current in the (m,n) cell of the structure can be written as a function of the current on the generating cell (the (0,0) cell):

$$\mathbf{J}_{mn}(\mathbf{r}') = \mathbf{J}_{00}(\mathbf{r}')\exp(-j(k_{0x}mT_x + k_{0y}nT_y)) = \mathbf{J}_0(\mathbf{r}')\exp(-j\mathbf{k}^i\cdot\mathbf{T}_{mn}) \tag{10.17}$$

where $\mathbf{T}_{mn}$ is the translation vector:

$$\mathbf{T}_{mn} = mT_x\mathbf{x} + nT_y\mathbf{y} \tag{10.18}$$

Substituting expression (10.17) into the general form of the EFIE (10.13), we obtain the following EFIE for periodic problems:

$$\mathbf{E}_t^i(\mathbf{r}) = \int_{S_0} G^F(\mathbf{r}-\mathbf{r}')\mathbf{J}_0(\mathbf{r}')\,ds' \qquad \text{for} \quad \mathbf{r},\mathbf{r}' \text{ on } S_0 \tag{10.19}$$

where S_0 is the surface of the generating cell of the FPS. The function G^F is a new Green's function that takes into account the periodicity, and is defined as

$$G^F(\mathbf{r},\mathbf{r}') = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} G(\mathbf{r},\mathbf{r}'+\mathbf{T}_{mn})\exp(-j\mathbf{k}^i\cdot\mathbf{T}_{mn}) \tag{10.20}$$

In the spectral domain, the Green's dyad has nearly the same general form as in (10.16). The difference, which is a consequence of periodicity, is that $\tilde{G}^F(\mathbf{k})$ vanishes except at discrete points in the spectral domain:

$$\tilde{G}^F(\mathbf{k}) = \frac{(2\pi)^2}{T_x T_y} \amalg(\mathbf{k}^{mn} - \mathbf{k}^i)\tilde{G}(\mathbf{k}) \tag{10.21}$$

where $\tilde{G}(\mathbf{k})$ is given by (10.16) and

$$\mathbf{k}^{mn} = \frac{2\pi m}{T_x}\mathbf{x} + \frac{2\pi n}{T_y}\mathbf{y} \tag{10.22}$$

Expression (10.21) can be obtained easily, as has been indicated in Chapter 2, by performing the CFT of (10.20).

10.3 DISCRETIZATION PROCESS

As in the previous chapter, we must discretize the EFIE (10.19) in order to solve it numerically. Rooftop and razor blade functions will be used as the basis and testing functions, respectively. In the scheme developed in the previous chapter, the MPIE was discretized in the spatial domain for the most part, although the discretization process was completed in the spectral domain. In the procedure presented here, we will pass to the spectral domain earlier than in the case of the MPIE.

First of all, the current is expressed in terms of rooftop functions:

$$J_\alpha(\mathbf{r}) \simeq \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} J_\alpha^D[m,n]\, T_{mn}^\alpha(x,y) \tag{10.23}$$

where α can be x or y, $J_\alpha^D[m,n]$ are the current coefficients, and T_{mn}^α are the rooftop basis functions in the α-direction with centers at the points r_{mn}^α defined in (9.45).

The EFIE (10.19) is discretized by performing a convolution over both terms with the test functions B_x and B_y, for the $\mathbf{x}$ and $\mathbf{y}$ components, respectively, and sampling with a train of delta functions. The functions B_x and B_y are the razor blade functions centered at the origin as defined in (5.30).

With this procedure, the following discrete equations can be obtained from (10.19):

$$E_x^{iD}(\mathbf{r}) = -E_{xx}^{SD}(\mathbf{r}) - E_{xy}^{SD}(\mathbf{r}) \qquad \text{for} \quad \mathbf{r} \text{ on } S_0 \tag{10.24a}$$

$$E_y^{iD}(\mathbf{r}) = -E_{yx}^{SD}(\mathbf{r}) - E_{yy}^{SD}(\mathbf{r}) \qquad \text{for} \quad \mathbf{r} \text{ on } S_0 \tag{10.24b}$$

where

$$E_{\alpha\beta}^{SD}(\mathbf{r}) = \amalg(\mathbf{r}_{mn}^\alpha)\{B_\alpha \otimes G_{\alpha\beta}^F \otimes J_\beta\} \tag{10.25}$$

We shall now transform the operator of the discrete relationships (10.24) into the spectral domain. This is done for two major reasons: first, because we have chosen to use the Green's dyad in this domain, and second, because the convolutions in (10.25) can be computed efficiently in the spectral domain by using the FFT. In the following paragraphs, we will develop the calculations with E_{xx}^{SD} and E_{xy}^{SD} in detail; the procedure for the other terms, E_{yx}^{SD} and E_{yy}^{SD}, are similar.

To compute E_{xx}^{SD}, we consider a new coordinate system, pictured in Figure 10.3. The new coordinates x and y are defined as follows:

$$x = x' - \Delta x \tag{10.26a}$$

$$y = y' - \frac{\Delta y}{2} \tag{10.26b}$$

where x' and y' are the old coordinates.

The discrete function E_{xx}^{SD} is obtained by sampling the result of the convolution operations, since it can be calculated with the help of the CFT:

$$E_{xx}^{SD}(\mathbf{r}) = \frac{1}{(2\pi)^2} \mathrm{CFT}^{-1}\{\mathrm{CFT}(\mathit{III}(\mathbf{r}_{mn}) \otimes \tilde{E}_{xx}^{Sa}\} \tag{10.27}$$

where

$$\mathbf{r}_{mn} = x^m\mathbf{x} + y^n\mathbf{y} = m\Delta x\mathbf{x} + n\Delta y\mathbf{y} \tag{10.28}$$

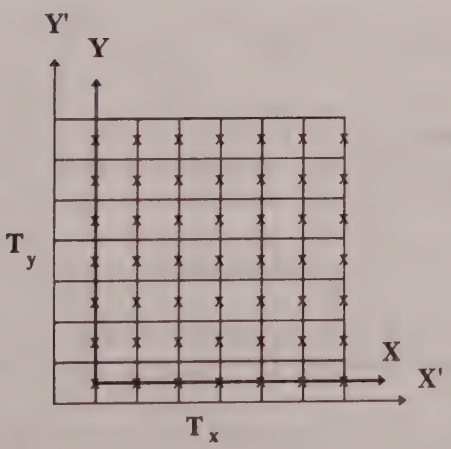


Figure 10.3 Coordinate system considered for the computation of E_{xx}^{SD}.

$$\tilde{E}_{xx}^{Sa}(\mathbf{k}) = \tilde{B}^{x}(\mathbf{k})\,\tilde{G}_{xx}^{F}(\mathbf{k})\,\tilde{J}_{x}(\mathbf{k}) \tag{10.29}$$

Taking into account the convolution properties of the sampling function, we can write

$$\tilde{E}_{xx}^{SD}(\mathbf{k}) = \frac{1}{\Delta x \Delta y} \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} \tilde{E}_{xx}^{Sa}(k_x - m'T_x^k, k_y - n'T_y^k) \tag{10.30}$$

where

$$T_x^k = \frac{2\pi}{\Delta x} \quad \text{and} \quad T_y^k = \frac{2\pi}{\Delta y}$$

To calculate the function $\tilde{E}_{xx}^{Sa}(\mathbf{k})$, we must know the Green's dyad in the spectral domain, which is obtained from (10.4), and the Fourier transform of the testing function:

$$\tilde{B}^{x}(\mathbf{k}) = \Delta x \,\mathrm{sinc}\left(\frac{k_x \Delta x}{2}\right) \tag{10.31}$$

Finally, it is necessary to know the function $\tilde{J}_x$. But we need to calculate it only at the points $\mathbf{k} = \mathbf{k}^{mn} - \mathbf{k}^{i}$, where $\tilde{G}^{F}$ does not vanish. It can be found that

$$\tilde{J}_x(\mathbf{k}^{mn} - \mathbf{k}^{i}) = \tilde{T}^{x}(\mathbf{k}^{mn} - \mathbf{k}^{i})\tilde{J}_x^{D}[m,n] \tag{10.32}$$

where $\tilde{T}^{x}(\mathbf{k})$ is the CFT of the rooftop basis functions centered at the origin:

$$\tilde{T}^{x}(\mathbf{k}) = \Delta x \Delta y \,\mathrm{sinc}^2\left(\frac{k_x \Delta x}{2}\right) \mathrm{sinc}\left(\frac{k_y \Delta y}{2}\right) \tag{10.33}$$

and the discrete function $\tilde{J}_x^{D}[m,n]$ is obtained as

$$\tilde{J}_x^{D}[m,n] = N_x N_y \,\mathrm{FFT}(J_x^{D}[i+1, j+1]) \tag{10.34}$$

The function $\tilde{E}_{xx}^{Sa}(\mathbf{k})$, is a discrete function because of the discrete nature of the Green's dyad of periodic structures. This function can be written as follows:

$$\tilde{E}_{xx}^{Sa}(\mathbf{k}), = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(k_x - (k_x^m - k_x^i))\delta(k_y - (k_y^n - k_y^i))\, W_{xx}^{a}[m,n]\, \tilde{J}_x^{D}[m,n] \tag{10.35}$$

where

$$W_{xx}^{a}[m,n] = (2\pi)^2 \Delta x \operatorname{sinc}^3\left(\frac{(k_x^m - k_x^i)}{2}\Delta x\right) \cdot \operatorname{sinc}\left(\frac{(k_y^n - k_y^i)}{2}\Delta y\right) \tilde{G}_{xx}^{F}(\mathbf{k}^{mn} - \mathbf{k}^i) \tag{10.36}$$

Finally, from (10.30) we can obtain the function $\tilde{E}_{xx}^{SD}(\mathbf{k})$:

$$\tilde{E}_{xx}^{SD}(\mathbf{k}) = \frac{1}{\Delta x \Delta y} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(k_x - (k_x^m - k_x^i)) \delta(k_y - (k_y^n - k_y^i)) W_{xx}^{p}[m,n] \tilde{J}_x^{D}[m,n] \tag{10.37}$$

where

$$W_{xx}^{p}[m,n] = \sum_{s=-\infty}^{\infty} \sum_{z=-\infty}^{\infty} W_{xx}^{a}[m - sN_x, n - zN_y] \tag{10.38}$$

Expressions (10.37) and (10.38) are deduced from (10.30) in a way similar to the deduction presented in the previous chapter (Section 9.6). We can conclude, as in that case, that $W_{xx}^{p}[m,n]$ is a discrete and periodic function of period (N_x,N_y), as is $\tilde{J}_x^{D}[m,n]$. Therefore, $\tilde{E}_{xx}^{SD}(\mathbf{k})$, is a periodic function whose period is (T_x^k, T_y^k). The series (10.38) converges rapidly and only a few terms must be taken into account.

Finally, we can compute the scattered field in the real domain $\tilde{E}_{xx}^{SD}(\mathbf{k})$ by performing the inverse CFT of (10.37) with the help of the FFT.

$$E_{xx}^{SD}(\mathbf{r}) = \frac{1}{(2\pi)^2} \sum_{m=0}^{N_x} \sum_{n=0}^{N_y} \delta(x - x^m)\delta(y - y^n) \operatorname{FFT}^{-1}(W_{xx}^{p} \tilde{J}_x^{D}) \tag{10.39}$$

The other terms in (10.24) can be obtained in a similar way. For instance, the $E_{xy}^{SD}(\mathbf{r})$ component of the total scattered field is calculated by considering a new coordinate system (Figure 10.4):

$$x = x' - \frac{\Delta x}{2} \tag{10.40a}$$

$$y = y' - \Delta y \tag{10.40b}$$

Then E_{xy}^{SD} is computed using the relationship

$$E_{xy}^{SD}(\mathbf{r}) = \frac{1}{(2\pi)^2} \operatorname{CFT}^{-1}\{\operatorname{CFT}\{\mathit{III}(\mathbf{r}^{mn} - \mathbf{d})\} \otimes \tilde{E}_{xy}^{Sa}\} \tag{10.41}$$

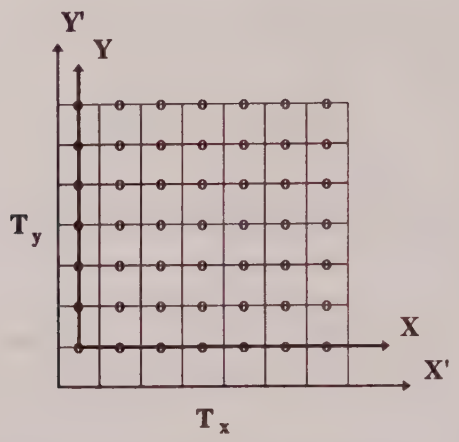


Figure 10.4 Coordinate system considered for the computation of E_{xy}^{SD}.

$$\mathbf{d} = \left(\frac{\Delta x}{2}\right)\mathbf{x} - \left(\frac{\Delta y}{2}\right)\mathbf{y} \tag{10.42}$$

$$\tilde{E}_{xy}^{Sa}(\mathbf{k}) = \tilde{B}_x(\mathbf{k})\,\tilde{G}_{xy}^{F}(\mathbf{k})\,\tilde{J}_y(\mathbf{k}) \tag{10.43}$$

Proceeding similarly to the case of $\tilde{E}_{xx}^{SD}(\mathbf{k})$, we obtain

$$E_{xy}^{SD}(\mathbf{r}) = \frac{1}{(2\pi)^2}\sum_{m=0}^{N_x}\sum_{n=0}^{N_y}\delta\left(x - \left(x^m + \frac{\Delta x}{2}\right)\right)\delta\left(y - \left(y^n - \frac{\Delta y}{2}\right)\right)\mathrm{FFT}^{-1}(W_{xy}^{p}\tilde{J}_y^{D}) \tag{10.44}$$

where

$$W_{xy}^{p}[m,n] = \exp\left(jk_x^m\frac{\Delta x}{2}\right)\exp\left(-jk_y^n\frac{\Delta y}{2}\right) \cdot \sum_{s=-\infty}^{\infty}\sum_{z=-\infty}^{\infty}(-1)^{-s}(-1)^{-z}\,W_{xy}^{a}[m - sN_x, n - zN_y] \tag{10.45}$$

$$W_{xy}^{a}[m,n] = (2\pi)^2\Delta x\,\mathrm{sinc}^2\left((k_x^m - k_x^i)\frac{\Delta x}{2}\right)\mathrm{sinc}^2\left((k_y^n - k_x^i)\frac{\Delta y}{2}\right) \cdot \tilde{G}_{xy}^{F}(\mathbf{k}^{mn} - \mathbf{k}^i) \tag{10.46}$$

10.4 SOME NUMERICAL RESULTS AND APPLICATIONS

As we mentioned in the introduction to this chapter, the dielectric materials in which FPSs are embedded are important not only because they strengthen and protect the devices, but also because they can be used to control some electromagnetic characteristics of the structure. In the following sections, we will attempt to illustrate some of the response variations that arise due to the influence of the dielectric.

10.4.1 Induced Currents

The induced currents on the metallic surface of the generating cell are the values initially obtained by applying the CG-FFT scheme previously presented. We will illustrate the variations in the currents due to the dielectric support of a periodic array of metallic patches whose geometrical parameters are shown in Figure 10.5.

The electric field is normally incident on the structure and has a value of 120π (V/m) polarized in the x-direction. Two cuts of the surface current in the x-direction J_x are shown in Figure 10.6. Note that the current goes to zero at the edges perpendicular to the flux direction, while it rises asymptotically at the edges parallel to the x-direction. This behavior is thoroughly explained in [28]. The values for the free-standing case that are obtained by using the method presented in the previous sections are compared with those obtained by Rubin and Bertoni by using an MM

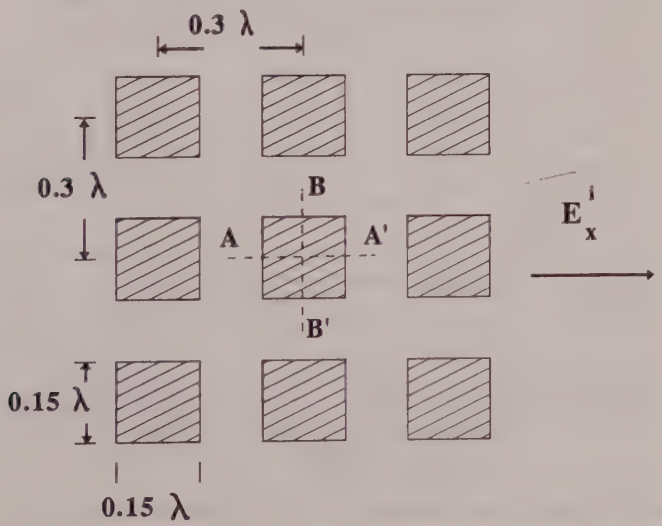


Figure 10.5 FPS composed of square metallic patches.

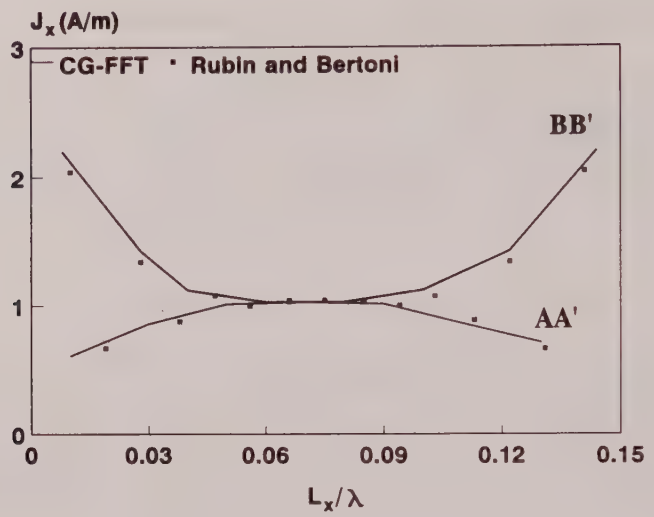


Figure 10.6 Two cuts of the surface current J_x for the free-standing case.

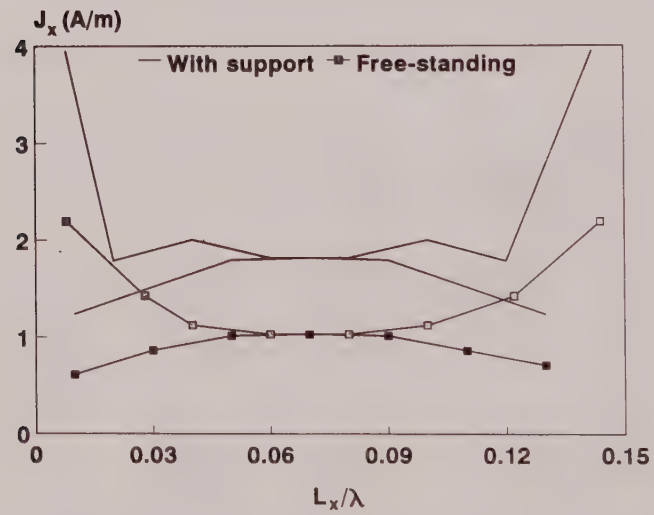


Figure 10.7 Induced current in the free-standing case compared with the current in a structure with dielectric support.

scheme [20]. Figure 10.7 presents a comparison of the previous results with those obtained when considering a dielectric support with $\varepsilon_r = 2.5$ and a thickness of 0.1λ. It can be observed that in this case the average value of the current increases.

10.4.2 Transmission and Reflection Coefficients

As we pointed out in the previous chapter, the transmission and reflection coefficients are the most significant values for characterizing the performance of an FPS. The dielectric surrounding the metallic array produces variations in these coefficients, thus providing a means of controlling the frequency response of the structure.

As an illustrative example, we present results for normal incidence corresponding to a structure proposed in [29], which is sketched in Figure 10.8. This figure also shows the power reflection coefficient of the structure without dielectrics. The power reflection coefficients for the same array on a dielectric support 0.3 cm thick, with relative dielectric constants of 2 and 4, are shown in Figures 10.9 and 10.10, respectively.

For the free-standing array, the reflection maximum appears when the wavelength is equal to the dipole length. It is interesting to observe how the resonance frequency is shifted to lower frequencies by increasing the relative dielectric constant.

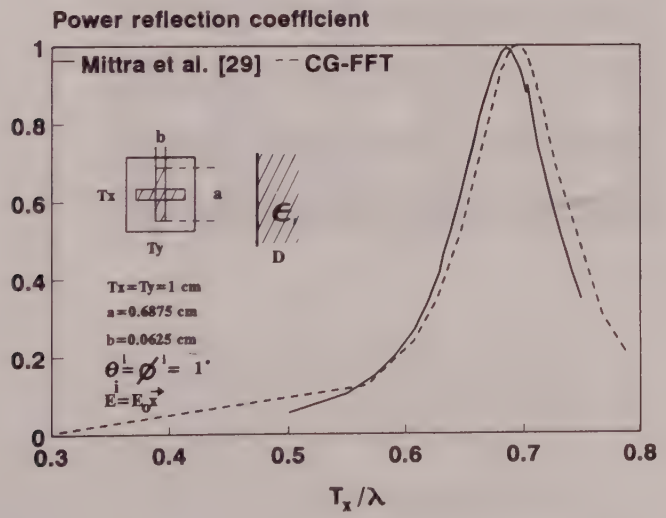


Figure 10.8 Power reflection coefficient of the structure in the sketch. Free-standing case.

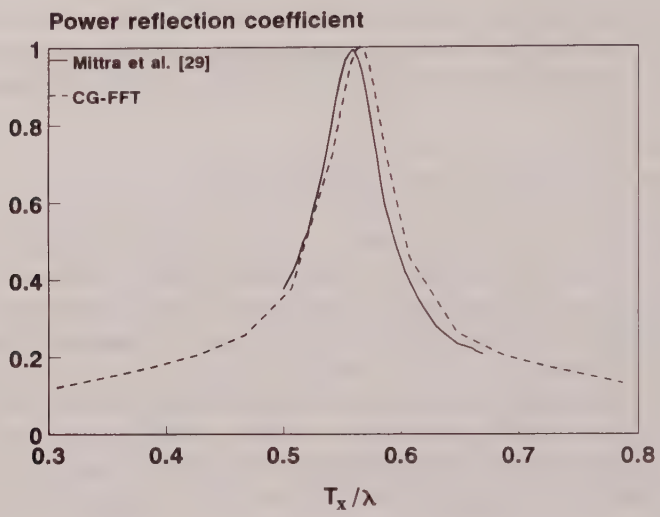


Figure 10.9 Power reflection coefficient of the array sketched in Figure 10.8 considering a dielectric support 0.3 cm thick, with a relative dielectric constant of 2.

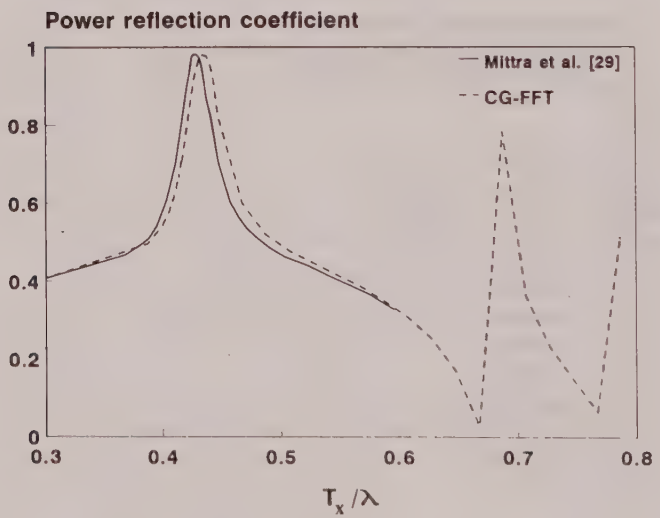


Figure 10.10 The same structure as Figures 10.8 and 10.9, but with relative dielectric constant of 4.

It should also be noted that there is good agreement between the results obtained with the CG-FFT method and the values in [29].

As an example of an inductive screen, we will consider a structure previously analyzed by Luebbers and Munk [4]. This structure consists of rectangular holes perforated in a metallic screen whose geometrical parameters are shown in Figure 10.11. The values of the reflection coefficient are depicted in Figure 10.12 for normal incidence and $D = 0.35$. Figure 10.13 shows the values for the case where $D = 0.7$ cm. It can be observed that when the dielectric is made thicker, the transmission bandwidth increases. This relation between dielectric thickness and bandwidth can be used to control the frequency response of the FPS.

Another point of interest is how the variations in bandwidth and resonance frequency depend on the incidence direction and polarization. This subject has been dealt with in a very interesting paper by Munk [6]. The author shows that the bandwidth can be stabilized by using a stratified dielectric medium in front of and behind the periodic screen.

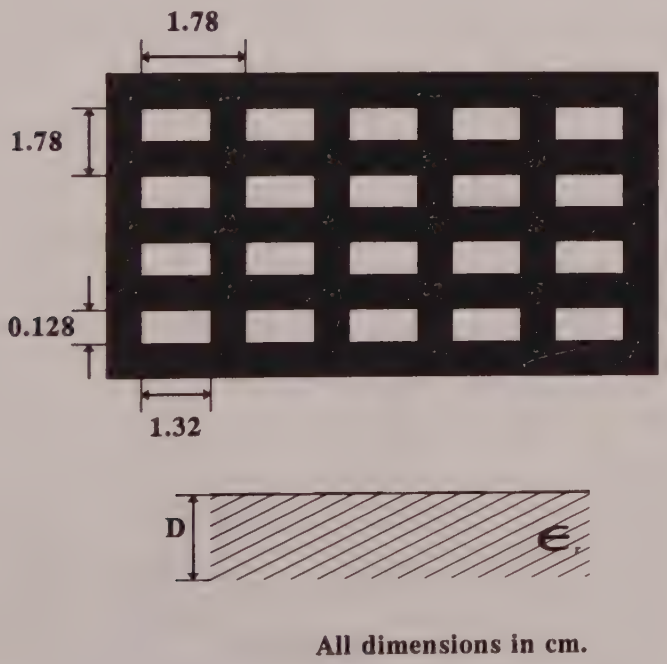


Figure 10.11 FPS composed of rectangular holes perforated in a metallic screen.

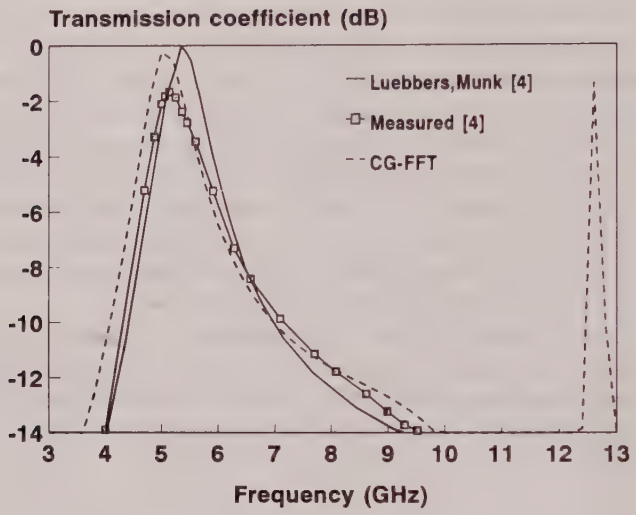


Figure 10.12 Reflection coefficient for normal incidence. $D = 0.35$ cm.

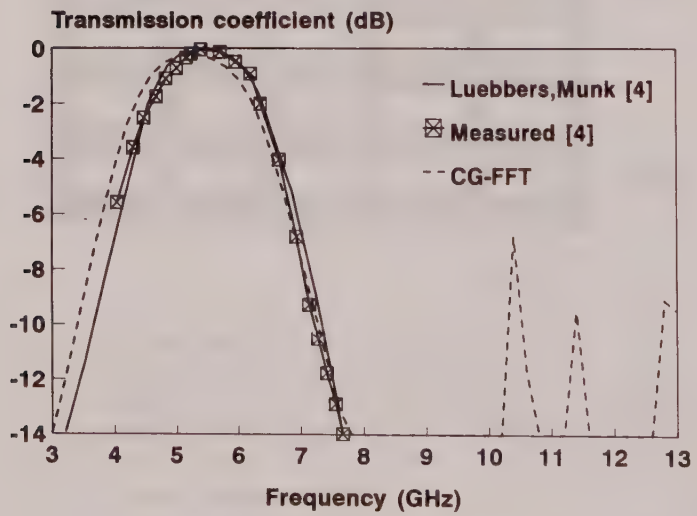


Figure 10.13 Reflection coefficient for normal incidence. $D = 0.7$ cm.

10.4.3 Some Comments on Convergence

The convergence of the CG method when solving periodic structure problems depends on the number of basis and testing functions, on the number of Floquet's harmonics considered, and perhaps on the particular implementation of the numerical method. Many authors have studied this subject (see, for example, [24,29–32]).

The convergence of the CG method to accurate values of currents depends on the convergence of the series $W^p_{\alpha\beta}[m,n]$ in (10.38) and (10.45). This spectral series converges quite rapidly when rooftop basis functions and razor blade testing functions are used. For instance, values of $s = -1,0,1$ and $z = -1,0,1$ are usually considered.

The number of basis functions that must be chosen depends on the geometry: the number of subdomains must be large enough to represent the geometry accurately. The electrical size of the cell must also be considered. As a rule of thumb, about eight basis functions per wavelength must be used. These rules are very simple and intuitive, but they allow the CG-FFT method to be used without having to do a convergence study for every case considered, which is very important from the user's point of view.

Finally, we consider the structure sketched in Figure 10.14 as an example. This structure has been analyzed by using 32 × 32 subdomain basis functions and the

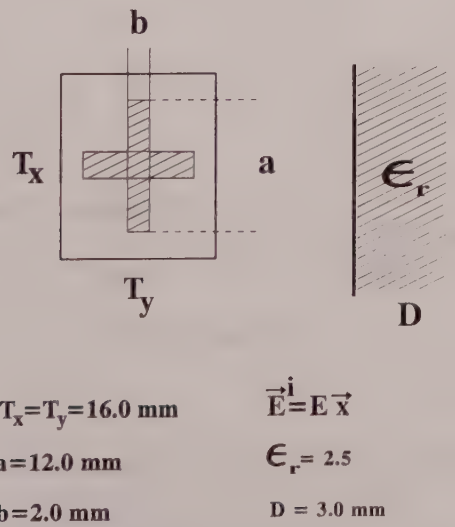


Figure 10.14 Geometrical parameters of an FPS composed of cross-dipoles photoetched on a dielectric support.

following values in the spectral series: $s = -1,0,1$ and $z = -1,0,1$. Note that the square subdomains have a length of 0.05 cm and that all the dimensions of the cell are integer multiples of that length. In a general case, this will not be true and the possible discretization error must be taken into account.

We have built this structure and used it for measurements, approximating the infinite array by an array of 15 × 21 cells. Figure 10.15 shows the calculated and measured values of the magnitude of the transmission coefficient, and Figure 10.16 depicts the values of the phase. The agreement between the numerical and measured values is within the margin of experimental error. Figure 10.17 shows the variation in the error versus the number of iterations for a frequency of 12 GHz. The number of iterations required to obtain an error below 1% is plotted versus frequency in Figure 10.18. It is worth noting how this number increases at frequencies near resonance.

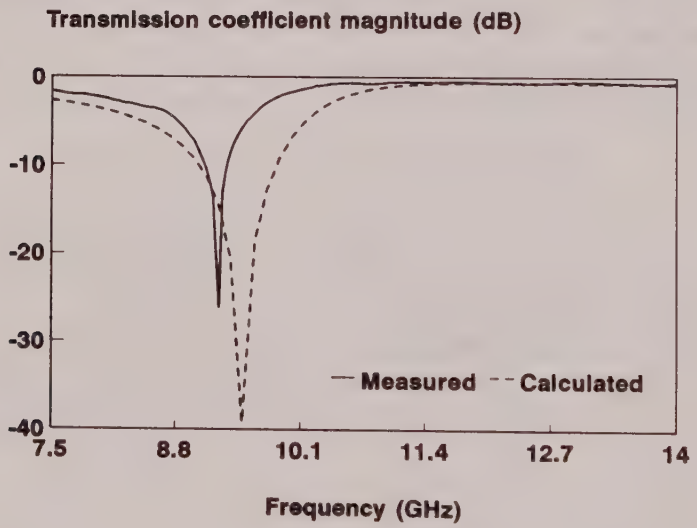


Figure 10.15 Magnitude of the calculated and measured transmission coefficient for the structure in Figure 10.14.

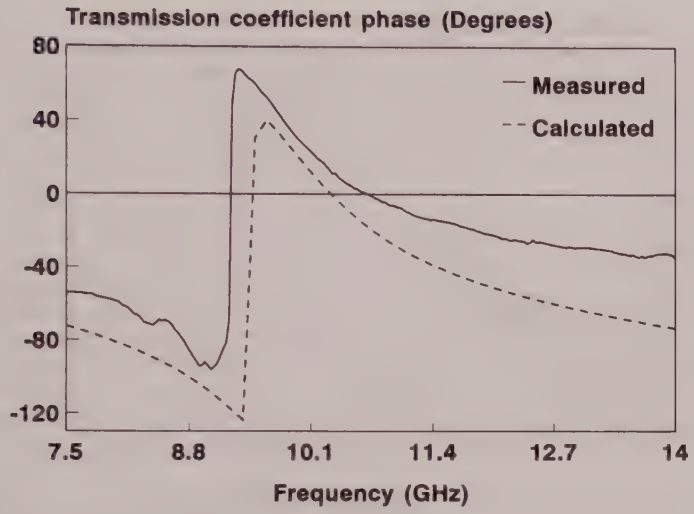


Figure 10.16 Phase of the transmission coefficients in Figure 10.15.

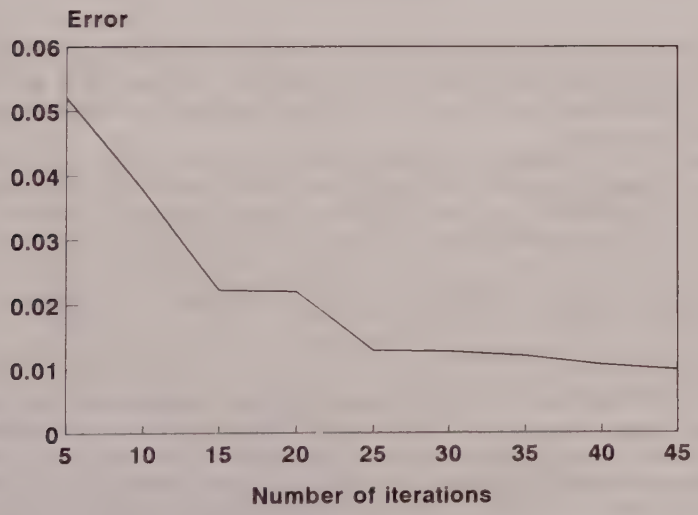


Figure 10.17 Error versus the number of iterations.

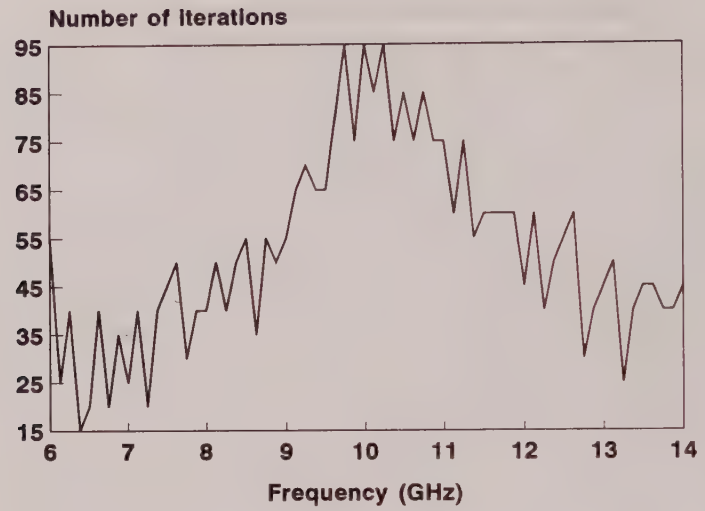


Figure 10.18 Number of iterations required for an error below 1% versus frequency.

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Chapter 11
Finite-Sized Conducting Patches in Multilayer Media

11.1 INTRODUCTION

The design of flat structures such as microstrip antennas, microwave integrated circuits (MIC) and microwave monolithic integrated circuits (MMIC), and feeding networks has become a very important topic in current electromagnetics. Due to the complexity of these devices, efficient numerical tools are required for their analysis. As in the case of the free-space structures studied in previous chapters, one of the most popular techniques for analyzing these devices is the MM, whose advantages and limitations are well known [1–3]. The CG-FFT method is a good and efficient alternative for the analysis of these kinds of structures [4,5]. Some other techniques that have been employed for their analysis are described in [6–8]. In this chapter, we use a spectral formulation together with the CG-FFT method to analyze multilayer structures having a metallic patch of arbitrary geometry located at the interface between two of the layers.

Chapters 3 and 5 underlined the importance of the Green's function problem and its solution in order to obtain a suitable equation for the original problem. In the present case, it is not easy to obtain analytical solutions of the Green's function problem in the real space. In the spectral domain, on the other hand, simple and general expressions of this function can be easily found [9,10]. Based on these expressions, it is possible to develop a general mathematical procedure for obtaining discrete forms of both the direct and adjoint operators. The process is similar in part to that described in Chapters 9 and 10 when developing formulations for periodic problems: that is, the use of CFTs and Floquet's theory to obtain approximate semi-analytical discrete expressions to compute the convolutions of the integrodifferential operator equations. However, in the present case, an additional smoothing will be performed in the spectral domain in order to limit the extension of the Green's function

in the space domain [11,12]. In this way, a space-limited problem can be analyzed by taking advantage of the expressions developed for periodic problems. These expressions will be represented in terms of DFTs, and the corresponding CG-FFT scheme will be obtained. With this procedure, a variety of multilayer problems can be analyzed by considering the analytical expressions of their Green's functions in the spectral domain.

Chapter 11 is organized as follows. In Section 11.2, the problem is presented in an integral operator form. Section 11.3 deals with the formulation of an equivalent periodic problem, which involves defining a Green's function that is windowed in the space domain so as to take into account the finite size of the metallic patch. Several window functions will be described in order to understand their role in the analytical and numerical implementation of the approach. Section 11.4 gives some details about the computation in the spectral domain of the equivalent periodic Green's functions, along with a discussion of how the window functions should be chosen. Finally, Sections 11.5 to 11.7 discuss results for current distributions, RCS, and scattering parameters in microstrip discontinuities.

11.2 FORMULATION OF THE PROBLEM

The problem to be solved is presented in Figure 11.1. An arbitrary-shaped, perfectly conducting patch of zero thickness is located at the interface between two of the layers in an arbitrary planar multilayer structure parallel to the $z = 0$ plane.

The metallization is enclosed in a rectangle of dimensions L_x and L_y, the maximum sizes of the patch in the x- and y-coordinates. The current $\mathbf{J}$ induced on the metallization by an arbitrary impressed field $\mathbf{E}^i$ can be obtained by solving the operator equation

$$\mathbf{n} \times \mathbf{E}^i = \mathbf{n} \times \int_S \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, ds' \qquad \mathbf{r} \in S \tag{11.1}$$

where S denotes the surface of the patch and G the dyadic space-domain Green's function for the structure in Figure 11.1. The unit vector $\mathbf{n}$ normal to the plane containing the metallization coincides with $\mathbf{z}$ in this formulation. Expression (11.1) has been formulated in the space domain as a result of applying the corresponding boundary conditions to the patch (tangential component of the total field null on S). Since G is known in the spectral domain, the operator will be evaluated in this domain, although we shall return to the space domain to satisfy the boundary conditions of (11.1). The approximations used in Section 5.1.1 for the free-space case are used again here, which means that $\mathbf{J}$ represents the total equivalent current as described there (sum of the top and bottom surface currents).

If (11.1) is compared to (10.13), it is evident that the equations are similar to each other. In fact, the only difference between the two problems is that the present

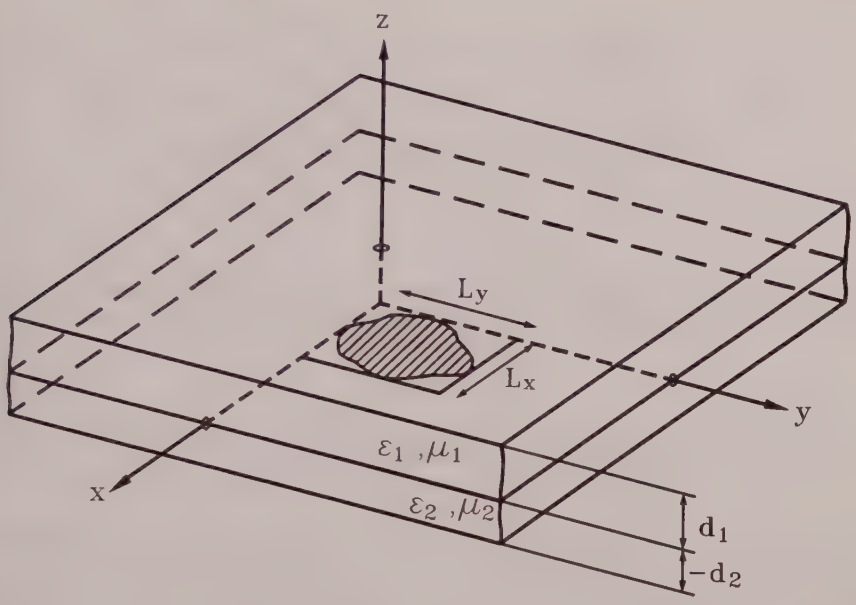


Figure 11.1 Schematic representation of a two-layer structure. The finite-sized patch lies in the $z = 0$ plane.

problem is finite-sized because all the currents can be enclosed by a rectangle of sides L_x and L_y, whereas in the problem of Chapter 10, the currents are periodic and therefore unlimited. In Chapter 10, we took advantage of the periodic nature of the problem. Since the CFT of the operator equation was discrete, this equation could be discretized quite accurately in the spectral domain, where simple analytical expressions are available for evaluating the Green's function.

Taking into account the similarity between the periodic and the finite-sized problems, we shall now analyze the problem under consideration in this chapter by adapting the technique developed for periodic structures in Chapter 10. This will be done in the next section, following the technique outlined in Chapter 3 for converting a finite-sized problem into an equivalent periodic one.

11.3 FORMULATION OF THE EQUIVALENT CONTINUOUS OPERATOR EQUATION

In order to illustrate graphically how an equivalent periodic problem can be obtained, let us consider the 1D real-domain convolution shown in Figure 11.2(a). If we are interested in evaluating the convolution

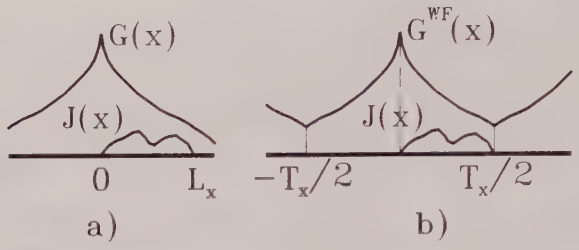


Figure 11.2 (a) Convolution between the source $J(x)$ and the unlimited Green's function $G(x)$; (b) convolution between the source $J(x)$ and the equivalent periodic Green's function $G^{WF}(x)$.

$$\phi(x) = \int_{-\infty}^{\infty} J(x')\, G(x - x')\, dx' \tag{11.2}$$

in the interval $0 < x < L_x$, where the source is located, we can use the equivalent periodic function G^{WF} indicated in Figure 11.2(b). This function is obtained by first multiplying the original Green's function $G(x)$ by a window function $W(x)$, which results in G^W:

$$G^W(x) = G(x)W(x) \tag{11.3}$$

Then, G^{WF} is formed by replaying $G^W(x)$ at regular intervals:

$$G^{WF}(x) = \sum_{k=-\infty}^{\infty} G^W(x - kT_x) \tag{11.4}$$

G^{WF} can also be obtained from a convolution of G^W with a train of delta functions:

$$G^{WF}(x) = G^W(x) \otimes \mathit{III}_{T_x}(x) \tag{11.5}$$

$$\mathit{III}_{T_x}(x) = \sum_{k=-\infty}^{\infty} \delta(x - kT_x) \tag{11.6}$$

where the symbol $\otimes$ indicates the convolution operator and III is a train of delta functions.

Figures 11.3 and 11.4 picture, in the real and spectral domains, respectively, all the steps required to obtain the equivalent periodic function G^{WF} from the original unlimited Green's function. It should be noted that $G(x)$ can be replaced by G^{WF} if

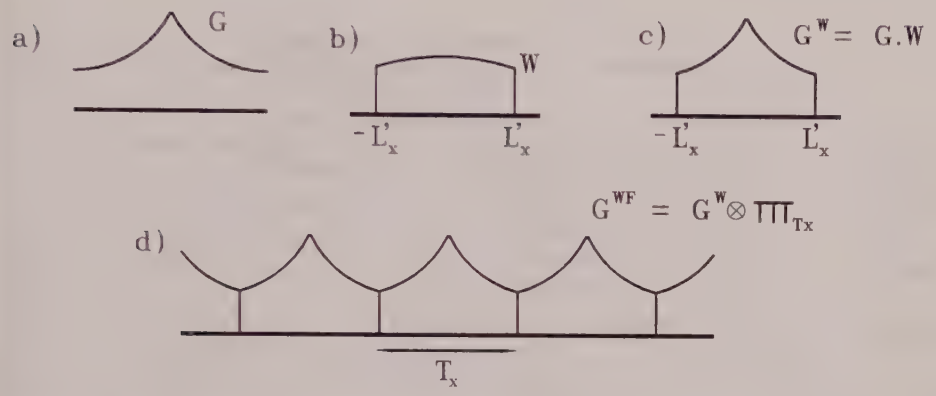


Figure 11.3 The equivalent periodic Green's function $G^{WF}(x)$ is obtained by multiplying $G(x)$, the unlimited Green's function of (a), by $W(x)$, the window function of (b), and by replaying G^W, the resulting space-limited function of (c), at regular points in (d).

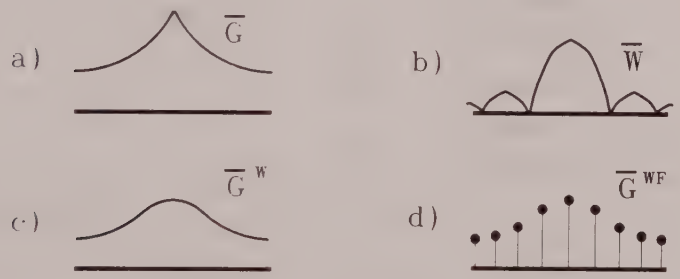


Figure 11.4 Spectral domain shape of the functions in Figure 11.3.

$2L'_x$, the length of $W(x)$, is greater than twice L_x, the length of the source region in which we are interested in evaluating the convolution. Moreover, $W(x)$ should ideally be equal to 1 in the interval $(-L_x,L_x)$ in order to compute the convolution accurately. If the value of $W(x)$ is not exactly 1 in this interval and 0 elsewhere, some error will be introduced in the computation of the convolution by means of an equivalent periodic function. Table 11.1 shows some of the most common 1D window functions. The real- and spectral-domain representations of these window functions are depicted in Figure 11.5. The convolution is computed exactly in the interval $(0,L_x)$ if $W(x)$ is made equal to the square function defined in Table 11.1.

Returning now to the 2D problem of Figure 11.1, and assuming that we are interested in evaluating the convolution operator only in the rectangle of dimensions

Table 11.1
Analytical Expressions of Some Normalized 1D Window Functions

Window	*Space Domain (x)*	*Spectral Domain (K)*
Squared	$1 \quad x \leq \lvert 1 \rvert$ $0 \quad$ otherwise	$2\,\mathrm{sinc}(K)$
Parzen	$1 - \lvert x \rvert \; x \leq \lvert 1 \rvert$ $0 \quad$ otherwise	$2\,\mathrm{sinc}^2(K)$
Hanning	$0.5(1 - \cos(\pi x + \pi)) \; x \leq \lvert 1 \rvert$ $0 \quad$ otherwise	$\mathrm{sinc}(K)$ $+0.5\,\mathrm{sinc}(K - \pi)$ $+0.5\,\mathrm{sinc}(K + \pi)$
Welch	$1 - \alpha x^2 \quad x \leq \lvert 1 \rvert$ $0 \quad$ otherwise	$2(1 - \alpha)\,\mathrm{sinc}(K)$ $+4\alpha \sin(K)/K^3$ $-4\alpha \cos(K)/K^2$

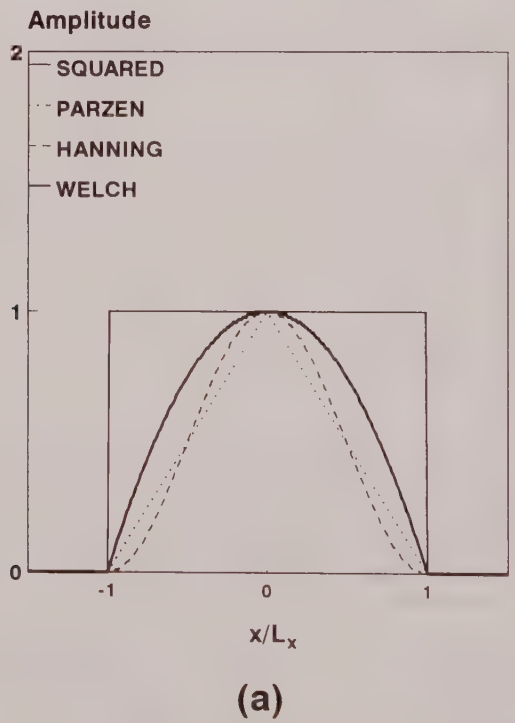


Figure 11.5 (a) Real-domain and (b) spectral-domain representations of the window functions of Table 11.1.

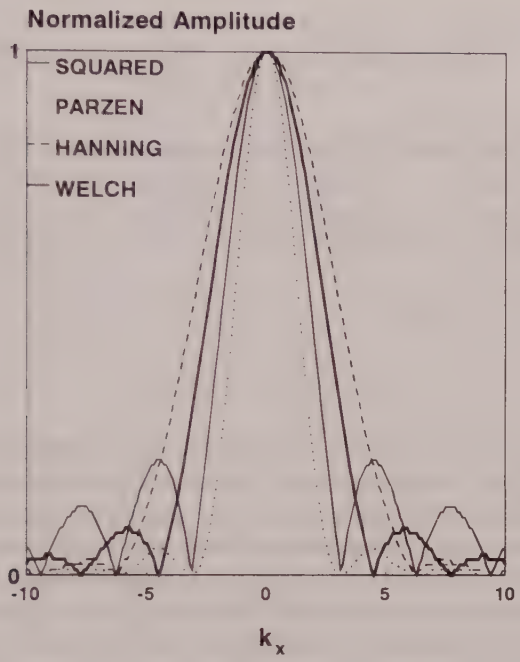


Figure 11.5 (Continued)

L_x and L_y enclosing the metallic patch, we can replace $G(\mathbf{r})$ in (11.1) with the equivalent periodic function $G^{WF}(\mathbf{r})$, which has a periodic cell of size T_xT_y defined as follows:

$$G^{WF}(\mathbf{r}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} G^{W}(\mathbf{r} - \mathbf{r}_{mn}) \tag{11.7}$$

where

$$\mathbf{r}_{mn} = mT_x\mathbf{x} + nT_y\mathbf{y} \tag{11.8}$$

$$T_x = 2L_x \tag{11.9a}$$

$$T_y = 2L_y \tag{11.9b}$$

and

$$G^W(\mathbf{r}) = W(\mathbf{r})\, G(\mathbf{r}) \tag{11.10}$$

The periodic function $G^{WF}(\mathbf{r})$ is now expressed in terms of the windowed function $G^W(\mathbf{r})$ defined in (11.10), where $W(\mathbf{r})$ is a 2D function that must be as close as possible to a 2D pulse (i.e., a function equal to 1 for $|x| \leq T_x$ and $|y| \leq T_y$, and to 0 elsewhere). If $W(\mathbf{r})$ is made equal to this 2D pulse, then (11.1) is exactly equivalent to

$$\mathbf{z} \times \mathbf{E}^i = \mathbf{z} \times \int_S \mathbf{J}(\mathbf{r}')\, G^{WF}(\mathbf{r},\mathbf{r}')\, ds' \qquad \mathbf{r} \in \mathbf{S} \tag{11.11}$$

This equation is very similar to those of (10.19), the fundamental equation relation in Chapter 10 for the study of periodic patches in a multilayer medium. In fact, (10.19) and (11.11) are both applied over the metallic patches of a single cell of a periodic lattice. The Green's functions in both cases are periodic functions, given by (10.20) and (11.7), respectively. The spectral representations of both functions are obtained by sampling a continuous function with a 2D train of delta functions, as can be seen in (10.21) and in the following expression for the equivalent periodic Green's function:

$$\tilde{G}_{xx}^{WF}(\mathbf{k}) = \frac{(2\pi)^2}{T_x T_y}\, \mathit{\Pi}(\mathbf{k}^{mn})\, \tilde{G}^W(\mathbf{k}) \tag{11.12}$$

where $\mathbf{k}^{mn}$ is given by (10.22).

Therefore, it seems evident that here in Chapter 11 we can use the same CG-FFT scheme as the one in Chapter 10 to analyze the problem. The only difference is that the spectral Green's function to be sampled here, $G^W(\mathbf{k})$ in (11.12), is somewhat more complex than the corresponding function in (10.21). In order to avoid unnecessary repetition in this chapter, we will describe only the special details involved in computing the windowed Green's function $G^W(\mathbf{k})$ in the spectral domain.

11.4 COMPUTATION OF THE WINDOWED GREEN'S FUNCTION

In the scheme proposed, the windowed Green's function G^W must be computed in the spectral domain. From (11.10) it can be seen that this computation involves a convolution particularized at the discrete points $\mathbf{k}^{mn}$. Mathematically,

$$\tilde{G}^W(\mathbf{k}^{mn}) = \frac{(2\pi)^2}{T_x T_y} \int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} \tilde{G}(\mathbf{k}^{mn} - \mathbf{k}) \tilde{W}(\mathbf{k})\, dk_x \tag{11.13}$$

If we consider a window function with a main lobe at the origin like those in Table 11.1, the convolution integral can be approximated by

$$\tilde{G}^W(\mathbf{k}^{mn}) = \frac{(2\pi)^2}{T_x T_y} \int_{-S_y}^{S_y} \mathrm{d}k_y \int_{-S_x}^{S_x} \tilde{G}(\mathbf{k}^{mn} - \mathbf{k})\tilde{W}(\mathbf{k})\, \mathrm{d}k_x \tag{11.14}$$

in which the integrals are reduced to the spectral bands over which the window function does not vanish.

For the present 2D case, we can choose a window function W defined by the product of a pair of 1D functions, such as those described in Table 11.1:

$$W(\mathbf{r}) = W_x(x) \cdot W_y(y) \tag{11.15}$$

In the spectral domain, these functions have a well-defined main lobe and a set of decreasing lateral lobes. If the square window is used, (11.1) and (11.11) have exactly the same solution, but in the spectral domain this function has lobes that decrease very slowly, which means that it is not feasible to perform a large truncation in the integration limits of (11.14). The triangle window has very quickly decreasing lobes but G^W deviates significantly from G and, consequently, the solution of (11.11) may not be approximately equal to that of the original equation (11.1). Hanning and Welch's windows represent a trade-off between functions with quickly decreasing secondary lobes and functions for which G^W and G can reasonably be considered as equal over nearly all the points where the window does not vanish. When using these functions, the limits of the main lobes can be taken as the integration limits S_x and S_y, and the results obtained are quite accurate, while, at the same time, the CPU time involved in computing (11.14) is very satisfactory. The authors have experimented with these functions, studying parameters such as (1) agreement of the CG-FFT results with reliable solutions, (2) convergence rates of the solutions, (3) stability of the solutions with the number of samples considered to represent the currents, and (4) CPU time. The best results were obtained when using Welch's function, considering the main lobe for the integral in (11.14) and using a Gauss quadrature technique for the numerical implementation of the integration. The number of quadrature points used in the integration was 2×2, 8×8, or 32×32, depending on whether the singularities of the modified spectral Green's function were far away from, close to, or within the integration band, respectively.

All results presented in this chapter were obtained by using Welch's function, whose expression may be recalled by referring to Table 11.1. With this function, the deformation of the original Green's function can be controlled by means of the parameter α. If $\alpha = 1$, the function behaves as is shown in Figure 11.5(a). If $\alpha = 0$, the square function is obtained. The spectral representation of Welch's function provides a good compromise compared with the other functions, since the side lobes decay quite fast and only the main lobe has to be considered when computing the integral in (11.14). Note that both the width of the main lobe and the height of the

lateral lobes depend on α ($\alpha = 1$ for the result shown in Figure 11.5(b)). All results in this chapter have been obtained with $\alpha = 1$.

11.5 SOME RESULTS FOR INDUCED CURRENTS

In order to verify the CG-FFT scheme proposed in this chapter, we have studied the currents on different conducting plate structures. The plates have been considered in free space, which is a particular case of a multilayer structure, because reliable results are available for other methods. There is good agreement between the results obtained here and those for identical structures in the literature, such as the results found for the square plate in [13] using an MM scheme, or those obtained for the circular disk and other curved-edge surfaces by means of a space-domain formulation (presented in Chapter 5). Since the results are so similar, we will only discuss the case of the square plate in free space. This case allows us to present an example of convergence rates as well as a brief discussion of the computation of the series in (10.38).

11.5.1 Square Plate in Free Space

Consider a square plate with sides $L_x = L_y = 1\lambda$ illuminated by a normal incident x-polarized plane wave ($\theta^i = 0$ deg, $\phi^i = -90$ deg), as shown in Figure 11.6.

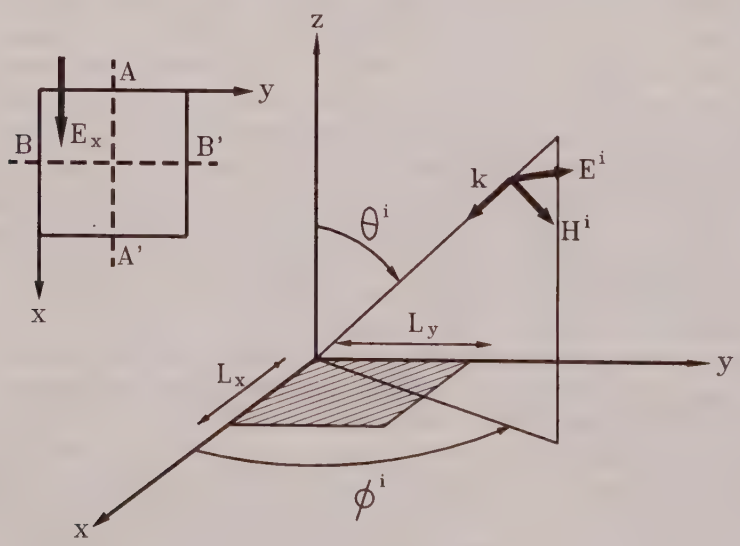


Figure 11.6 Schematic representation of the square plate problem.

Figure 11.7 shows the induced currents obtained by using the CG-FFT scheme suggested in this chapter in terms of the number of spectral terms considered to compute a series like the one of (10.38) in the following approximate form:

$$W_{xx}^{P}(m,n) = \sum_{p=N_{pa}}^{N_{pb}} \sum_{q=N_{qa}}^{N_{qb}} W_{xx}^{a}(m - pN_x, n - qN_y) \tag{11.16}$$

These results show that the ripple in the current shape can be avoided reasonably well by using only four terms in the summation (i.e., $N_{pa} = N_{qa} = 0$, $N_{pb} = N_{qb} = 1$), and the resulting CPU time is very short. If a larger number of terms is used, 36 for instance ($N_{pa} = N_{qa} = -2$, $N_{pb} = N_{qb} = 3$), the improvement is not great, and the CPU time becomes so large that the scheme is inefficient in practice. The results shown in Figure 11.8 (using $N_x = N_y = 7$) were computed using four terms, which lends support to the preceding discussion. Finally, Figure 11.9 presents a convergence study of the relative error versus number of iterations in terms of the number of samples. Note that these curves are very close to those obtained in Chapter 5.

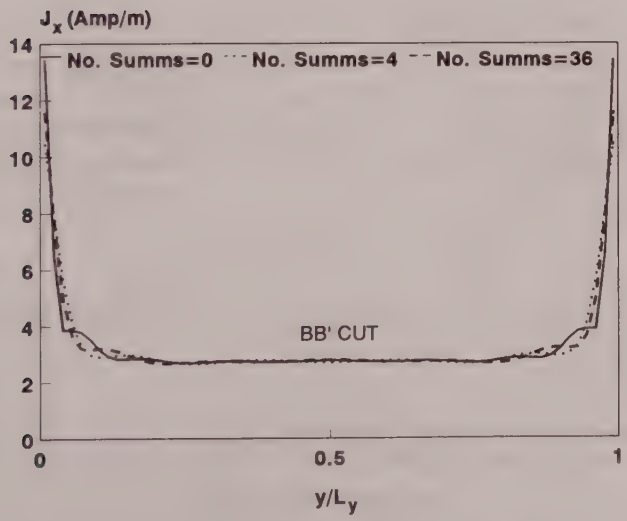


Figure 11.7 Current density in terms of the number of spectral sums in (10.38).

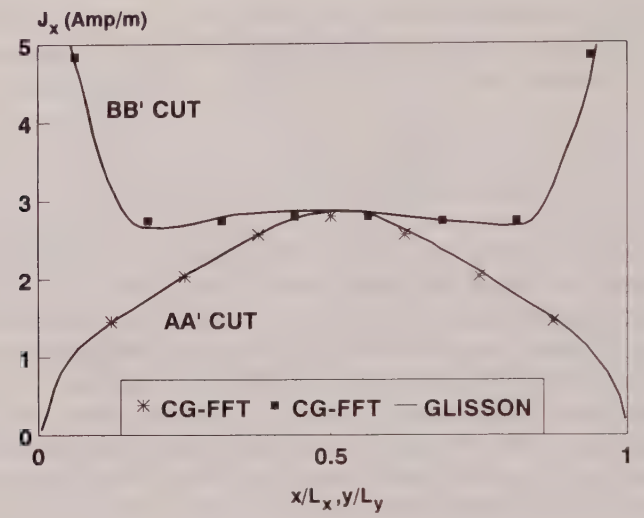


Figure 11.8 Comparison of results for the currents obtained with the CG-FFT and with the MM.

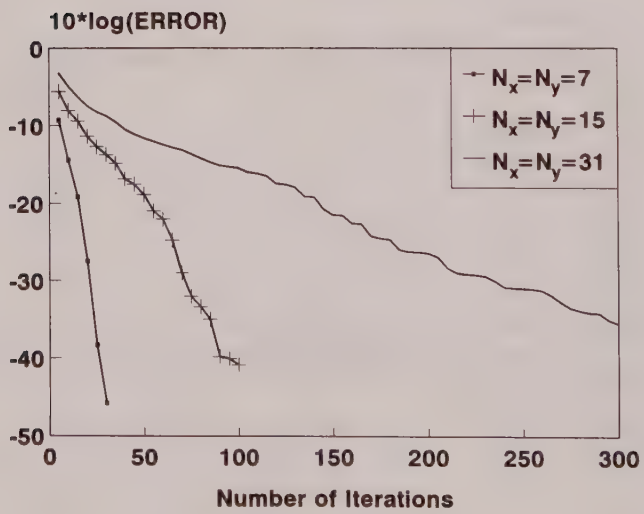


Figure 11.9 Relative error versus number of iterations for the square plate problem.

11.6 APPLICATION TO SCATTERING PROBLEMS

For the analysis of scattering and RCS problems, the metallic patches are assumed to be on the upper face of the multilayer medium; in other words, they are on the boundary between an upper vacuum half space and the first dielectric layer. Assuming the following incident wave in the upper half space,

$$\mathbf{E}^{\text{inc}}(r) = \mathbf{E}_0^{\text{inc}} \exp(-jk_x^{\text{inc}} x - jk_y^{\text{inc}} y - jk_z^{\text{inc}} z) \tag{11.17}$$

we can obtain the impressed field in the plane of the patch by solving the problem of a plane wave incident on a multilayer medium without metallic patches. This impressed field is given by

$$\mathbf{E}^i(x,y) = \mathbf{E}_0^i \exp(-jk_x^{\text{inc}} x - jk_y^{\text{inc}} y) \tag{11.18}$$

where

$$\mathbf{E}_0^i = \mathbf{E}_0^{\text{inc}} \frac{2Z_L(k^{\text{inc}})}{Z_L(k^{\text{inc}}) + Z_0} \tag{11.19}$$

with $Z_L(k^{\text{inc}})$ being the impedance that the multilayer media presents to the incident wave in the plane of the patch, and Z_0 being the characteristic impedance of vacuum.

Using the formulation outlined in this chapter, we can obtain the induced current on the patch and, from this current, the scattering fields $\mathbf{E}^s$. These fields can be obtained easily in the spectral domain. Next we will show how the radiated fields can be computed as a function of the electric fields in the patch plane.

By using the well-known equivalence and image theorems, we can compute the scattering fields in the upper half space from the equivalent magnetic currents in the patch plane $\mathbf{M}^s$. These currents can be written in terms of the scattered electric field $\mathbf{E}^s$ in the plane patch as

$$\mathbf{M}^s = \mathbf{E}^s \times \mathbf{z} \tag{11.20}$$

The radiated fields can be expressed in terms of the magnetic vector potential $\mathbf{F}$:

$$\mathbf{F} = \frac{\varepsilon_0}{4\pi} \int_S \mathbf{M}^s(\mathbf{r}') \frac{\exp(-jk_0|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} \, ds' \tag{11.21}$$

Equation (11.21) can now be written in terms of the scattered electric field and by using the far-field approximation as

$$\mathbf{F} = \frac{\varepsilon_0}{2\pi} \frac{\exp(-jk_0 r)}{r} \int_S \mathbf{E}^s(\mathbf{r}') \exp(jk_0 \mathbf{r}\mathbf{r}')\, ds' \times \mathbf{z} \tag{11.22}$$

Using the coordinate system shown in Figure 11.10,

$$\mathbf{r}' = x'\mathbf{x} + y'\mathbf{y} \tag{11.23}$$

$$\mathbf{r} = \sin\theta\cos\phi\mathbf{x} + \sin\theta\sin\phi\mathbf{y} + \cos\theta\mathbf{z} \tag{11.24}$$

$$\mathbf{r}\mathbf{r}' = \sin\theta\cos\phi x' + \sin\theta\sin\phi y' \tag{11.25}$$

the integral in (11.22) (renamed $\mathbf{I}^s$) can be written as

$$\mathbf{I}^s = \int_S \mathbf{E}^s(\mathbf{r}') \exp(jk_0(\sin\theta\cos\phi x' + \sin\theta\sin\phi y'))\, dx'\, dy' \tag{11.26}$$

Recalling the definition of the CFT, (11.26) can be identified with the Fourier transform of the scattered fields in the metallization plane,

$$\mathbf{I}^s = \tilde{E}_x^s(k_x^s, k_y^s)\mathbf{x} + \tilde{E}_y^s(k_x^s, k_y^s)\mathbf{y} \tag{11.27}$$

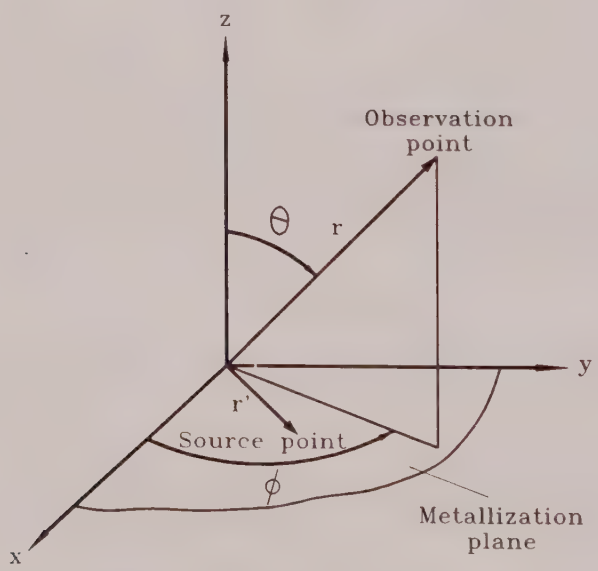


Figure 11.10 Coordinate system used to represent the scattering fields.

where

$$\tilde{E}^s_\alpha(k_x,k_y) = \int_S E^s_\alpha(x',y') \exp(-jk_x x') \exp(-jk_y y')\, dx'\, dy' \tag{11.28}$$

and

$$k^s_x = -k_0 \sin\theta \cos\phi \tag{11.29a}$$

$$k^s_y = -k_0 \sin\theta \sin\phi \tag{11.29b}$$

Bearing these considerations in mind, (11.22) becomes

$$\mathbf{F} = \frac{\varepsilon_0}{2\pi} \frac{\exp(-jk_0 r)}{r} [\tilde{E}^s_y(k^s_x,k^s_y)\mathbf{x} - \tilde{E}^s_x(k^s_x,k^s_y)\mathbf{y}] \tag{11.30}$$

By using the cartesian spherical coordinate transformation and the relationship between electric fields and the magnetic vector potential, the following scattered field components are obtained:

$$E_\theta = \frac{jk_0}{2\pi} \frac{\exp(-jk_0 r)}{r} (\tilde{E}^s_y(k^s_x,k^s_y) \sin\phi + \tilde{E}^s_x(k^s_x,k^s_y) \cos\phi) \tag{11.31a}$$

$$E_\phi = \frac{jk_0}{2\pi} \frac{\exp(-jk_0 r)}{r} \cos\theta(\tilde{E}^s_y(k^s_x,k^s_y) \cos\phi - \tilde{E}^s_x(k^s_x,k^s_y) \sin\phi) \tag{11.31b}$$

The scattered fields in the patch plane can be obtained in the spectral domain simply by calculating the CFT of the induced current on the metallic plates and the CFT of the Green's function

$$\tilde{E}^s_x(k_x,k_y) = \tilde{G}_{xx}(k_x,k_y)\tilde{J}_x(k_x,k_y) + \tilde{G}_{xy}(k_x,k_y)\tilde{J}_y(k_x,k_y) \tag{11.32a}$$

$$\tilde{E}^s_y(k_x,k_y) = \tilde{G}_{yx}(k_x,k_y)\tilde{J}_x(k_x,k_y) + \tilde{G}_{yy}(k_x,k_y)\tilde{J}_y(k_x,k_y) \tag{11.32b}$$

where the Green's function components are given by (10.16) and the CFT of the currents in terms of the rooftop-based discretization procedure can be obtained by using the numerical approach proposed in this chapter. The RCS can also be defined by using the values of E_θ and E_ϕ in (11.31a,b):

$$\sigma = 10 \log\left[4\pi r^2 \frac{|E_\theta|^2 + |E_\phi|^2}{|E^i_0|^2} \right] \tag{11.33}$$

We will now present examples of the RCS results obtained by applying this procedure to a free-space and microstrip structure.

11.6.1 Curved-Edge Plate in Free Space

The structure to be analyzed is depicted in Figure 11.11. The radius of the curved edge is $r = 11.43$ cm, and the total dimensions of the plate are $L_x = 25.3238$ cm and $L_y = 22.86$ cm. The frequency $f_0 = 2.9$ GHz, and the incident plane wave is x-polarized. The number of samples used to represent the currents is $N_x = N_y = 31$. The results plotted in Figures 11.12 and 11.13 correspond to the values of the $\phi\phi$ and $\theta\theta$ components of the backscatter RCS of the plate. Both results are compared with numerical and measured data from [14].

11.6.2 Microstrip Patch

Consider the microstrip structure with a rectangular patch of dimensions $L = 1$ cm and $W = 1.5$ cm shown in Figure 11.14.

The thickness of the dielectric layer, denoted by d, is equal to 0.127 cm. The relative dielectric constant of the layer is $\varepsilon_r = 10.2$. The incident field is normal ($\theta = 0°$) to the metallization plane and θ-polarized, with an angle $\phi = 45°$. The results in Figure 11.15 for the backscatter RCS (θ,θ) are compared with those by Pozar [15]. Note that the resonant frequencies of the structure [16] are well resolved, even

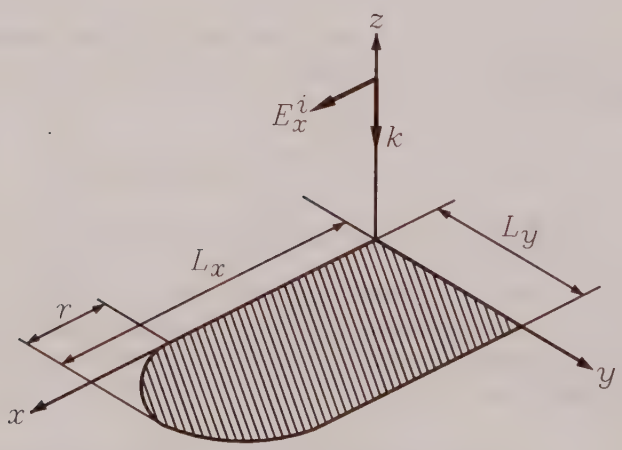


Figure 11.11 Schematic representation of the curved-edge plate problem.

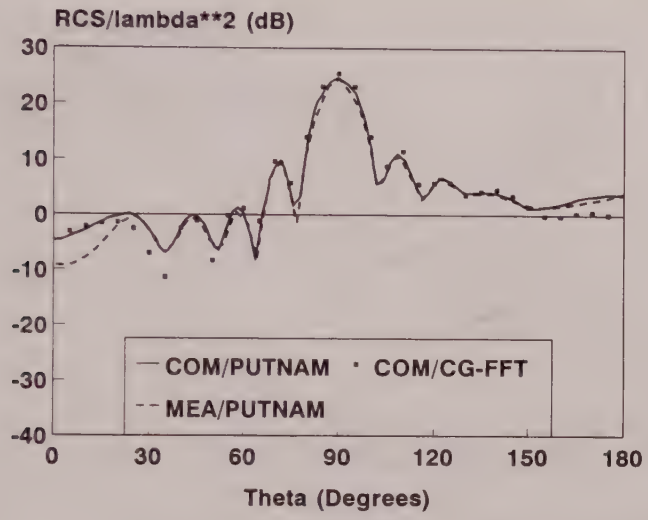


Figure 11.12 Backscatter RCS($\sigma_{\phi\phi}$) of a curved-edge plate.

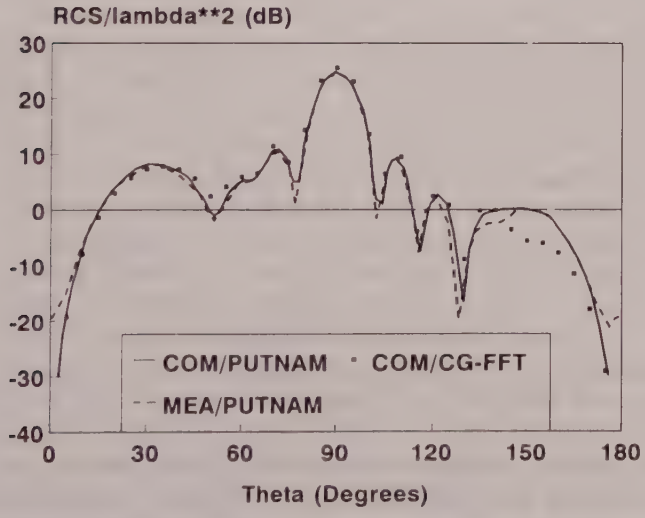


Figure 11.13 Backscatter RCS($\sigma_{\theta\theta}$) of a curved-edge plate.

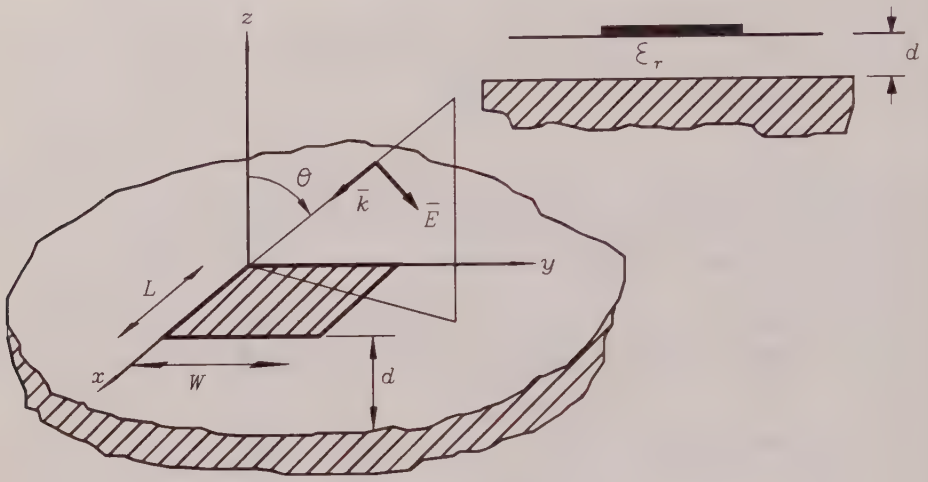


Figure 11.14 Schematic representation of the microstrip patch problem.

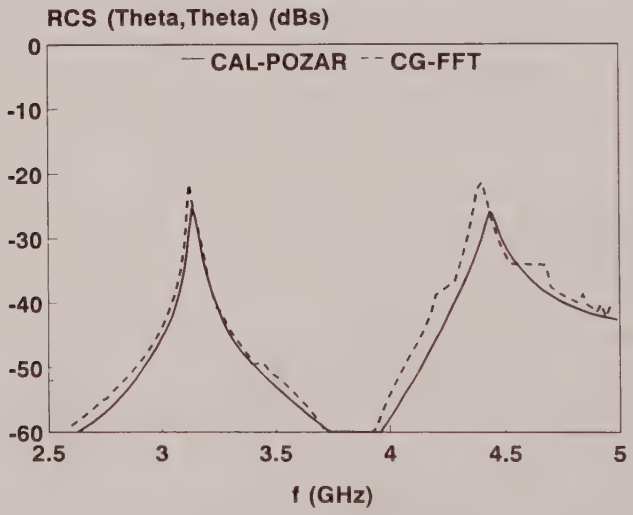


Figure 11.15 Backscatter RCS($\sigma_{\theta\theta}$) for the microstrip problem.

with the small number of samples used to represent the currents ($N_x = N_y = 15$ with only 9 of them in each direction within the boundaries of the metallization patch surface). Figure 11.16 shows the convergence rate for this kind of structure (in this example, $f_0 = 3{,}120$ MHz).

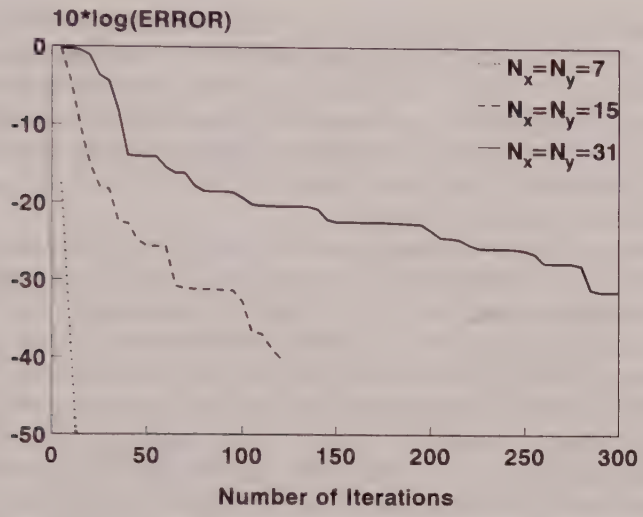


Figure 11.16 Relative error versus number of iterations for the microstrip structure in Figure 11.14.

11.7 APPLICATION TO *S*-PARAMETER ANALYSIS OF OPEN MICROSTRIP STRUCTURES

Microstrip structures are basic elements of many flat devices, such as antennas and their feeding networks and MICs. The key to the analysis of these types of configurations is the feeding system. Basically, two kinds of systems are employed in practice: (1) coaxial feeding systems connected directly at some point of the metallization, which are used for antennas, and (2) microstrip-line fed systems, which are used for both antennas and MICs. An example using the first type of feeding system has been proposed [8] in which a spectral domain formulation is combined with the CG method (called the spectral conjugate gradient algorithm). In this section, we will concentrate on the second type of feeding system, and more specifically on the direct extraction of the scattering (S) parameters of microstrip discontinuities. Microstrip structures have been studied by many authors using various approaches. MM formulations that combine subsectional basis functions with precomputed sinusoidal expansion currents for the input and output lines have been used to simulate open microstrip structures [3,17]. Harokopus and Katehi, who have done considerable work in this field, used an MM formulation in the space domain, together with delta gap generators [18,19]. In another work [20], Wertgen and Jansen used an MPIE formulation in conjunction with the Galerkin method to analyze packaged

structures. The Fourier sine and cosine transforms that appear implicitly in this formulation are computed efficiently by means of FFTs. For our purposes, one of the most important features of [20] is the feeding system: an impressed current with a longitudinal current distribution similar to that of the fundamental mode in the feed strips. In all the works cited above, S-parameters were computed from the stationary wave patterns that formed in the long transition regions between the feeding current and the set of discontinuities under study.

The feeding system used here is very similar to those systems in [20]. It circumvents the problem of long transition regions and is suitable for the direct computation of S-parameters with the CG-FFT scheme presented in this chapter. The structures that are analyzed consist of arbitrary metallizations located at the interface between two media fed by line sections (microstrip in the cases presented here, but other multilayer structures can be treated by adequately modifying the formulation). The feeding system is represented by considering the propagation of the fundamental mode on the feeding lines with a longitudinal current distribution similar to that of the fundamental mode (see Figure 11.17(a)). Its phase variation can be computed by using approximate expressions like those given in [21] for the effective wave number (see Figure 11.17(b)).

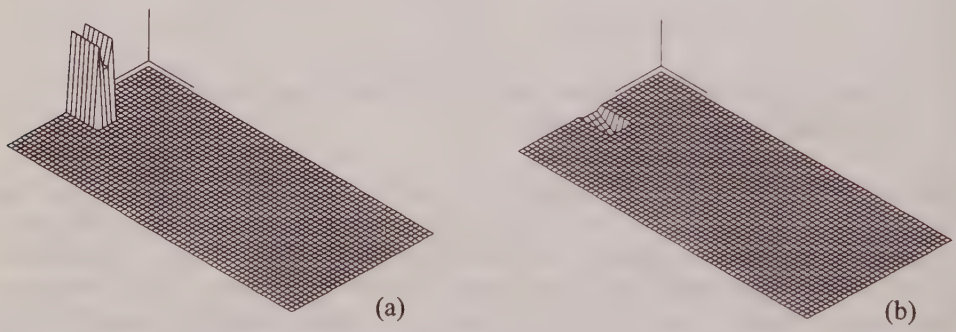


Figure 11.17 (a) Amplitude and (b) phase of the feeding current.

11.7.1 Feeding and Loading System

The development of the proposed CG-FFT scheme with the feeding system described above requires the definition of the scattering parameters as a function of the incident (a_i) and reflected (b_i) waves at each one of the ports (assuming N ports) of the structure

$$S_{ji} = \frac{b_j}{a_i} \quad \text{with} \quad a_k = 0 \quad \text{if} \quad k \neq i \tag{11.34}$$

That is to say, the parameter S_{ji} is obtained as the ratio between the outgoing wave

at port j and the incident one at port i with the other ports matched. Thus, in order to obtain the S-parameters, we must develop a method of computing the transmitted and reflected waves at any port. For the sake of simplicity, this problem will be discussed for the two-port case only. Several regions will be defined in the metallization, which are shown in Figure 11.18.

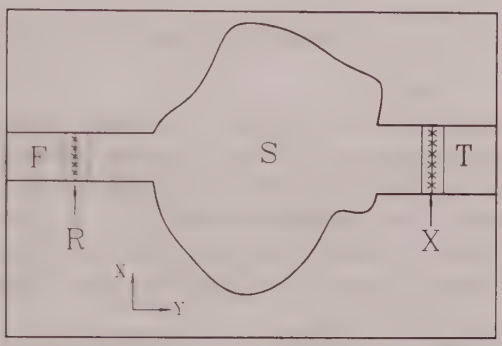


Figure 11.18 Different regions of the metallization. F: feeding and reflected-wave area; S: scattering area (the region being analyzed); T: transmitted-wave area, R: reflected-wave generation area; X: transmitted-wave generation area.

Region S is the area under analysis, in which only induced currents will exist. Boundary conditions are applied to this region. The impressed current is placed on F, although the reflected wave will also exist here. Region T is used to simulate the transmitted wave (matching). In region R, impressed and induced currents exist simultaneously. In both R and X, boundary conditions are imposed and the induced currents obtained by using the CG-FFT are assumed to be the reflected current in region F and the transmitted current in region T, with a phase variation given by the effective wave numbers in each line. When applying boundary conditions, fields produced by the currents of all regions must be taken into account.

With this feeding system, the following currents must be considered when applying the direct operator

$$
\begin{aligned}
J_x \approx & \sum_{i=1}^{N_x} \sum_{k=N_{\text{ref}}}^{N_{\text{tra}}} J_x^D(i,k) T_{ik}^x(\mathbf{r}) \\
& + \sum_{i=1}^{N_x} \sum_{k=1}^{N_{\text{ref}}-1} J_x^D(i,N_{\text{ref}}) \exp(jk_{\text{in}}(k - N_{\text{ref}})\Delta y)\, T_{ik}^x(\mathbf{r}) \qquad (11.35a) \\
& + \sum_{i=1}^{N_x} \sum_{k=N_{\text{tra}}+1}^{N_y+1} J_x^D(i,N_{\text{tra}}) \exp(-jk_{\text{out}}(k - N_{\text{tra}})\Delta y)\, T_{ik}^x(\mathbf{r})
\end{aligned}
$$

$$J_y \approx \sum_{i=1}^{N_x+1} \sum_{k=N_{\text{ref}}}^{N_{\text{tra}}} J_y^D(i,k) T_{ik}^y(\mathbf{r})$$

$$+ \sum_{i=1}^{N_x+1} \sum_{k=1}^{N_{\text{ref}}-1} J_y^D(i,N_{\text{ref}}) \exp(jk_{\text{in}}(k - N_{\text{ref}})\Delta y)\, T_{ik}^y(\mathbf{r}) \qquad (11.35b)$$

$$+ \sum_{i=1}^{N_x+1} \sum_{k=N_{\text{tra}}+1}^{N_y} J_y^D(i,N_{\text{tra}}) \exp(-jk_{\text{out}}(k - N_{\text{tra}})\Delta y)\, T_{ik}^y(\mathbf{r})$$

where N_{ref} and N_{tra} denote the beginning and end positions along y where the boundary conditions are applied (regions R and X). The exponential terms represent the phase variation of the fundamental mode at the input and output lines (F and T), and k_{in} and k_{out} are their effective wave numbers.

The fields computed by using the adjoint operator $E_{\alpha\beta}^{sDA}$ have the form

$$E_{\alpha\beta}^{sDA}(\mathbf{r}) = \sum_{i=1}^{N_x} \sum_{j=N_{\text{ref}}}^{N_{\text{tra}}} \delta(x - x_\alpha^i)\delta(y - y_\alpha^j) V_{\alpha\beta}^A(i,j) \qquad (11.36)$$

where

$$V_{\alpha\beta}^A(i,j) = \begin{cases} V_{\beta\alpha}^*(i,j) & N_{\text{ref}} < j < N_{\text{tra}} \\ \displaystyle\sum_{s=1}^{N_{\text{ref}}} V_{\beta\alpha}^*(i,s) & j = N_{\text{ref}} \\ \displaystyle\sum_{s=N_{\text{tra}}}^{N_y} V_{\beta\alpha}^*(i,s) & j = N_{\text{tra}} \end{cases} \qquad (11.37)$$

and $V_{\alpha\beta}^*$ is the complex conjugate of the direct voltage defined as

$$E_{\alpha\beta}^{sD}(\mathbf{r}) = \sum_{i=1}^{N_x} \sum_{j=N_{\text{ref}}}^{N_{\text{tra}}} \delta(x - x_\alpha^i)\, \delta(y - y_\alpha^j)\, V_{\alpha\beta}(i,j) \qquad (11.38)$$

After the desired error bound is reached, the S-parameters can be obtained directly by integrating the longitudinal current along the transverse section of the corresponding port. With this formulation, there must be a certain length of line between the reference plane and the problem in order to avoid the undesired evanescent waves that are always associated with discontinuities. This modification does not result in an excessive increase in the size of the problem.

An arbitrary load Z_L, placed sufficiently far from the discontinuities, can be simulated in a similar way. In this case, the reflection coefficient at the end of region $T(N_y\Delta y)$ is given by

$$\Gamma(N_y\Delta y) = \frac{Z_L - Z_C + j(Z_L - Z_C)tg(k_{out}N_y\Delta y)}{Z_L + Z_C - j(Z_L + Z_C)tg(k_{out}N_y\Delta y)} \tag{11.39}$$

and the current along T can be obtained by using

$$J_\alpha^D(i,k) = J_\alpha^D(i,N_{tra})$$

$$\frac{\exp(-jk_{out}(k - N_{tra} - N_y)\Delta y) - \Gamma(N_y\Delta y)\exp(jk_{out}(k - N_{tra} - N_y)\Delta y)}{\exp(jk_{out}N_y\Delta y) - \Gamma(N_y\Delta y)\exp(-jk_{out}N_y\Delta y)} \tag{11.40}$$

where Z_C is the characteristic impedance of the line in region T.

11.7.2 Results for Equivalent Currents

To demonstrate the behavior of the feeding and loading system, we have calculated the currents for the matching section sketched in Figure 11.19. The widths of the input and output lines are $L_1 = 20$ mm, and the length (L_3) and width (L_2) of the matching section are 50 and 40 mm, respectively. The thickness of the dielectric layer is 10 mm and the relative dielectric constant is 9.9. The feeding current is shown in Figure 11.17. Figures 11.19(a,c) show the amplitude of the longitudinal and transverse components of the current distribution for a frequency of 14 GHz, while 11.19(b,d) depict the phase of these components. Note that the transverse component is almost null except near the discontinuities, as is to be expected, and its phase is mainly due to numerical noise. The stationary wave in the input line due to the incident and reflected waves should also be noted.

11.7.3 Results for Scattering Parameters

Two simple structures have been analyzed to illustrate the behavior of the method when dealing with S-parameters. The first one is the asymmetrically fed patch shown in Figure 11.20. The values of the geometrical parameters are $L_1 = 16$ mm, $L_2 = 12.45$ mm, $W = 2.46$ mm, and $d = 2.09$ mm; the dielectric data are $\varepsilon_r = 2.2$ and thickness 0.794 mm. In this case, the results obtained for the parameter S_{11} are compared with experimental data obtained in [22]. The small discrepancy observed at the resonant frequencies is due to the uniform discretization method, which does not allow an exact geometrical representation of the structure (simulated values: $L_1 =$

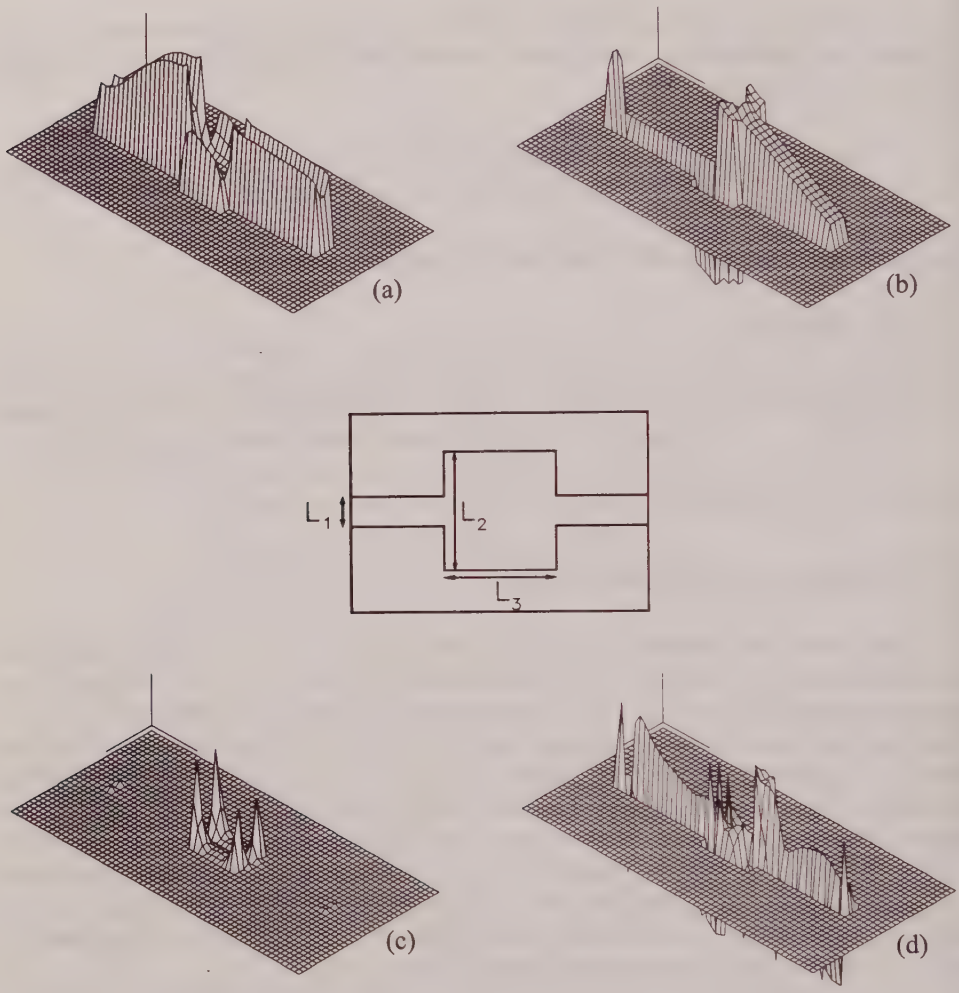


Figure 11.19 Current distributions for the microstrip matching section shown at 14 GHz: (a) modulus of longitudinal current; (b) phase of longitudinal current; (c) modulus of transverse current; (d) phase of transverse current.

16 mm, $L_2 = 12.448$ mm, $W = 2.334$ mm, $d = 2.334$ mm). The number of samples used for discretization of the currents was $N_x = N_y = 63$.

Figure 11.21 shows the second structure, a stub with dimensions $W_1 = W_2 = 1.40$ mm and $L = 2.2$ mm, and the results for the amplitude of the transmission parameter S_{21}. The substrate has a relative dielectric constant of 10.65 and a thickness of 1.27 mm. The results are compared with the MM and experimental data obtained in [17].

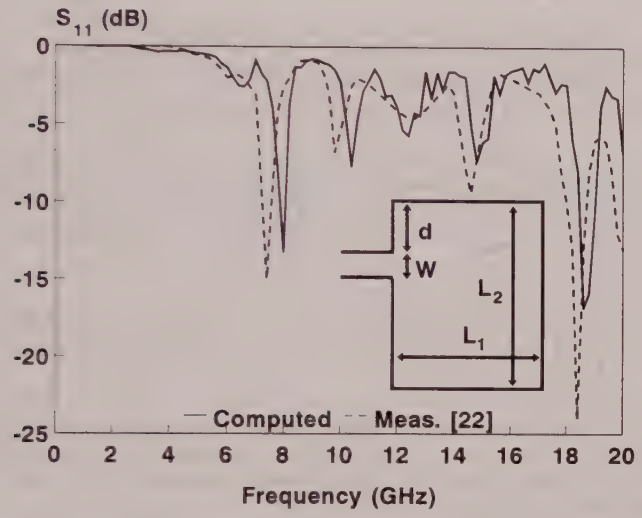


Figure 11.20 S_{11} parameter for the asymmetrically fed patch problem.

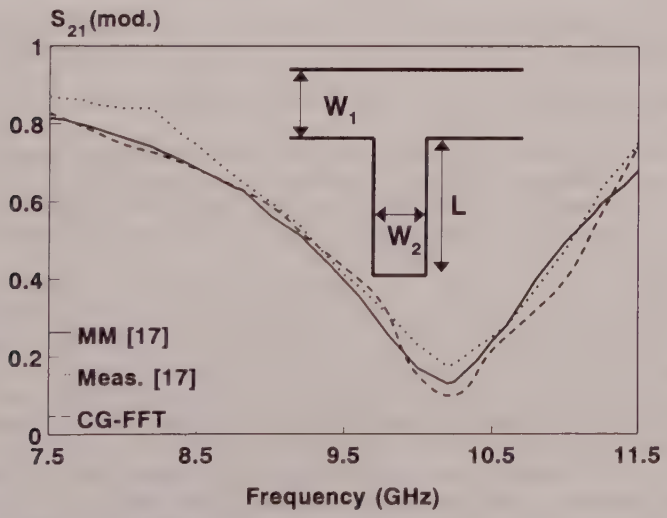


Figure 11.21 S_{21} parameter for the stub problem shown.

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Chapter 12
Volumetric Analysis of 3D Bodies That Are Periodic in One Direction

12.1 INTRODUCTION

In this last chapter, we will discretize the operator equation in both the spatial and spectral domains. Two main techniques have been used in this book to discretize the operator equation to be solved. In the first technique, the operator is discretized in the spatial domain, and we obtain a discrete operator equation in which the known and unknown functions are discrete periodic functions related by a cyclic convolution. In this convolution, the Green function (or impulse response in signal processing terminology) is also a discrete periodic function. In this case, the convolution is performed in a natural way, with the aid of the FFT, without any approximation. In addition, the derivatives that appear in the original continuous operator are transformed into differences that are easily taken into account by means of the shift property of the DFT. With this sampling technique, nearly all errors are introduced in the discretization process, if we disregard the rounding and truncation errors due to the use of a binary floating-point representation with finite-length words. All the problems considered in Chapters 5 through 8 were solved by using this sampling technique.

In the other discretization approach, the functions are considered to be continuous and the convolutions are computed by means of CFTs that are approximated with the FFT. The derivatives of the operators are transformed in the spectral domain into scalar products. In fact, when using this technique, the discretization process is carried out to a large extent in the approximation to the CFT with the FFT. This approach is quite suitable for periodic problems such as frequency-selective surfaces (Chapters 9 and 10), where the CFT of the operator equation functions (including the Green's function) are discrete. For nonperiodic problems, the application of this technique can be troublesome, because windowing techniques must be introduced to

avoid aliasing problems. However, if the operator (or Green's function) can be easily evaluated only in the spectral domain, this sampling approach may be more advantageous than a sampling in the spatial domain. In Chapter 11, one kind of finite structure (an individual microstrip patch) was analyzed by using CFTs and windowing techniques.

In this chapter, we will study 3D bodies that are periodic in one direction (z-axis) and finite-sized in the other two (x- and y-axes). The acronym VODIPEB (volumetric 1D periodic body) will be used to refer to these bodies. Figure 12.1 shows an example of a body with these features. It is worth noting that bodies with symmetry of translation (e.g., cylinders) are particular cases of VODIPEBs. It will be shown that the CPU time and memory requirements for these bodies are no greater for a periodic 3D formulation than for a 2D formulation, because it is only necessary to compute the finite problem (x,y) in one sampling plane (z = constant). Thus, the 3D formulation is in fact converted into a 2D formulation.

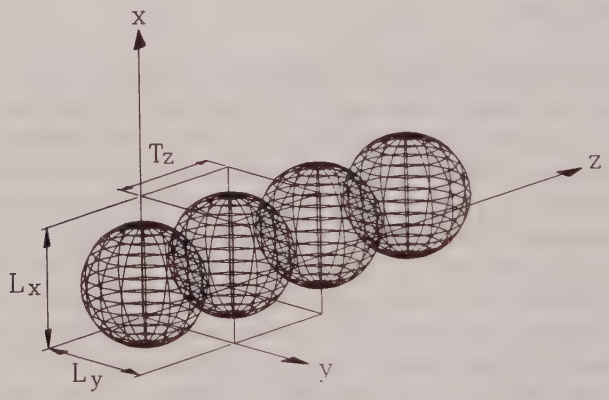


Figure 12.1 Example of a VODIPEB.

The EFIE, which uses the same volumetric currents as in Chapter 6, will be considered as the starting point in the formulation of the continuous operator equation to be solved. In addition, the discretization process will be carried out with the same basis (volumetric rooftops) and weighting (razor blades) functions as in Chapter 6.

Since sampling in the spatial domain is well suited to finite-sized problems, while discretization in the spectral domain is better for periodic structures, it appears that VODIPEBs should be analyzed with a combination of both approaches. The operator should be discretized in the spatial domain in the directions in which they are of finite length (x- and y-axes), while the sampling in the periodic direction should be performed in the spectral domain.

Briefly, the procedure is based on splitting the 3D convolutions of the problem into two separate convolutions: a 1D convolution for the coordinate in which the VODIPEB is periodic (z-coordinate) and a 2D cyclic convolution for the other two coordinates (x and y). For the 1D convolution, a spectral expansion in the z-coordinate is used to take advantage of the analytical and discrete nature of the Fourier transform of the Green's function for this variable. The convolution for the other two coordinates is performed by transforming in the spatial domain the EFIE linear convolution into a cyclic convolution. All of these convolutions can be computed efficiently by using the FFT.

It is of great practical interest to have a tool available for the analysis of structures of this type. For instance, many components of optical or microwave circuits are VODIPEBs whose transverse section can be quite arbitrary in terms of both geometry and material composition. In RCS analysis, too, there are many parts of targets that can be treated as bodies with symmetry of translation. If the shape of these parts is modified by introducing periodicity into their geometries, the scattered energy can be redistributed in various directions, thereby reducing the RCS.

Among the works that have been published on VODIPEBs or similar structures, the following ones are particularly relevant to the present discussion. In a study by Bertoni et al. [1] periodic arrays of dielectric rectangular cylinders are analyzed. In order to represent the fields in these arrays, the authors considered the modes in infinite parallel layers of the same material and with the same thickness as the rectangular cylinders. In another work related to VODIPEBs, Jorgenson and Mittra [2] developed a periodic moment method (PMM) formulation to study slab structures with 2D periodicity. More recently, the analysis and design of absorbent panels for anechoic chambers have motivated PMM formulations for the analysis of 3D lossy materials such as those of [3,4], which used surface and volumetric equivalent currents, respectively.

This chapter is organized as follows. Section 12.2 introduces the problem and the continuous equation with an operator that takes into account the periodicity of the structure. The discretization process for this continuous operator is explained in Section 12.3. In Section 12.4, the convolution integrals that appear in the discrete operator are computed by using FFTs. This part of the development is quite important, because the efficiency of the CG-FFT method is based on the computation of the operator convolutions by means of FFTs. Finally, in Section 12.5, results for RCS, induced equivalent currents, the total field inside bodies, and convergence rates are presented and compared with analytical solutions available in the literature.

12.2 FORMULATION OF THE CONTINUOUS OPERATOR EQUATION FOR A VODIPEB

Figure 12.1 shows an example of a VODIPEB composed of a volumetric reference cell repeated periodically in the z-direction. The reference cell can be inscribed in a

parallelepiped of dimensions (L_x, L_y, L_z). A plane wave with a time-varying factor $\exp(j\omega t)$ is incident upon the structure and has a field given by

$$\mathbf{E}^i(\mathbf{r}) = \mathbf{E}_0^i \exp(-jk_{0x}x - jk_{0y}y - jk_{0z}z) \tag{12.1}$$

The volumetric equivalent current in the reference cell can be obtained by solving the following operator equation:

$$\mathbf{E}^i(\mathbf{r}) = C(\mathbf{r})\mathbf{J}(\mathbf{r}) - \mathbf{E}^s(\mathbf{r}) \qquad \text{for } \mathbf{r} \text{ in } V \tag{12.2}$$

where V is the volume of the reference cell of the VODIPEB, and $C(\mathbf{r})$, which will be called the material admittance, is related to the permittivity ε and the conductivity σ by

$$C(\mathbf{r}) = \frac{1}{j\omega(\varepsilon(\mathbf{r}) - \varepsilon_0) + \sigma(\mathbf{r})} \tag{12.3}$$

and $\mathbf{E}^s(\mathbf{r})$ is the scattered field

$$\mathbf{E}^s(\mathbf{r}) = -k^c \int_V G^F(\mathbf{r} - \mathbf{r}')\mathbf{J}(\mathbf{r}')\, dv' - k^q \nabla \int_V \nabla \mathbf{J}(\mathbf{r}')\, G^F(\mathbf{r} - \mathbf{r}')\, dv' \tag{12.4}$$

with $k^c = -j\omega\mu_0$ and $k^q = 1/(j\omega\varepsilon_0)$. G^F is a Green's function that takes into account Floquet's harmonics:

$$G^F(\mathbf{r}) = \sum_{m=-\infty}^{\infty} G(\mathbf{r} - mL_z\mathbf{z})\, \mathbf{e}^{-jmk_{0z}L_z} \tag{12.5}$$

where G is the Green's function for free space.

Expression (12.3) is a volumetric version of the EFIE. It is quite similar to the one considered as the starting point in Chapter 6 or in [5], but with the difference that, in this case, the range is limited to the volume of the reference cell, and that the Green's function, as in Chapters 9 and 10, implicitly takes into account the periodic behavior of the current in the z-direction.

Expressing the induced current in the reference cell as

$$\mathbf{J}(\mathbf{r}) = \mathbf{J}^0(x,y) \exp(-jk_{0z}z) \tag{12.6}$$

we can write (12.2) in a more detailed form as

$$\begin{aligned} E_x^i &= E_{xx}^S + E_{xy}^{SC} + E_{xz}^{SC} \\ E_y^i &= E_{yx}^{SC} + E_{yy}^S + E_{yz}^{SC} \qquad \text{for } \mathbf{r} \text{ in } V \\ E_z^i &= E_{zx}^{SC} + E_{zy}^{SC} + E_{zz}^S \end{aligned} \tag{12.7}$$

where

$$E^{S}_{\alpha\alpha} = E^{SI}_{\alpha\alpha} + E^{SC}_{\alpha\alpha} \tag{12.8}$$

As in Chapter 6, the components of the induced field are written in terms of inductive and capacitive parts. For the inductive part, we have

$$E^{SI}_{\alpha\alpha} = L^{I}_{\alpha\alpha} J^{0}_{\alpha} \tag{12.9}$$

and for the capacitive part,

$$E^{SC}_{\alpha\beta} = L^{C}_{\alpha\beta} J^{0}_{\beta} \tag{12.10}$$

where

$$L^{I}_{\alpha\alpha} = C(\mathbf{r}) + k^{c} \int_{V} G^{F}(\mathbf{r} - \mathbf{r}') \exp(-jk_{0z} z)\, dv' \tag{12.11}$$

$$L^{C}_{\alpha\beta} = k^{q} \frac{\partial}{\partial \alpha} \int_{V} G^{F}(\mathbf{r} - \mathbf{r}') \exp(-jk_{0z} z)\, L^{d}_{\beta}\, dv' \tag{12.12}$$

$$L^{d}_{\beta} = \frac{\partial}{\partial \beta} - jk'_{0\beta} \tag{12.13}$$

with

$$k'_{0\beta} = \begin{cases} k_{0\beta} & \text{if} \quad \beta = z \\ 0 & \text{otherwise} \end{cases} \tag{12.14}$$

12.3 FORMULATION OF THE DISCRETE OPERATOR EQUATION

To apply the CG-FFT method, (12.7) must be sampled. The sampling procedure will be presented in two steps. First, the approximations for the geometry and for the induced currents will be examined. Then we will describe the approximations that appear as a result of sampling the continuous fields of the problem.

12.3.1 Geometry and Current Approximations

As in Chapter 6, the parallelepiped whose dimensions are L_x, L_y, and L_z is divided by a set of $(N_x + 1)(N_y + 1)(N_z + 1)$ planes parallel to the coordinate axes into a

set of $N_xN_yN_z$ small parallelepipeds of volume $\Delta x \Delta y \Delta z$, with $\Delta\alpha = L_\alpha/(N_\alpha + 1)$. The material admittance of the body is assumed to be constant in each one of these small parallelepipeds. The induced current J^0 is expanded in terms of 3D rooftop functions T^x_{ijk}, T^y_{ijk}, T^z_{ijk}, as follows:

$$J^0_\alpha(\mathbf{r}) = \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} J^D_\alpha[i,j,k]\, T^\alpha_{ijk}(\mathbf{r}) \quad \text{for} \quad \alpha = x \quad \text{or} \quad y \quad \text{or} \quad z \tag{12.15}$$

where J^D_α is a discrete function and the rooftop functions T^α_{ijk} are defined as in Chapter 6. The relative positions of the rooftops are maintained. For instance, the center of the rooftop T^x_{ijk} is located at the point

$$\mathbf{r}^{ijk}_x = i\Delta x\mathbf{x} + (j - 1/2)\,\Delta y\mathbf{y} + (k - 1/2)\,\Delta z\mathbf{z} \tag{12.16}$$

and the centers of T^y_{ijk} and T^z_{ijk} are defined analogously.

Using rooftops to represent the current, we can approximate the derivatives $\partial/\partial\beta$ ($\beta = x$, y, or z) that appear in (12.13) in terms of 3D pulses. For instance, for $\partial J_x/\partial x$, we have

$$\frac{\partial J^0_x(\mathbf{r})}{\partial x} = \frac{1}{\Delta x} \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} J^{D\delta}_x[i,j,k]\, P_{ijk}(\mathbf{r}) \tag{12.17}$$

where

$$J^{D\delta}_x[i,j,k] = J^D_x[i,j,k] - J^D_x[i-1,j,k] \tag{12.18}$$

$$P_{ijk}(\mathbf{r}) = P(\mathbf{r} - \mathbf{r}^{ijk}) \tag{12.19}$$

$$P(\mathbf{r}) = \begin{cases} 1 & \text{for} \quad |\alpha| \le \dfrac{\Delta\alpha}{2}, \ (\alpha = x, y, \text{or } z) \\ 0 & \text{otherwise} \end{cases} \tag{12.20}$$

$$\mathbf{r}^{ijk} = (i - 1/2)\,\Delta x\mathbf{x} + (j - 1/2)\,\Delta y\mathbf{y} + (k - 1/2)\,\Delta z\mathbf{z} \tag{12.21}$$

It can be seen that for the three components of the current, we obtain the same 3D pulses P_{ijk}, which can also be used to represent the material admittance of the body

$$C(\mathbf{r}) \simeq \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} C_d[i,j,k]\, P_{ijk}(\mathbf{r}) \tag{12.22}$$

All the discrete functions defined in this chapter are considered periodic with a reference cell of size $(2N_x, 2N_y, N_z)$. The period lengths $2N_x$ and $2N_y$ are obtained by padding with zero the discrete functions that appear for the x- and y-components. It will be seen later that this padding is required for the cyclic convolutions to be performed for these components.

12.3.2 Approximations and Sampling of the Fields

The razor blade functions W^x, W^y, and W^z are employed to smooth the fields. For the x-component, the razor blade function is defined as

$$W^x(\mathbf{r}) = \begin{cases} 1 & \text{for} \quad |x| \leq \dfrac{\Delta x}{2};\ y = z = 0 \\ 0 & \text{otherwise} \end{cases} \tag{12.23}$$

W^y and W^z are defined analogously.

Discrete functions are obtained from the continuous field components by first performing a convolution of these fields with their corresponding razor blade and then sampling the resulting continuous functions at discrete points. For instance, for the α-component of the impressed field, we have

$$E_\alpha^{iD}(\mathbf{r}) = \mathit{\Pi\!I}(\mathbf{r}_\alpha^{mnp}) \int_V E_\alpha^i(\mathbf{r}')\, W^\alpha(\mathbf{r} - \mathbf{r}')\, \mathrm{d}v' \tag{12.24}$$

where $\mathit{\Pi\!I}(\mathbf{r}_\alpha^{mnp})$ indicates the 3D sampling function:

$$\mathit{\Pi\!I}(\mathbf{r}_\alpha^{mnp}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \sum_{p=-\infty}^{\infty} \delta(x - x_\alpha^{mnp})\delta(y - y_\alpha^{mnp})\delta(z - z_\alpha^{mnp}) \tag{12.25}$$

with δ being the impulse function.

For the plane wave excitation defined in (12.1), we have

$$E_\alpha^{iD}(\mathbf{r}) = \mathit{\Pi\!I}(\mathbf{r}_\alpha^{mnp}) \left[\exp(-jk_{0x} - jk_{0y}y - jk_{0z}z)\, \Delta\alpha\, E_{0\alpha}^i\, \mathrm{sinc}\left(k_{0\alpha}' \frac{\Delta\alpha}{2} \right) \right] \tag{12.26}$$

Applying this smoothing and sampling process to the continuous equations of (12.7), we can obtain a discrete system of equations that can be written (for the x-component, for example) as

$$E_x^{SD} = E_{xx}^{SD} + E_{xy}^{SDC} + E_{xz}^{SDC} \tag{12.27}$$

where

$$E_{xx}^{SD} = E_{xx}^{SDI} + E_{xx}^{SDC} \tag{12.28}$$

Similar expressions can be obtained for the y- and z-components.

Next, we will derive the term $E_{\alpha\alpha}^{SDI}$ for the inductive part and $E_{\alpha\alpha}^{SDC}$ for the capacitive part. For $E_{\alpha\alpha}^{SDI}$, we have

$$\begin{aligned} E_{\alpha\alpha}^{SDI}(\mathbf{r}) &= \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \int_V W^\alpha(\mathbf{r}-\mathbf{r}') E_{\alpha\alpha}^{SI}(\mathbf{r}')\, dv' \\ &\approx \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \Delta\alpha E_{\alpha\alpha}^{SI}(\mathbf{r}) = \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \Delta\alpha L'_{\alpha\alpha} J_\alpha^0(\mathbf{r}') \approx \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \Delta\alpha \exp(-jk_{0z}z) \\ &\cdot \left[C(\mathbf{r}) \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} J_\alpha^D[i,j,k]\, P_{ijk}^\alpha(\mathbf{r}') \right. \\ &\left. + k^c \int_V G^F(\mathbf{r}-\mathbf{r}') \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} J_\alpha^D[i,j,k]\, P_{ijk}^\alpha(\mathbf{r}')\, dv' \right] \end{aligned} \tag{12.29}$$

where $P_{ijk}^\alpha(\mathbf{r})$ is a 3D pulse centered at r_α^{ijk}, which has the same dimensions as the pulse P defined in (12.20). To obtain the last expression for the inductive term, we have approximated $E_{\alpha\alpha}^{SDI}$ by its value at the center point of W^α, and the rooftop T_{ijk}^α by the pulse P_{ijk}^α, which has the same volume and center point as T_{ijk}^α. The foundations for these two approximations are discussed in [6].

The term $E_{\alpha\beta}^{SDC}$ is derived as follows:

$$\begin{aligned} E_{\alpha\beta}^{SDC}(\mathbf{r}) &= \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \int_V W^\alpha(\mathbf{r}-\mathbf{r}') E_{\alpha\beta}^{SC}(\mathbf{r}')\, dv' \\ &= \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \int_{\mathbf{r}_\alpha^-}^{\mathbf{r}_\alpha^+} \frac{\partial}{\partial\alpha} V(\mathbf{r}) \exp(-jk_{0z}z)\, d\alpha \\ &= \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \left[V(\mathbf{r}_\alpha^+) \exp(-jk_{0z}z_\alpha^+) - V(\mathbf{r}_\alpha^-) \exp(-jk_{0z}z_\alpha^-) \right] \\ &\approx \Pi\Pi(\mathbf{r}_\alpha^{mnp}) \exp(-jk_{0z}z) \left[V(\mathbf{r}_\alpha^+) - V(\mathbf{r}_\alpha^-) - jk'_{0\alpha} \frac{\Delta\alpha}{2} \left(V(\mathbf{r}_\alpha^+) + V(\mathbf{r}_\alpha^-) \right) \right] \end{aligned} \tag{12.30}$$

where

$$\mathbf{r}_\alpha^{\pm} = \mathbf{r} + \frac{\Delta\alpha}{2}\boldsymbol{\alpha} \tag{12.31}$$

$$V(\mathbf{r}) = k^q \exp(+jk_{0z}z) \int_V G^F(\mathbf{r}-\mathbf{r}') \exp(-jk_{0z}z') \left(\frac{\partial}{\partial\beta} - jk'_{0z}\right) J_\beta^0(\mathbf{r}')\, dv'$$

$$\approx k^q \frac{\exp(+jk_{0z}z)}{\Delta\beta} \int_V G^F(\mathbf{r}-\mathbf{r}') \exp(-jk_{0z}z') \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{k=1}^{N_z+1} P_{ijk}(\mathbf{r}') \tag{12.32}$$

$$\cdot \left[J_\beta^{D\delta}[i,j,k] - jk'_{0\beta} \frac{\Delta\beta}{2} \left(J_\beta^D[i,j,k] + J_\beta^D[i,j,k-1]\right)\right] dv'$$

12.4 COMPUTATION OF CONVOLUTION INTEGRALS USING FFT

The 3D integrals required to compute the scattered fields of (12.7) have been written in terms of convolution integrals in (12.29) and (12.32). In these integrals, the convolution product is formed by the Green's function and a source function that is expanded in terms of 3D pulse functions. The integrals are sampled by trains of delta functions centered on the same points as the pulses of the corresponding source functions: r_α^{mnp} in (12.29) and r^{mnp} in (12.30) through (12.32), where r_α^{mnp} and r^{mnp} were defined in (12.16) and (12.21), respectively. All these integrals can be written in the same general form:

$$E^\delta(\mathbf{r}) = \Pi\Pi(\mathbf{r}^{mnp}) \int_V G^F(\mathbf{r}-\mathbf{r}')\, S(\mathbf{r}')\, dv' \tag{12.33}$$

where $\Pi\Pi(r^{mnp})$ is a 3D train of delta functions centered at r^{mnp}. $S(r)$ represents any one of the source functions that appear in (12.29) and (12.32), which have been approximated in terms of 3D pulse functions:

$$S(\mathbf{r}) = \exp(-jk_{0z}z) \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \sum_{l=1}^{N_z+1} S^D[i,j,l]\, P_0^{ijl}(\mathbf{r}) \tag{12.34}$$

where $S^D[i,j,l]$ is a discrete function and P_0^{ijl} is a 3D pulse of volume $\Delta x \Delta y \Delta z$ centered at r^{ijl}.

All these convolution integrals can be computed by using the same general procedure. The 3D integrals are split into two convolutions: a 1D convolution for the z-variable integration, and a 2D convolution for the integration in the variables x and y. The convolution for the variable z is performed by taking advantage of the

periodicity of the Green's function G^F, which has a Fourier transform that is discrete for the variable z. The convolution for the other two coordinates (x and y) is approximated by a cyclic convolution of two discrete functions.

12.4.1 Convolution for the Variable z

Expression (12.33) can be ordered as follows:

$$E^{\delta}(\mathbf{r}) = \mathit{III}(\boldsymbol{\rho}^{mn}) \int_{x'=0}^{L_x} \int_{y'=0}^{L_y} W^{\delta}(\boldsymbol{\rho},\boldsymbol{\rho}',z)\, dx'\, dy' \tag{12.35}$$

where $\mathit{III}(\rho^{mn})$ is a 2D train of delta functions centered at ρ^{mn},

$$\boldsymbol{\rho}^{mn} = m\Delta x\mathbf{x} + n\Delta y\mathbf{y} = x_m\mathbf{x} + y_n\mathbf{y} \tag{12.36}$$

and $\boldsymbol{\rho}$ and $\boldsymbol{\rho}'$ are the projections of $\mathbf{r}$ and $\mathbf{r}'$ in the plane $z = 0$. W^{δ} contains the convolution for the z-coordinate for which the body is periodic:

$$W^{\delta}(\boldsymbol{\rho},\boldsymbol{\rho}',z) = \mathit{III}(z_p) \int_{z'=0}^{L_z} G^F(\mathbf{r} - \mathbf{r}')S(\mathbf{r}')\, dz' \tag{12.37}$$

where $\mathit{III}(z_p)$ is a 1D train of delta functions centered at

$$z_p = p\Delta z \tag{12.38}$$

Taking into account the properties of the sampling function [7–9], we can compute the CFT of (12.37) from

$$\tilde{W}^{\delta}(\boldsymbol{\rho},\boldsymbol{\rho}',k_z) = \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} 2\pi P_z^{ij}(\boldsymbol{\rho}') \sum_{p=-\infty}^{\infty} \delta(k - k_z^p - k_{0z})\, \tilde{G}^p(\boldsymbol{\rho} - \boldsymbol{\rho}', p)\, \tilde{S}^D[i,j,p] \tag{12.39}$$

where P_z^{ij} is a 2D pulse function with area $\Delta x\Delta y$ centered at ρ^{ij} and

$$k_z^p = p\frac{T_z^k}{N_z} \tag{12.40}$$

T_z^k is the period in the spectral domain of the train of delta functions that results from performing the Fourier transform of $\mathit{III}(z_p)$:

$$T_z^k = \frac{2\pi}{\Delta z} \tag{12.41}$$

$\tilde{G}^p$ is a periodic function for the integer variable p with a period of N_z,

$$\tilde{G}^p(\boldsymbol{\rho} - \boldsymbol{\rho}', p) = \sum_{p'=-\infty}^{\infty} \operatorname{sinc}\left(\frac{p - p'N_z}{N_z}\right) \tilde{G}(\boldsymbol{\rho} - \boldsymbol{\rho}', k_z^p - p'T_z^k - k_{0z}) \tag{12.42}$$

and $\tilde{G}$ is the CFT of the Green's function for free space. As was established in [9,10], $\tilde{G}$ can be expressed analytically as follows:

$$\tilde{G}(\boldsymbol{\rho}, k_z) = \begin{cases} H_0^{(2)}\left(\rho\sqrt{k_0^2 - k_z^2}\right) & \text{for} \quad k_0^2 > k_z^2 \\ K_0\left(\rho\sqrt{k_0^2 - k_z^2}\right) & \text{otherwise} \end{cases} \tag{12.43}$$

where $H_0^{(2)}$ is the Hankel function of the second kind of zero order, K_0 is the modified Bessel function of the second kind of zero order, and k_0 is the wave number for a vacuum.

Finally, $\tilde{S}^D$ is the DFT of S^D:

$$\tilde{S}^D[i,j,p] = N_z \, \mathrm{DFT}_z(S^D[i,j,l]) \tag{12.44}$$

The symbol DFT_z denotes the discrete Fourier transform for the variable l, which corresponds to the continuous variable z.

As was indicated in [7] and in Chapters 3 and 9, a good approximation of $\tilde{G}^p$ is obtained by considering only the first six terms of the series ($p' = -2$ to 3). With this number of terms, the ripple that appears in the Fourier inverse of $\tilde{W}^\delta$ is practically negligible. In fact, if only the first two terms are considered ($p' = 0, 1$), only small ripples can be observed in $\tilde{W}^\delta$.

12.4.2 Convolutions for the Variables (x,y)

Performing the CFT for the z-variable in (12.35), we obtain the following expression:

$$\tilde{E}^\delta(\boldsymbol{\rho}, k_z) = \Pi\Pi(\boldsymbol{\rho}^{mn}) \int_{x'=0}^{L_x} \int_{y'=0}^{L_y} \tilde{W}^\delta(\boldsymbol{\rho}, \boldsymbol{\rho}', k_z) \, dx' \, dy' = \sum_{p=-\infty}^{\infty} \delta(k - k_z^p - k_{0z}) \Phi^\delta(\boldsymbol{\rho}, p) \tag{12.45}$$

where

$$\Phi^{\delta}(\boldsymbol{\rho},p) = \mathit{III}(\boldsymbol{\rho}^{mn}) 2\pi \int_{x'=0}^{L_x} \int_{y'=0}^{L_y} \tilde{G}^{p}(\boldsymbol{\rho} - \boldsymbol{\rho}', p) \left\{ \sum_{i=1}^{N_x+1} \sum_{j=1}^{N_y+1} \tilde{S}^{D}[i,j,p]\, P_z^{ij}(\boldsymbol{\rho}') \right\} dx'\, dy' \tag{12.46}$$

This last double integral is a 2D convolution that can be computed efficiently with FFTs by using techniques similar to those in Chapters 3 and 5. To compute this 2D convolution, (12.46) is transformed into a discrete cyclic convolution. For $0 \leq x \leq L_x$ and $0 \leq y \leq L_y$, we can obtain Φ^{δ} from

$$\Phi^{\delta}(\boldsymbol{\rho},p) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} \delta(x - x_m)\, \delta(y - y_n)\, V^{D}[m,n,p] \tag{12.47}$$

where

$$V^{D}[m,n,p] = \frac{2\pi}{(2N_x + 2)(2N_y + 2)} \mathrm{DFT}_{xy}^{-1} \left[\mathrm{DFT}_{xy}(\tilde{G}^{pd}[m,n,p])\, \mathrm{DFT}_{xy}(\tilde{S}^{pd}[m,n,p]) \right] \tag{12.48}$$

$\mathrm{DFT}_{xy}(\mathrm{DFT}_{xy}^{-1})$ indicates the direct (inverse) discrete Fourier transform for the variables (m,n) associated with the (x,y) coordinates. These discrete transforms operate over the functions $\tilde{G}^{pd}$ and $\tilde{S}^{pd}$, which have a period of $(2N_x + 2)(2N_y + 2)$:

$$\tilde{G}^{pd}[m,n,p] = \int_{-\Delta y/2}^{\Delta y/2} \int_{-\Delta x/2}^{\Delta x/2} \tilde{G}^{p}(\boldsymbol{\rho}^{mn} - \boldsymbol{\rho}', p)\, dx'\, dy' \quad \text{for} \quad \begin{cases} |m| \leq N_x + 1 \\ |n| \leq N_y + 1 \end{cases} \tag{12.49}$$

$$\tilde{S}^{pd}[m;n,p] = \begin{cases} \tilde{S}^{D}[m,n,p] & \text{for} \quad \begin{cases} 0 \leq m \leq N_x \\ 0 \leq n \leq N_y \end{cases} \\ 0 & \text{for} \quad \begin{cases} N_x + 1 \leq m \leq 2N_x + 1 \\ N_y + 1 \leq n \leq 2N_y + 1 \end{cases} \end{cases} \tag{12.50}$$

Since $\Phi^{\delta}(\boldsymbol{\rho},p)$ is periodic for the variable p (with a period of N_z), $E^{\delta}(\mathbf{r})$ can finally be obtained by using an inverse DFT_z. For points in the reference volume, we have

$$E^{\delta}(\mathbf{r}) = \sum_{m=1}^{N_x+1} \sum_{n=1}^{N_y+1} \sum_{p=1}^{N_z+1} \delta(x - x_m)\, \delta(y - y_n)\, \delta(z - z_p)\, \mathrm{DFT}_z^{-1}\, (V^{D}[m,n,p]) \tag{12.51}$$

where the inverse DFT_z is applied to the variable p.

This completes the description of the procedure for computing the integral convolution of (12.33). This procedure is highly efficient because it is mainly based on the DFTs of (12.44), (12.48), and (12.51), which can be computed by using an FFT algorithm. An additional improvement in efficiency can be obtained by using the shift property of the DFT to calculate the discrete differences that appear in (12.30) and (12.32), as has already been indicated in Chapters 5 and 6 and in [5].

12.5 RESULTS

Four types of results are presented in this section:

- BCS for perfectly conducting and dielectric cylinders;
- Convergence rates for these cylinders;
- Volumetric equivalent currents in a model of a muscle;
- Total fields in two muscle models: one for a single muscle and the other for a layered muscle.

The problems analyzed are particular cases of 3D periodic structures (infinitely long cylinders with arbitrary longitudinal sections). These problems provide an initial test of the validity of the approach using symmetric structures. Results for more general VODIPEBs (those that are nonsymmetric for the z-variable) are not presented because of the lack of results in the literature to compare them with. Due to the translation symmetry of the problems considered, only one sample is needed to represent the current for the z-direction. All the results presented in this chapter have been obtained with only one sample for the z-direction. In all cases, the results have a relative error of less than 10^{-3}.

12.5.1 BCS

Perfectly Conducting Cylinder

Figure 12.2 shows BCS results for a circular cylinder (ka = 0.2) for the incident field indicated in the sketch. The numerical results are compared with analytical values taken from [11].

Dielectric Cylinders

Figures 12.3 and 12.4 show BCS values for two lossless dielectric cylinders with ka = 0.7 and relative dielectric constants of 2.56 and 50, respectively. The incidence is the same as in the previous cases. All the results have been compared with the analytical solution published in [12]. It should be noted that good accuracy is obtained even when the dielectric constant is very high, as in Figure 12.4.

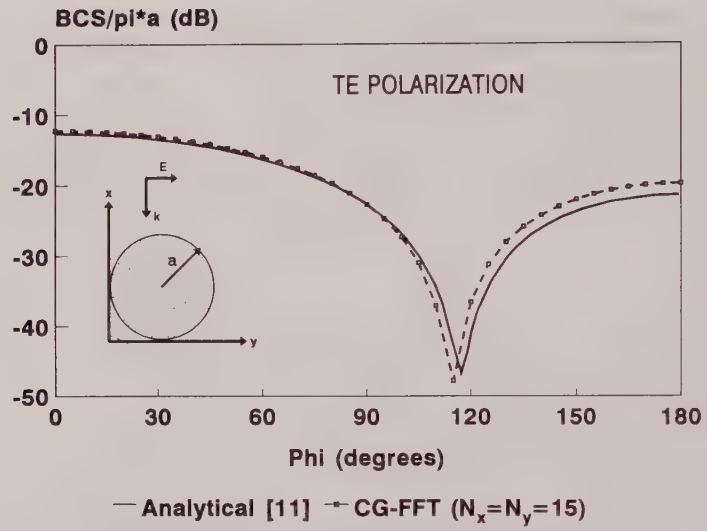


Figure 12.2 BCS of a perfectly conducting cylinder with ka = 0.2.

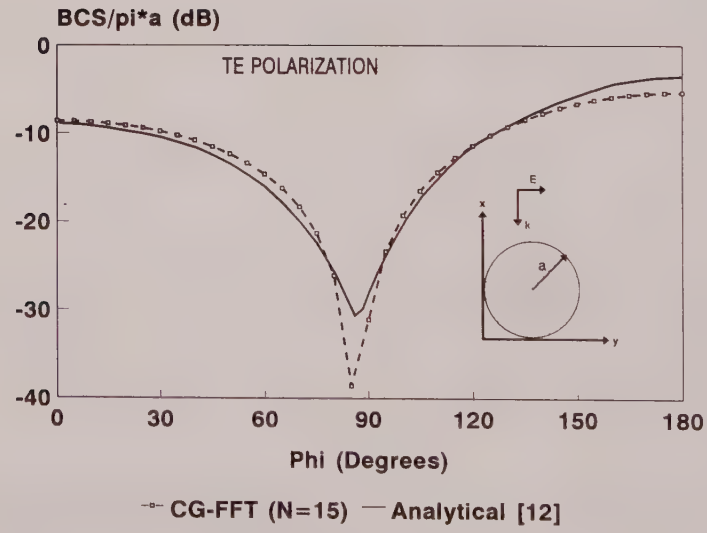


Figure 12.3 BCS of a dielectric cylinder with $\varepsilon_r = 2.56$, $\sigma = 0$, and ka = 0.7.

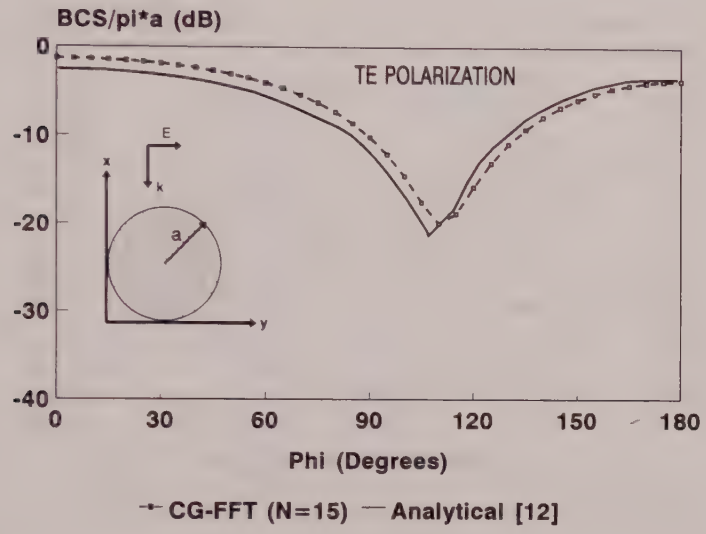


Figure 12.4 BCS of a dielectric cylinder with $\varepsilon_r = 50.0$, $\sigma = 0$, and ka = 0.7.

12.5.2 Convergence Rates

Figures 12.5 and 12.6 present studies of the relative error as a function of the number of iterations and the number of samples for the cylinders of Figures 12.3 and 12.4. The convergence obtained for the cylinder with low dielectric constant ($\varepsilon_r = 2.56$) is noteworthy. The results obtained with different numbers of samples are very similar, and with only 15 iterations, the relative error is less than 10^{-5}. In the second case ($\varepsilon_r = 50$), the convergence rate is more similar to typical convergence curves obtained when using the CG method.

12.5.3 Current Distributions

The currents shown in Figure 12.7 have been obtained for a biological loss model of a muscle ($\sigma = 1.4$ S/m, $\varepsilon_r = 54$). The incident field is indicated in the figure. The current distributions are plotted along the AA′ cut. Several mesh densities have been considered.

12.5.4 Fields Inside Muscle Models

Two different cases have been analyzed.

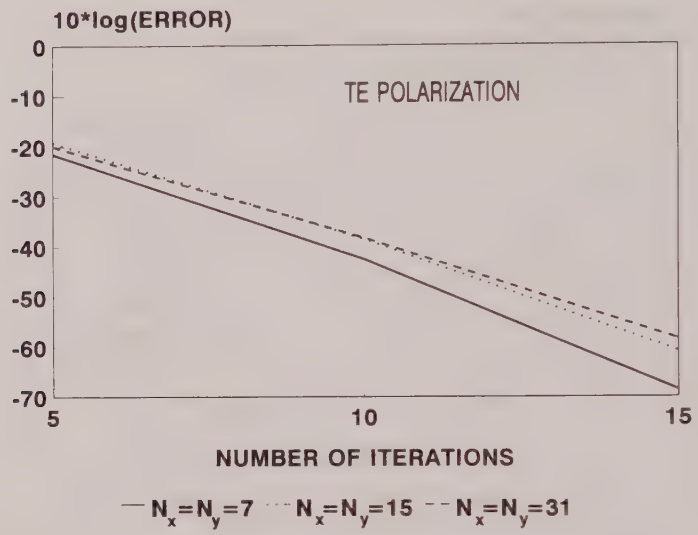


Figure 12.5 Relative error versus number of iterations as a function of the number of samples for the cylinder presented in Figure 12.3.

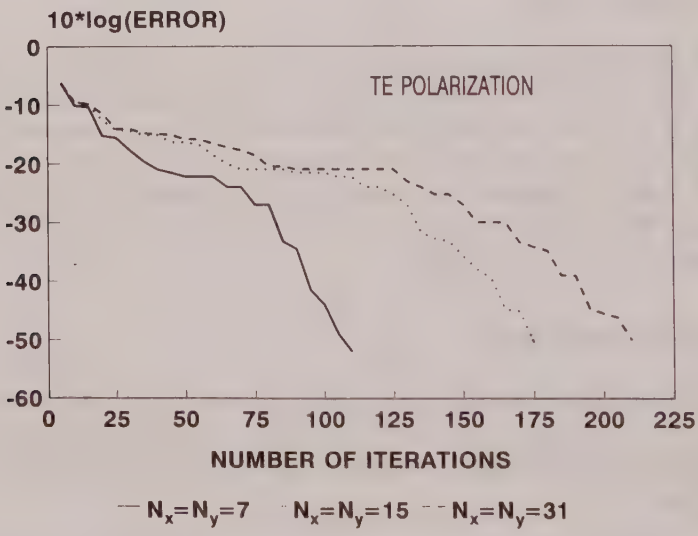


Figure 12.6 Relative error versus number of iterations as a function of the number of samples for the cylinder shown in Figure 12.4.

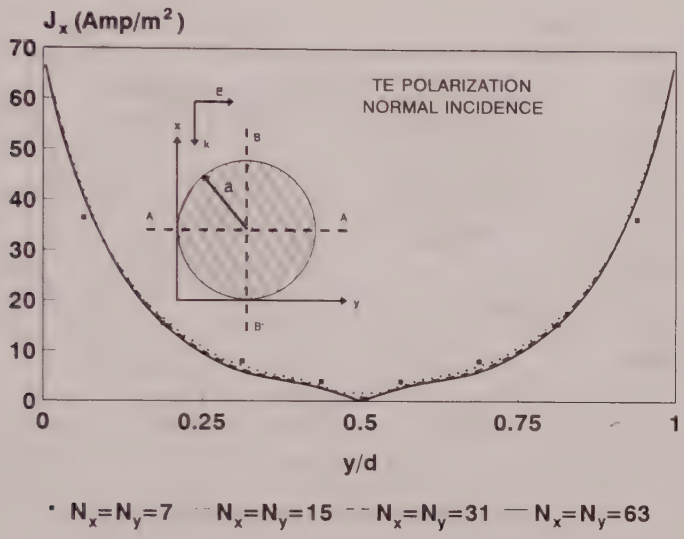


Figure 12.7 The x-component of the current density along the AA′ cut for a biological model of a muscle (infinitely long cylinder with $\varepsilon_r = 54$, $\sigma = 1.4$ S/m, $a = 15$ cm, and $f = 300$ MHz).

1. Biological model of a single muscle: Figure 12.8 shows computed values along the AA′ cut for the total electric field inside the structure indicated in Figure 12.7. The number of samples employed to represent the current is $N_x = N_y = 31$. The computed results are compared with analytical values obtained in [13,14].
2. Biological model of a layered muscle: This case can be considered an example of bodies consisting of composite materials. Figures 12.9 and 12.10 show the results for the y- and x-components, respectively, of the field along the AA′ cut, using 15 samples in each one of these directions. These results are compared with analytical ones from [14]. The data for this structure are $a_1 = 7.9$ cm, $\varepsilon_1 = 72$, $\sigma_1 = 0.9$ S/m, $a_2 = 15$ cm, $\varepsilon_2 = 7.5$, $\sigma_2 = 0.048$ S/m, and $f = 100$ MHz. It can be observed that the numerical results are quite accurate, even at the interface between the two media.

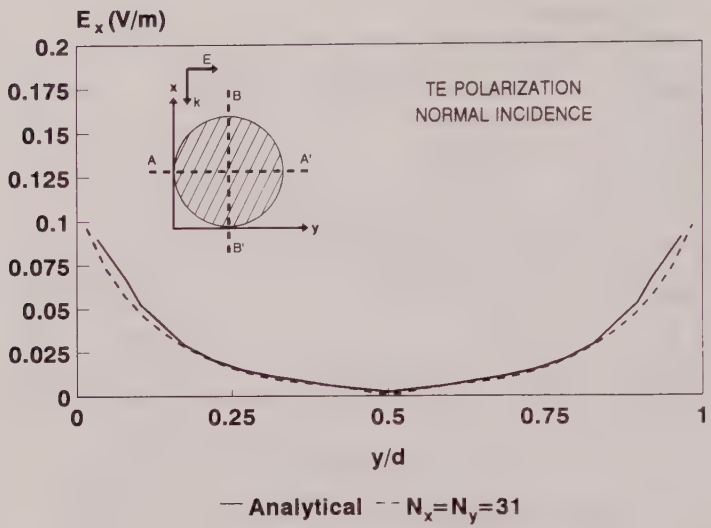


Figure 12.8 The x-component of the field along the AA′ cut inside the cylinder of Figure 12.7.

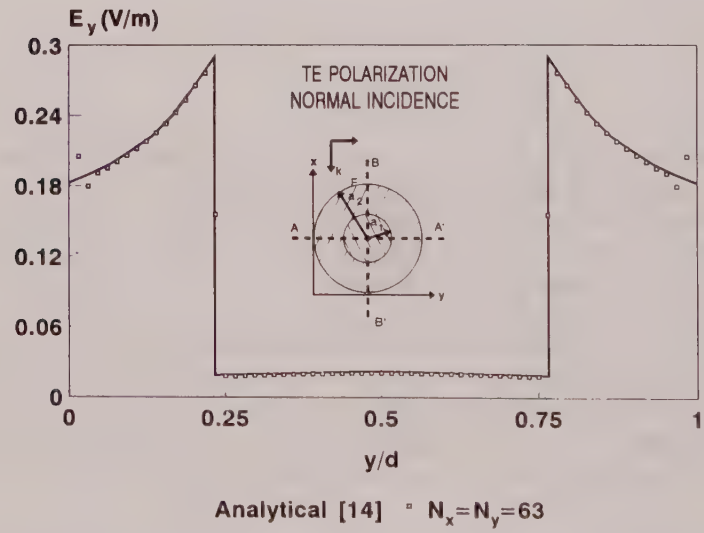


Figure 12.9 The y-component of the field inside a biological model of a layered muscle along the AA′ cut (cylinder with $a_1 = 7.9$ cm, $\varepsilon_r = 72$, $\sigma = 0.9$ S/m, $a_2 = 15$ cm, $\varepsilon_r = 7.5$, $\sigma = 0.048$ S/m, and $f = 100$ MHz).

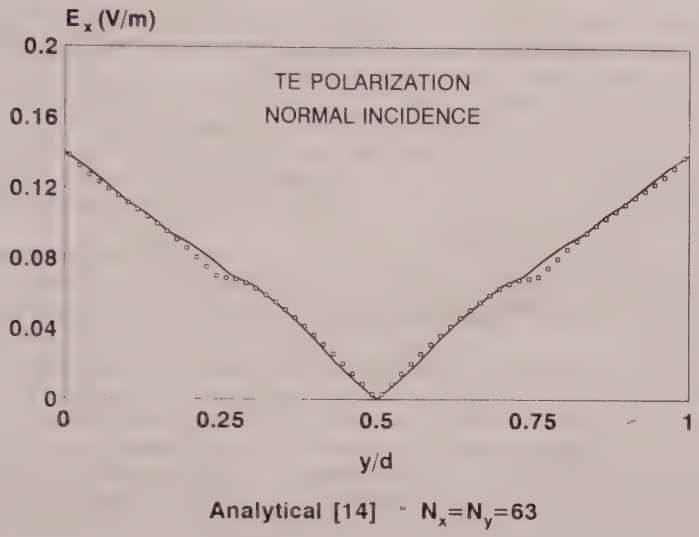


Figure 12.10 The x-component of the field along the AA′ cut for the structure in Figure 12.9.

12.6 CONCLUSIONS

This chapter has described a formulation in which the Green's function is discretized in both the spatial and spectral domains. The resulting scheme proves to be efficient in terms of CPU time and memory requirements, and the accuracy of the solutions obtained is good. The CPU time used by the developed programs varies with the material constants and the geometry of the structures being studied, but, in general, it can be assumed that in a μVAX, the computing time will be about 1 minute for each 15 or 20 iterations. Finally, it should be pointed out that the generality of the formulation allows a great variety of structures to be analyzed, including 3D periodic structures, which have not received much attention in the literature.

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About the Authors

Manuel F. Cátedra

Manuel Cátedra received his M.S. and Ph.D. degrees in Telecommunications Engineering from the Polytechnic University of Madrid (UPM) in 1977 and 1982, respectively. From 1976 to 1989 he taught and conducted research in the Radio Communication and Signal Processing Department of the UPM. He became a full professor at the University of Cantabria in 1989.

He has worked in about 25 research projects, solving problems of electromagnetic compatibility in radio and telecommunication equipment, antennas, microwave components, RCS, and mobile communications. He has developed and applied CAD tools for radio equipment systems used in warships, aircraft, helicopters, and satellites, for contractors such as Spanish or European institutions such as CASA, ALCATEL, BAZAN, the Spanish Defense Department, and the French company MATRA.

He has directed about 10 Ph.D. dissertations, has published about 20 papers (published by *IEEE*, *Electronics Letters*, and others) and 10 chapters in different books, and has made around a hundred presentations at international symposia.

Rafael P. Torres

Rafael Torres was born in Málaga, Spain, in 1961. He received his M.S. degree in Physics from the University of Granada in 1986 and his Ph.D. in Telecommunications Engineering from the UPM in 1990. From 1986 to 1990 he was with the Radio Communication and Signal Processing Department of the UPM as a research assistant. He became an associate professor in the Electronics Department of the University of Cantabria in 1990.

He has worked in about 10 research projects, applying numerical and high-frequency techniques to antennas, microwave devices, and RCS. His research in-

terests include numerical and high-frequency methods in electromagnetics, radio-propagation, and mobile communications, and microwave, millimeter-wave, and integrated optics components.

José Basterrechea

José Basterrechea was born in Santander, Spain, in 1963. He received his M.S. and Ph.D. degrees in Electronics from the University of Cantabria in 1988 and 1992, respectively. He joined the Electronics Department at the University of Cantabria in 1987, where he was a research assistant while pursuing his Ph.D. In 1993 he became an associate professor at the same university, where he carries out his research and academic activities. He has been involved in several research projects, solving problems of electromagnetic compatibility in telecommunications equipment, microwaves, and integrated optics components. His research interests include numerical methods in electromagnetics, electromagnetic compatibility, and the design of microwave and millimeter-wave passive devices and microstrip antennas.

Emilio Gago

Emilio Gago was born in La Bañeza (León), Spain, in 1964. He received his M.S. and Ph.D. degrees in Electrical Engineering from the UPM in 1988 and 1990, respectively. He joined the Radiation Group at the UPM in 1986. From 1991 to 1993 he worked with the research group led by Professor Leopold B. Felsen at the Polytechnic University and the Weber Research Institute (New York). He is currently working as an associate professor at the University of Valladolid (Spain), where he carries out his research and academic activities. These activities are centered on analytical and numerical methods applied to electromagnetic problems (wave problems in general) extending from low-frequency to very-high-frequency techniques.

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+44 (0) 71-973-8077
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Telex: 951-659
email: bookco@artech.demon.co.uk

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